Chapter 1

Integration on Measure Spaces

A measure space is a triple consisting of a non-empty set X, a σ-algebra (a subset of the power set of X), and a measure defined on the σ-algebra. Measurable functions can be defined whenever a σ-algebra is given on X, and the integration of a non-negative measurable function is possible whenever there is a measure on the σ-algebra. In the first three sections of this chapter we study measure spaces and measurable functions. As passing to infinity for functions is very common in analysis, it is necessary to consider functions taking infinity as their values, so the extended real number system is introduced in Section 2. Sections 4 and 5 are devoted to the integration theory. Starting with the integration of simple functions, we next integrate non-negative measurable functions and end up with arbitrary measurable functions. Properties of the integral and the behavior of sequences of functions under integration are studied too. In Section 6 we investigate the relations among four notions of convergence of functions. They are, namely, pointwise almost everywhere convergence, uniform convergence, L^1-convergence and convergence in measure.

1.1 Measurable Functions

Throughout these notes X always denotes a non-empty set. A *σ-algebra* $\mathcal{M}$ is a subset of the power set $\mathcal{P}_X$ of X satisfying (i) $X \in \mathcal{M}$, (ii) $A' \in \mathcal{M}$ whenever $A \in \mathcal{M}$, and (iii) $\bigcup_{j=1}^{\infty} A_j \in \mathcal{M}$, whenever $A_j \in \mathcal{M}, j \geq 1$. Here A' denotes the complement of A in X. From (i)–(iii) we immediately deduce the following properties

- $\phi \in \mathcal{M}$;
- $B \backslash A \in \mathcal{M}$ for $A, B \in \mathcal{M}$, and

- $\bigcap_{j=1}^{\infty} A_j \in \mathcal{M}$ for $A_j \in \mathcal{M}, \quad j \geq 1$.

Consequently, a σ-algebra is closed under taking complement, and countable union and intersection. For any given X, there is at least one σ-algebra, namely, $\mathcal{P}_X$, the power set itself. Observe that the intersection of any collection of σ-algebras is still a σ-algebra. Thus for every subset $\mathcal{S}$ of $\mathcal{P}_X$, there is a minimal σ-algebra containing S called the *σ-algebra generated by S*. Taking $X = \mathbb{R}^n$, the n-dimensional Euclidean space with the topology induced by the Euclidean norm and $\mathcal{S}$ the collection of all open sets, the σ-algebra generated by $\mathcal{S}$ is called the *Borel σ-algebra* and its elements are called *Borel sets*. We will study Borel sets in Chapter 2.

With a σ-algebra $\mathcal{M}$ on X, one can talk about measurable functions. A real-valued function f is called *measurable* or, more precisely, *measurable with respect to $\mathcal{M}$* or *$\mathcal{M}$-measurable*, if $f^{-1}(G) \in \mathcal{M}$ for every open set G in $\mathbb{R}$. Recall that a set G is open if for every $x \in G$, there exists an open interval I containing x that is contained in G. Therefore, every open set can be written as a union of open intervals. In fact, one can show that the union can be taken to be a countable one. (Prove it.) Let $G = \bigcup_n I_n$ where $I_n, n \geq 1$, are open intervals. Using $f^{-1}(\bigcup_n I_n) = \bigcup_n f^{-1}(I_n)$ and the fact that a σ-algebra is closed under countable unions, we see that f is measurable if and only if $f^{-1}(a, b) \in \mathcal{M}$ for all $a, b, a < b$. In fact, we have

Proposition 1.1. *f is measurable if and only if one of (a)–(d) holds:*

(a) $f^{-1}(a, \infty) \in \mathcal{M}, \quad \forall a \in \mathbb{R}$,

(b) $f^{-1}[a, \infty) \in \mathcal{M}, \quad \forall a \in \mathbb{R}$,

(c) $f^{-1}(-\infty, a) \in \mathcal{M}, \quad \forall a \in \mathbb{R}$,

(d) $f^{-1}(-\infty, a] \in \mathcal{M}, \quad \forall a \in \mathbb{R}$.

Proof. (a) First claim $f^{-1}[a, \infty) \in \mathcal{M}$, for all $a \in \mathbb{R}$. Indeed, we have

$$f^{-1}[a, \infty) = \bigcap_{n=1}^{\infty} f^{-1}(a - 1/n, \infty) \qquad \text{(check!)}$$

As $f^{-1}(a - 1/n, \infty)$ is measurable for all n, $f^{-1}[a, \infty) \in \mathcal{M}$ is measurable. Using

$$f^{-1}(a, b) = f^{-1}(a, \infty) \backslash f^{-1}[b, \infty) \in \mathcal{M},$$

$f^{-1}(a, b)$ is measurable. The proofs of (b)–(d) are similar. □

Next we examine how measurability is preserved feasibly under composition of functions, algebraic operations, and in the process of passing to limits.

Proposition 1.2. *Let $\Phi : \mathbb{R} \to \mathbb{R}$ be continuous. For every measurable f, $\Phi \circ f$ is measurable.*

Recall that Φ is continuous means for each $x \in \mathbb{R}$ and $\{x_n\}$ such that $x_n \to x$, $f(x_n) \to f(x)$. It can be shown that this is equivalent to, $f^{-1}(G)$ is open for every open G.

Proof. For every open set G, $\Phi^{-1}(G)$ is open in $\mathbb{R}$. From $(\Phi \circ f)^{-1}(G) = f^{-1}(\Phi^{-1}(G)) \in \mathcal{M}$, so $\Phi \circ f$ is measurable. □

By choosing the continuous functions $\Phi(z) = z^2$ in this proposition, we deduce that f^2 is measurable provided f is measurable. By choosing other suitable Φ's, we conclude that f_+, f_-, and $|f|$ are measurable when f is measurable. We point out that the assumption $\Phi : \mathbb{R} \to \mathbb{R}$ can be relaxed to $\Phi : V \to \mathbb{R}$ where V is an open set in $\mathbb{R}$ and the same conclusion still holds. It is because $\Phi^{-1}(G)$ is now open in V. However, as V is open, it is also open in $\mathbb{R}$.

Proposition 1.3. *(a) All measurable functions form a vector space.*
(b) fg is measurable whenever f and g are measurable.
(c) f/g is measurable provided f and g are measurable and $g(x) \neq 0$ for every x.

Proof. (a) Clearly, a constant multiple of a measurable function is measurable. It suffices to show that $f + g$ is measurable for measurable f and g, but this follows from the formula

$$(f+g)^{-1}(a,\infty) = \bigcup_{\substack{t+s>a \\ t,s\in\mathbb{Q}}} f^{-1}(t,\infty) \cap g^{-1}(s,\infty) \in \mathcal{M}.$$

To verify this formula, let's consider a point x satisfying $(f + g)(x) > a$. We can always choose two rational numbers t and s such that $f(x) > t, g(x) > s$ and $t + s > a$. It follows that $x \in f^{-1}(t,\infty) \bigcup g^{-1}(s,\infty)$, and we have one side inclusion. The other side inclusion is immediate.
(b) From

$$fg = \frac{1}{4}\left[(f+g)^2 - (f-g)^2\right],$$

we conclude that fg is measurable from (a).
(c) It suffices to show that $1/g$ is measurable. This follows from the remark right after the proof of Proposition 1.2. We simply take $\Phi(z) = 1/z$. An alternate, elementary proof runs as follows: For $a, b > 0$, we have $(1/g)^{-1}(a, b) = g^{-1}(1/b, 1/a)$ is measurable. A similar result holds when $a, b < 0$. When $a < 0 < b$, it suffices to observe that

$$\left(\frac{1}{g}\right)^{-1}(a,b) = \left(\frac{1}{g}\right)^{-1}(a,0) \bigcup \left(\frac{1}{g}\right)^{-1}(0,b)$$

and $(1/g)^{-1}(a,0) = \bigcup_n (1/g)^{-1}(a,-1/n)$. □

Proposition 1.4. *Let f_k, $k \geq 1$, be measurable. Then*

$$\sup_k f_k(x), \quad \inf_k f_k(x), \quad \overline{\lim_{k\to\infty}} f_k(x) \text{ and } \underline{\lim_{k\to\infty}} f_k(x)$$

are all measurable provided they are finite everywhere.

Proof. Letting

$$g(x) = \sup_k f_k(x),$$

we have

$$g^{-1}(a,\infty) = \bigcup_{k=1}^{\infty} f_k^{-1}(a,\infty) \in \mathcal{M},$$

so g is measurable. Similarly, $\inf_k f_k$ is measurable. On the other hand, by definition

$$\overline{\lim_{k\to\infty}} f_k(x) = \inf_k \sup_{j\geq k} f_j(x),$$

and

$$\underline{\lim_{k\to\infty}} f_k(x) = \sup_k \inf_{j\geq k} f_j(x),$$

so they are also measurable.

□

1.2 Extened Real Numbers

In real analysis we need to pass to limit all the time. It will be convenient to take infinity as the value of functions. For this reason, we need to extend the real number system to accommodate the infinity. The *extended real number system* $\overline{\mathbb{R}}$ is obtained from $\mathbb{R}$ by adding ∞ and $-\infty$, that is, $\overline{\mathbb{R}} = \mathbb{R} \cup \{\infty, -\infty\}$. It carries a metric structure given by, for every $x, y \in \mathbb{R}$,

$$\begin{aligned}
&d(x,y) = |\tan^{-1} x - \tan^{-1} y|,\\
&d(\infty,x) = d(x,\infty) = \left|\frac{\pi}{2} - \tan^{-1} x\right|,\\
&d(-\infty,x) = d(x,-\infty) = \left|\tan^{-1} x + \frac{\pi}{2}\right|,\\
&d(\infty,-\infty) = d(-\infty,\infty) = \pi.
\end{aligned}$$

The induced topology of this metric structure on $\mathbb{R}$ coincides with the usual topology. Next, we extend the algebraic structure a little bit by setting

$$\begin{aligned}
&x + \infty = \infty + x = \infty, \\
&x - \infty = -\infty + x = -\infty, \\
&x \cdot \infty = \infty, \quad \forall x \in (0, \infty], \\
&x \cdot \infty = -\infty, \quad \forall x \in [-\infty, 0), \\
&0 \cdot \infty = 0, \quad 0 \cdot (-\infty) = 0, \\
&x \cdot (-\infty) = -\infty, \quad \forall x \in (0, \infty], \\
&x \cdot (-\infty) = \infty, \quad \forall x \in [-\infty, 0).
\end{aligned}$$

Note that $\infty - \infty$ is undefined. By a routine argument, one can show that the associative, distributive and commutative laws are valid as long as both sides of the formulas are well-defined. Moreover, $(x, y) \to x + y$ is continuous from $\overline{\mathbb{R}}/\{(\infty, -\infty), (-\infty, \infty)\}$ to $\overline{\mathbb{R}}$, and $(x, y) \to xy$ is continuous from $\overline{\mathbb{R}} \setminus \{(\pm\infty, 0), (0, \pm\infty)$ to $\overline{\mathbb{R}}$.

We also extend the order structure of $\mathbb{R}$ to $\overline{\mathbb{R}}$ by defining

$$-\infty \le x \le \infty, \qquad \forall x \in \mathbb{R}.$$

The extended ordering is still a total ordering.

From now on the notation $\overline{\mathbb{R}}$ stands for the extended real numbers with the topological, algebraic and order structure described above.

We note that in calculus, a sequence $\{x_n\}$ tends to infinity means for each $M > 0$, there exists some n_0 such that

$$x_n > M, \qquad \forall n \ge n_0$$

and the notation $x_n \to \infty$ is used. In $\overline{\mathbb{R}}$, $x_n \to \infty$ allows another interpretation; now ∞ is a point in $\overline{\mathbb{R}}$. It means

$$d(x_n, \infty) \to 0 \quad \text{as } n \to \infty.$$

It is easy to see that both interpretations are consistent.

An extended real-valued function f is *measurable* if $f^{-1}(G) \in \mathcal{M}$ for every open G open $\overline{\mathbb{R}}$. As every open set in $\overline{\mathbb{R}}$ can be written as the countable union of intervals of the form (a, b), $(a, \infty]$, $[-\infty, a)$, $a, b \in \mathbb{R}$, f is measurable if and only if $f^{-1}(a, b)$, $f^{-1}(a, \infty]$, $f^{-1}[-\infty, a)$ are in $\mathcal{M}$. In fact, it suffices to check $f^{-1}(a, b)$, $f^{-1}(\infty)$, $f^{-1}(-\infty) \in \mathcal{M}$. Corresponding to Propositions 1.2 and 1.4, we have

Proposition 1.2'. Let $\Phi : \overline{\mathbb{R}} \to \overline{\mathbb{R}}$ be continuous. For every measurable, extended real-valued f, $\Phi \circ f$ is measurable.

Proposition 1.4'. Let $f_k, k \geq 1$, measurable, extended real-valued functions. Then

$$\sup_k f_k(x), \quad \inf_k f_k(x), \quad \overline{\lim_{k\to\infty}} f_k(x), \text{ and } \underline{\lim_{k\to\infty}} f_k(x)$$

are measurable, extended real-valued functions.

As $\infty - \infty$ does not make sense, extended real-valued functions do not form a vector space.

1.3 Measure Space

Let $\mathcal{M}$ be a σ-algebra on X. A *measure* μ is a function from $\mathcal{M}$ to $[0, \infty]$ satisfying

(i) $\mu(\phi) = 0$ and

(ii) (countable additivity)

$$\mu\Big(\bigcup_{k=1}^{\infty} A_k\Big) = \sum_{k=1}^{\infty} \mu(A_k),$$

if $A_k \in \mathcal{M}$, $k \geq 1$, are mutually disjoint.

From (i) and (ii) one deduces, for every $A, B \in \mathcal{M}$,

- $\mu(B) = \mu(A) + \mu(B \backslash A)$.
- $\mu(A) \leq \mu(B)$, $\forall A \subset B$; and
- (countable subadditivity)

$$\mu(\bigcup_{j=1}^{\infty} A_j) \leq \sum_{j=1}^{\infty} \mu(A_j), \quad A_j \in \mathcal{M}, \ j \geq 1.$$

We recall the proof. Let $B_1 = A_1, B_2 = A_2 \setminus A_1, B_3 = A_3 \setminus (A_1 \bigcup A_2), \cdots$. Then B_j's are disjoint, $B_j \subset A_j$, and $\bigcup_j B_j = \bigcup_j A_j$. By countable additivity,

$$\mu(\bigcup_j A_j) = \mu(\bigcup_j B_j) = \sum_j \mu(B_j) \leq \sum_j \mu(A_j).$$

The triple $(X, \mathcal{M}, \mu)$ is called a *measure space*.

Example 1.1. For any subset E in X, define $c(E)$ to be the number of elements in E. Set $c(E)$ to be ∞ if E is an infinite set. Then $(X, \mathcal{P}_X, c)$ is called the *counting measure* on X.

Example 1.2. Let B_N be the collection of all sequences of the form $a_1a_2\cdots a_N$ where $a_j \in \{0,1\}, j = 1,\cdots,N$. This is a finite set consisting of 2^N elements. For each singleton $\{x\}$ in the power set of B_N we assign $\mu(\{x\}) = 1/2^N$. The measure space $(B_N, \mathcal{P}_{B_N}, \mu)$ is called the *N-Bernoulli space.* In general, a measure space $(X, \mathcal{M}, \mu)$ satisfying $\mu(X) = 1$ a *probability measure.* Each element in B_N represents the outcome of N many times of tossing a coin where 0 and 1 indicate a head and a tail respectively.

In general, a measure space $(X, \mathcal{M}, \mu)$ satisfying $\mu(X) = 1$ is called a *probability measure.* The Bernoulli spaces are probability spaces. A measure space is the setting for probability theory.

Example 1.3. Fix a point a in X and define $\delta_a(E) = 1$ or 0 according to whether $a \in E$ or not. The triple $(X, \mathcal{P}_X, \delta_a)$ is called the *Dirac measure* at a.

Example 1.4. We have learned that the Lebesgue measure $\mathcal{L}^1$ is a measure on $\mathcal{M}$, the σ-algebra containing all Lebesgue measurable sets in $\mathbb{R}$, which satisfies $\mathcal{L}^1\big((a,b)\big) = b - a$ for $a \le b$. It is known that $\mathcal{M}$ is a proper subset of $\mathcal{P}_{\mathbb{R}}$ which strictly contains all Borel sets. Remember the existence of non-measurable sets !

We have the following basic result.

Proposition 1.5. *Let $(X, \mathcal{M}, \mu)$ be a measure space.*

(a) Let A_k, $k \ge 1$, be an ascending sequence of measurable sets. We have

$$\mu\Big(\bigcup_{k=1}^{\infty} A_k\Big) = \lim_{n\to\infty} \mu(A_n).$$

(b) Let A_k, $k \ge 1$, be a descending sequence of measurable sets. We have

$$\mu\Big(\bigcap_{k=1}^{\infty} A_k\Big) = \lim_{n\to\infty} \mu(A_n),$$

provided $\mu(A_1) < \infty$.

We call a sequence of sets A_k *ascending* (resp. *descending*) if $A_k \subset A_{k+1}$ (resp. $A_{k+1} \subset A_k$) for all k.

Proof. (a) Let $B_1 = A_1$, $B_2 = A_2 \backslash A_1$, $B_3 = A_3 \backslash A_1 \cup A_2$, $\cdots$ etc. Then $\{B_k\}$ is mutually disjoint and $\bigcup_{k=1}^N B_k = \bigcup_{k=1}^N A_k$, $N \le \infty$. By countable additivity,

$$\begin{aligned}
\mu\Big(\bigcup_{k=1}^{\infty} A_k\Big) &= \mu\Big(\bigcup_{k=1}^{\infty} B_k\Big) = \sum_{k=1}^{\infty} \mu(B_k) \\
&= \lim_{n\to\infty} \sum_{k=1}^{n} \mu(B_k) = \lim_{n\to\infty} \mu\Big(\bigcup_{k=1}^{n} B_k\Big) \\
&= \lim_{n\to\infty} \mu(A_n).
\end{aligned}$$

(b) Let $C_k = A_1 \setminus A_k$, $k \geq 2$. Then $\{C_k\}$ is ascending and by (a),

$$\begin{aligned}\mu\Big(\bigcup_{k=1}^{\infty} C_k\Big) &= \lim_{n\to\infty} \mu(C_n) \\ &= \lim_{n\to\infty} \mu(A_1 \setminus A_k) \\ &= \lim_{n\to\infty} \big(\mu(A_1) - \mu(A_k)\big) \\ &= \mu(A_1) - \lim_{n\to\infty} \mu(A_k).\end{aligned}$$

As $\bigcup_{k=1}^{\infty} C_k = \bigcup_{k=1}^{\infty}(A_1 \bigcap A_k') = A_1 \bigcap \bigcup_{k=1}^{\infty} A_k' = A_1 \setminus \bigcap_{k=1}^{\infty} A_k$, $\mu\big(\bigcup_{k=1}^{\infty} C_k\big) = \mu\big(A_1 \setminus \bigcap_{k=1}^{\infty} A_k\big) = \mu(A_1) - \mu\big(\bigcap_{k=1}^{\infty} A_k\big)$ and the result follows as $\mu(A_1) < \infty$. □

We point out that the assumption $\mu(A_1) < \infty$ in Proposition 1.5 (b) is needed. To see this let $X = \mathbb{R}$ and consider the descending intervals $A_k = [k, \infty), k \geq 1$. Then $\bigcap_{k=1}^{\infty} A_k = \phi$ and $\mathcal{L}^1\big(\bigcap_{k=1}^{\infty} A_k\big) = 0$. However, on the other hand, $\mathcal{L}^1(A_k) = \infty$ for all k.

1.4 Simple Functions

Let $(X, \mathcal{M}, \mu)$ be a measure space. A *simple function* is a measurable, real-valued function in X whose range is a finite set. It can be expressed in the form

$$s(x) = \sum_{j=1}^{N} \alpha_j \chi_{A_j}(x)$$

where $\alpha_1 < \alpha_2 < \cdots < \alpha_N$ and $A_j = \{x \in X : s(x) = \alpha_j\}, j \geq 1$, are mutually disjoint. It is clear that such representation of a simple function is unique. In general, a function of the form $s(x) = \sum_{j=1}^{N} \gamma_j \chi_{E_j}$ where $\gamma_j \in \mathbb{R}$ and $E_j, j = 1, \cdots, N$, are not necessarily mutually disjoint measurable sets, is a simple function since its range is a finite set. It can always be expressed in the form $\sum_{j=1}^{M} \alpha_j \chi_{F_j}$ where α_j are distinct and in ascending order and F_j are mutually disjoint and measurable. We may call it the "standard form" of s.

Simple functions are not to be confused with step functions. Recall that a step function is a function defined on $[a, b]$ which can be expressed in the form $\sum_k \alpha_k \chi_{I_k}$, where the sum is finite and I_k's are bounded intervals. A step function is always a simple functions (for the Lebesgue measure), but not every simple function is a step function. Moreover, step functions can only be defined on $\mathbb{R}$.

The following result demonstrates the importance of simple functions.

Theorem 1.6. *Let f be a non-negative, measurable extended real-valued function in $(X, \mathcal{M}, \mu)$. There exists a sequence of non-negative simple functions $\{s_k\}$, $k \geq 1$, such that for all x,*

(a) $s_k(x) \le s_{k+1}(x)$, $\forall k \ge 1$; *and*

(b) $s_k(x) \to f(x), \quad as \;\; k \to \infty$.

Proof. We define, for each k, a function φ_k on $[0,\infty]$ as follows. Divide $[0,\infty)$ into subintervals $[j/2^k,(j+1)/2^k), j = 0,1,\ldots,k2^k-1$, and $[k,\infty)$ and define $\varphi_k(t) = j/2^k$ for $t \in [j/2^k,(j+1)/2^k)$ and $\varphi(t) = k$ for $t \ge k$. Then $\varphi_k \le \varphi_{k+1}$ for all k. Next we define $s_k(x) = \varphi_k(f(x))$, $\forall x \in X$. From the definition of φ_k we know that

$$s_k(x) \le f(x) \le s_k(x) + \frac{1}{2^k}, \qquad \text{if } f(x) \in [0,k)$$

and $f(x) \ge s_k(x)$ if $f(x) \ge k$. It follows that $s_k(x) \to f(x)$, $\forall x$. The last thing we need to check is that s_k is measurable. But this follows from the fact that φ_k is a step function where the inverse image of an interval under φ_k is an interval. Check it. □

1.5 Integration Theory

In this section we discuss how to integrate a measurable function over a measurable space $(X,\mathcal{M},\mu)$. We will accomplish this in three steps, first for for simple functions, next for non-negative measurable functions and finally for measurable functions.

In the following we sometimes call a function measurable in X without referring to the σ-algebra and the measure when it is easily identified in the context.

We briefly describe the ideas behind the abstract integration theory in light of Lebesgue and Riemann integrals. In fact, for a non-negative function f defined on $[0,1]$, say, its integral over $[0,1]$ is the area between the x-axis and the graph of f bounded by the two vertical lines $x = a, b$. Consider first the function $f_1(x) = 1$. Clearly its area is equal to 1. Next consider $f_2(x)$ which is equal to 0 when $x = 1/2^k, k \ge 1$, and 1 otherwise. The area of f_2 exists and is equal to 1. To see why the area is equal to 1, for each small $\varepsilon > 0$, consider the rectangles $R_k = [1/2^k - \varepsilon/2^{k+1}, 1/2^k + \varepsilon/2^{k+1}] \times [0,1], k \ge 1$. The total sum of area of these rectangles are bounded by $\sum_k \varepsilon/2^k = \varepsilon$ (Note that some R_k may overlap.) As f_1 is equal to 1 outside $[1/2^k - \varepsilon/2^{k+1}, [1/2^k + \varepsilon/2^{k+1}]$, the area of f_2 is bounded between $1-\varepsilon$ and 1. As ε can be arbitrarily small, the area of f_2 is equal to 1. Now, let $\{x_k\}$ be a sequence in $[0,1]$ and let f_3 be the function which is equal to 0 at x_k and 0 elsewhere. The same argument above shows that the area of f_3 should be equal to 1. As all rational numbers form a countable set, we can take $\{x_k\}$ to be the set of all rational numbers in $[0,1]$. Then f_3 becomes the function $f_4(x)$ which is equal to 0 when x is rational and to 0 when x is irrational. Although we have argued and been convinced that the area of f_4 is equal to 1, it is well-known

that f_4 is not Riemann integral. While our argument is quite reasonable, there must be something wrong with Riemann integral. In fact, the defect of Riemann integral lies on its definition where it is required the Riemann sums converge to the same number as their lengths shrink to 0. Since there are rational and irrational numbers in any interval, the Riemann sums for any partition could be 1 or 0 depending on the tags, and thus do not converge to a definite value. This undesirable situation can be remedied by considering more general sums rather than the Riemann sums. Indeed, it is more reasonable to define the area of f over $[0,1]$ to be

$$\inf\Big\{\sum_j \alpha_j|E_j| :\ s(x)=\sum_j \alpha_j\chi_{E_j}(x)\le f(x), \forall x\in[0,1], E_j\subset[0,1]\Big\},$$

where $|E|$ denotes the "length" of E. Note that in Riemann integral one essentially takes E_j's to be disjoint intervals. In the example above, the area of f_4 is attained by taking $s(x)=1\times\chi_{[0,1]\setminus\mathbb{Q}}+0\times\chi_{\mathbb{Q}\cap[0,1]}$. Thus integration theory is built upon measure theory and measure theory was developed in an endeavor to understand the meaning of length for any set E.

Let s be a non-negative simple function given in standard form $\sum_{j=1}^N \alpha_j\chi_{A_j}$. We define the *integral* of s over the measurable set E to be, in notation,

$$\int_E s\,d\mu=\sum_{j=1}^N \alpha_j\mu(E\cap A_j).$$

Although the values of a simple function are finite, $\mu(E\cap A_j)$ could be infinite. From this definition we see that the value of the integral lies in $[0,\infty]$.

Recall that a function of the form $\sum_{j=1}^N \beta_j\chi_{E_j}$ where E_j's are measurable is a simple function. We first show that its integral can be expressed in terms of β_j and E_j without requiring them to be distinct or mutually disjoint.

Proposition 1.7. *Let $s=\sum_{j=1}^N \beta_j\chi_{E_j}$ be a non-negative simple function. Then*

$$\int_E s\,d\mu=\sum_{j=1}^N \beta_j\mu(E\cap E_j),\qquad \forall E\in\mathcal{M}.$$

Proof. First of all, we observe that this formula holds when β_j's are not necessarily distinct and E_j's are mutually disjoint.

To treat the general case, we can decompose $\{E_j\}$ into a mutually disjoint family of non-empty sets $\{F_i\}$ such that each F_i is contained in some E_j and each E_j is the union of those F_i's lying inside E_j. Indeed, each F_i is of the form $A_1\cap A_2\cdots\cap A_N$ where A_k is either E_k or E_k' (but excluding $E_1'\cap E_2'\cdots\cap E_N'$).

There are at most $2^n - 1$ of them. We have

$$\begin{aligned} s &= \sum \beta_j \chi_{E_j} \\ &= \sum_j \beta_j \sum_{F_i \subset E_j} \chi_{F_i} \\ &= \sum_i \Big(\sum_{j,\ F_i \subset E_j} \beta_j \Big) \chi_{F_i} \\ &= \sum_i \gamma_i \chi_{F_i}, \quad \gamma_i \equiv \sum_{j,\ F_i \subset E_j} \beta_j. \end{aligned}$$

As $\{F_i\}$ are mutually disjoint but γ_i's are not necessarily distinct. By the remark beginning in this proof,

$$\begin{aligned} \int_E s \, d\mu &= \sum \gamma_i \mu(E \cap F_i) \\ &= \sum_i \sum_{j,\ \text{all } F_i \subset E_j} \beta_j \mu(E \cap F_i) \\ &= \sum_j \beta_j \sum_{i,\ F_i \subset E_j} \mu(E \cap F_i) \\ &= \sum_j \beta_j \mu(E \cap E_j), \end{aligned}$$

and the result follows. □

Next, for a measurable function $f : X \to [0, \infty]$ and a measurable set E, we define its *integral over* E to be, in notation,

$$\int_E f \, d\mu = \sup \left\{ \int_E s \, d\mu : s \le f \text{ a.e., } s \text{ is a non-negative simple function} \right\}.$$

Here $s \le f$ a.e. means that s is less than or equal to f almost everywhere, in other words, there exists a measurable set N of measure zero such that $s \le f$ in $X \setminus N$. The definition is consistent with the principle that integration over a set of measure zero is equal to zero even though the integrand reaches infinity somewhere. Alternatively one could define the integral to be

$$\int_E f \, d\mu = \sup \left\{ \int_E s \, d\mu : s \le f, \ s \text{ is a non-negative simple function} \right\}.$$

It is easily seen that both definitions are the same. Besides, when f is a simple function, these definitions coincide with the integral of a simple function defined before.

Here are some elementary properties of the integral.

Proposition 1.8. *Let f and g be measurable functions from X to $[0,\infty]$. We have*

(a) $\displaystyle\int_E f\,d\mu = \int_X f\chi_E\,d\mu, \quad \forall E \in \mathcal{M},$

(b) $\displaystyle\int_X g\,d\mu \geq \int_X f\,d\mu$ *if* $g \geq f$ *a.e.. When* $\displaystyle\int_X g\,d\mu$ *is finite, equality holds if and only if* $g = f$ *a.e.,*

(c) $\displaystyle\int_{E_1} f\,d\mu \leq \int_{E_2} f\,d\mu$ *for measurable* $E_1, E_2,\ E_1 \subset E_2,$

(d) $\displaystyle c\int_X f\,d\mu = \int_X cf\,d\mu, \quad \forall c \in [0,\infty).$

In the following we drop the subscript X when the domain of integration is over the whole space.

Proof. (a) For any $s = \sum_k \alpha_k\chi_k \leq f$ a.e., we have

$$\int_E s d\mu = \sum_k \alpha_k \mu(E \cap E_k) = \int_X s\chi_E d\mu,$$

and the conclusion follows from the definition of the integral.
(b) Obviously, when $f \leq g$ a.e.,

$$\int f\,d\mu \leq \int g\,d\mu.$$

Now, suppose that $g > f$ on some $A \in \mathcal{M}$, $\mu(A) > 0$. Letting $A_n = \{x \in A : g(x) > f(x) + 1/n\}$, we have $A = \bigcup_{n=1} A_n$. By Proposition 1.5(a) there exists some A_{n_0} with positive measure such that $g > f + 1/n_0$ on A_{n_0}. For any simple $s \leq f$ a.e.,

$$g \geq s + \frac{1}{n_0}\chi_{A_{n_0}} \text{ a.e.},$$

so

$$\int g\,d\mu \geq \int s\,d\mu + \frac{1}{n_0}\mu(A_{n_0}),$$

by Proposition 1.7. (b) follows from this inequality after taking supremum over s.
(c) It follows from combining (a) and (b).
(d) Use the definition of the integral.

□

Next we have

Proposition 1.9 (**Markov's Inequality**). *Let f be a non-negative measurable function from X to $[0,\infty]$ with finite integral. Then*

$$\mu\big(\{x \in X : \ f(x) \geq M\}\big) \leq \frac{1}{M}\int_X f d\mu,$$

for all $M > 0$. Consequently

(a) Every integrable function is finite a.e..

(b) $f = 0$ a.e. if f is integrable and $\int f = 0$.

We leave the proof of this proposition as a good exercise.

We will present further properties of the integral in Propositions 1.12 and 1.13. Now a detour.

Given a sequence of measurable functions which converges to some function almost everywhere, we would like to study when their integrals converge to the integral of the limit function. However, examples show that this is not always true. Here are some typical situations.

Example 1.5 (Mass leaking at peaks). Take $\varphi_k = 0$ in $(1/k, 1]$, $\varphi_k(0) = k$ and linear between $(0, 1/k)$. Then $\varphi_k \to 0$ except at $x = 0$, so

$$\int \lim_{k\to\infty} \varphi_k \, d\mathcal{L}^1 = 0 \quad \text{but} \quad \int \varphi_k \, d\mathcal{L}^1 = \frac{1}{2}, \ \ \forall k.$$

Example 1.6 (Mass vanishing at infinity). Take $f_k = \chi_{[k,k+1]}$. Then $\lim\limits_{k\to\infty} f_k(x) = 0$, so

$$\int \lim_{k\to\infty} f_k \, d\mathcal{L}^1 = 0 \quad \text{but} \quad \int f_k \, d\mathcal{L}^1 = 1, \ \forall k.$$

Example 1.7 (Mass dispersing away). Take $g_k = k^{-1}\chi_{[0,k]}$, $k \geq 1$. Then $\lim_{k\to\infty} g_k(x) = 0$ for all x, so

$$\int \lim_{k\to\infty} f_k \, d\mathcal{L}^1 = 0 \quad \text{but} \quad \int g_k \, d\mathcal{L}^1 = 1, \ \forall k.$$

In light of these examples we need to impose further assumptions on the sequence in order to achieve this goal. The following theorem, making use of a monotonicity assumption, is one of the most frequently used results in integration theory.

Theorem 1.10 (**Lebesgue's Monotone Convergence Theorem**). *Let $f, f_k, k \geq 1$, be extended real-valued measurable functions in the measure space $(X, \mathcal{M}, \mu)$. Suppose that there exists a measurable set N of zero measure such that f, f_k are non-negative and satisfy $f_k(x) \uparrow f(x)$ for $x \in X \setminus N$. Then*

$$\lim_{k\to\infty} \int f_k \, d\mu = \int f \, d\mu.$$

Proof. As $f \geq f_k$ a.e., by Proposition 1.8(b),

$$\int f \, d\mu \geq \int f_k \, d\mu, \quad \forall k,$$

which implies

$$\int f \, d\mu \geq \lim_{k\to\infty} \int f_k \, d\mu.$$

The limit on the right hand side exists because $\{f_k\}$ is increasing. On the other hand, fix some $\delta \in (0,1)$. For any $s \leq f$ a.e., there exists a set of measure zero N_1 such that $s \leq f$ in $X \setminus N_1$. Let $Y = X \setminus (N \cup N_1)$ and $E_k = \{x \in Y : f_k(x) \geq \delta s(x)\}$. Clearly, E_k is ascending. We claim that $\bigcup_{k=1}^{\infty} E_k = Y$. For, if $x \in Y$ and $f(x) = 0$, then $s(x) = 0$ and $x \in E_k$ for all k. If $\infty > f(x) > 0$, then $\delta s(x) \leq \delta f(x) < f_k(x)$ for all large k. When $f(x) = \infty, f_k(x) \to \infty$, so $\delta s(x) < f_k(x)$ for all large k too.

Consequently, for these k,

$$\begin{aligned}
\int f_k \, d\mu &\geq \int_{E_k} f_k \, d\mu \\
&\geq \delta \int_{E_k} s \, d\mu \\
&= \delta \sum_{j=1}^{N} \alpha_j \mu(E_k \cap A_j), \qquad \text{if } s = \sum \alpha_j \chi_{A_j}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\lim_{k\to\infty} \int f_k \, d\mu &\geq \delta \sum_{j=1}^{N} \alpha_j \lim_{k\to\infty} \mu(E_k \cap A_j) \\
&= \delta \sum_{j=1}^{N} \alpha_j \mu(Y \cap A_j) \qquad \text{(Proposition 1.5(a))} \\
&= \delta \sum_{j=1}^{N} \alpha_j \mu(A_j) \quad (\because \ \mu(N \cup N_1) = 0) \\
&= \delta \int s \, d\mu \ .
\end{aligned}$$

Taking supremum over s,

$$\lim_{k\to\infty} \int f_k \, d\mu \geq \delta \int f \, d\mu$$

and

$$\lim_{k\to\infty}\int f_k\,d\mu \geq \int f\,d\mu$$

holds after letting $\delta \uparrow 1$.

□

The next result is also a frequently used one.

Theorem 1.11 (**Fatou's Lemma**). *Let $\{f_k\}$ be a sequence of extended real-valued measurable functions which are non-negative a.e.. Then*

$$\int \varliminf_{k\to\infty} f_k\,d\mu \leq \varliminf_{k\to\infty}\int f_k\,d\mu.$$

Proof. Set $g_k(x) = \inf_{j\geq k} f_j(x)$. Then each g_k is measurable and $\{g_k\}$ is an increasing sequence. The function g given by

$$g(x) \equiv \lim_{k\to\infty} g_k(x) = \sup_k g_k(x) = \varliminf_{k\to\infty} f_k(x) \geq 0$$

is measurable. By Lebesgue's monotone convergence theorem,

$$\begin{aligned}\int \varliminf_{k\to\infty} f_k\,d\mu &\equiv \int g\,d\mu \\ &= \lim_{k\to\infty}\int g_k\,d\mu \\ &= \varliminf_{k\to\infty}\int g_k\,d\mu \\ &\leq \varliminf_{k\to\infty}\int f_k\,d\mu\ .\end{aligned}$$

□

Previous examples show that strict inequality could occur in Fatou's lemma. Now we establish the linearity of the integral.

Proposition 1.12. *Let f and g be extended real-valued measurable in X and $\alpha, \beta \geq 0$. Then*

$$\int(\alpha f + \beta g)\,d\mu = \alpha\int f\,d\mu + \beta\int g\,d\mu.$$

Proof. By Theorem 1.6, we can find sequences of simple functions $\{s_j\}$, $\{t_j\}$,

$s_j \uparrow f$, $t_j \uparrow g$ in X. Thus $\alpha s_j + \beta t_j \uparrow f + g$ and

$$\begin{aligned}\int(\alpha f + \beta g)\, d\mu &= \lim_{j\to\infty} \int(\alpha s_j + \beta t_j)\, d\mu \quad \text{(monotone convergence theorem)} \\ &= \lim_{j\to\infty} \left(\int \alpha s_j \, d\mu + \int \beta t_j \, d\mu \right) \quad \text{(Proposition 1.7)} \\ &= \int \alpha f d\mu + \int \beta g d\mu \\ &= \alpha \int f \, d\mu + \beta \int g \, d\mu. \quad \text{(Proposition 1.8)}\end{aligned}$$

□

So far we have been considering non-negative functions. Now we come to functions which may change sign. For a measurable function f, its positive and negative parts f_+ and f_- are non-negative and measurable. The integrals

$$\int f_+ \, d\mu \quad \text{and} \quad \int f_- \, d\mu$$

both make sense. If either one of these integrals is finite, we can define

$$\int f \, d\mu = \int f_+ \, d\mu - \int f_- \, d\mu.$$

An extended real-valued measurable function f in X is called *integrable* if both $\int f_+ \, d\mu$ and $\int f_- \, d\mu$ are finite. Using Proposition 1.12, when a function f is integrable, we have

$$\begin{aligned}\int |f| d\mu &= \int \big(f_+ + f_-\big) d\mu \\ &= \int f_+ d\mu + \int f_- d\mu < \infty,\end{aligned}$$

so $|f|$ is also integrable. Moreover, if one of the sets $A = \{x : f(x) = \infty\}$ or $B = \{x : f(x) = -\infty\}$ is positive, let's say, $\mu(A) > 0$, then

$$\int |f| d\mu = \int \big(f_+ + f_-\big) d\mu \geq \infty \times \mu(A) = \infty,$$

which shows that any integrable function must be finite almost everywhere.

Proposition 1.13. *Let f and g be integrable in X and α, $\beta \in \mathbb{R}$ such that $\alpha f + \beta g$ is well-defined in X. Then $\alpha f + \beta g$ is integrable and*

$$\int(\alpha f + \beta g)\, d\mu = \alpha \int f \, d\mu + \beta \int g \, d\mu.$$

Moreover,

$$\left|\int f\,d\mu\right| \leq \int |f|\;d\mu.$$

Since f and g are extended real-valued and integrable, they are finite almost everywhere. The combination $\alpha f + \beta g$ may not be defined over a null set. When this happens, we may assign ∞ (or any other fixed value) to $\alpha f + \beta g$ at all points in this set. Under this convention $\alpha f + \beta g$ is a measurable function defined everywhere and this proposition still holds.

Proof. Observe that $(f+g)_+$, $(f+g)_- \leq |f+g| \leq |f| + |g|$ so

$$\int (f+g)_\pm\, d\mu \leq \int (|f| + |g|)\, d\mu = \int |f|\;d\mu + \int |g|\;d\mu < \infty.$$

We conclude that $f + g$ is integrable.

Using $f+g = (f+g)_+ - (f+g)_-$ on one hand and $f+g = f_+ - f_- + g_+ - g_-$ on the other hand, we have

$$(f+g)_+ + f_- + g_- = (f+g)_- + f_+ + g_+ \quad \text{on } X.$$

Therefore,

$$\int (f+g)_+\, d\mu + \int f_-\, d\mu + \int g_-\, d\mu = \int (f+g)_-\, d\mu + \int f_+\, d\mu + \int g_+\, d\mu$$

by Proposition 1.12. It follows that

$$\begin{aligned}
\int (f+g)\, d\mu &= \int (f+g)_+ d\mu - \int (f+g)_- d\mu \\
&= \int f_+ d\mu - \int f_- d\mu + \int g_+ d\mu - \int g_- d\mu \\
&= \int f\, d\mu + \int g\, d\mu.
\end{aligned}$$

Next, we need to show

$$\alpha \int f\, d\mu = \int \alpha f\, d\mu, \quad \forall \alpha \in \mathbb{R}.$$

When $\alpha \geq 0$, this is contained Proposition 1.12. To show it for $\alpha < 0$ it suffices to show

$$-\int f\, d\mu = \int (-f)\, d\mu.$$

Indeed, we have

$$\int f\,d\mu + \int (-f)\,d\mu = \int \big[f + (-f)\big]\,d\mu = 0.$$

Finally, we have

$$\begin{aligned}\left|\int f\,d\mu\right| &= \left|\int f_+\,d\mu - \int f_-\,d\mu\right| \\ &\le \int f_+\,d\mu + \int f_-\,d\mu \\ &= \int |f| d\mu\ .\end{aligned}$$

□

The following result ensures when pointwise convergence implies convergence in the corresponding integrals. The required condition is the existence of a dominator for the sequence of functions under consideration. While Lebesgue's monotone convergence theorem and Fatou's lemma apply to non-negative functions, Lebesgue's dominated convergence theorem deals with functions that may change sign. It is more general.

Theorem 1.14 (**Lebesgue's Dominated Convergence Theorem**). *Let f, f_k, $k \ge 1$, be extended real-valued measurable in X satisfying $f_k \to f$ a.e. and $|f_k| \le g$ a.e. for some integrable g. Then f is integrable and*

$$\lim_{k\to\infty} \int |f_k - f|\ d\mu = 0.$$

Proof. By assumption, the function $2g - |f_k - f|$ is non-negative a.e.. We can apply Fatou's lemma to get

$$\begin{aligned}\int 2g\,d\mu &= \int \varliminf_{k\to\infty} (2g - |f_k - f|)\ d\mu \\ &\le \varliminf_{k\to\infty} \int (2g - |f_k - f|)\ d\mu \\ &= \int 2g\,d\mu - \varlimsup_{k\to\infty} \int |f_k - f|\ d\mu.\end{aligned}$$

As $\int g\,d\mu < \infty$, we can cancel it from both sides to get

$$\varlimsup_{k\to\infty} \int |f_k - f|\ d\mu \le 0.$$

□

Note that $\int |f_k - f|\, d\mu \to 0$ implies $\int f_k\, d\mu \to \int f\, d\mu$.

Corollary 1.15 (**Tonelli's Theorem**). *Let $a_k, k \geq 1$, be extended real-valued measurable and*

$$\sum_{k=1}^{\infty} \int |a_k(x)|\, d\mu < \infty.$$

Then $\sum_{k=1}^{\infty} a_k$ is integrable and

$$\int \sum_{k=1}^{\infty} a_k\, d\mu = \sum_{k=1}^{\infty} \int a_k\, d\mu < \infty\ .$$

Proof. Set

$$g_k(x) = \sum_{j=1}^{k} |a_j(x)|\,, \quad g(x) = \sum_{j=1}^{\infty} |a_j(x)|\,,$$

$$h_k(x) = \sum_{j=1}^{k} a_j(x), \quad h(x) = \sum_{j=1}^{\infty} a_j(x).$$

As $g_k \uparrow g$, by the monotone convergence theorem

$$\int g\, d\mu = \int \lim_{k\to\infty} g_k\, d\mu = \sum_{j=1}^{\infty} \int |a_j(x)|\, d\mu < \infty,$$

whence g is integrable. Now the corollary follows from using g as a dominator for h_k and applying the dominated convergence theorem. □

1.6 Convergence of Measurable Functions

For a sequence of measurable functions from the measure space $(X, \mathcal{M}, \mu)$ to $\mathbb{R}$ or $\overline{\mathbb{R}}$, the following four notions of convergence make sense.

- $\{f_k\}$ converges to f almost everywhere, that is, there exists a null set N such that $f_k(x) \to f(x)$ for all $x \in X \setminus N$ as $k \to \infty$.
- $\{f_k\}$ converges to f uniformly on a subset Y, that is, for each $\varepsilon > 0$, there exists some k_0 such that $|f_k(x) - f(x)| < \varepsilon$, for all $k \geq k_0$ and $x \in Y$. (Assuming that $f_k(x)$ and $f(x)$ do not taking ∞ or $-\infty$ simultaneously.)
- $\{f_k\}$ converges in f in L^1-*sense*, that is, $\lim_{k\to\infty} \int |f_k - f|\, d\mu = 0$.

- $\{f_k\}$ converges to f *in measure*, that is, for each $\rho > 0$,

$$\lim_{k\to\infty} \mu\big(\{x \in X : |f_k(x) - f(x)| \geq \rho\}\big) = 0.$$

  It is implicitly assumed that $f_k(x) - f(x)$ is well-defined almost everywhere. Note that the set $\{x \in X : |f_k(x) - f(x)| \geq \rho\}$ is measurable.

In this section we discuss some relations among them.

Theorem 1.16 (**Egorov Theorem**). *Let $f, f_k, k \geq 1$, be extended real-valued measurable functions in X which are finite a.e.. Suppose that $\mu(X)$ is finite and $f_k \to f$ a.e. as $k \to \infty$. Then for each $\varepsilon > 0$, there exists a measurable A, $\mu(A) < \varepsilon$, such that $f_k \to f$ uniformly on $X \setminus A$ as $k \to \infty$.*

Proof. WLOG assume that f_k and f are finite everywhere. Set

$$A_j^i = \bigcup_{k\geq j} \Big\{x : |f_k(x) - f(x)| \geq \frac{1}{2^i}\Big\}.$$

For each fixed i, $\{A_j^i\}$ is a descending family in $\mathcal{M}$. As $\mu(X) < \infty$, by Proposition 1.5 we have

$$\mu\Big(\bigcap_{j=1}^{\infty} A_j^i\Big) = 0.$$

We can find a large $J(i)$ such that

$$\mu\Big(\bigcap_{j=1}^{J(i)} A_j^i\Big) < \frac{\varepsilon}{2^i}.$$

The set $A = \bigcup_{i=1}^{\infty} B_i, B_i \equiv \bigcap_{j=1}^{J(i)} A_j^i$, satisfies

$$\mu(A) \leq \sum_{i=1}^{\infty} \mu\big(B_i\big) < \sum_{i=1}^{\infty} \frac{\varepsilon}{2^i} = \varepsilon.$$

We claim that $f_k \to f$ uniformly on $X \setminus A$. For, if $x \notin A$, x does not belong to $A_{J(i)}^i$ for all i. In other words,

$$|f_k(x) - f(x)| < \frac{1}{2^i}, \quad \forall k \geq J(i).$$

Given $\varepsilon > 0$, we can find a large i such that $2^{-i} < \varepsilon$. Then

$$|f_k(x) - f(x)| < \varepsilon, \quad \forall x \in X \setminus A, \quad \forall k \geq J(i).$$

□

Consider $f_k(x) = x^k,\ x \in [0,1]$. Clearly, $\{f_k\}$ converges pointwisely to $\chi_{\{1\}}$ as $k \to \infty$. Moreover, it converges uniformly to 0 in every interval $[0, 1-\varepsilon]$. However, if we extend f_k as a periodic function of period 1 to $\mathbb{R}$, $\{f_k\}$ converges pointwisely to the periodic extension of $\chi_{\{1\}}$, but the conclusion of Egorov theorem does not hold. It shows that the condition $\mathcal{L}^1(X) < \infty$ cannot be removed.

Proposition 1.17. *If $\{f_k\}$ converges to f in measure, then it admits a subsequence $\{f_{k_j}\}$ converging to f a.e. as $k_j \to \infty$.*

Proof. As $\{f_k\}$ converges to f in measure, we can pick a subsequence $\{f_{k_j}\}$ such that

$$\mu\Big(\Big\{x : \big|f_{k_j}(x) - f(x)\big| \geq \frac{1}{j}\Big\}\Big) < \frac{1}{2^j}.$$

Set

$$B_n = \bigcup_{j \geq n} \Big\{|f_{k_j} - f| \geq \frac{1}{j}\Big\},$$

where $\big\{\big|f_{k_j} - f\big| \geq 1/j\big\}$ is the short form for $\big\{x \in X : \big|f_{k_j}(x) - f(x)\big| \geq 1/j\big\}$. Then $\{B_n\}$ is descending and

$$\begin{aligned}
\mu(B_1) &= \mu\Big(\bigcup_{j=1}^{\infty} \Big\{\big|f_{k_j} - f\big| \geq \frac{1}{j}\Big\}\Big) \\
&\leq \sum_{j=1}^{\infty} \mu\Big(\Big\{\big|f_{k_j} - f\big| \geq \frac{1}{j}\Big\}\Big) \\
&\leq \sum_{j=1}^{\infty} \frac{1}{2^j} = 1.
\end{aligned}$$

Applying Proposition 1.5, we have

$$\begin{aligned}
\mu\Big(\bigcap_{n=1}^{\infty} B_n\Big) &= \lim_{n\to\infty} \mu(B_n) \\
&\leq \lim_{n\to\infty} \sum_{j=1}^{\infty} \frac{1}{2^{n+j}} = 0.
\end{aligned}$$

The set $N \equiv \bigcap_{n=1}^{\infty} B_n$ is of measure zero. We claim that $\{f_{k_j}\}$ converges to f on $X \setminus N$. Indeed, for $x \in X \setminus N$, $x \in \bigcup_{n=1}^{\infty} B_n'$. There exists some n_1 such that $x \in B_{n_1}'$. From the definition of B_{n_1},

$$\big|f_{k_j}(x) - f(x)\big| < \frac{1}{j}, \quad \forall j \geq n_1,$$

so $\{f_{k_j}(x)\}$ converges to $f(x)$ for each $x \in X \setminus N$.

□

Proposition 1.18. *Let $f, f_k, k \geq 1$, be measurable functions from X to $\overline{\mathbb{R}}$ which are finite a.e.. Suppose that $\mu(X)$ is finite and $f_k \to f$ a.e. as $k \to \infty$. Then $\{f_k\}$ converges to f in measure.*

Proof. By Egorov theorem, for $\varepsilon > 0$, there exists some measurable $A, \mu(A) < \varepsilon$, such that $f_k \to f$ uniformly on $X \setminus A$. Therefore, for $\rho > 0$, we can find k_0 such that $|f_k(x) - f(x)| < \rho$ for all $k \geq k_0$ and $x \in X \setminus A$. It follows that $\big\{x : \ |f_k(x) - f(x)| \geq \rho, \ x \in X, \forall k \geq k_0\big\} \subset A$ and so $\mu\big(\big\{x : \ |f_k(x) - f(x)| \geq \rho, \ x \in X, \forall k \geq k_0\big\}\big) < \varepsilon$. □

Proposition 1.19. *If $\{f_k\}$ L^1-converges to f, then $\{f_k\}$ converges to f in measure.*

Proof. For $\rho > 0$, let $E_k = \big\{x \in X : |f_k(x) - f(x)| \geq \rho\big\}$. As in the proof of Markov's inequality,

$$\int |f_k - f| \, du \geq \int_{E_k} |f_k - f| \, d\mu \geq \rho\mu(E_k),$$

which implies

$$\mu(E_k) \leq \frac{1}{\rho} \int |f_k - f| d\mu \to 0, \qquad k \to \infty.$$

□

Combining this proposition with Proposition 1.17, we arrive at the following useful fact.

Proposition 1.20. *If $\{f_k\}$ L^1-converges to f, then it has a subsequence converging almost everywhere to f.*

It is a good exercise to prove this fact directly.

Comments on Chapter 1. With differences only in some details, we largely follow [R] in this chapter. It is amazing that an integration theory can be fully developed based on the simple setting of a measure space. All results in this chapter are of fundamental nature, and will be used again and again in the subsequent chapters. Lebesgue's monotone convergence theorem, Fatou's lemma and Lebesgue's dominated convergence theorem are the corner stones for integration theory. You should understand them well. By the way, in these notes we deduce Fatou's lemma from Lebesgue's monotone convergence theorem. Alternatively one can prove Fatou's lemma first and then use it to deduce Lebesgue's monotone convergence theorem, see, for instance, [HS].

In [EG], a measurable function is called integrable if either $\int f_+$ or $\int f_-$ is finite, and it is called summable if both $\int f_+$ and $\int f_-$ are finite. Thus summable in [EG] means integrable in [R] as well as in these notes. Be careful of this discrepancy.

We do not consider complex-valued functions in these notes. Extending results from real-valued to complex-valued functions are usually straightforward and could be established on site when one really needs it.

Finally, *from now on a measurable function always means an extended real-valued measurable function. A null set is always a measurable set with zero measure.*

Chapter 2

Outer Measures

There are two ways to construct measure spaces, namely, Caratheodory's approach via outer measures and Riesz representation theorem. In Section 1 we discuss outer measures and show there is always a measure space associated to an outer measure. Outer measures require less stringent conditions than measures, so they are easier to construct. Next we study Borel measures. These are measures in which continuous functions are measurable. We will work on a topological space. Basic notions in point set topology and metric spaces are reviewed and, Borel sets, the σ-algebra generated by open sets, are discussed in Section 2. Sections 3 and 4 are devoted to the Riesz representation theorem. The setting is a locally compact Hausdorff topological space where continuous functions are abundant. We prove the Riesz representation theorem by constructing an outer measure for every positive functional. In Section 5 we prove Lusin's theorem, which is about how to approximate measurable functions by continuous functions. Finally, in Section 6 we briefly discuss two classes of special measurable function with respect to a Borel measure. These functions are often used to approximate a general measurable function in various context.

2.1 Outer Measures

Let X be a non-empty set. An *outer measure* on X is a function μ from $\mathcal{P}_X$ to $[0, \infty]$ satisfying

(i) $\mu(\phi) = 0$,

(ii) (countable subadditivity)

$$\mu(A) \leq \sum_{j=1}^{\infty} \mu(A_j) \ ,$$

whenever $A \subset \cup_{j=1}^{\infty} A_j, \ A_j \subset X, j \geq 1$.

Taking $A_1 = B$ and $A_j = \phi, j \geq 2$ in (ii), we have $\mu(A) \leq \mu(B)$ whenever $A \subset B$. A well-known outer measure is the Lebesgue measure on $\mathbb{R}$, which is defined by

$$\mathcal{L}^1(A) = \inf\Big\{ \sum_{j=1}^{\infty} |I_j| : \text{ All } \{I_j\} \text{ satisfying } A \subset \cup_j I_j \Big\},$$

where $I_j = [a_j, b_j], j \geq 1$, are closed intervals and $|I_j| = b_j - a_j$. It is readily verified that $\mathcal{L}^1$ is an outer measure on $\mathbb{R}$. The construction of the Lebesgue is a typical one. It is worthwhile to generalize it in the following way. We call the pair $(\mathcal{G}, \varphi)$ where $\mathcal{G} \subset \mathcal{P}_X$ and $\varphi : \mathcal{G} \to [0, \infty]$ a "gauge" if

(a) $\inf_{G \in \mathcal{G}} \varphi(G) = 0$, and

(b) $\bigcup_j G_j = X$, for some $\{G_j\} \subset \mathcal{G}$.

With a gauge, an outer measure can be defined as follows

$$\mu(A) = \inf\left\{ \sum_{j=1}^{\infty} \varphi(G_j) : \ A \subset \cup_{j=1}^{\infty} G_j, \ G_j \in \mathcal{G} \right\}.$$

To see how (i) is satisfied, observe that the empty set ϕ is contained in any G, so by (a) $\mu(\phi) = 0$. Next, by (b) every set A can be covered by the countable union of some G_j in $\mathcal{G}$, so the set for which the infimum is taken is non-empty and $\mu(A)$ is well-defined. Let $A \subset \bigcup_{j=1}^{\infty} A_j$. Suppose that $\sum_j \mu(A_j) < \infty$ (otherwise there is nothing to prove). For each $\varepsilon > 0$, we can find $G_k^j, k \geq 1$, in $\mathcal{G}$ such that $A_j \subset \bigcup_k G_k^j$ and $\sum_k \varphi(G_k^j) \leq \mu(A_j) + \varepsilon/2^j$. As $\{G_k^j\}$ covers A, we have

$$\begin{aligned} \mu(A) &\leq \sum_{j,k} \varphi(G_k^j) \\ &\leq \sum_j \sum_k \varphi(G_k^j) \\ &\leq \sum_j \Big(\mu(A_j) + \frac{\varepsilon}{2^j} \Big) \\ &\leq \sum_j \mu(A_j) + \varepsilon, \end{aligned}$$

and (ii) holds after letting ε tend to 0.

Given an outer measure μ, call a set E *measurable w.r.t.* μ, or *μ-measurable*, or simply *measurable* when the context is clear if

$$\mu(C) = \mu(C \cap E) + \mu(C \setminus E), \quad \forall C \subset X.$$

By countable subadditivity, $\mu(C) \leq \mu(C \cap E) + \mu(C \setminus E)$ always holds. Thus to establish the measurability of E it is sufficient to show that the one-sided

inequality $\mu(C) \geq \mu(C \cap E) + \mu(C \setminus E)$. Denote the collection of all measurable sets by $\mathcal{M}_C$. We know that it contains at least two elements, namely, ϕ and X. In fact, it forms an σ-algebra, and this is the content of the following theorem.

Theorem 2.1. *$\mathcal{M}_C$ is a σ-algebra and $(X, \mathcal{M}_C, \mu)$ forms a measure space.*

Here we do not distinguish μ and its restriction on $\mathcal{M}_C$. This theorem is due to Caratheodory and sometimes called Caratheodory's construction of measures. The subscript "C" in $\mathcal{M}_C$ refers to his name.

Proof. First of all, from the definition of the measurability of a set we know that the complement of E, E', is measurable whenever E is measurable. Next, we claim that $E_1 \cup E_2 \in \mathcal{M}_C$ for $E_1, E_2 \in \mathcal{M}_C$. Indeed, for $C \subset X$,

$$\begin{aligned}
\mu(C) &= \mu(C \cap E_1) + \mu(C \setminus E_1) \\
&= \mu(C \cap E_1 \cap E_2) + \mu(C \cap E_1 \setminus E_2) + \mu((C \setminus E_1) \cap E_2) + \mu((C \setminus E_1) \setminus E_2) \\
&= \mu(C \cap E_1 \cap E_2) + \mu(C \cap E_1 \cap E_2') + \mu(C \cap E_1' \cap E_2) + \mu(C \setminus (E_1 \cup E_2)) \\
&\geq \mu(C \cap (E_1 \cup E_2)) + \mu(C \setminus E_1 \cup E_2),
\end{aligned}$$

where in the last step subadditivity has been used. Our claim holds.

By induction, for any $n \geq 2$, $\bigcup_{j=1}^{n} E_j \in \mathcal{M}_C,\ E_j \in \mathcal{M}_C$.

Using $E_1 \bigcap E_2 = \left(E_1' \bigcup E_2'\right)'$, and so on we know that $\bigcap_{j=1}^{n} E_j \in \mathcal{M}_C$ for $E_j \in \mathcal{M}_C$.

Now, given $\{E_j\}, j \geq 1$, in $\mathcal{M}_C$, we want to show that $\bigcup_{j=1}^{\infty} E_j \in \mathcal{M}_C$. We assume E_j's are mutually disjoint first. For $C \subset X$,

$$\begin{aligned}
\mu(C \cap A_n) &= \mu(C \cap A_n \cap E_n) + \mu(C \cap A_n \setminus E_n) \\
&= \mu(C \cap E_n) + \mu(C \cap A_{n-1})
\end{aligned}$$

where

$$A_n = \bigcup_{j=1}^{n} E_j \in \mathcal{M}, \text{ and } A = \bigcup_{j=1}^{\infty} E_j.$$

Repeating n many times, we get

$$\mu(C \cap A_n) = \sum_{j=1}^{n} \mu(C \cap E_j), \tag{2.1}$$

Using (2.1),

$$\begin{aligned}
\mu(C) &= \mu(C \cap A_n) + \mu(C \setminus A_n) \\
&\geq \mu(C \cap A_n) + \mu(C \setminus A) \\
&= \sum_{1}^{n} \mu(C \cap E_j) + \mu(C \setminus A).
\end{aligned}$$

Letting $n \to \infty$,

$$\begin{aligned}\mu(C) &\geq \sum_1^\infty \mu(C \cap E_j) + \mu(C \setminus A) \\ &\geq \mu(C \cap A) + \mu(C \setminus A),\end{aligned}$$

whence $A \in \mathcal{M}_C$. Taking $C = A$ in this inequality, we obtain

$$\mu(A) \geq \sum_1^\infty \mu(E_j) \quad \text{whenever } A = \bigcup_{j=1}^\infty E_j, \;\; E_j \in \mathcal{M}_C, \text{ are mutually disjoint.}$$

We have shown that μ is countably additive and hence a measure on $\mathcal{M}_C$.

Finally, when $E_j \in \mathcal{M}_C$ may not be disjoint, we set $F_1 = E_1, F_2 = E_2 \setminus E_1, F_3 = E_3 \setminus \big(E_1 \bigcup E_2\big), \cdots$. Then F_j's are mutually disjoint and belong to $\mathcal{M}_C$. Using $\bigcup_j E_j = \bigcup_j F_j$, we conclude that $\bigcup_j E_j$ belongs to $\mathcal{M}_C$, so $\mathcal{M}_C$ is closed under countable union. The proof of Theorem 2.1 is completed.

□

A measure space $(X, \mathcal{M}, \mu)$ is *complete* if every subset of a null set is measurable (and hence) a null set. In Exercise 1 you are asked to show that every measure space $(X, \mathcal{M}, \mu)$ admits a completion $(X, \overline{\mathcal{M}}, \overline{\mu})$. In fact, $\overline{\mathcal{M}}$ is the σ-algebra generated by $\mathcal{M}$ and subsets N of null sets and $\overline{\mu}(A \cup N) = \overline{\mu}(A \setminus N) = \mu(A)$.

We point out that the measure space associated to an outer measure $(X, \mathcal{M}_C, \mu)$ is always complete. To see this, let B be a subset of null set A. Let us verify

$$\mu(C) \geq \mu(C \cap B) + \mu(C \setminus B), \;\; \forall C \subset X.$$

Indeed, from $\mu(C \cap B) \leq \mu(C \cap A)$ and $C \setminus B \subset (C \setminus A) \cup (A \setminus B)$ we have

$$\begin{aligned}\mu(C \setminus B) &\leq \mu(C \setminus A) + \mu(A \setminus B) \\ &= \mu(C \setminus A),\end{aligned}$$

so

$$\begin{aligned}\mu(C) &\geq \mu(C \cap A) + \mu(C \setminus A) \\ &\geq \mu(C \cap B) + \mu(C \setminus B).\end{aligned}$$

Now, an interesting question arises. Given a complete measure space $(X, \mathcal{M}, \mu)$, it is clear that $(\mathcal{M}, \mu)$ as a gauge and we can use it to define an outer measure by

$$\widetilde{\mu}(E) = \inf\left\{\sum_j \mu(E_j) : \quad E \subset \bigcup_j E_j, \;\; E_j \in \mathcal{M}\right\}.$$

Let $\mathcal{M}_C$ be the σ-algebra of $\widetilde{\mu}$-measurable sets. Is $\mathcal{M}_C = \mathcal{M}$? Not quite, but we have

Theorem 2.2. *Let $(X, \mathcal{M}, \mu)$ be a complete measure space and $\widetilde{\mu}$ and $\mathcal{M}_C$ as described above.*

(a) $\mathcal{M} \subset \mathcal{M}_C$ and $\widetilde{\mu} = \mu$ on $\mathcal{M}$.

(b) $\mathcal{M} = \mathcal{M}_C$ provided $(X, \mathcal{M}, \mu)$ is σ-finite.

A measure space is *σ-finite* if there exist $X_j \in \mathcal{M},\ \mu(X_j) < \infty,\ j \geq 1$, such that $X = \bigcup_{j=1}^{\infty} X_j$. The Lebesgue measure is σ-finite on $\mathbb{R}$ as $\mathbb{R} = \bigcup_j [-j, j]$ and $\mathcal{L}^1([-j, j]) = 2j < \infty$.

Proof. First, $\widetilde{\mu}$ coincides with μ on $\mathcal{M}$. For, let $E \in \mathcal{M}$. Suppose that $E \subset \bigcup_j E_j, E_j \in \mathcal{M}$. Then

$$\mu(E) \leq \sum_j \mu(E_j)$$

by countable subadditivity. Taking supremum over all these $\{E_j\}$, we have

$$\mu(E) \leq \widetilde{\mu}(E).$$

On the other hand, from $E \subset E$ we have $\widetilde{\mu}(E) \leq \mu(E)$.

Next, for $E \in \mathcal{M}$, we wish to show

$$\widetilde{\mu}(C) \geq \widetilde{\mu}(C \cap E) + \widetilde{\mu}(C \setminus E), \quad \forall C \subset X.$$

Clearly we could assume $\widetilde{\mu}(C) < \infty$. First observe from the definition of $\widetilde{\mu}$ that we can find a descending family $\{A_n\}$ in $\mathcal{M}$ such that $C \subset A_n$ for each n and $\widetilde{\mu}(C) = \widetilde{\mu}(A_\infty)$ where $A_\infty = \bigcap_{n=1}^{\infty} A_n \in \mathcal{M}$. We have

$$\begin{aligned}
\widetilde{\mu}(C) &= \widetilde{\mu}(A_\infty) \\
&= \mu(A_\infty) \\
&= \mu(A_\infty \cap E) + \mu(A_\infty \setminus E) \\
&= \widetilde{\mu}(A_\infty \cap E) + \widetilde{\mu}(A_\infty \setminus E) \\
&\geq \widetilde{\mu}(C \cap E) + \widetilde{\mu}(C \setminus E),
\end{aligned}$$

done.

To show (b), assume that $\mu(X) < \infty$ first. For $E \in \mathcal{M}_C$, there exists some $A \in \mathcal{M}$ such that $E \subset A$ and $\widetilde{\mu}(E) = \mu(A)$. Thus $\widetilde{\mu}(A \setminus E) = \mu(A) - \widetilde{\mu}(E) = 0$. Using the definition of $\widetilde{\mu}$ again, we can find some $N \in \mathcal{M}, A \setminus E \subset N$ satisfying $\mu(N) = \widetilde{\mu}(A \setminus E) = 0$. It shows that $A \setminus E$ is a subset of a set of μ-measure zero. By the completeness of μ, $A \setminus E$ is measurable, so is E as it can be expressed as $A \setminus \big(A \setminus E\big)$.

When X is σ-finite, we can find measurable sets X_j with finite measure such that $X = \cup_j X_j$. For $E \in \mathcal{M}_C$, $E \cap X_j$ has finite measure for each j. By the proof above, $E \cap X_j \in \mathcal{M}$, so $E = \cup_j \big(E \cap X_j\big) \in \mathcal{M}$ too. $\square$

2.2 Topological and Metric Spaces

A subset τ of $\mathcal{P}_X$ is called a topology on X if it satisfies the following conditions:

(i) $\phi,\ X \in \tau$,

(ii) $A_\alpha \in \tau \ \Rightarrow\ \cup_\alpha A_\alpha \in \tau$, and

(iii) $A_j \in \tau,\ j = 1, \ldots, N \ \Rightarrow\ \cap_{j=1}^N A_j \in \tau$.

Note that in (ii) the union is taken over any index set, but in (iii) the intersection must be over a finite set. Any element in τ is called an open set and the pair (X, τ) is called a topological space. We recall

- F is called a closed set if its complement F' is open.
- X and ϕ are both open and closed.
- N is a neighborhood of x if there exists an open set G such that $x \in G \subset N$.
- x is an interior point of a set $A \subset X$ if there exists an open set G such that $x \in G \subset A$. Denote all interior points of A by A°. It is the largest open set contained in A. More precisely, A° contains all open subsets of A.
- For $A \subset X$, the closure of A, $\overline{A}$, is the intersection of all closed sets containing A. One can show that $(\overline{A})' = (A')^\circ$. The closure of A is the smallest closed set containing A. In other words, $\overline{A}$ is contained in any closed set containing A. The boundary of A, ∂A, is given by $\overline{A} \cap \overline{A'}$.
- The set A is compact if whenever $A \subset \cup_\alpha G_\alpha$ for a family of open sets $\{G_\alpha\}$, there exist $\alpha_1, \ldots, \alpha_N$ such that $A \subset \cup_{j=1}^N G_{\alpha_j}$.

We now consider metric spaces. A function d from $X \times X \to [0, \infty)$ is called a metric if it satisfies, for $\forall x, y, z \in X$,

(1) $d(x, y) = 0$ if and only if $x = y$,

(2) $d(x, y) = d(y, x)$,

(3) $d(x, y) \le d(x, z) + d(z, y)$.

The pair (X, d) is called a metric space. The metric induces a topology on X as follows. Call a set G open if for each $x \in G$, there exists $B_r(x) \subset G$ where $B_r(x) = \{y : d(y, x) < r\}$ is the metric ball of radius r centered at x $\{y : d(y, x) < r\}$. The collection of all open sets forms a topology on X. For example, the standard topology in $\mathbb{R}^n$ is induced by the Euclidean metric.

Let (X, τ) be a topological space. We use $\mathcal{B}$ to denote the σ-algebra generated by τ. Elements in $\mathcal{B}$ are called *Borel sets*. All open and closed sets are Borel sets. There are more, for instance, a set A is called a G_δ*-set* if $A = \bigcap_{j=1}^{\infty} G_j$, G_j open and it is called an F_σ*-set* if $A = \bigcup_{j=1}^{\infty} F_j$, F_j closed. Both G_δ and F_σ sets are Borel sets. For instance, the closed-open interval

$$\begin{aligned}[a, b) &= \bigcup_{n=1}^{\infty} [a, b - \frac{1}{n}] \\ &= \bigcap_{n=1}^{\infty} (a - \frac{1}{n}, b)\end{aligned}$$

is a F_σ- and a G_δ-set in $\mathbb{R}$ at the same time. However, there are G_δ-sets which are not F_σ-sets, and vice versa in $\mathbb{R}$.

Proposition 2.3. *Let $(X, \mathcal{M}, \mu)$ be a measure space where X is a topological space and $\mathcal{B} \subset \mathcal{M}$. Every continuous function $f : X \to \overline{\mathbb{R}}$ is measurable.*

Proof. That f is continuous means $f^{-1}(G)$ is open for all open $G \subset \overline{\mathbb{R}}$. So $f^{-1}(G) \in \mathcal{B} \subset \mathcal{M}$. □

A measure μ on a measure space $(X, \mathcal{M}, \mu)$ is called a *Borel measure* if $\mathcal{M}$ contains all Borel sets. An outer measure μ is called a Borel measure if μ is a Borel measure on $(X, \mathcal{M}_C, \mu)$.

The following Caratheodory's criterion is very useful in verifying Borel measurability when the underlying space is a metric space. It will be used in the next chapter when we discuss Hausdorff measures.

Theorem 2.4 (**Caratheodory's Criterion**). *Let (X, d) be a metric space and μ an outer measure on X satisfying*

$$\mu(A \cup B) = \mu(A) + \mu(B)$$

whenever dist$(A, B) \equiv \inf \{d(x, y) : \ x \in A, \ y \in B\} > 0$. Then μ is a Borel measure.

Note that $\mathcal{M}_C$ may be larger than $\mathcal{B}$. We know that this is the case for the Lebesgue measure on $\mathbb{R}$.

Proof. It suffices to show that all closed sets are measurable. Let A be closed. We would like to show

$$\mu(C) \geq \mu(C \cap A) + \mu(C \setminus A), \quad \forall C \subset X.$$

As usual, we may assume $\mu(C) < \infty$. Let

$$A_n = \left\{ x \in X : \quad d(x, A) \leq \frac{1}{n} \right\}.$$

Then $C \cap A$ and $C \setminus A_n$ have positive distance. We have

$$\begin{aligned} \mu(C) &\geq \mu(C \cap A \cup C \setminus A_n) \\ &= \mu(C \cap A) + \mu(C \setminus A_n). \end{aligned}$$

Let

$$R_k = \left\{ x \in X : \quad \frac{1}{k+1} < d(x, A) \leq \frac{1}{k} \right\}.$$

We claim that $A_n = A \cup \cup_{k=n}^{\infty} R_k$. For, if $x \in A_n$ satisfies $d(x, A) = 0$, then $x \in A$ because A is a closed set. On the other hand, if $d(x, A_n) > 0$, there exists some $k \geq n$ such that $x \in R_k$. It follows that $A = A_n \setminus \cup_{k=n}^{\infty} R_k$ and

$$\begin{aligned} C \setminus A &= (C \setminus A_n) \cup \left(C \cap \cup_{k=n}^{\infty} R_k \right) \\ &= (C \setminus A_n) \cup \left(\cup_{k=n}^{\infty} \left(R_k \cap C \right) \right) \end{aligned}$$

holds and therefore,

$$\begin{aligned} \mu(C \setminus A) &\leq \mu(C \setminus A_n) + \mu\left(\cup_{k=n}^{\infty} \left(R_k \cap C \right) \right) \\ &\leq \mu(C \setminus A_n) + \sum_{k=n}^{\infty} \mu(R_k \cap C). \end{aligned}$$

If we can show that

$$\sum_{k=1}^{\infty} \mu(R_k \cap C) < \infty, \tag{2.2}$$

then $\sum_{k=n}^{\infty} \mu(R_k \cap C) \to 0$ as $n \to \infty$. As a result,

$$\begin{aligned} \mu(C \setminus A) &\leq \lim_{n \to \infty} \mu(C \setminus A_n) + \lim_{n \to \infty} \sum_{k=n}^{\infty} \mu(R_k \cap C) \\ &\leq \lim_{n \to \infty} \mu(C \setminus A_n), \end{aligned}$$

and

$$\mu(C) \geq \mu(C \cap A) + \lim_{n \to \infty} \mu(C \setminus A_n) \geq \mu(C \cap A) + \mu(C \setminus A),$$

the theorem follows.

To prove (2.2), we split the sum into

$$\sum_{k=1}^{\infty} \mu(R_{2k} \cap C) \text{ and } \quad \sum_{k=1}^{\infty} \mu(R_{2k+1} \cap C).$$

We claim that the distance between R_k and R_{k+2} is at least $1/(k+1)(k+2)$. For, let $y \in R_{k+2}$. For $\varepsilon > 0$, there exists some $z \in A$ such that $d(y,z) \leq 1/(k+2)+\varepsilon$. Using $d(x,y) + d(y,z) \geq d(x,z) \geq d(x,A) \geq 1/(k+1)$, we have $d(x,y) + \varepsilon \geq 1/(k+1) - 1/(k+2) = 1/(k+1)(k+2) > 0$, and the claim follows by letting $\varepsilon \to 0$. As the distance between any two of $R_2, R_4, \ldots, R_{2N}$ is positive,

$$\mu(\cup_{k=1}^{N} R_{2k} \cap C) = \sum_{k=1}^{N} \mu(R_{2k} \cap C).$$

Letting $N \to \infty$,

$$\sum_{k=1}^{\infty} \mu(R_{2k} \cap C) = \mu\left(\cup_{k=1}^{\infty} R_{2k} \cap C\right) \leq \mu(C) < \infty.$$

and similarly, one can show that

$$\sum_{k=1}^{\infty} \mu(R_{2k+1} \cap C) < \infty.$$

□

2.3 Locally Compact Hausdorff Spaces

Two types of topological spaces are usually employed as the platform for performing integration. The first is metric spaces and the second is locally compact Hausdorff spaces. Both spaces have a large supply of continuous functions.

A topological space is a Hausdorff space if any two distinct points in it can be separated by disjoint open sets, that is, for $x \neq y$ in X there exist open G_1 and G_2 containing x and y respectively such that $G_1 \cap G_2 = \phi$. A topological space is locally compact if each point has a compact neighborhood N. We refer to p.35-40, [R] for further information on topological spaces. For those who do not want to go in topology, you may simply assume the topological space is a Euclidean space $\mathbb{R}^n$.

The following are two fundamental properties of a locally compact Hausdorff space we will use from time to time.

Proposition 2.5. *Let $K \subset G$ where K is compact and G is open in a locally*

compact Hausdorff space. Then there exists an open set V with $\overline{V}$ compact such that $K \subset V \subset \overline{V} \subset G$.

We will use the notation $V \subset\subset G$ to denote the situation that $\overline{V}$ is compact and contained in G.

Theorem 2.6 (**Urysohn's Lemma**). *Let $K \subset G$ where G is open and K is compact in a locally compact Hausdorff space. Then there exists a continuous function f with compact support in G such that $0 \leq f \leq 1$ in X and $f \equiv 1$ on K.*

Since a single point is a compact set, this proposition implies that for every point x in an open set G, there always exists a continuous function compactly supported in G and satisfies $f(x) = 1$. Proposition 2.5 and Theorem 2.6 are proved in [R]. A simpler proof can be constructed when the space is $\mathbb{R}^n$. In fact, there are several versions of the Urysohn's lemma and you may google for more information.

The *support* of a function f defined to be

$$\mathrm{spt} f \equiv \overline{\{x : f(x) \neq 0\}}.$$

So it is a closed set by definition. Let x and y be two distinct points in X. Taking K to be x and G to be the complement of y, one concludes from the Urysohn's lemma that there is always a continuous function which is equal to 1 at x and 0 at y. This demonstrates that there are many continuous functions in X.

The notation $f < G$ means $f \in C_c(X), 0 \leq f \leq 1$, and $\mathrm{spt} f \subset G$.

The following assertion on the partition of unity is a useful technical tool.

Theorem 2.7. *Let X be a locally compact Hausdorff space and $K \subset \bigcup_{j=1}^{N} G_j$ where K is compact and G_j's are open. There exist $\varphi_j < G_j, j = 1, \cdots, N$, satisfying $\sum_{j=1}^{N} \varphi_j \equiv 1$ on K.*

Proof. (Following [R]) For each $x \in K$, there is an open set W_x containing x such that $\overline{W_x}$ is compact, and is contained in one of these G_j. There is a finite covering of K by $W_{x_1}, \ldots, W_{x_m}$. Let H_j be the union of all those $\overline{W_{x_k}}$ contained in G_j. Then H_j is a compact set in G_j and $K \subset \bigcup_{j=1}^{N} H_j$. By Urysohn's lemma, there exist $f_j \in C(X)$, $0 \leq f_j \leq 1$, $f_j \equiv 1$ on H_j and $\mathrm{spt} f_j \subset G_j$. We set

$$\begin{aligned}
\varphi_1 &= f_1 \\
\varphi_2 &= (1 - f_1) f_2 \\
&\vdots \\
\varphi_N &= (1 - f_1)(1 - f_2) \cdots (1 - f_{N-1}) f_N.
\end{aligned}$$

By a routine induction one can show that

$$\varphi_1 + \varphi_2 + \cdots + \varphi_N = 1 - (1 - f_1)(1 - f_2) \cdots (1 - f_N),$$

so $\sum_1^N \varphi_j \equiv 1$ on K. Clearly, $\varphi_j \in [0, 1]$ and spt $\varphi_j \subset G_j$. □

2.4 Riesz Representation Theorem

We denote the set of all continuous functions with compact support in a topological space X by $C_c(X)$. It carries the structure of a vector space. Recall that a linear functional is a linear map from a vector space to $\mathbb{R}$. A linear functional Λ on $C_c(X)$ is called *positive* if

$$\Lambda f \geq 0, \quad \forall f \geq 0 \text{ in } C_c(X).$$

Given a Borel measure μ which is finite on compact sets, we can associate it with a positive functional Λ on $C_c(X)$ given by

$$f \mapsto \int f d\mu.$$

Indeed, let K be the support of f, we have

$$\begin{aligned} \int |f| \, d\mu &= \int_K |f| d\mu \\ &\leq \sup_x |f(x)| \mu(K) \\ &< \infty, \end{aligned}$$

whence $\int |f| d\mu$ is finite. It shows that every function in $C_c(X)$ is integrable. In particular, Λ is finite on $C_c(X)$. It is not at all clear that the construction can be reversed. This is the content of the Riesz representation theorem. In fact, we construct an outer measure from a positive functional as follows. Let G be an open set in the locally compact Hausdorff space X. By Urysohn's lemma there exists at least one continuous function with values in $[0, 1]$ compactly supported in G. The set $\{ f \in C_c(X) : \ f < G \}$ is non-empty and we define

$$\mu_0(G) = \sup \big\{ \Lambda f : \ f < G \big\},$$

and $\mu_0(\phi) = 0$. For $E \subset X$, define

$$\mu(E) = \inf \big\{ \mu_0(G) : E \subset G, \ G \text{ is open} \big\}.$$

Theorem 2.8 (**Riesz Representation Theorem**). *Let X be a locally compact Hausdorff space and Λ a positive linear functional on $C_c(X)$. Then μ defined*

above is a Borel (outer) measure which is finite on compact sets and

$$\Lambda f = \int f \, d\mu, \quad \forall f \in C_c(X),$$

holds.

Proof. It is clear from the definition of μ_0 that $\mu_0(G_1) \leq \mu_0(G_2$ for open sets $G_1, G_2, G_1 \subset G_2$. It follows from the definition of μ that μ coincides with μ_0 on open sets. We divide the proof of the representation theorem into four steps.

1. μ is an outer measure.
2. All Borel sets are μ-measurable.
3. μ is finite on compact sets.
4. Identifying Λ with the integral w.r.t. μ.

Step 1. Let $E \subset \bigcup_{j=1}^{\infty} E_j, E_j \subset X, j \geq 1$. We claim

$$\mu(E) \leq \sum_{1}^{\infty} \mu(E_j).$$

We can assume $\sum_1^\infty \mu(E_j) < \infty$, for otherwise there is nothing to prove. For $\varepsilon > 0$, there is some open G_j containing E_j such that

$$\mu(E_j) + \frac{\varepsilon}{2^j} \geq \mu(G_j).$$

The set $G = \bigcup_j G_j$ is open and contains E. Let $f < G$ and consider the compact set $K = \text{spt} f$. There are finitely many G_j's such that $K \subset \bigcup_{j=1}^N G_{n_j}$. Let φ_j be a partition of unity subordinate to $\{G_{n_j}\}$. Then $\sum_j \varphi_j = 1$ on K. We have

$$\begin{aligned}
\Lambda f &= \sum_{1}^{N} \Lambda(f\varphi_j) \\
&\leq \sum_{j=1}^{N} \mu(G_{n_j}) \quad (\text{since } f\varphi_j < G_{n_j}) \\
&\leq \sum_{j=1}^{\infty} \mu(G_j) \\
&\leq \sum_{j=1}^{\infty} \mu(E_j) + \varepsilon.
\end{aligned}$$

Taking supremum over all these f's, we have

$$\begin{aligned}\mu(E) &\le \mu(G)\\ &= \sup\{\Lambda f : f < G\}\\ &\le \sum_{j=1}^{\infty}\mu(E_j) + \varepsilon,\end{aligned}$$

and claim follows by letting $\varepsilon \to 0$.
Step 2. It suffices to show for every open E,

$$\mu(C) \ge \mu(C \cap E) + \mu(C \setminus E), \quad \forall C \subset X.$$

From the definition of μ, it suffices to show, for every open E,

$$\mu(G) \ge \mu(G \cap E) + \mu(G \setminus E), \quad \forall \text{ open } G. \tag{2.3}$$

For $\varepsilon > 0$, pick $\varphi < G \cap E$ such that

$$\Lambda\varphi + \varepsilon \ge \mu(G \cap E).$$

Let $K \equiv \text{spt}\varphi \subset G \cap E$ and $\psi < G \setminus K$. Then $\varphi + \psi < G$ and

$$\begin{aligned}\mu(G) &\ge \Lambda(\varphi + \psi) = \Lambda\varphi + \Lambda\psi\\ &\ge \mu(G \cap E) - \varepsilon + \Lambda\psi.\end{aligned}$$

Taking supremum over all ψ,

$$\begin{aligned}\mu(G) &\ge \mu(G \cap E) - \varepsilon + \mu(G \setminus K)\\ &\ge \mu(G \cap E) - \varepsilon + \mu(G \setminus E)\end{aligned}$$

and (2.3) holds after letting $\varepsilon \to 0$.

Step 3. We show that for every compact set K ,

$$\mu(K) = \inf\{\Lambda f : K < f\}.$$

In particular, it implies that μ is finite on compact sets. Here $\{K < f\}$ means $f \in C_c(X), 0 \le f \le 1$ and $f \equiv 1$ on K. Such functions exist by Urysohn's lemma. For $\alpha \in (0,1)$, $G_\alpha = \{x : f(x) > \alpha\}$ is open and $K \subset G_\alpha$. For any $\varphi < G_\alpha$, we have $\varphi \le f/\alpha$. As $\mu(G_\alpha) = \sup \Lambda\varphi$,

$$\mu(K) \le \mu(G_\alpha) \le \frac{1}{\alpha}\Lambda f.$$

Letting $\alpha \uparrow 1$, we obtain $\mu(K) \le \Lambda f$. By taking infimum over all f, $f < K$, we

have

$$\mu(K) \leq \inf \{\Lambda f : K < f\}.$$

On the other hand, for $\varepsilon > 0$, we can find an open set G containing K such that

$$\mu(K) + \varepsilon \geq \mu(G).$$

By Urysohn's lemma, there exists some $\varphi_1,\ K < \varphi_1 < G$. We have

$$\mu(K) + \varepsilon \geq \mu(G) \geq \Lambda\varphi_1,$$

which implies

$$\mu(K) + \varepsilon \geq \inf \{\Lambda\varphi : K < \varphi\},$$

and

$$\mu(K) \geq \inf \{\Lambda\varphi : K < \varphi\},$$

after letting $\varepsilon \downarrow 0$.
Step 4. (Following [R]) It suffices to show

$$\Lambda f \leq \int f \, d\mu, \quad \forall f \in C_c(X).$$

For, if this is true, the reverse inequality follows from replacing f by $-f$ in this inequality. Let $f(X) \in [a, b]$. For $\varepsilon > 0$, pick

$$y_0 < a < y_1 < \cdots < y_n = b,$$

so that $y_j - y_{j-1} < \varepsilon$. Let $E_j = f^{-1}((y_{j-1}, y_j]) \cap K$, $K \equiv \text{spt} f$. Then $\{E_j\}$ are measurable, mutually disjoint and $K = \bigcup_{j=1}^n E_j$. As $\mu(K) < \infty$ by Step 3, $\mu(E_j) < \infty,\ \forall j$. We can find an open set G_j containing E_j such that

$$\mu(E_j) + \frac{\varepsilon}{n} \geq \mu(G_j).$$

We could further require G_j to satisfy

$$y_{j-1} - \varepsilon < f(x) < y_j + \varepsilon, \quad \forall x \in G_j.$$

As $K \subset \bigcup_{j=1}^n G_j$, we fix a partition of unity $\{\varphi_j\}$ on K subordinate to $\{G_j\}$.

Then $f = \sum_j f\varphi_j$ in X and

$$\begin{aligned}
\Lambda f &= \sum \Lambda(f\varphi_j) \\
&\le \sum (y_j + \varepsilon)\Lambda\varphi_j = \sum (|a| + y_j + \varepsilon)\Lambda\varphi_j - |a| \sum \Lambda\varphi_j \\
&\le \sum (|a| + y_j + \varepsilon)\mu(G_j) - |a| \sum \Lambda\varphi_j \\
&\quad (|a| + y_j + \varepsilon \ge 0) \\
&\le \sum (y_{j-1} + |a| + 2\varepsilon)\Big(\mu(E_j) + \frac{\varepsilon}{n}\Big) - |a| \sum \Lambda\varphi_j \\
&= \frac{\varepsilon}{n} \sum (y_{j-1} + |a| + 2\varepsilon) + \sum y_{j-1}\mu(E_j) + \sum (|a| + 2\varepsilon)\mu(E_j) \\
&\quad - |a| \sum \Lambda\varphi_j \\
&\le \varepsilon(|a| + b + 2\varepsilon) + \sum y_{j-1}\mu(E_j) \\
&\quad + 2\varepsilon\mu(K) + |a| \Big(\sum \mu(E_j) - \sum \Lambda\varphi_j \Big) \\
&\le \varepsilon(|a| + b + 2\varepsilon) + \sum y_{j-1}\mu(E_j) + 2\varepsilon\mu(K) \\
&\quad + |a| \Big(\mu(K) - \Lambda\big(\sum \varphi_j \big) \Big) \\
&\le \varepsilon(|a| + b + 2\varepsilon) + \sum y_{j-1}\mu(E_j) + 2\varepsilon\mu(K),
\end{aligned}$$

where $\mu(K) - \Lambda\Big(\sum \varphi_j \Big) \le 0$ has been used in the last step. On the other hand,

$$\int f \, d\mu = \sum_j \int_{E_j} f \, d\mu \ge \sum_j y_{j-1}\mu(E_j).$$

Hence

$$\Lambda f \le \varepsilon(|a| + b + 2\varepsilon) + \int f \, d\mu + 2\varepsilon\mu(K),$$

and the desired result follows after letting $\varepsilon \to 0$.

□

The measure μ may be written as μ_Λ to emphasize its dependence on Λ. We may call it the "Riesz measure" associated to Λ.

The Riesz measure has further regularity properties. Here regularity means how close we can approximate a set by open sets from outside or by compact sets from inside. To state them we need to introduce some terminologies.

Let X be a topological space and μ a Borel measure on it. A set $E \subset X$ is called *outer regular* (w.r.t. μ) if for every $\varepsilon > 0$, there exists some open set $G, E \subset G$, such that $\mu(G) \le \mu(E) + \varepsilon$. It is *inner regular* (w.r.t. μ) if for every $\varepsilon > 0$, there exists a compact $K \subset E$ such that $\mu(E) \le \mu(K) + \varepsilon$.

Proposition 2.9. *Let μ_Λ be the Riesz measure of Λ. Under the setting of Theorem 2.7,*

(a) Every set is outer regular w.r.t. μ_Λ.

(b) Every open set is inner regular w.r.t. μ_Λ.

(c) Every measurable set with finite measure is inner regular w.r.t. μ_Λ.

Proof. (a) It follows immediately from the definition of the Riesz measure.
(b) We have

$$\begin{aligned}\mu(G) &= \sup\{\Lambda f : f < G\} \\ &= \sup\left\{\int f\,d\mu : f < G\right\} \qquad \text{(Theorem 2.7)} \\ &\le \sup\{\mu(K) :\ K \subset G \text{ is compact }\}\end{aligned}$$

since

$$\int f\,d\mu \le \mu(K) \text{ where } K \equiv \mathrm{spt} f \subset G.$$

(c) Let A be measurable, $\mu(A) < \infty$. Given $\varepsilon > 0$, we find an open $G \supset A$ such that $\mu(G) \le \mu(A) + \varepsilon$. By countable additivity, $\mu(G \setminus A) < \varepsilon$. Using (a), there is some open G_1 contained in G such that $G \setminus A \subset G_1$ with $\mu(G_1) < 2\varepsilon$. Then $G \setminus G_1 \subset A$. Noting $A \subset G \setminus G_1 \cup (A \cap G_1)$, we have $\mu(A) \le \mu(G \setminus G_1) + \mu(A \cap G_1) \le \mu(G \setminus G_1) + 2\varepsilon$, so $\mu(G \setminus G_1) + 2\varepsilon \ge \mu(A)$. As G is open, by (b) there exists compact $K \subset G$ such that $\mu(G \setminus K) < \varepsilon$. Then $K_1 \equiv K \setminus G_1$ is compact, $K_1 \subset G \setminus G_1 \subset A$. Writing $G \setminus G_1 = K_1 \cup (G \setminus K) \setminus G_1$, we have

$$\mu(K_1) + 3\varepsilon \ge \mu(A).$$

□

Examples (see Exercise) show that some measurable sets of infinite measure are not inner regular. To remedy this undesirable situation we need to put extra assumption on the space or the measure. Recall that a measure on a topological space X is σ-finite if there exist measurable $\{X_j\}$ such that $X = \bigcup_j X_j$. Without loss of generality one may assume that X_j's are mutually disjoint.

Proposition 2.10. *Let μ be a Riesz measure on a locally compact Hausdorff space X which is σ-finite with respect to μ. Then*

(a) For every measurable set E and given $\varepsilon > 0$, there exist an open set G and a closed F such that $F \subset E \subset G$ and $\mu(G \setminus F) < \varepsilon$.

(b) For each measurable E, there exist a G_δ set A and an F_σ set B such that $A \subset E \subset B$ and $\mu(B \setminus A) = 0$. Consequently, $\mathcal{M}_C$ is the completion of $\mathcal{B}$.

(c) Every measurable set is inner regular.

Proof. (a) Let E be measurable and $E_j = E \cap X_j$ where $X = \bigcup_j X_j$ in a σ-finite decomposition of X. By outer regularity of the Riesz measure, for each $\varepsilon > 0$, there exists an open set G_j containing E_j such that $\mu(G_j \setminus E_j) = \mu(G_j) - \mu(E_j) \leq \varepsilon/2^j$ for all $j \geq 1$. It follows that $\mu(G \setminus E) \leq \sum_j \mu(G_j \setminus E_j) \leq \varepsilon$ where $G = \bigcup_j G_j$ is open after using $G \setminus E = (\bigcup_j G_j) \setminus (\bigcup_k E_k) = \bigcup_j (G_j \setminus \bigcup_k E_k) \subset \bigcup_j (G_j \setminus E_j)$. Next, we apply this result to the complement of E, E', to get an open G_1 such that $E' \subset G_1$ and $\mu(G_1 \setminus E') < \varepsilon$. Then the closed set $F = G_1'$ is contained in E and satisfies $\mu(E \setminus F) = \mu(G_1 \setminus E') < \varepsilon$.

(b) For each j, there exists an open G_j containing E such that $\mu(G_j \setminus E) < 1/j$. Letting $A = \bigcap_j G_j$, A is a G_δ set containing E and satisfies $\mu(A \setminus E) \leq \mu(G_j \setminus E) < 1/j$ for all j. Letting $j \to \infty$, we conclude that $\mu(A \setminus E) = 0$. Applying this result to E' instead of E, we obtain a G_δ set A_1 containing E' with $\mu(A_1 \setminus E') = 0$. Therefore, the F_σ-set $B = A_1'$ is contained in E and $\mu(E \setminus B) = \mu(A_1 \setminus E') = 0$ holds.

(c) In view of Proposition 2.9(c), it suffices to show that

$$\sup \{ \mu(K) : \quad K \subset E \text{ is compact } \} = \infty$$

when E is a measurable set with infinite measure. Assuming X_j's are mutually disjoint in a σ-finite decomposition of X and $E_j = E \cap X_j$ as above, we have $\infty = \mu(E) = \sum_j \mu(E_j)$ and therefore for each $M > 0$, we can find some J such that $\sum_{j=1}^J \mu(E_j) \geq M - 1$. For each j, let K_j be a compact set in E_j such that $\mu(K_j) \geq \mu(E_j) - 1/2^j$. Then the compact set $K = \cup_{j=1}^J K_j$ is contained in E and $\mu(K) = \sum_{j=1}^J \mu(K_j) \geq M$, and the desired result holds.

□

If one prefers to impose assumptions on the space itself rather than the measure, one may use the notion of σ-compactness. Indeed, a topological space is *σ-compact* if it can be written as a countable union of compact subsets. The Euclidean space is an example of a σ-compact space. Apparently, every Riesz measure on such a space is σ-finite.

Finally, we have the following rather simple characterization of Riesz measures in $\mathbb{R}^n$.

Proposition 2.11. *Let λ be a Borel measure on $\mathbb{R}^n$ which is finite on compact sets. There exists a Riesz measure μ such that λ and μ coincide on $\mathcal{B}$.*

I leave the proof as an exercise.

2.5 Lusin's Theorem

Here we study how to approximate measurable functions by continuous functions. This is possible only if the measure has some regularity properties. The following Lusin's theorem is a typical example. You may consult [EG] or google for more.

Theorem 2.12. *Let μ be a Riesz measure on a locally compact Hausdorff space X and let f be a real-valued measurable function vanishing outside some measurable set A, $\mu(A) < \infty$. For every $\varepsilon > 0$, there exists some $g \in C_c(X)$, such that*

$$\mu\big(\{x : \ f(x) \neq g(x)\}\big) < \varepsilon.$$

Moreover, when f is bounded, g can be chosen to satisfy

$$\sup_{x\in X} |g(x)| \leq \sup_{x\in X} |f(x)|.$$

Proof. (Following [R]) First assume that $0 \leq f < 1$ and A is compact. Recalling in the proof of Theorem 1.6, there is a sequence of simple functions $\{s_j\}$ converging to f in a monotone way. Letting $t_j = s_j - s_{j-1}$ and $t_1 = s_1$, we can write

$$f(x) = \sum_{j=1}^{\infty} t_j(x), \quad \forall x \in X.$$

One can verify that $2^j t_j$ is the characteristic function of some T_n in A. Let G be an open set with compact closure containing A. By the regularity properties of the Riesz measure, we can fix compact sets K_j and open sets G_j satisfying $K_j \subset T_j \subset G_j \subset G$ and $\mu(G_j \setminus K_j) \leq \varepsilon/2^j$. By Urysohn's lemma, we can find h_j, $K_j < h_j < G_j$ and so define

$$g(x) = \sum_{j=1}^{\infty} \frac{h_j(x)}{2^j}.$$

By Weierstrass M-test, the series defining g is uniformly convergent, hence g is continuous. It is clear that the support of g is inside G. Moreover, as h_j is equal to $2^j t_j$ in K_j, the set g and f differ has measure less than $\sum_j \mu(G_j \setminus K_j) \leq \sum_j \varepsilon/2^j \leq \varepsilon$.

By an approximation argument, one can remove the compactness of A and the boundedness of f as long as it is finite a.e..

Finally, when f is bounded, we set $\varphi(z)$ to be z for $z \in [-M, M]$ where $M = \sup_X |f|, M$ and $-M$ respectively for $z > M$ and $z < M$. Then the composite function $g_1(x) = \varphi(g(x))$ meets our additional requirement. □

Corollary 2.13. *Setting as above, let f be a measurable function satisfying $|f| \leq 1$. Then there exists a sequence of functions $\{g_j\}$ in $C_c(X), |g_j| \leq 1$, such that $\lim_{j\to\infty} g_j(x) = f(x)$ a.e..*

Proof. Applying Lusin's theorem to get $g_j \in C_c(X)$ which is equal to f except in a measurable set E_j with $\mu(E_j) \leq 2^{-j}$. Then $\sum_j \mu(E_j) \leq 1$. By Borel-Cantelli lemma, the set E consisting of points which belong to infinitely many E_j's is a null set. For each x not in this set, it could only belong to finitely many E_j's. Letting the largest j be j_0, $g_j(x) = f(x)$ for all $j > j_0$. □

2.6 Semicontinuous and Borel Functions

Let X be a topological space and $x_0 \in X$. The function $f : X \to (-\infty, \infty]$ is *lower semicontinuous at* x_0 if for every $\varepsilon > 0$, there is a neighborhood N of x_0 such that

$$f(x) > f(x_0) - \varepsilon, \quad \forall x \in N.$$

When $f(x_0) = \infty$, the definition should be modified to: For every $M > 0$, there exists a neighborhood N of x_0 such that

$$f(x) > M, \quad \forall x \in N.$$

We define *upper semicontinuity at* x_0 for $f : X \to [-\infty, \infty)$ as requiring $-f$ to be lower semicontinuous at x_0. A function is lower semicontinuous on a set E, $E \subset X$, if it is so at every point of E.

Proposition 2.14. *A function $f : X \to (-\infty, \infty]$ is lower semicontinuous everywhere if and only if $\{x \in X : f(x) > a\}$ is open for every $a \in \mathbb{R}$.*

Proof. $\Rightarrow$) Let $x_0 \in \{x \in X : f(x) > a\}$. Then $f(x_0) > a$. For $\varepsilon = (f(x_0) - a)/2$, there exists a neighborhood N of x_0 such that

$$\begin{aligned} f(x) &\geq f(x_0) - \frac{1}{2}(f(x_0) - a) \\ &= \frac{1}{2} f(x_0) + \frac{a}{2} \\ &> a, \quad \forall x \in N, \end{aligned}$$

hence $N \subset \{x \in X : f(x) > a\}$ and $\{x \in X : f(x) > a\}$ is an open set.
$\Leftarrow$) Taking $a = f(x_0) - \varepsilon$, then $\{x : f(x) > f(x_0) - \varepsilon\}$ is an open set and contains x_0. □

Note the following elementary facts:

- If f is lower semicontinuous at x_0, then

$$\underline{\lim}_{x \to x_0} f(x) \geq f(x_0).$$

If f is upper semicontinuous at x_0, then

$$\overline{\lim_{x \to x_0}} f(x) \leq f(x_0).$$

- Every continuous function is lower and upper semicontinuous.
- The characteristic function χ_G is lower semicontinuous when G is open and χ_F is upper semicontinuous when F is closed.
- The supremum $f = \sup_\alpha f_\alpha$ where f_α's are lower semicontinuous is lower semicontinuous. The infimum $f = \inf_\alpha f_\alpha$ where f_α's are upper semicontinuous is upper semicontinuous.

See Exercise 5 for more.

One can use lower and upper semicontinuous functions to approximate measurable functions. The following Vitali-Carathéodory theorem is a sample.

Theorem 2.15. *Let μ be a Borel outer measure in which all measurable sets with finite measure are regular. For each integrable function f and $\varepsilon > 0$, there exist an upper continuous u which is bounded above and a lower semicontinuous v which is bounded from below such that $u \leq f \leq v$ and*

$$\int (v - u) d\mu < \varepsilon.$$

See [R] for a proof.

An extended real-valued function f is a *Borel function* if $f^{-1}(G)$ is a Borel set for every open set G in $\overline{\mathbb{R}}$. Lower and upper semicontinuous functions constitute a special class of Borel measurable functions. For a Borel measure, we have the following inclusion relations

$$\{\text{continuous functions}\} \subset \{\text{lower/upper semicontinuous functions}\}$$

$$\subset \{\text{Borel functions}\} \subset \{\text{measurable functions}\}.$$

We now study how to approximate a measurable function by Borel functions. When the measure is a Riesz measure, a measurable function is in fact equal to a Borel function modulo a null set. To see this, we start with a general observation.

Lemma 2.16. *Let $(X, \mathcal{M}, \mu)$ and $(X, \mathcal{M}_1, \mu_1)$ be two measure spaces where $\mathcal{M}_1 \subset \mathcal{M}$ and μ coincides with μ_1 on $\mathcal{M}_1$. Suppose that for each $A \in \mathcal{M}$, there exists some $A_1 \in \mathcal{M}_1$ such that $A \subset A_1$ and $\mu(A_1 \setminus A) = 0$. Given any extended real-valued, $\mathcal{M}$-measurable function f from X to $\overline{\mathbb{R}}$, there exists an $\mathcal{M}_1$-measurable g such that $g = f$ μ-a.e.*

Proof. By Theorem 1.6 we can find an increasing sequence of simple functions $\{s_k\}$ satisfying $\lim_{k \to \infty} s_k(x) = f(x)$ for every $x \in X$. Each s_k is of the form

$\sum_j \alpha_j^k \chi_{E_j^k}$ where $E_j^k \in \mathcal{M}$ are mutually disjoint for each fixed k. By assumption, we can find $F_j^k \in \mathcal{M}_1$ such that $E_j^k \subset F_j^k$ and $\mu(F_j^k \setminus E_j^k) = 0$. We set $t_k = \sum_j \alpha_j^k \chi_{F_j^k}$ and $g(x) = \underline{\lim}_{k\to\infty} t_k(x)$. As each t_k is $\mathcal{M}_1$-measurable, so is g. Furthermore, each t_k is equal to s_k μ-a.e.. It follows that g is equal to f μ-a.e.. □

Proposition 2.17. *Let μ be a Borel measure on a topological space X in which every measurable set is outer regular. Suppose that μ is σ-finite. Then for every measurable function from X to $\overline{\mathbb{R}}$, there exists a Borel function g which coincides with f μ-a.e..*

Proof. It suffices to prove the proposition assuming that $\mu(X) < \infty$. By outer regularity, for each measurable set E and n, there exists a Borel set B_n satisfying $E \subset B_n$ and $\mu(B_n \setminus E) = \mu(B_n) - \mu(E) < 1/n$. It follows that the Borel set $B = \bigcap_n B_n$ contains E and $\mu(B) = \mu(E)$. Now we can apply the lemma above to draw the desired conclusion.

□

Comments on Chapter 2. In this chapter we discuss outer measures following [EG] and use it to prove the Riesz representation theorem despite many details are from [R]. A full version of Riesz representation theorem which associated a bounded linear functional on $C_c(X)$ to a signed measure will be discussed in Chapter 5.

The modern theory of integration was due to Lebesgue (1902) who defined the Lebesgue integration on the Euclidean space using basically the outer measure approach. Instead of using closed rectangles to cover a set, he used open rectangles, but this is not essential. On the other hand, Riesz (1909) established the unexpected result that every positive linear functional on $[0, 1]$ is a Riemann-Stieltjes integral. Nowadays it is known that each Riemann-Stieltjes integral corresponds to the integral with respect to some Borel measure. Lebesgue's approach was generalized by Caratheodory (1918) in his construction of outer measures. Since then these have been the two main ways to construct measure spaces. In [R] the approach by Riesz representation theorem is preferred and outer measures are not mentioned at all. However, in the past several decades, the advances in analysis (PDE's and dynamical systems) and geometry (fractals and geometric measure theory) have shown the importance of Caratheodory's approach over the approach by the representation theorem. One reason is the restriction of the latter: The measures obtained by the representation are finite on compact sets. However, there are many interesting measures, for instance, fractional Hausdorff measures, are not finite on compact sets and consequently cannot be associated to any positive linear functionals. To make our discussion updated and balanced,

in this chapter we discuss outer measures first and use it to prove Riesz representation theorem. Nevertheless, you should not be left with the impression that I am downplaying the role of Riesz representation theorem in real analysis. In fact, both Caratheodory's construction and Riesz representation theorem will be used in our subsequent development.

Chapter 3

Lebesgue and Hausdorff Measures

In the first two chapters we have studied measures, outer measures and Borel measures in a general manner without much attention to particular cases. In this chapter we examine two principal Borel measures—Lebesgue and Hausdorff measures. In Section 1 we define the Lebesgue measure as an outer measure and subsequently show that it coincides with the measure obtained by applying the Riesz representation theorem to the Riemann integral. Various properties of this measure are discussed. In Section 2 we focus on the Lebesgue measure on the real line. First we review non-measurable sets and the Cantor set. Next we use them to illustrate various delicate properties of the Lebesgue measure, especially its incompatibility with the topology on $\mathbb{R}$. In Section 3 we prove the fundamental Brunn-Minkowski inequality. Hausdorff measures and Hausdorff dimension are introduced in Section 4. A main result in this section is a discussion on the equivalence between the n-dimensional Hausdorff measure and the n-dimensional Lebesgue measure on $\mathbb{R}^n$. In the last section we study how Hausdorff measures change under Lipschitz maps, and as an application, we determine the Hausdorff dimension of the Cantor set.

3.1 The Lebesgue Measure

Let $R = \Pi_j[a_j, b_j]$ be a closed rectangle in $\mathbb{R}^n$ and $|R| = \Pi_j(b_j - a_j)$ be its volume. It is a closed cube when $b_j - a_j$ are all equal for $j = 1, 2 \cdots, n$. In this chapter rectangles are referred to those that are parallel to the coordinate planes. We define the n-dimensional *Lebesgue measure* $\mathcal{L}^n$ on $\mathbb{R}^n$ by

$$\mathcal{L}^n(E) = \inf\Big\{ \sum_{j=1}^{\infty} |R_j| : E \subset \bigcup_{j=1}^{\infty} R_j,\ R_j \text{ closed cubes} \Big\}.$$

In other words, observe $(\mathcal{R}, |\cdot|)$ where $\mathcal{R}$ is the collection of all closed cubes in $\mathbb{R}^n$ and $|R|$ the volume of the cube R forms a gauge on $\mathbb{R}^n$. The Lebesgue measure is the outer measure resulting from this gauge. It is not hard to see that we could also use

$$\inf\Big\{\sum_1^\infty |R_j| : E \subset \bigcup_1^\infty R_j,\ R_j \text{ are open cubes}\Big\}$$

to define $\mathcal{L}^n(E)$.

We note the following basic facts on the Lebesgue measure.

- The Lebesgue measure $\mathcal{L}^n$ is an outer measure whose measurable sets include the Borel σ-algebra.
- $\mathcal{L}^n(R) = |R|$ for any cube R.
- Every set is outer regular, that is, for any $E \subset \mathbb{R}^n$, $\mathcal{L}^n(E) = \inf\{\mathcal{L}^n(G) : E \subset G,\ G \text{ open }\}$. Every Lebesgue measurable set is inner regular, that is, for any measurable E, $\mathcal{L}^n(E) = \inf\{\mathcal{L}^n(K) :\ K \subset E \text{ compact }\}$.
- The Lebesgue σ-algebra is the completion of the Borel σ-algebra.
- The Lebesgue measure is translational invariant, that is, $\mathcal{L}^n(E + x) = \mathcal{L}^n(E)$, for every $E \subset \mathbb{R}^n$.

Most properties have been assigned as problems in Ex 3. Inner regularity is contained in Proposition 2.10 (b) and (c0.

The Riesz representation suggests another way of defining the Lebesgue measure. We could use the Riemann integral to play the role of the positive linear functional in the theorem. The Riemann integral on $[a, b]$ is discussed in elementary analysis, and the extension to $\mathbb{R}^n$ for continuous functions of compact support is immediate.

Let P be a partition of $\mathbb{R}^n$: $\cdots < x^j_{-1} < x^j_0 < x^j_1 < x^j_2 < \cdots$, $j = 1, \ldots, n$, and $\|P\| = \max\big\{x^j_k - x^j_{k-1} : j = 1, \ldots, n,\ k \in \mathbb{Z}\big\}$. For a bounded function f with compact support in $\mathbb{R}^n$, define its Riemann sum of f w.r.t. the tagged partition $\dot{P}$ to be

$$R(f, \dot{P}) = \sum_J^\infty f(z_J)\Delta x_J,$$

where z_J is a tag point in $R_J = [x^1_{k-1}, x^1_k] \times \cdots \times [x^n_{l-1}, x^n_l]$, $J = (k, \cdots, l)$, and

$$\Delta x_J = (x^1_k - x^1_{k-1}) \times \cdots \times (x^n_l - x^n_{l-1}), \quad J \in \mathbb{Z}^n.$$

Note that the summands in the Riemann sum are zero except for finitely many

terms. The Darboux upper and lower sums are given by

$$\overline{R}(f,P) = \sum_J \sup_{R_J} f\,\Delta x_J, \qquad \text{and}$$
$$\underline{R}(f,P) = \sum_J \inf_{R_J} f\,\Delta x_J.$$

As in the one dimensional case, f is called *Riemann integrable* if there exists $L \in \mathbb{R}$ such that for every $\varepsilon > 0$, there is some δ such that

$$\left| R(f,\dot{P}) - L \right| < \varepsilon, \quad \forall P,\ \|P\| < \delta.$$

We will use the notation

$$L = \int f\,dx$$

to denote the Riemann integral of f. The same as in the one dimensional case, it can be shown that f is Riemann integrable if and only if

$$\lim_{n\to\infty} \overline{R}(f,P_n) = \lim_{n\to\infty} \underline{R}(f,P_n)$$

for some P_n, $\|P_n\| \to 0$. We have

- Every f in $C_c(X)$ is Riemann integrable.
- $\displaystyle\int (\alpha f + \beta g)\,dx = \alpha \int f\,dx + \beta \int g\,dx, \qquad \forall f,g \in C_c(\mathbb{R}^n),\ \alpha,\beta \in \mathbb{R}.$
- $\displaystyle\int f\,dx \geq 0$ if $f \geq 0, f \in C_c(\mathbb{R})$.

In view of these,

$$\Lambda_R f = \int f\,dx$$

defines a positive linear functional on $C_c(\mathbb{R}^n)$. By Theorem 2.7, there exists a Borel (outer) measure μ_R such that

$$\int f\,dx = \int f\,d\mu_R, \quad \forall f \in C_c(X).$$

As $\mathbb{R}^n$ can be written as a countable union of compact sets and μ_R is finite on compact sets, $\mathbb{R}^n$ is σ-finite with respect to μ_R. By Proposition 2.8, μ_R enjoys the same regularity properties as the Lebesgue measure. In fact, we have

Proposition 3.1. *The measure μ_R coincides with $\mathcal{L}^n$.*

Proof. According to the properties of the Riesz measure, μ_R is an outer Borel measure where all sets are outer regular. Furthermore, let G be an open set and $f < G, f \in C_c(\mathbb{R}^n)$. For a fixed $x_0 \in \mathbb{R}^n$, the function $g(x) = f(x + x_0$ satisfies $g < G + x_0$. By looking at the Riemann sums and then passing to limit, we see that

$$\int_G f dx = \int_{G+x_0} g dx.$$

Using the characterization

$$\mu_R(G) = \sup\{ \int_G f dx : \ f < G\},$$

and a similar one for $\mu_R(G+x_0)$ we deduce that $\mu_R(G+x_0) = \mu_R(G)$ for all open sets. By outer regularity, it is also true on all sets, so μ_R is translational invariant. By a problem on the characaterization of translational invariant measures in Ex 5, μ_R is equal to a constant multiple of the Lebesgue measure on Borel sets and hence on all sets by outer regularity. To complete the proof it remains to show that the constant is equal to 1. For this purpose, let R_1 be the open unit cube $(0,1)^n$ and $R_\varepsilon = (\varepsilon, 1-\varepsilon)^n$ for small $\varepsilon > 0$. For every $f, \overline{R_\varepsilon} < f < R_1$, we have

$$\begin{aligned} \mu_R(R_1) &\geq \int f d\mu_R \\ &= \int f dx \\ &\geq |\overline{R_\varepsilon}| \quad \text{(by the definition of the Riemann integral)} \\ &= (1-2\varepsilon)^n, \end{aligned}$$

which implies $\mu_R(R_1) \geq 1$ after letting $\varepsilon \downarrow 0$. On the other hand, by considering cubes $(-\varepsilon, 1+\varepsilon)^n$ enclosing R_1, a similar argument shows that $\mu_R(R_1) \leq 1$. It follows that $\mu_R(R_1) = 1$. As $\mathcal{L}^n(R_1) = |R_1| = 1$, we conclude that the constant is equal to 1. □

Every Euclidean motion, or rigid motion, on $\mathbb{R}^n$ is a finite composition of translations, reflections and rotations. The Lebesgue measure has a nice scaling property which implies that it is invariant under all Euclidean motions.

Proposition 3.2. *Let T be a linear transformation from $\mathbb{R}^n$ to itself. For every measurable set E, TE is measurable and*

$$\mathcal{L}^n(TE) = \Delta(T)\mathcal{L}^n(E),$$

where $\Delta(T)$ is a nonnegative constant depending only T and it is equal to 1 when T is a rotation or reflection.

Proof. Assume that T is a nonsingular linear transformation and so it has a

continuous inverse T^{-1}. For $E \subset \mathbb{R}^n$, set

$$\mu(E) = \mathcal{L}^n(TE).$$

It is easy to see that μ is an outer measure. Using

$$|x - y| \leq \|T^{-1}\||Tx - Ty|,$$

we see that $d(TA, TB) > 0$ whenever $A, B \subset \mathbb{R}^n$ with $d(A, B) > 0$, . It follows from Caratheodory's criterion that μ is a Borel measure. Furthermore, it is straightforward to verify that it is outer regular and translational invariant. According to a problem in Ex 5 concerning the characterization of translational invariant measures, μ is a constant multiple of the Lebesgue measure.

When T is singular, it maps $\mathbb{R}^n$ into a subspace of less dimension and hence of measure zero. It follows that TE is a null set for all E. The measure $\mu(E) = \mathcal{L}^n(TE)$ is always equal to 0, so we can take $\Delta(T)$ to be 0. When T is non-singular, ΔT is positive.

To show that $\Delta(T) = 1$ for a rotation or a reflection it suffices to take E to be a ball B centered at the origin, which is unchanged under a rotation or a reflection. Therefore, $\Delta(T) = \mu(B)/\mathcal{L}^n(B) = \mathcal{L}^n(TB)/\mathcal{L}^n(B) = 1$. □

For a general linear transformation T, $\Delta(T)$ is in fact the absolute value of the determinant of the matrix representation of T, see [R].

In the paragraphs above we developed Riemann integral as a tool to define a positive linear functional on $C_c(\mathbb{R}^n)$. In fact, one can mimic the one dimensional situation to define a concept of Riemann integrability for bounded functions with compact support in $\mathbb{R}^n$. We will not give the rather straightforward details, but simply point out that Lebesgue's theorem characterizing Riemann integrability holds in all dimensional, namely, a bounded, Lebesgue measurable function with compact support is Riemann integrable if and only if its set of discontinuity forms a null set in $\mathbb{R}^n$. When this happens, its Riemann integral is equal to its Lebsegue integral. Note that a Riemann integrable function can be approximated by simple functions and hence must be Lebesgue integrable.

3.2 Lebesgue Measure on $\mathbb{R}$

Next we discuss some properties of the Lebesgue measure on the real line. Corresponding results can be established in higher dimensions by using Fubini's theorem. Since Fubini's theorem will not be discussed until Chapter 8, we will focus on the one dimensional case. Many properties have been examined in an undergraduate real analysis. In view of this, we will be sketchy in some of the discussion below.

There are two basic constructions on the real line.

First, non-measurable sets. Introduce a relation $\sim$ on $\mathbb{R}$ by $x \sim y$ if and only if $x - y \in \mathbb{Q}$. "$\sim$" is easily seen to be an equivalence relation. Then $\mathbb{R} = \bigcup_\alpha \mathcal{E}_\alpha$, $\mathcal{E}_\alpha$ its equivalence classes. By the axiom of choice, we pick x_α from $\mathcal{E}_\alpha$ to form $N = \{x_\alpha\}$. Depending on which points you pick, there are many different N.

Proposition 3.3. *Every measurable set of $\mathbb{R}$ with positive measure contains a non-measurable subset.*

Proof. (Following [R]) It is equivalent to proving: If every subset of a measurable set A is measurable, then $\mathcal{L}^1(A) = 0$.

We note that

$$\mathbb{R} = \bigcup_k N + z_k, \qquad z_k \in \mathbb{Q}$$

is a disjoint union. Letting $A_k = A \bigcap_k (N + z_k)$, as $\mathcal{L}^1(A) = \sum_{k=1}^{\infty} \mathcal{L}^1(A_k)$, it suffices to show $\mathcal{L}^1(A_k) = 0$ for each k.

Let $K \subset A_k$ be compact. We will show that $\mathcal{L}^1(K) = 0$, then $\mathcal{L}^1(A_k) = 0$ by the inner regularity of the Lebesgue measure. Let $H = \bigcup_j K + r_j$, $r_j \in \mathbb{Q} \cap [0, 1]$, be a disjoint union. As K is bounded, H is a bounded set. Then

$$\begin{aligned} \infty > \mathcal{L}^1(H) &= \sum_j \mathcal{L}^1(K + r_j) \\ &= \sum_j \mathcal{L}^1(K), \quad \text{(translational invariance)} \end{aligned}$$

which forces $\mathcal{L}^1(K) = 0$. □

The second construction is the Cantor set. It can be described as follows: Let $I_k^n, 1 \le k \le 2^{n-1}, n \ge 1$, be open intervals obtained from the removal of a repeating trisection procedure, for instance, $I_1^1 = (1/3, 2/3), I_1^2 = (1/3^2, 2/3^2), I_2^2 = (7/3^2, 8/3^2)$. Then $\mathcal{C}_n = [0, 1] \setminus \bigcup_{k,m \le n} I_k^m$ is compact and so is the Cantor set $\mathcal{C} = \bigcap_{n=1}^{\infty} \mathcal{C}_n$. It can be shown that

- C is an uncountable set.

In fact, $x \in \mathcal{C}$ if and only if $x = 0.a_1a_2a_3\cdots$, $a_j \in \{0, 2\}$ in its ternary representation. The Cantor set has the following topological properties

- $\mathcal{C}$ is compact in $[0, 1]$.
- $\mathcal{C}$ is a nowhere dense set.

- $\mathcal{C}$ is a perfect set.

Recall that a set in a topological space is nowhere dense if its closure does not contain any nonempty, open set. Since the Cantor set C is closed, it means C does not contain any non-empty, open interval, but this is evident from its construction. A set E is called perfect if for every $x \in E$ and open set V containing x, $V \cap E$ contains infinitely many elements. For the Cantor set this is also clear from its construction.

We also recall the measure-theoretic property:

- $\mathcal{C}$ is a null set.

It is possible to modify the construction of the Cantor set to obtain a Cantor-like set which satisfies all properties of $\mathcal{C}$ listed above except the last one. In other words, these Cantor-like sets are of positive measure, see exercise 5.

We use these two constructions to illustrate various points.

First of all, the existence of non-measurable sets shows that the inclusion of the Lebesgue σ-algebra $\mathcal{M}_C$ in $\mathcal{P}_{\mathbb{R}}$ is proper. How about the inclusion of $\mathcal{B}$ in $\mathcal{M}_C$? We know that for each $E \in \mathcal{M}_C$, there exists $E_1, E_2 \in \mathcal{B}$ such that $E_1 \subset E \subset E_2$ and $\mathcal{L}^1(E) = \mathcal{L}^1(E_1) = \mathcal{L}^1(E_2)$. But this does not answer our question. In fact, the Borel σ-algebra, by definition the smallest σ-algebra containing all open intervals, is not easy to described explicitly. Using G_δ and F_σ to denote respectively the collections of all countable intersection of open sets and countable union of closed sets, one can show that sets such as half-open and half-closed intervals are G_δ and F_σ sets. We may continue this process to construction, in self-evident notation, $G_{\delta\sigma}, F_{\sigma\delta}, G_{\delta\sigma\delta}, F_{\sigma\delta\sigma}, \cdots$, many many other Borel sets. But to get all Borel sets we need to go beyond countably many steps. Nevertheless, by transfinite induction (see section 10, chapter 3 in [HS]), one can show that the cardinal number of the Borel σ-algebra on $\mathbb{R}$ is equal to $\mathfrak{c}$, the same as the cardinal of $\mathbb{R}$. On the other hand, the Cantor set is of measure zero, so all its subsets are also of measure zero. As the Lebesgue measure is complete, all these subsets are measurable. As the cardinal number of $\mathcal{C}$ is $\mathfrak{c}$, all subsets of $\mathcal{C}$ have the cardinal number $2^{\mathfrak{c}}$. So $|\mathcal{M}_C| \geq$ |subsets of $\mathcal{C}| = 2^{\mathfrak{c}} > \mathfrak{c} = |\mathcal{B}|$. So there are many more Lebesgue measurable sets than Borel measurable sets.

Next, the unit interval $[0, 1]$ carries both a topological structure and a measure-theoretic structure, namely the Lebesgue measure restricted to this interval. The measure of size in a measure space is easy to describe. For instance, a set is small if its measure is zero and is large or full if its measure is equal to one. However, there are various ways to describe largeness in a topological space X. For instance, a set $E \subset X$ is dense if every n'd of any point in X contains some point in E. It is nowhere dense if its closure does not contain open set. It is easy to see that a set of full measure must be dense, but a dense set can be null (for example,

the set of all rational numbers in $[0,1]$). Also, one can produce an open dense of in $[0,1]$ whose measure is equal to any prescribed number in $(0,1)$. Moreover, by modifying the construction of the Cantor set slightly to obtain a Cantor-like set which is closed, nowhere dense with measure equal to any number in $(0,1)$.

A more useful topological concept for size is the category. A set E is of the first category if it can be written as a countable union of nowhere dense sets. It is of the second category if it is not of the first category. Baire theorem states that a complete metric space is of the second category. A set of the first category is regarded small in topological setting. However, category and Lebesgue measure are still not compatible. Let us display a set S of full measure in $[0,1]$ which is of first category. Indeed, let C_j be Cantor-like sets of measure $1-1/k$ for each $k \geq 1$. Setting $S = \bigcup_k C_k$, we have $\mathcal{L}^1(S) \geq \mathcal{L}^1(C_j)$ and hence $\mathcal{L}^1(S) = 1$. But, as each C_k is nowhere dense, S is of the first category. On the other, the Cantor set C is closed, so it is a complete metric space under the Euclidean metric. By Baire category theorem, it is of the second category. However, its measure is equal to zero.

The incompatibility between the topological and Lebesgue measure-theoretic properties on the real line can be further reflected from the behavior of mappings between intervals.

First of all, we construct the Cantor function g from the Cantor set. Define, for each $n \geq 1$, $g_n : [0,1] \to [0,1]$ to a continuous, piecewise linear function that satisfies $g_n(0) = 0$, $g_n(x) = (2k-1)/2^m$, $x \in I_k^m$, $m \leq n$, $k = 1, 2, \ldots, 2^{m-1}$, and $g_n(1) = 1$. From the definition of g_n it is not hard to see that $|g_{n+1} - g_n| < 1/2^{n+1}$. By Weierstrass M-test, the series $\{h_n = g_{n+1} - g_n\}$ converges uniformly to some continuous h. It follows that $g = \lim_{n\to\infty} g_n = \lim_{n\to\infty} \sum_{k=1}^{n} h_n + g_1 = h + g_1$ is also continuous. The function g is called the *Cantor function*. It is increasing, continuous and constant on each $I_k^n, k, n \geq 1$.

The Cantor function maps $\mathcal{C}$, a set of measure zero, to a set of full measure. For, first we have $g([0,1]) = [0,1]$. On the other hand, $\bigcup_{n,k} g(I_k^n) = \{1/2, 1/4, 3/4, \ldots\}$ is a countable set, so $\mathcal{L}^1(g(\mathcal{C})) = 1$. So a continuous map could send a null set to a set of full measure.

We modify g to $h(x) = x + g(x) : [0,1] \to [0,2]$. Then h is strictly increasing and maps $[0,1]$ onto $[0,2]$. Its inverse h^{-1} is a continuous function from $[0,2]$ to $[0,1]$, so h is a homeomorphism between $[0,1]$ and $[0,2]$. Since g is constant on each I_k^n and $\bigcup_{n,k} I_k^n$ has measure 1, it is not hard to see $h(\mathcal{C})$ has measure equal to 1. By Proposition 3.3, there exists some non-measurable $A \subset h(\mathcal{C})$. But then $h^{-1}(A) \subset \mathcal{C}$ is measurable. We have shown that a non-measurable set can be mapped to a measurable set by a homeomorphism.

3.3 Brunn-Minkowski Inequality

The Euclidean space carries not only a topological structure induced by the Euclidean metric but also a vector space. The translational invariance of the Lebesgue measure reflects the interaction between the measure-theoretic and algebraic properties of the Euclidean space. The Brunn-Minkowski inequality is an inequality of this nature.

Proposition 3.4.

(a) If A and B are open in $\mathbb{R}^n$, $A+B$ is open.

(b) If A and B are compact in $\mathbb{R}^n$, $A+B$ is compact.

(c) If A and B are closed in $\mathbb{R}^n$, $A+B$ is an F_δ-set.

Proof. (a) As

$$A+B=\bigcup_{b\in B} A+b, \quad A+b=\{a+b : a\in A\},$$

$A+B$ is open if we can show that each $A+b$ is open, since the union of open sets is still open. Consider the continuous map $\varphi:\mathbb{R}^n\to\mathbb{R}^n$ given by $\varphi(x)=x+b$ whose inverse given by $\varphi^{-1}(x)=x-b$ is also continuous. so φ is an open map, i.e., it maps open sets to open sets. In particular, $A+b$ is open whenever A is open.

(b) Consider the continuous function $\Phi:\mathbb{R}^n\times\mathbb{R}^n\to\mathbb{R}^n$ given by $\Phi(x,y)=x+y$. When A and B are compact, $A\times B$ is compact in $\mathbb{R}^n\times\mathbb{R}^n$. (This is a general fact in topology, or you can check it directly). So $A+B=\Phi(A,B)$ is compact. (This is again a general fact from topology; the image of any compact set under a continuous map is compact.)

(c) Since every closed set can be expressed as a countable union of compact sets in $\mathbb{R}^n$, (c) follows from (b).

□

Examples show that $A+B$ may not be measurable when A and B are measurable, see exercise in the Chapter 8.

Theorem 3.5 (**Brunn-Minkowski Inequality**). *Let A, B, and $A+B$ be measurable in $\mathbb{R}^n$. Then*

$$\mathcal{L}^n(A+B)^{\frac{1}{n}}\geq\mathcal{L}^n(A)^{\frac{1}{n}}+\mathcal{L}^n(B)^{\frac{1}{n}}.$$

One also has certain characterization on the equality case. For instance, equality sign holds if A and B are convex and B is homothetic to A. But this is not necessary.

Proof. We will divide the proof in five steps. The second step is decisive.
Step 1. Let A and B be two rectangles, A having sides $a_1, \ldots, a_n$ and B sides $b_1, \ldots, b_n$. Then $A + B$ is again a rectangle of sides $a_1 + b_1, \ldots, a_n + b_n$. In this special case Brunn-Minkowski inequality becomes

$$\Big(\prod_j (a_j + b_j)\Big)^{\frac{1}{n}} \geq \Big(\prod_j a_j\Big)^{\frac{1}{n}} + \Big(\prod_j b_j\Big)^{\frac{1}{n}}.$$

Divide both sides to get

$$1 \geq \Big(\prod_j \frac{a_j}{a_j + b_j}\Big)^{\frac{1}{n}} + \Big(\prod_j \frac{b_j}{a_j + b_j}\Big)^{\frac{1}{n}}.$$

By AM-GM inequality,

$$\Big(\prod_j \frac{a_j}{a_j + b_j}\Big)^{\frac{1}{n}} \leq \frac{1}{n} \sum_j \frac{a_j}{a_j + b_j}, \qquad \text{and}$$
$$\Big(\prod_j \frac{b_j}{a_j + b_j}\Big)^{\frac{1}{n}} \leq \frac{1}{n} \sum_j \frac{b_j}{a_j + b_j}.$$

Therefore,

$$\Big(\prod_j \frac{a_j}{a_j + b_j}\Big)^{\frac{1}{n}} + \Big(\prod_j \frac{b_j}{a_j + b_j}\Big)^{\frac{1}{n}} \leq \frac{1}{n} \sum_j \frac{a_j}{a_j + b_j} + \frac{1}{n} \sum_j \frac{b_j}{a_j + b_j} = 1,$$

done.
Step 2. Let A and B both consist of finitely many almost disjoint rectangles. (Almost disjoint means their interiors are mutually disjoint.) We use induction on the total number of rectangles in A and B. When $N = 2$, it is done in Step 1. Now, assume it has been proved for $N \geq 2$ and we prove it for $N + 1$ rectangles.

Pick two rectangles in A (or B) we can find a coordinate hyperplane to separate their interiors. WLOG we may assume the hyperplane is $\{(x_1, \ldots, x_n) : x_n = 0\}$. We let

$$A_+ = \{R \cap H_+ : R \text{ is a rectangle in } A\},$$
$$A_- = \{R \cap H_- : R \text{ is a rectangle in } A\},$$

where $H_+ = \{x = (x_1, \ldots, x_n) : x_n \geq 0\}$ is the upper half space and $H_- = \{x : x_n \leq 0\}$ the lower half space. Then A_+ and A_- have at most $m - 1$ many rectangles where m is the number of rectangles in A. Now we translate B up and

down so that

$$\frac{\mathcal{L}^n(A_+)}{\mathcal{L}^n(A)} = \frac{\mathcal{L}^n(B_+)}{\mathcal{L}^n(B)} \quad \left(\text{so } \frac{\mathcal{L}^n(A_-)}{\mathcal{L}^n(A)} = \frac{\mathcal{L}^n(B_-)}{\mathcal{L}^n(B)}\right).$$

Here

$$B_+ = \{R \cap H_+ : R \text{ is a rectangle in } B\},$$
$$B_- = \{R \cap H_- : R \text{ is a rectangle in } B\},$$

where B is the B after translation. Note that translations of A and B do not change the values of $\mathcal{L}^n(A)$, $\mathcal{L}^n(B)$ and $\mathcal{L}^n(A+B)$.

Now, each of A_+ and B_+ has at most N many rectangles, so have A_- and B_-. By induction hypothesis,

$$\mathcal{L}^n(A_+ + B_+) \geq \left(\mathcal{L}^n(A_+)^{\frac{1}{n}} + \mathcal{L}^n(B_+)^{\frac{1}{n}}\right)^n$$
$$\mathcal{L}^n(A_- + B_-) \geq \left(\mathcal{L}^n(A_-)^{\frac{1}{n}} + \mathcal{L}^n(B_-)^{\frac{1}{n}}\right)^n.$$

As the interiors of $(A_+ + B_+)$ and $(A_- + B_-)$ are disjoint (they belong to the upper and lower half-spaces) and $(A_+ + B_+) \cup (A_- + B_-) \subset A + B$, we have

$$\begin{aligned}
\mathcal{L}^n(A+B) &\geq \mathcal{L}^n(A_+ + B_+) + \mathcal{L}^n(A_- + B_-) \\
&\geq \left(\mathcal{L}^n(A_+)^{\frac{1}{n}} + \mathcal{L}^n(B_+)^{\frac{1}{n}}\right)^n + \left(\mathcal{L}^n(A_-)^{\frac{1}{n}} + \mathcal{L}^n(B_-)^{\frac{1}{n}}\right)^n \\
&= \mathcal{L}^n(A_+)\left[1 + \left(\frac{\mathcal{L}^n(B_+)}{\mathcal{L}^n(A_+)}\right)^{\frac{1}{n}}\right]^n + \mathcal{L}^n(A_-)\left[1 + \left(\frac{\mathcal{L}^n(B_-)}{\mathcal{L}^n(A_-)}\right)^{\frac{1}{n}}\right]^n \\
&= (\mathcal{L}^n(A_+) + \mathcal{L}^n(A_-))\left[1 + \left(\frac{\mathcal{L}^n(B)}{\mathcal{L}^n(A)}\right)^{\frac{1}{n}}\right]^n \\
&= \left(\mathcal{L}^n(A)^{\frac{1}{n}} + \mathcal{L}^n(B)^{\frac{1}{n}}\right)^n.
\end{aligned}$$

By induction, the inequality holds when A and B are finite union of rectangles.

Step 3. A and B are open sets. WLOG $\mathcal{L}^n(A+B) < \infty$. Then $\mathcal{L}^n(A)$, $\mathcal{L}^n(B) < \infty$. It is an exercise to show that every open set can be decomposed into countably many almost disjoint closed cubes. Given $\varepsilon > 0$, there is some A_1 consisting of finitely many almost disjoint cubes such that $A_1 \subset A$ and $\mathcal{L}^n(A \setminus A_1) < \varepsilon$. Similarly, we have $B_1 \subset B$ and $\mathcal{L}^n(B \setminus B_1) < \varepsilon$. Then $A_1 + B_1 \subset A + B$ and

$$\begin{aligned}
\mathcal{L}^n(A+B) \geq \mathcal{L}^n(A_1 + B_1) &\geq \left(\mathcal{L}^n(A_1)^{\frac{1}{n}} + \mathcal{L}^n(B_1)^{\frac{1}{n}}\right)^n \\
&\geq \left[(\mathcal{L}^n(A) - \varepsilon)^{\frac{1}{n}} + (\mathcal{L}^n(B) - \varepsilon)^{\frac{1}{n}}\right]^n,
\end{aligned}$$

and the inequality follows after letting $\varepsilon \to 0$.

Step 4. A, B compact. Let $A^\varepsilon = \{x \in \mathbb{R}^n : \operatorname{dist}(x, A) < \varepsilon\}$ and $B^\varepsilon = \{x \in \mathbb{R}^n : \operatorname{dist}(x, B) < \varepsilon\}$ be open and $A^\varepsilon \downarrow A$, $B^\varepsilon \downarrow B$ as $\varepsilon \to 0$. So $\mathcal{L}^n(A^\varepsilon) \to \mathcal{L}^n(A)$ and $\mathcal{L}^n(B^\varepsilon) \to \mathcal{L}^n(B)$ as $\varepsilon \to 0$ (see Ex 6). We have $A + B$ is compact and $A + B \subset A^\varepsilon + B^\varepsilon \subset (A + B)^{2\varepsilon}$. So, assuming $\mathcal{L}^n(A + B) < \infty$, for every $\rho > 0$, there exists ε such that

$$\begin{aligned}
\rho + \mathcal{L}^n(A + B) &\geq \mathcal{L}^n\left((A + B)^{2\varepsilon}\right) \\
&\geq \mathcal{L}^n\left(A^\varepsilon + B^\varepsilon\right) \\
&\geq \left[\mathcal{L}^n(A^\varepsilon)^{\frac{1}{n}} + \mathcal{L}^n(B^\varepsilon)^{\frac{1}{n}}\right]^n \\
&\geq \left[\mathcal{L}^n(A)^{\frac{1}{n}} + \mathcal{L}^n(B)^{\frac{1}{n}}\right]^n,
\end{aligned}$$

and the inequality follows after letting $\rho \to 0$.
Step 5. Now, let A, B and $A + B$ be measurable. WLOG $\mathcal{L}^n(A + B) < \infty$. (So $\mathcal{L}^n(A), \mathcal{L}^n(B) < \infty$.) Given $\varepsilon > 0$, by inner regularity pick $K_1 \subset A$, $K_2 \subset B$, such that $\mathcal{L}^n(A \setminus K_1), \mathcal{L}^n(B \setminus K_2) < \varepsilon$. Then $K_1 + K_2 \subset A + B$, and

$$\begin{aligned}
\mathcal{L}^n(A + B) &\geq \mathcal{L}^n(K_1 + K_2) \\
&\geq \left[\mathcal{L}^n(K_1)^{\frac{1}{n}} + \mathcal{L}^n(K_2)^{\frac{1}{n}}\right]^n \\
&\geq \left[(\mathcal{L}^n(A) - \varepsilon)^{\frac{1}{n}} + (\mathcal{L}^n(B) - \varepsilon)^{\frac{1}{n}}\right]^n.
\end{aligned}$$

Letting $\varepsilon \to 0$, we conclude the proof of Brunn-Minkowski inequality.

□

As an application of Brunn-Minkowski inequalityl, we discuss how to use it to deduce the isoperimetric inequality.

The isoperimetric inequality asserts that for every set E in $\mathbb{R}^n$,

$$A_n(E) \geq c_n \mathcal{L}^n(E)^{\frac{n-1}{n}}, \qquad c_n = \frac{A_n(B)}{\mathcal{L}^n(B)^{(n-1)/n}},$$

where $A_n(E)$ denotes the "surface area" or "perimeter" of E and B is a ball. It seems that this inequality is never proved in full generality. In history, it was first established for convex sets, next for sets with differentiable boundaries and finally for domains bounded by rectifiable boundaries.

In the following we will use $|E|$ to denote the n-dimensional Lebsegue measure of a set E in $\mathbb{R}^n$ instead of $\mathcal{L}^n(E)$.

In fact, for an arbitrary set, there is a very general notion of perimeter replaced by the Minkowski content. It is based on the geometric observation that for a

smooth hypersurface Σ in $\mathbb{R}^n$, its δ-tube

$$\Sigma^\delta = \left\{x \in \mathbb{R}^2 : d(x, \Sigma) < \delta\right\}$$

satisfies

$$\lim_{\delta \downarrow 0} \frac{\left|\Sigma^\delta\right|}{2\delta} = \text{the surface area of } \Sigma.$$

Motivated by this, for any set E in $\mathbb{R}^n$, we set

$$A_n(E) = \lim_{\delta \to 0} \frac{|(\partial E)^\delta|}{2\delta},$$

provided the limit exists.

Theorem 3.6. *Let Ω be a bounded open set in $\mathbb{R}^n$ such that $A_n(\Omega)$ exists. Then*

$$A_n(\Omega) \geq n\omega_n^{1/n} |\Omega|^{\frac{n-1}{n}},$$

where ω_n is the volume of the unit ball in $\mathbb{R}^n$. Equality sign in this inequality holds when Ω is a ball.

Proof. Here we set $\Gamma = \partial\Omega$. For a positive δ, let

$$\Omega_+(\delta) = \left\{x \in \mathbb{R}^n : d(x, \overline{\Omega}) < \delta\right\},$$

and

$$\Omega_-(\delta) = \{x \in \mathbb{R}^n : d(x, \Omega') \geq \delta\}$$

and

$$\Gamma^\delta = \{x \in \mathbb{R}^n : d(x, \partial\Omega) < \delta\}.$$

It is readily checked that

$$\Omega_+(\delta) = \Omega_-(\delta) \cup \Gamma^\delta (\text{ disjoint union}),$$

$$\Omega + B_\delta(0) \subset \Omega_+(\delta)$$

and

$$\Omega_-(\delta) + B_\delta(0) \subset \Omega.$$

Applying Brunn-Minkowski inequality to the open set $\Omega + B_\delta(0)$, we have

$$|\Omega_+(\delta)|^{\frac{1}{n}} \geq |\Omega + B_\delta(0)|^{\frac{1}{n}} \geq |\Omega|^{\frac{1}{n}} + |B_\delta(0)|^{\frac{1}{n}}.$$

Using binomial expansion we have

$$|\Omega_+(\delta)| \geq |\Omega| + n\,|\Omega|^{\frac{n-1}{n}}\,|B_\delta(0)|^{\frac{1}{n}}$$

Similarly,

$$|\Omega| \geq |\Omega_-(\delta)| + n\,|\Omega_-(\delta)|^{\frac{n-1}{n}}\,|B_\delta(0)|^{\frac{1}{n}}\,.$$

By adding up these two inequalities and using $\Omega_+(\delta) \setminus \Omega_-(\delta) = \Gamma^\delta$,

$$\begin{aligned}\left|\Gamma^\delta\right| &= |\Omega_+(\delta)| - |\Omega_-(\delta)| \\ &\geq n\left(|\Omega|^{\frac{n-1}{n}} + |\Omega_-(\delta)|^{\frac{n-1}{n}}\right)|B_\delta(0)|^{\frac{1}{n}}\,.\end{aligned}$$

Using $|B_\delta(0)| = \omega_n \delta^n$, we deduce

$$\frac{|\Gamma_\delta|}{2\delta} \geq \frac{n}{2}\left(|\Omega|^{\frac{n-1}{n}} + |\Omega_-(\delta)|^{\frac{n-1}{n}}\right)\omega_n^{\frac{1}{n}}.$$

Now the desired inequality follows readily by letting δ go to 0.

Let us verify that it becomes equality when Ω is a ball of radius r. In fact, in this case

$$|\Omega| = \omega_n r^n$$

and

$$A_n(\Omega) = n\omega_n r^{n-1}.$$

The left hand side of this inequality becomes $n\omega_n R^{n-1}$ and its right hand side becomes

$$n\omega_n^{\frac{1}{n}}\left(\omega_n r^n\right)^{\frac{n-1}{n}} = n\omega_n r^{n-1},$$

so equality holds at a ball. □

3.4 Hausdorff Measures

Let $0 \leq s < \infty$ and $A \subset \mathbb{R}^n$. For $0 < \delta \leq \infty$, define

$$\mathcal{H}^s_\delta(A) = \inf\left\{\sum_j d(C_j)^s : A \subset \bigcup_j C_j,\ d(C_j) < \delta, C_j \subset \mathbb{R}^n\right\}$$

where $d(C)$ is the diameter of C, that is, $d(C) = \sup\{|x-y| : x, y \in C\}$ and $|x-y|$ is the Euclidean distance in $\mathbb{R}^n$. The s-dimensional Hausdorff measure (or more precisely, unnormalized Hausdorff measure) of A is defined as

$$\begin{aligned}\mathcal{H}^s(A) &= \lim_{\delta \to 0} \mathcal{H}^s_\delta(A) \\ &= \sup_{\delta > 0} \mathcal{H}^s_\delta(A).\end{aligned}$$

We note that $\mathcal{H}^s(A) \in [0, \infty]$.

Theorem 3.7. *(a) $\mathcal{H}^s$ is a Borel measure on $\mathbb{R}^n$.*

(b) For every $A \subset \mathbb{R}^n$,

$$\mathcal{H}^s(A) = \inf\{\mathcal{H}^s(G) : A \subset G, \quad G \text{ open }\}.$$

(c) For every open set G, $\mathcal{H}^s(G) = \sup\{\mathcal{H}^s(K) : K \subset G \text{ is compact}\}$.

(d) For every measurable set A with $\mathcal{H}^s(A) < \infty$, given any $\varepsilon > 0$, there exists a compact set $K \subset A$ such that $\mathcal{H}^s(A \setminus K) < \varepsilon$.

Proof. Step 1. $\mathcal{H}^s_\delta$ and $\mathcal{H}^s$ are outer measures. As $\phi \subset \{x\}$ for every $x \in \mathbb{R}^n$ and $d(\{x\}) = 0$, $\mathcal{H}^s_\delta(\phi) = 0$ for all $\delta > 0$ and $\mathcal{H}^s(\phi) = 0$. Next, let $A \subset \bigcup_j A_j$, we want to show

$$\mathcal{H}^s_\delta(A) \le \sum_{j=1}^{\infty} \mathcal{H}^s_\delta(A_j). \tag{3.1}$$

WLOG, $\mathcal{H}^s_\delta(A_j) < \infty, \forall j$. For $\varepsilon > 0$, there exist C^j_k, $A_j \subset \bigcup_k C^j_k$, $d(C^j_k) < \delta$, such that

$$\mathcal{H}^s_\delta(A_j) + \frac{\varepsilon}{2^j} \ge \sum_k d(C^j_k)^s.$$

Thus

$$\begin{aligned}
\sum_j \mathcal{H}^s_\delta(A_j) + \varepsilon &\ge \sum_j \sum_k d(C^j_k)^s \\
&= \sum_{j,k} d(C^j_k)^s \\
&\ge \mathcal{H}^s_\delta(A) \qquad (\text{since } A \subset \bigcup_{j,k} C^j_k),
\end{aligned}$$

and (3.1) follows by letting $\varepsilon \to 0$. Now, from (3.1)

$$\begin{aligned}
\mathcal{H}^s_\delta(A) &\le \sum_{j=1}^{\infty} \mathcal{H}^s_\delta(A_j) \\
&\le \sum_{j=1}^{\infty} \mathcal{H}^s(A_j),
\end{aligned}$$

and $\mathcal{H}^s(A) \le \sum_j \mathcal{H}^s(A_j)$ holds after letting $\delta \to 0$.

Step 2. $\mathcal{H}^s$ is a Borel measure. Here we use Caratheodory's criterion. Letting $A, B \subset \mathbb{R}^n$ with $d(A, B) = \delta_0 > 0$, we want to show that

$$\mathcal{H}^s(A \cup B) = \mathcal{H}^s(A) + \mathcal{H}^s(B) \tag{3.2}$$

On one hand, $\mathcal{H}^s(A \cup B) \le \mathcal{H}^s(A) + \mathcal{H}^s(B)$ is evident by countable subadditivity.

To show the reverse inequality, let $\{C_j\}$ be a covering of $A \cup B$ with $d(C_j) < \delta < \delta_0/4$. Each C_j can only intersect either A or B but not both. We can divide $\{C_j\}$ into two classes $\{C_j'\}$ and $\{C_j''\}$ such that $A \subset \bigcup_j C_j'$ and $B \subset \bigcup_j C_j''$. Then

$$\begin{aligned}\sum_j d(C_j)^s &= \sum_j d(C_j')^s + \sum_j d(C_j'')^s \\ &\geq \mathcal{H}^s_\delta(A) + \mathcal{H}^s_\delta(B).\end{aligned}$$

Taking infimum over all $\{C_j\}$, we conclude

$$\mathcal{H}^s_\delta(A \cup B) \geq \mathcal{H}^s_\delta(A) + \mathcal{H}^s_\delta(B), \quad \forall\ 0 < \delta < \frac{\delta_0}{4}.$$

Letting $\delta \to 0$, we get (3.2).

Step 3. For any set $C \subset \mathbb{R}^n$, let $C^\varepsilon = \{x : \operatorname{dist}(x, C) < \varepsilon\}$. Then C^ε is open and $d(C^\varepsilon) \leq d(C) + 2\varepsilon$ (check!). If $\mathcal{H}^s(A) = \infty$, (b) holds by taking $B = \mathbb{R}^n$. When $\mathcal{H}^s(A) < \infty$, for any given k, we fix $\{C_j\}$, $A \subset \bigcup_j C_j$, $d(C_j) < 1/k$, such that

$$\mathcal{H}^s_{\frac{1}{k}}(A) + \frac{1}{k} \geq \sum_j d(C_j)^s.$$

Pick a small ε_j such that $d(C_j^{\varepsilon_j}) \leq 2/k$ and

$$d(C_j)^s \geq d\left(C_j^{\varepsilon_j}\right)^s - \frac{1}{k2^j}.$$

Then

$$\mathcal{H}^s_{\frac{1}{k}}(A) + \frac{2}{k} \geq \sum_j d\left(C_j^{\varepsilon_j}\right)^s.$$

The set $G_k = \bigcup_j C_j^{\varepsilon_j}$ is open, and we have

$$\begin{aligned}\mathcal{H}^s(A) + \frac{2}{k} &\geq \mathcal{H}^s_{\frac{1}{k}}(A) + \frac{2}{k} \\ &\geq \mathcal{H}^s_{\frac{2}{k}}(G_k) \\ &\geq \mathcal{H}^s_{\frac{2}{k}}(B), \quad B = \bigcap_{k=1}^\infty G_k,\ E \in \mathcal{B}.\end{aligned}$$

Letting $k \to \infty$, $\mathcal{H}^s(A) \geq \mathcal{H}^s(B)$. As the sets $B_n = \bigcap_{k=1}^n G_k$ are open, $\mathcal{H}^s(B_1) < \infty$, and $\lim_{n\to\infty} \mathcal{H}^s(B_k) = \mathcal{H}^s(B) = \mathcal{H}^s(A)$, (b) holds.

Step 4. Let $K_j = \{x \in G : \operatorname{dist}(x, G) \leq 1/j,\ |x| \leq j\}$. Then $K_j, j \geq 1$, are compact and $K_j \uparrow G$. By Proposition 1.4, (c) holds.

Step 5. You are referred to Lemma 1 (i) on p. 6 in [EG] for a proof of (d). Originally I tried to pack their proof into a few lines, but it turns out to be unreadable. □

When $s \in (0, n)$ is not an integer, it can be shown that $\mathcal{H}^s(R) = \infty$ on every cube R. It implies that Hausdorff measures $\mathcal{H}^s, s < n$, are not finite on compact sets. In particular, they cannot be constructed by Riesz representation theorem.

Every Euclidean motion is of the form

$$Tx = Ax + b, \quad b \in \mathbb{R}^n,$$

where A is an $n \times n$-orthogonal matrix. The following immediate proposition shows that the Hausdorff measures are geometric measures in the sense that they are invariant under all Euclidean motions.

Proposition 3.8. *Let $A \subset \mathbb{R}^n$.*

(a) $\mathcal{H}^s(TA) = \mathcal{H}^s(A)$, where T is a Euclidean motion.

(b) $\mathcal{H}^s(\lambda A) = \lambda^s \mathcal{H}^s(A), \qquad \forall \lambda > 0.$

We now proceed to define the Hausdorff dimension of a set.

Proposition 3.9. *Let $A \subset \mathbb{R}^n$.*

(a) $\mathcal{H}^s(A) = 0$, $\forall s > n$.

(b) If $\mathcal{H}^s(A) < \infty$ for some $s \in [0, \infty)$, then $\mathcal{H}^t(A) = 0$, $\forall t > s$.

(c) If $\mathcal{H}^s(A) \in (0, \infty]$ for some $s \in [0, \infty)$, then $\mathcal{H}^t(A) = \infty$, $\forall t < s$.

Proof. (a) Divide the unit cube Q into subcubes of side $1/k$. There are k^n many such subcubes with diameters equal to $\sqrt{n}/k$. We have,

$$\begin{aligned} \mathcal{H}^s_{\frac{\sqrt{n}}{k}}(Q) &\le \sum_j d(Q_j)^s = k^n \left(\frac{\sqrt{n}}{k}\right)^s \\ &= \sqrt{n}^s \frac{1}{k^{s-n}} \to 0, \qquad \text{as } k \to \infty, \end{aligned}$$

so $\mathcal{H}^s(Q) = 0$ and $\mathcal{H}^s(\mathbb{R}^n) = 0$ for $s > n$.

(b) As $\mathcal{H}^s(A)$ is finite, for each $\delta > 0$, there exists $\{C_j\}$, $d(C_j) < \delta$, such that $A \subset \bigcup_j C_j$ and

$$\sum_j d(C_j)^s \le \mathcal{H}^s(A) + 1.$$

As $d(C_j)^s = d(C_j)^t d(C_j)^{s-t} \geq \delta^{s-t} d(C_j)^t$,

$$\begin{aligned} \mathcal{H}^t_\delta(A) &\leq \sum_j d(C_j)^t \\ &\leq \delta^{t-s}\left(\mathcal{H}^s(A)+1\right) \to 0, \qquad \text{as } \delta \to 0. \end{aligned}$$

(c) It follows from (b). □

From this proposition we know that for every $A \subset \mathbb{R}^n$, $\mathcal{H}^s(A) = 0$ for $s > n$. Lowering s passing through n, we will first hit a point $s_0 \in [0, n]$ such that $\mathcal{H}^s(A) = 0$ for all $s > s_0$ and $\mathcal{H}^s(A) = \infty$ for all $s < s_0$. This point is characterized by

$$s_0 = \inf\{s : \mathcal{H}^s(A) = 0\}$$

and is called the *Hausdorff dimension* of A.

Proposition 3.10.

(a) $\mathcal{H}^0$ *is the counting measure in* $\mathbb{R}^n$.

(b) $\mathcal{H}^1 = \mathcal{L}^1$ *on* $\mathbb{R}$.

(c) $\mathcal{H}^n_N = \mathcal{L}^n$ *on* $\mathbb{R}^n$ *for* $n \geq 2$.

Here the *normalized Hausdorff measure* is given by

$$\mathcal{H}^s_N(A) = \sigma(s)\mathcal{H}^s(A) \quad \forall A \subset \mathbb{R}^n,$$

where

$$\sigma(s) = \frac{\pi^{s/2}}{2^s \Gamma(s/2+1)}, \quad \Gamma(s) = \int_0^\infty e^{-x} x^{s-1} dx.$$

When $s = n$, we have $\sigma(n) = \mathcal{L}^n(B)/2^n$ where B is the unit ball in $\mathbb{R}^n$.

Proof. (a) Exercise.

(b) Let $A \subset \mathbb{R}$ and $\delta > 0$. We have

$$\begin{aligned}
\mathcal{L}^1(A) &= \inf\left\{\sum_j |I_j| : A \subset \bigcup_j I_j\right\} \\
&= \inf\left\{\sum_j |I_j| : A \subset \bigcup_j I_j,\ d(I_j) < \delta\right\} \\
&\quad \text{(every interval can be chopped up to subintervals of } d(I_j) < \delta) \\
&= \inf\left\{\sum_j d(I_j) : A \subset \bigcup_j I_j,\ d(I_j) < \delta\right\} \\
&= \inf\left\{\sum_j d(C_j) : A \subset \bigcup_j C_j,\ d(C_j) < \delta\right\} \\
&= \mathcal{H}^1_\delta(A),
\end{aligned}$$

so $\mathcal{L}^1(A) = \mathcal{H}^1(A)$. Here we have used the fact that for each set C in $\mathbb{R}$ there is an interval I satisfying $d(I) = d(C)$.

(c) To show that $c \geq 1/\sigma(n)$ we need the isodiametric inequality from [EG]: For $A \subset \mathbb{R}^n$,

$$\mathcal{L}^n(A) \leq \sigma(n) d(A)^n.$$

It asserts that the ball has the largest volume among all sets with the same diameter. Suppose $A \subset \bigcup_j C_j$, $d(C_j) < \delta$, then

$$\begin{aligned}
\mathcal{L}^n(A) &\leq \sum_j \mathcal{L}^n(C_j) \\
&\leq \sum_j \sigma(n) d(C_j)^n \\
&= \sigma(n) \sum_j d(C_j)^n.
\end{aligned}$$

Taking infimum over all $\{C_j\}$, we get

$$\begin{aligned}
\mathcal{L}^n(A) &\leq \sigma(n)\mathcal{H}^n_\delta(A) \\
&\leq \sigma(n)\mathcal{H}^n(A),
\end{aligned}$$

hence $c \geq 1/\sigma(n)$.

To show the reverse inequality, we first claim that there exists some constant C such that

$$\mathcal{H}^n(A) \leq C\mathcal{L}^n(A), \quad \forall A \subset \mathbb{R}^n \tag{3.3}$$

Assume $\mathcal{L}^n(A) < \infty$. For every $\varepsilon > 0$, we can find congruent cubes $\{R_j\}$, $A \subset \bigcup_j R_j$, such that

$$\mathcal{L}^n(A) + \varepsilon \geq \sum_j |R_j| .$$

Using $|R_j| = \left(\frac{d(R_j)}{\sqrt{n}}\right)^n$,

$$\begin{aligned} \mathcal{L}^n(A) + \varepsilon &\geq \left(\frac{1}{\sqrt{n}}\right)^n \sum_j d(R_j)^n \\ &\geq \left(\frac{1}{\sqrt{n}}\right)^n \mathcal{H}^n_l(A), \end{aligned}$$

where l is the side length of R_j. Letting first $l \to 0$ and then $\varepsilon \to 0$,

$$\mathcal{L}^n(A) \geq \left(\frac{1}{\sqrt{n}}\right)^n \mathcal{H}^n(A).$$

At last, for $\varepsilon > 0$, there are R_j, $A \subset \bigcup_j R_j$, $d(R_j) < \delta$, such that

$$\mathcal{L}^n(A) + \varepsilon \geq \sum_j |R_j| .$$

Using the fact that each R_j can be written as a disjoint union of balls B^j_k (p. 28 in [EG]) such that

$$\mathcal{L}^n\left(R_j \setminus \bigcup_k B^j_k\right) = 0,$$

we have

$$\begin{aligned} \mathcal{L}^n(A) + \varepsilon &\geq \sum_{j,k} \mathcal{L}^n\left(B^j_k\right) \\ &= \sigma(n) \sum d\left(B^j_k\right)^n \\ &\geq \sigma(n) \mathcal{H}^n_\delta \left(\bigcup_{j,k} B^j_k\right). \end{aligned}$$

As $\mathcal{L}^n\left(R_j \setminus \bigcup_k B^j_k\right) = 0$, by (3.3), $\mathcal{H}^n\left(R_j \setminus \bigcup_k B^j_k\right) = 0$, so $\mathcal{H}^n\left(\bigcup_j R_j \setminus \bigcup_k B^j_k\right) =$

0 and

$$\begin{aligned}\mathcal{H}^n_\delta\left(\bigcup_{j,k} B^j_k\right) &= \mathcal{H}^n_\delta\left(\bigcup_j R_j\right) - \mathcal{H}^n_\delta\left(\bigcup_j R_j \setminus \bigcup_k B^j_k\right)\\ &= \mathcal{H}^n_\delta\left(\bigcup_j R_j\right)\\ &\geq \mathcal{H}^n_\delta(A).\end{aligned}$$

That is,

$$\mathcal{L}^n(A) + \varepsilon \geq \sigma(n)\mathcal{H}^n_\delta(A),$$

and $\mathcal{L}^n(A) \geq \mathcal{H}^n_N(A)$ holds. □

3.5 Hausdorff Dimension

We present a preliminary study on the mapping properties of the Hausdorff measures.

A map $f : A \subset \mathbb{R}^n \to \mathbb{R}^m$ is *Hölder continuous* with exponent $\alpha \in (0, 1)$ if there exists a constant M such that

$$|f(x) - f(y)| \leq M\,|x - y|^\alpha, \qquad \forall x, y \in A.$$

The constant M is called a *Hölder constant.* The map f is called *Lipschitz continuous* when $\alpha = 1$ in this condition and M is called a *Lipschitz constant.* Lipschitz and Höler continuous functions are very much different.

Proposition 3.11. *Let $f : A \subset \mathbb{R}^n \to \mathbb{R}^m$ be Hölder or Lipschitz continuous. We have*

$$\mathcal{H}^{\frac{s}{\alpha}}(f(A)) \leq M^{\frac{1}{\alpha}}\mathcal{H}^s(A).$$

Proof. Let $A \subset \bigcup_j C_j$, $d(C_j) < \delta$, satisfy $\mathcal{H}^s(A) + \varepsilon \geq \sum_j d(C_j)^s$. We have

$$d(f(C_j)) \leq M d(C_j)^\alpha.$$

Thus

$$\frac{1}{M^{\frac{1}{\alpha}}} d(f(C_j))^{\frac{1}{\alpha}} \leq d(C_j),$$

and

$$\mathcal{H}^s(A) + \varepsilon \geq \frac{1}{M^{\frac{1}{\alpha}}} \sum_j d(f(C_j))^{\frac{s}{\alpha}}.$$

As $f(A) \subset \bigcup_j f(C_j)$ and $d(f(C_j)) \leq \rho \equiv M\delta^\alpha$,

$$M^{\frac{1}{\alpha}}\left(\mathcal{H}^s(A) + \varepsilon\right) \geq \mathcal{H}^s_\rho(f(A)),$$

and the result follows by letting $\delta \to 0$. $\square$

Let us examine an example. Let $f : [0,1] \to \mathbb{R}^2$ be Lipschitz continuous and its graph is given by

$$C = \{(x, f(x)) : x \in [0,1]\} \subset \mathbb{R}^2.$$

By Proposition 3.9, we have

$$\mathcal{H}^1(C) \leq M\mathcal{H}^1([0,1]) = M.$$

On the other hand, $g : C \to [0,1]$ given by $g(x,y) = x$ is the inverse of f. From $|g(x,y) - g(x_1,y_1)| \leq |x - x_1| \leq |(x,y) - (x_1,y_1)|$, we see that it is Lipschitz continuous with Lipschitz constant $M = 1$. By Proposition 3.9,

$$1 = \mathcal{H}^1[0,1] \leq \mathcal{H}^1(C).$$

We conclude that

$$1 \leq \mathcal{H}^1(C) \leq M.$$

Indeed, one can show that

$$\mathcal{H}^1(C) = \int_0^1 \sqrt{1 + f'^2(x)}\, dx$$

when f is Lipschitz continuous. This formula is a special case of the area formula (see chapter 3 of [EG]). It shows that the Hausdorff measures in special cases such as $\mathcal{H}^1$ in $\mathbb{R}^2$, $\mathcal{H}^1$ and $\mathcal{H}^2$ in $\mathbb{R}^3$, really coincide with the length of curves and surface area of surfaces defined in advanced calculus.

We end this section by determining the Hausdorff dimension of the Cantor set $\mathcal{C}$.

Proposition 3.12. *The Hausdorff dimension of $\mathcal{C}$ is* $\log 2/\log 3$. *In fact,*

$$\mathcal{H}^\gamma(\mathcal{C}) \in (0,1], \quad \gamma = \frac{\log 2}{\log 3}.$$

Proof. Observe that $\mathcal{C} = \bigcap_n \mathcal{C}_n$ where $\mathcal{C}_n$ consists of 2^n many intervals of length $1/3^n$. So

$$\begin{aligned}\mathcal{H}^\gamma_{\frac{1}{3^n}}(\mathcal{C}) &\leq \sum d(\mathcal{C}_n)^\gamma = 2^n \left(\frac{1}{3^n}\right)^\gamma, \quad 3^\gamma = 2 \\ &= \left(\frac{2}{3^\gamma}\right)^n = 1.\end{aligned}$$

On the other hand, the Cantor function g is the uniform limit of $\{g_n\}$ and we

have

$$|g_{n+1}(x) - g_n(x)| \le \frac{1}{2^n}.$$

Therefore,

$$|g_l(x) - g_n(x)| \le \sum_{j=0}^{l-n} \frac{1}{2^{n+j}} < \sum_{0}^{\infty} \frac{1}{2^{n+j}} = \frac{1}{2^{n-1}}.$$

Letting $l \to \infty$,

$$|g(x) - g_n(x)| \le \frac{1}{2^{n-1}}.$$

On the other hand, each g_n has slope at most $(3/2)^n$, so

$$|g_n(x) - g_n(y)| \le \left(\frac{3}{2}\right)^n |x - y|.$$

As a result,

$$\begin{aligned} |g(x) - g(y)| &\le |g(x) - g_n(x)| + |g_n(x) - g_n(y)| + |g_n(y) - g(y)| \\ &\le \left(\frac{3}{2}\right)^n |x - y| + \frac{4}{2^n} \\ &= \frac{1}{2^n} \left(3^n |x - y| + 4\right), \quad \forall n \ge 1. \end{aligned}$$

For each $x, y \in [0, 1]$, we pick n so that

$$\frac{1}{3^n} \le |x - y| < \frac{1}{3^{n-1}}.$$

Then $1 \le 3^n |x - y| < 3$ and

$$|g(x) - g(y)| \le \frac{7}{2^n} = \frac{7}{3^{n\gamma}}.$$

As $3^n |x - y| \ge 1$, $\dfrac{1}{3^{n\gamma}} \le |x - y|^\gamma$ and

$$|g(x) - g(y)| \le 7 |x - y|^\gamma.$$

As $g(\mathcal{C}) = [0, 1]$ by Proposition 3.9, $1 = \mathcal{H}^1(g(\mathcal{C})) \le 7^{\frac{1}{\gamma}} \mathcal{H}^\gamma(\mathcal{C})$. We have shown that the Hausdorff dimension of $\mathcal{C}$ is γ. □

Comments on Chapter 3. Lebesgue measure on the real line was introduced by Lebesgue in 1901/02 together with his integration theory. Nowadays, at least three ways to define the Lebesgue measure are known. First, it is an

outer measure using the cubes as a gauge. Second, it is the measure obtained by the positive linear functional defined by the Riemann integral via the representation theorem. Third, it is the n-times product measure of the one dimensional Lebesgue measure. (This is the approach adapted in [EG]. We will discuss it in Chapter 8.) We follow [SS] by employing the first approach, which is intuitive clear, while the second approach is used in [R]. Both approaches can be used to construct the invariant Haar measure on a locally compact topological group, see Nachbin's book, The Haar Integral. Another application of the representation theorem is, on a Riemannian manifold one can use the metric to define an n-form called the volume form on the manifold. Then the analogue of Riemann integral with respect to this volume form is well-defined on the manifold. By the theorem there exists a canonical volume measure defined on the manifold which is invariant under all isometries.

Note that in some books the Lebesgue outer measure and Lebesgue measure are distinguished and used notations like m^* and m. Here Lebesgue measure is always referred to the outer measure and the notation $\mathcal{L}^n$ is applied to measurable and non-measurable sets alike.

The proof of the scaling property of the Lebesgue measure under a linear transformation using the characterization of translational invariant measures is taken from [R]. Different proofs can be found in [SS] either by Fubini's theorem (problem 4 in chapter 2) or in exercise 26, chapter 3. But Rudin's proof is most elegant. In the second part of this chapter we will show that the n-dimensional Lebesgue measure is equal to the n-dimensional Hausdorff measure on $\mathbb{R}^n$. As it is obvious from the definition that the latter is Euclidean invariant, there we will have another proof of the Euclidean invariance of the Lebesgue measure. Any Euclidean-invariant measure is called a *geometric measure*. The Hausdorff measures and Lebesgue measures are geometric measures. The book by Federer, Geometric Measure Theory, is a classic on this topic. You may find many examples of geometric measures in it.

Nonmeasurable sets were first found by Vitali(1905) and every book I know follows his construction. The nonmeasurable sets $N \cap [0,1]$ are usually called Vitali sets. The existence of nonmeasurable sets depends essentially on the axiom of choice. It destroys the hope to have a geometric measure on all sets in $\mathbb{R}$ and $\mathbb{R}^n$ as well. One may make concession by asking whether there is a translational, finitely additive "measure" on all sets in $\mathbb{R}^n$. Here the requirement of countably additivity has been relaxed to finite additivity. It turns out the answer is affirmative. By using Hahn-Banach theorem, of which proof also depends on the axiom of choice, one can establish the existence of such a measure. To go one step beyond, we may ask if there is a Euclidean invariant finitely additive "measure" on all sets in $\mathbb{R}^n$. The answer is yes for $n = 1, 2$ by Banach(1920) and no for $n \geq 3$ as

a consequence of Banach-Tarski paradox (1923). This striking paradox (theorem) asserts that given any two bounded sets S and T in $\mathbb{R}^n$ with non-empty interior, there is a partition of S into $S_1, S_2, \cdots, S_k$ and a partition of T into $T_1, T_2, \cdots, T_k$ so that S_i and T_i are congruent for $i = 1, 2, \cdots, k$, where the number k depends on S and T. In the special case where S is the unit ball B_1 in $\mathbb{R}^3$ and T is the union of two copies of 1, it was known that k is equal to 5. Specifically that means we can find a partition of 1 into five pieces A, B, C, D and E so that A and B can be put up by Euclidean motions to form a copy of 1 while C, D and F form another B_1 in a similar way. It shows that finitely additive Euclidean invariant "measure" does not exist for $\mathbb{R}^3$. For, If there is such a nontrivial μ, we will have $\mu(B_1) = \mu(A \cup B \cup C \cup D \cup E) = \mu(A \cup B) + \mu(C \cup D \cup E) = 2\mu(B_1)$, which is impossible. The same idea works for all $n > 3$. You may google for more information.

There are many fine, delicate properties of the one dimensional Lebesgue measure we do not cover. The interested reader may consult the books by Halmos' Measure Theory, Oxtoby's Measure and Category, and many others.

The Brunn-Minkowski inequality is taken from [SS]. It is strange that this fundamental result was not included in text books on real analysis until [SS]. In Chapter 7 we will apply it to prove the isoperimetric inequality. We did not discuss the equality case of the inequality. In fact, this inequality was first proved by Brunn for convex sets in the plane and subsequently his teacher, Minkowski established it for convex sets in all dimensions. He also showed that the equality in this inequality holds only in the following cases, first, the two sets are homothetic (they coincide after a translation and dilation), or second, both are degenerate and sit inside two parallel affine sets. The proof presented here is not restricted to convex sets and it was found by Hadwiger and Ohmann in 1956. The recent survey by R. J. Gardner, The Brunn-Minkowski Inequality in v.39, 355-405, Bulletin of the American Mathematical Society, 2002, gives a comprehensive account on this inequality as well as many other related inequalities.

For their importance applications and intricate properties, nowadays Hausdorff measures have become an extensively studied subject. The theory of fractals is devoted to the study of the fine properties of sets of fractional dimension. One may consult, for example, [SS] and the books by Falconer's Fractal Geometry, Mattila's Geometry of Sets and Measures in the Euclidean Spaces, etc, for more. Our exposition merely touches the tip of the iceberg. It is worthwhile to point out that the space filling curve discovered by Peano, a continuous map from $[0, 1]$ onto the unit square, is a fractal map, see [SS].

Chapter 4

The Lebesgue Spaces

In this chapter we study L^p-integrable functions as a function space. Knowledge on functional analysis required for our study is briefly reviewed in the first two sections. In Section 1 the notions of normed and inner product spaces and their properties such as completeness, separability, the Heine-Borel property and especially the so-called projection property are discussed. Section 2 is concerned with bounded linear functionals and the dual space of a normed space. The L^p-space is introduced in Section 3, where its completeness and various density assertions by simple or continuous functions are covered. The dual space of the L^p-space is determined in Section 4 where the key notion of uniform convexity is introduced and established for the L^p-spaces. Finally, we study strong and weak convergence of L^p-sequences respectively in Sections 5 and 6. Both are important for applications.

4.1 Normed Spaces

In this and the next section we review essential elements of functional analysis that are relevant to our study of the L^p-spaces. You may look up any book on functional analysis or my notes on this subject attached in this webpage.

Let X be a vector space over $\mathbb{R}$. A norm on X is a map from $X \to [0, \infty)$ satisfying the following three "axioms": For $\forall x, y, z \in X$,

(i) $\|x\| \geq 0$ and is equal to 0 if and only if $x = 0$;

(ii) $\|\alpha x\| = |\alpha| \, \|x\|, \qquad \forall \alpha \in \mathbb{R}$; and

(iii) $\|x + y\| \leq \|x\| + \|y\|$.

The pair $(X, \|\cdot\|)$ is called a normed vector space or normed space for short. The norm induces a metric on X given by

$$d(x, y) = \|x - y\| .$$

Hence, a normed space is always a metric space and a topological one. We can talk about convergence, continuity, etc. in a normed space. The notion of a Cauchy sequence also makes sense in a metric space, that is, a sequence $\{x_n\}$ in a metric space (X, d) is called a Cauchy sequence if for every $\varepsilon > 0$, there is an n_0 such that $d(x_n, x_m) < \varepsilon$, for all $n, m \geq n_0$. Recall that a metric is complete if every Cauchy sequence is convergent. A complete normed space is called a Banach space. A general result asserts that every normed space is a dense subspace of some Banach space, and this Banach space is called the completion of the normed space.

Here are some examples of Banach spaces.

- $\mathbb{R}^n$ is a Banach space under the Euclidean norm $\|x\| = \sqrt{\sum x_j^2}$.
- $\ell^p = \left\{x = (x_1, x_2, \dots) : \sum_j |x_j|^p < \infty,\ x_j \in \mathbb{R}\right\}$, $1 \leq p < \infty$, under the ℓ^p-norm: $\|x\|_p = \left(\sum_j |x_j|^p\right)^{1/p}$.
- $C([0,1]) = \{\text{continuous functions on } [0,1]\}$ under the uniform norm (or sup-norm)

$$\begin{aligned}\|f\|_\infty &= \sup\{|f(x)| : x \in [0,1]\} \\ &= \max\{|f(x)| : x \in [0,1]\}.\end{aligned}$$

- $C(K)$ where K is a compact subset in $\mathbb{R}^n$ or a compact metric space. It is also a normed space under the sup-norm. It becomes $C([0,1])$ when $K = [0,1]$.

Let X_1 be a subspace of the normed space $(X, \|\cdot\|)$. Then $(X_1, \|\cdot\|)$ is again a normed space. The followings are some subspaces in $C([0,1])$:

$$\begin{aligned}P([0,1]) &= \{\text{The restrictions of all polynomials on } [0,1]\}, \\ C^1([0,1]) &= \{\text{All continuously differentiable functions on } [0,1]\}, \\ C_0([0,1]) &= \{\text{All continuous functions with } f(0) = f(1) = 0\}, \\ C_c((0,1)) &= \{\text{All continuous functions on } (0,1) \text{ vanishing near 0 and 1}\}.\end{aligned}$$

Among these four subspaces, only the third one is a Banach space.

The properties of finite dimensional and infinite dimensional normed spaces are very different. Let us look at the finite dimensional one first.

The Euclidean space $(\mathbb{R}^n, \|\cdot\|)$ is a typical finite dimensional normed space. It has the following four basic properties. Later we will contrast them with those in infinite dimensional spaces.

- *Completeness.* Every finite dimensional normed space is a Banach space.

- *Separability.* Recall that a topological space is separable if it has a countable, dense subset. $\mathbb{R}^n$ is separable as it possesses the countable dense subset $\mathbb{Q}^n$.

- The *Heine-Borel Property.* A normed space has the Heine-Borel property if every bounded sequence in X has a convergent subsequence. It is well-known that $\mathbb{R}^n$ has this property.

- *Projection Property.* Let X_1 be a proper, closed subspace of the normed space X and x_0 not in X_1. X is called to have the projection property if there exists some $z \in X_1$ such that

$$\|z - x_0\| = \inf \{\|x - x_0\| : x \in X_1\}.$$

  In other words, there is a point on X_1 realizing the distance from x_0 to X_1. $\mathbb{R}^n$ has this property.

In fact, it is instructive to see how this is proved. Let $\{x_j\}$ be a minimizing sequence of d, i.e., $\|x_j - x_0\| \to d$ as $j \to \infty$. Taking $\varepsilon = 1$, there is some j_0 such that $\|x_j - x_0\| \leq d+1$, $\forall j \geq j_0$. It follows that $\{x_j\}$ is a bounded sequence: $\|x_j\| \leq \|x_j - x_0\| + \|x_0\| \leq d + 1 + \|x_0\|$. By the Heine-Borel property, there is a convergent subsequence $\{x_{j_k}\}$, $x_{j_k} \to z$ for some $z \in X_1$. By the continuity of the norm, $\|x_{j_k} - x_0\| \to \|z - x_0\|$. On the other hand, $\|x_{j_k} - x_0\| \to d$ as $\{x_j\}$ is minimizing. We conclude that $\|z - x_0\| = d$.

An inner product on the normed space X over $\mathbb{R}$ is a map : $X \times X \to \mathbb{R}$ satisfying, for all $x, y, z \in X$,

(i) $\langle x, x\rangle \geq 0$, and equal to 0 if and only if $x = 0$,

(ii) $\langle x, y\rangle = \langle y, x\rangle$,

(iii) $\langle \alpha x + \beta y, z\rangle = \alpha \langle x, z\rangle + \beta \langle y, z\rangle$, for all $\alpha, \beta \in \mathbb{R}$,

Note it follows that

$$\langle x, \alpha y + \beta z\rangle = \alpha \langle x, y\rangle + \beta \langle x, z\rangle.$$

The pair $(X, \langle\ ,\ \rangle)$ is called in inner product space. The Euclidean product, or the dot product, on $\mathbb{R}^n$ is given by

$$\langle x, y\rangle_e = \sum_{j=1}^{n} x_j y_j.$$

It makes $\mathbb{R}^n$ an inner product space. In general, an inner product induces a norm by

$$\|x\| = \sqrt{\langle x, x\rangle_e}.$$

For instance, the Euclidean norm comes from the Euclidean product. An inner product space is called a Hilbert space if it is a Banach space in the induced norm.

Using the inner product structure on the Euclidean space, we have the following characterization of the point that minimizes the distance between x_0 and X_1 in the projection property, namely, z is the projection of x_0 onto X_1 if and only it satisfies

$$\langle x, x_0 - z \rangle = 0, \quad \forall x \in X_1. \tag{4.1}$$

Now I compare $C([0,1])$, or $C[0,1]$ for simplicity, with the Euclidean space. It is easy to show that $C[0,1]$ is an infinite dimensional vector space.

- *Completeness.* Yes, $C[0,1]$ is complete under the sup-norm. In elementary analysis we proved that the uniform limit of a sequence of continuous functions is continuous.

- *Separability.* Yes. By the Weierstrass approximation theorem, every continuous function on $[0,1]$ can be approximated by polynomials with rational coefficients. Thus the collection of all these polynomials forms a countable, dense set in $C([0,1])$. More generally, it can be shown that $C(K)$ is separable when K is a compact metric space.

- The *Heine-Borel Property.* Not any more. In fact, let f_n be the piecewise linear function which is equal to 0 at 0 and $[1/n, 1]$ and $f_n(1/2n) = 1$. Clearly $f_n(x) \to 0$ $\forall x \in [0,1]$. However, it does not contain any uniformly convergent subsequence. Suppose on the contrary that there exists some $\{f_{n_j}\}$ converging to some continuous f. Then f must be the zero function. For every $\varepsilon > 0$, there is some j_0 such that

  $$\left\| f_{n_j} - 0 \right\|_\infty < \varepsilon, \quad j \geq j_0.$$

  Taking $\varepsilon = 1/2$, $\left\| f_{n_j} \right\|_\infty < 1/2$ for all large j. But $\left\| f_{n_j} \right\|_\infty = 1$, contradiction holds. Hence the Heine-Borel property no longer holds in $C([0,1])$. It is a bit striking that this property characterize finite dimensionality.

  Theorem 4.1. *The Heine-Borel property holds on a normed space X if and only if X is finite dimensional.*

  See, for instance, Theorem 2.1.2 in my notes on functional analysis.

- *Projection Property.* No. But the construction of an example is not easy. I just want to point out that very often a property so obvious for a finite dimensional space may not hold in an infinite dimensional space. If it holds, more effort is need to established it. Fortunately, this property holds in Hilbert spaces.

Theorem 4.2. *Let X be a Hilbert space and $x_0 \notin X_1$ where X_1 is a proper closed subspace. There exists a unique $z \in X_1$ such that*

$$\|z - x_0\| = \inf \{\|x - x_0\| : x \in X_1\},$$

and (4.1) holds.

Proof. First of all, by expanding $\|x \pm y\|^2 = \langle x \pm y, x \pm y\rangle$, we have the parallelogram rule

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

which is true on every inner product space. Now, let $\{y_k\}$ be a minimizing sequence for $d = \inf \{\|y - x_0\| : y \in X_1\}$, that is, $\|y_k - x_0\| \to d$ as $k \to \infty$. Using the parallelogram rule, we have

$$\|y_k - y_m\|^2 = 2(\|y_k - x_0\|^2 + \|y_m - x_0\|^2) - 2\|\frac{y_k + y_m}{2} - x_0\|^2.$$

As $(y_k + y_m)/2$ belongs to X_1, $\|(y_k + y_m)/2 - x_0\| \geq d$. Given $\varepsilon > 0$, there exists some n_0 such that $\|y_n - x_0\|^2 - d^2 < \varepsilon$ for all $n \geq n_0$. It follows that

$$\|y_k - y_m\| \leq 2d^2 + 2\varepsilon - 2d^2 = 2\varepsilon,$$

for all $k, m \geq n_0$. Thus $\{y_k\}$ is a Cauchy sequence in X_1. As X_1 is closed, this sequence converges to some z in X_1 which satisfies $\|z - x_0\| = d$.

In case there is another $z_1 \in X_1$ satisfying $\|z_1 - x_0\| = d$, we have

$$\begin{aligned} \|z_1 - z\|^2 &= 2(\|z_1 - x_0\|^2 + \|z - x_0\|^2 - 2\|\frac{z_1 + z}{2} - x_0\|^2 \\ &= 2d^2 - 2\|\frac{z_1 + z}{2} - x_0\|^2 \leq 0, \end{aligned}$$

which forces $z_1 = z$. □

4.2 Bounded Linear Functionals

A linear functional on a vector space is a linear map from X to $\mathbb{R}$. It is called *bounded* if there is some M such that

$$|\Lambda x| \leq M \|x\|, \quad \forall x \in X. \tag{4.2}$$

It means Λ maps bounded sets in X to bounded intervals.

Proposition 4.3. *A linear functional on the normed space X is bounded if and only if it is continuous.*

All bounded linear functionals on X form a normed vector space under the *operator norm*:

$$\|\Lambda\|_{op} = \sup\{|\Lambda x| : \|x\| \leq 1\}.$$

Furthermore, it is complete regardless of the completeness of X. This Banach space is called the dual space of X and is denoted by $(X', \|\cdot\|_{op})$.

The dual space of $\mathbb{R}^n$ can be identified with itself.

Theorem 4.4. *There is a bijective, norm-preserving linear map between* $((\mathbb{R}^n)', \|\cdot\|_{op})$ *and* $(\mathbb{R}^n, \|\cdot\|)$.

In view of this, the Euclidean space is self-dual. Indeed, let $\Lambda \in (\mathbb{R}^n)'$. For $x \in \mathbb{R}^n$, $x = \sum_j \alpha_j e_j$ where $\{e_j\}$ is the canonical basis of $\mathbb{R}^n$. We have

$$\Lambda x = \Lambda\left(\sum \alpha_j e_j\right) = \sum \alpha_j \Lambda(e_j).$$

The map $\Phi : (\mathbb{R}^n)' \to \mathbb{R}^n$ given by $\Phi(\Lambda) = (\Lambda e_1, \ldots, \Lambda e_n)$ is the desired bijective, norm-preserving linear map.

A fundamental result concerning linear functionals is the Hahn-Banach theorem.

Theorem 4.5. *Let* X_1 *be a subspace of the vector space* X. *Suppose* p *is a sublinear function defined on* X. *Any linear functional* Λ *on* X_1 *satisfying* $|\Lambda x| \leq p(x)$, $\forall x \in X_1$, *has an extension to a linear functional* Λ' *on* X *satisfying* $|\Lambda' x| \leq p(x)$, $\forall x \in X$.

A function $p : X \to [0, \infty)$ is sublinear if for all $x, y \in X$, $\alpha \geq 0$,

$$\begin{aligned} p(x+y) &\leq p(x) + p(y), \quad \text{and} \\ p(\alpha x) &= \alpha p(x), \ \alpha \geq 0 \end{aligned}$$

hold. The norm is of course a sublinear function on X. Taking the sublinear function to be a constant multiple of the norm, we have the following version of Hahn-Banach theorem that applies to normed spaces.

Theorem 4.6. *Let* X_1 *be a subspace of the normed space* X. *Every bounded linear functional* Λ *on* X_1 *satisfying* $|\Lambda x| \leq M\|x\|$, $\forall x \in X_1$, *has an extension to a bounded linear functional* Λ' *on* X *satisfying* $|\Lambda' x| \leq M\|x\|$.

We conclude this section by a discussion on the "codimension one" property of the kernel of Λ.

It is easy to see that $\ker \Lambda = \{x : \Lambda x = 0\}$ is a closed subspace of the normed space X whenever $\Lambda \in X'$.

Proposition 4.7. *Let* $\Lambda \neq 0$ *be a non-zero linear functional on* X *and* $x_1 \notin \ker \Lambda$. *For each* $x \in X$, *there exist* $\alpha \in \mathbb{R}$ *and* $x_2 \in \ker \Lambda$ *such that*

$$x = \alpha x_1 + x_2.$$

Proof. Let $x_2 = x - \alpha x_1$ where $\alpha = \Lambda x / \Lambda x_1$. Then

$$\begin{aligned} \Lambda x_2 &= \Lambda(x - \alpha x_1) \\ &= \Lambda x - \alpha \Lambda x_1 \\ &= \Lambda x - \frac{\Lambda x}{\Lambda x_1} \Lambda x_1 = 0. \end{aligned}$$

□

This proposition tells us that the space is spanned by the kernel of a non-zero bounded linear functional together with a one dimensional subspace spanned any fixed vector lying outside the kernel. Consequently, the kernel is of "codimension one".

Corollary 4.8. *Let Λ_1 and Λ_2 be two linear functionals on X such that $\ker \Lambda_1 = \ker \Lambda_2$. Then $\Lambda_2 = c\Lambda_1$ for some non-zero constant c.*

Proof. Pick $x_0 \notin \ker \Lambda_1$. From $x = (\Lambda_1 x / \Lambda_1 x_0) x_0 + y, y \in \ker \Lambda_1 = \ker \Lambda_2$, we have $\Lambda_2 x = (\Lambda_1 x / \Lambda_1 x_0) \Lambda_2 x_0$, hence $\Lambda_2 x = \alpha \Lambda_1 x$ where $\alpha = \Lambda_2 x_0 / \Lambda_1 x_0$. □

The codimension one property has the following interesting consequence for Hilbert spaces.

Theorem 4.9. *Let $\Lambda \in X'$ where X is a Hilbert space. There exists a unique $w \in X$ such that $\Lambda x = \langle x, w \rangle$, $\forall x \in X$.*

With further effort, one can show that $\Lambda \mapsto w$ is a bijective, norm-preserving linear map. Generalizing Theorem 4.4, every Hilbert space is self-dual.

Proof. Assume that the functional Λ is non-zero. We pick $x_0 \notin \ker \Lambda$ and let $z \in \ker \Lambda$ satisfy $\langle x, x_0 - z \rangle = 0$ for all $x \in \ker \Lambda$. Therefore, the linear functional Λ' given by $\Lambda' x = \langle x, x_0 - z \rangle$ has the same kernel as Λ and so these two functionals differ by a multiplying constant. In fact, from the last corollary we have

$$\begin{aligned} \Lambda' x &= \frac{\Lambda' x_0}{\Lambda x_0} \Lambda x \\ &= \frac{\langle x_0, x_0 - z \rangle}{\Lambda x_0} \Lambda x \\ &= \frac{\|x_0 - z\|^2}{\Lambda x_0} \Lambda x, \end{aligned}$$

which implies $\Lambda x = \langle x, w \rangle$ where

$$w = \frac{(\Lambda x_0)(x_0 - z)}{\|x_0 - z\|^2}.$$

□

4.3 Lebesgue Spaces

Let $(X, \mathcal{M}, \mu)$ be a measure space and $0 < p < \infty$. A measurable function f is called a *p-integrable function* if $|f|^p$ is integrable.

Proposition 4.10 (**Hölder's Inequality**). *Let f and g be measurable. For $1 < p < \infty$,*

$$\int_X |fg| \, d\mu \leq \left(\int_X |f|^p \, d\mu \right)^{\frac{1}{p}} \left(\int_X |g|^q \, d\mu \right)^{\frac{1}{q}},$$

where q is conjugate to p. Moreover, when the right hand side of this inequality is finite, equality sign holds if and only if either (a) f or g vanishes a.e. or (b) there are some $\alpha, \beta \geq 0$, $\alpha\beta \neq 0$, such that $\alpha |f|^p = \beta |g|^q$ a.e. .

Recall that for $p, q \in [1, \infty]$, p and q are conjugate if $1/p + 1/q = 1$.

Proposition 4.11 (**Minkowski's Inequality**). *Let f and g be measurable. For $1 \leq p < \infty$,*

$$\left(\int_X |f + g|^p \, d\mu \right)^{\frac{1}{p}} \leq \left(\int_X |f|^p \, d\mu \right)^{\frac{1}{p}} + \left(\int_X |g|^p \, d\mu \right)^{\frac{1}{p}}.$$

Minkowski's inequality could be deduced from Hölder's inequality, and Hölder's inequality follows from Young's inequality or Jensen's inequality. You are referred to [R] for the proofs of these inequalities.

Let

$$L^p(X, \mathcal{M}, \mu) = \{\text{all } p\text{-integrable functions on } (X, \mathcal{M}, \mu)\}.$$

Letting

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{\frac{1}{p}}, \quad 1 \leq p < \infty,$$

it follows from Minkowski's inequality that

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p, \qquad f, g \in L^p(X, \mathcal{M}, \mu).$$

It is almost a norm except that $\|f\|_p = 0$ implies $f = 0$ almost everywhere but not necessarily everywhere. To obtain a normed space, we introduce an equivalence relation $\sim$ on $L^p(X, \mathcal{M}, \mu)$ by setting

$$f \sim g \text{ if and only if } f - g = 0 \text{ a.e.}$$

This equivalence relation partitions all L^p-functions into equivalent classes $\widetilde{f}$. Define

$$\|\widetilde{f}\|_p = \|f\|_p, \quad f \in \widetilde{f}.$$

Then $(\widetilde{L}^p(X, \mathcal{M}, \mu), \|\cdot\|_p)$ becomes a normed space. In practise, people do not

distinguish $\|\cdot\tilde{\|}_p$ from $\|\cdot\|_p$. They simply write $L^p(X, \mathcal{M}, \mu)$ or even $L^p(\mu)$ for $(\widetilde{L}^p(X, \mathcal{M}, \mu), \|\cdot\tilde{\|}_p)$.

Theorem 4.12. *Let $\{f_n\}$ be a Cauchy sequence in $L^p(\mu)$. There exists $f \in L^p(\mu)$ such that $\|f_n - f\|_p \to 0$ as $n \to \infty$. Consequently, $\widetilde{L}^p(\mu)$ is a Banach space for $1 \le p < \infty$.*

Proof. Let $\{f_n\}$ be a Cauchy sequence of p-integrable functions. For $\varepsilon = 1/2^j$, there exists n_j such that $\|f_n - f_m\|_p < 1/2^j \ \forall n, m \ge n_j$. We could choose n_j such that $n_j \uparrow \infty$. Then

$$\left\|f_{n_{j+1}} - f_{n_j}\right\|_p < \frac{1}{2^j}, \qquad \forall j \ge 1.$$

Pick a null set N_1 so that all f_{n_j} are finite in $X_1 = X \setminus N_1$. Set

$$g = \sum_{j=1}^{\infty} \left|f_{n_{j+1}} - f_{n_j}\right|$$

on X_1 and $g = 0$ in N_1. We have

$$\begin{aligned}
\int_X g^p \, d\mu &= \int_X \Big(\sum_{j=1}^{\infty} \big| f_{n_{j+1}} - f_{n_j} \big| \Big)^p \, d\mu \\
&= \int_X \Big(\lim_{k\to\infty} \sum_{j=1}^{k} \big| f_{n_{j+1}} - f_{n_j} \big| \Big)^p \, d\mu \\
&= \int_X \lim_{k\to\infty} \Big(\sum_{j=1}^{k} \big| f_{n_{j+1}} - f_{n_j} \big| \Big)^p \, d\mu \\
&\quad \text{(continuity of } z \mapsto z^p\text{)} \\
&= \lim_{k\to\infty} \int_X \Big(\sum_{j=1}^{k} \big| f_{n_{j+1}} - f_{n_j} \big| \Big)^p \, d\mu \\
&\quad \text{(monotone convergence theorem)} \\
&= \varliminf_{k\to\infty} \int_X \Big(\sum_{j=1}^{k} \big| f_{n_{j+1}} - f_{n_j} \big| \Big)^p \, d\mu \\
&\le \varliminf_{k\to\infty} \Big[\sum_{j=1}^{k} \Big(\int_X \big| f_{n_{j+1}} - f_{n_j} \big|^p \, d\mu \Big)^{\frac{1}{p}} \Big]^p \\
&\quad \text{(Minkowski inequality)} \\
&\le \varliminf_{k\to\infty} \Big(\sum_{j=1}^{\infty} \frac{1}{2^j} \Big)^p \\
&\le 1.
\end{aligned}$$

Hence $g \in L^p(\mu)$ and in particular is finite almost everywhere. Let $N_2 \equiv \{x \in X_1 : g(x) = \infty\}$ and $X_2 = X_1 \setminus N_2 = X \setminus (N_1 \cup N_2)$ and set

$$h_k = \sum_{j=1}^{k} \left(f_{n_{j+1}} - f_{n_j} \right).$$

For $x \in X_2$,

$$|h_k(x) - h_l(x)| \le \sum_{j=l+1}^{k} |f_{n_{j+1}} - f_{n_j}|(x) \to 0, \quad \text{as } k, l \to \infty,$$

since g is finite in X_2. We conclude that $\{h_k(x)\}$ is a numerical Cauchy sequence and hence

$$h = \sum_{j=1}^{\infty} \left(f_{n_{j+1}} - f_{n_j} \right)$$

is well-defined in X_2. We may set $h(x) = 0$ elsewhere so that it is a measurable function defined in X. From $|h_k| \leq g$ we also know that $h \in L^p(X)$ as a result of Lebesgue's dominiated convergence theorem. By the same reason, using $|h - h_k| \leq |h| + |h_k| \leq 2g$, we conclude $\|h - h_k\|_p \to 0$ as $k \to \infty$. It follows that $\|f - f_{n_{k+1}}\|_p = \|h - h_k\|_p \to 0$ as $k \to \infty$. (In fact, we also have $f_{n_{k+1}}(x) - f(x) = h_k(x) - h(x) \to 0$ at every $x \in X_2$, that is, the subsequence $\{f_{n_k}\}$ converges to f almost everywhere.) We conclude that $f_{n_k} \to f$ in L^p-norm and almost everywhere as $n_k \to \infty$.

Now, for $\varepsilon > 0$, there is some n_0 such that

$$\|f_n - f_m\|_p < \frac{\varepsilon}{2}, \quad \forall n, m \geq n_0.$$

On the other hand, there is some k_0 such that

$$\|f_{n_k} - f\|_p < \frac{\varepsilon}{2}, \quad \forall k \geq k_0.$$

We pick $n_k \geq n_0$ and $k \geq k_0$, then

$$\|f_n - f\|_p \leq \|f_n - f_{n_k}\|_p + \|f_{n_k} - f\|_p < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

whence $\{f_n\}$ converges to f in $L^p(\mu)$.

Finally, for a Cauchy sequence $\{\widetilde{f}_n\}$ in $\widetilde{L}^p(\mu)$, pick $f_n \in \widetilde{f}_n$. Then $\{f_n\}$ is a Cauchy sequence of L^p-integrable functions. From what we have just proved, there exists a L^p-integrable function f such that $\|f_n - f\|_p \to 0$. Then $\|\widetilde{f}_n - \widetilde{f}\|_p = \|f_n - f\|_p \to 0$ once we take $\widetilde{f}$ to be the equivalence class containing f. □

From the above proof we also have the following useful information.

Corollary 4.13. *If $f_n \to f$ in $L^p(\mu)$, there exists a subsequence $\{f_{n_k}\}$ converging to f a.e. .*

This result is already known for $p = 1$, see the last section in Chapter 1.

After ascertaining that $L^p(\mu)$ is a Banach space, we study the approximation properties of these spaces.

First, we consider the density of simple functions in $L^p(\mu)$. Recall that a simple function s is in the form

$$s(x) = \sum_{j=1}^{n} \alpha_j \chi_{E_j}(x), \quad \alpha_j \neq 0 \in \mathbb{R},\ E_j \text{ measurable.}$$

Let

$$\mathcal{S} = \{s : s \text{ is a simple function with } \mu(E_j) < \infty,\ \forall j = 1, \ldots, n\}.$$

Proposition 4.14. *The set $\{s_n\} \subset \mathcal{S}$ is dense in $L^p(\mu)$ for $p \in [1, \infty)$.*

Proof. By writing $f = f^+ - f^-$ and noting $|f|^p = |f^+|^p + |f^-|^p$, it suffices to prove the proposition for non-negative f. By Theorem 1.6, there exists a sequence of simple functions $\{s_k\} \uparrow f$. Using $0 \leq f - s_k \leq f$, we can apply the Lebesgue's dominated convergence theorem to get

$$\lim_{k\to\infty} \int |f - s_k|^p, d\mu = \int \lim_{k\to\infty} |f - s_k|^p, d\mu = 0.$$

It remains to verify $s_k \in \mathcal{S}, n \geq 1$, but this is evident. As $s_k = \sum_{j=1}^{n} \alpha_j \chi_{E_j}$ where $\alpha_j > 0$. As $s_k \leq f$, $s_k^p \leq |f|^p$ implies $\alpha_j^p \mu(E_j) \leq \int |f|^p \, d\mu < \infty$ for each j. So $\mu(E_j) < \infty$ and $s_k \in \mathcal{S}$ for all k. □

When the underlying space X has a topological structure, one should study the density of continuous functions. We have

Proposition 4.15. *Let X be a locally compact Hausdorff space and μ a Riesz measure on X. Then $C_c(X)$ is dense in $L^p(\mu)$ for $1 \leq p < \infty$.*

Proof. Using Proposition 4.13 and the definition of $\mathcal{S}$, it suffices to show that for every measurable E, $\mu(E) < \infty$, we can find a sequence of continuous functions $\{\varphi_j\} \subset C_c(X)$ such that $\varphi_j \to \chi_E$ in L^p-norm. By the outer and inner regularity of E, there are descending open sets $\{G_j\} \downarrow E$ and ascending compact sets $\{K_j\} \uparrow E$ with $\mu(G_1) < \infty$. As X is locally compact Hausdorff, by Urysohn's lemma, for each j there is some $\varphi_j \in C_c(X)$, $\varphi_j \equiv 1$ on K_j, spt$\varphi_j \subset G_j$, $0 \leq \varphi_j \leq 1$ on X. Using $\varphi_j \leq \chi_{G_1}$ and $\mu(G_1) < \infty$, one can apply the Lebesgue's dominated convergencr theorem to conclude $\varphi_j \to \chi_E$ in $L^p(\mu)$. □

Proposition 4.16. *The space $L^p(\mathbb{R}^n)$ is separable for $1 \leq p < \infty$.*

Here $L^p(\mathbb{R}^n)$ stands for the L^p-space with respect to $\mathcal{L}^n$, the n-dimensional Lebesgue measure.

Proof. Let $\mathcal{P}_n$ be the collection of the restrictions of all polynomials with rational coefficients on the ball B_n centered at the origin (setting them to be zero outside the ball) and and $\mathcal{P} = \bigcup_n \mathcal{P}_n$. Then $\mathcal{P}$ is a countable subset in $L^p(\mathbb{R}^n)$. For $f \in L^p(\mathbb{R}^n)$, given $\varepsilon > 0$, there exists some $g \in C_c(\mathbb{R}^n)$ such that $\|f - g\|_p < \varepsilon/2$. Let n be large so that spt$g \subset B_n$. By Weierstrass approximation theorem, there is a polynomial $p \in \mathcal{P}_n$ such that $\|g - p\|_\infty < \varepsilon/2 \, |\mathcal{L}^n(B_n)|^{1/q}$. Using

$$\|g - p\|_p = \left(\int |g - p|^p \, d\mathcal{L}^n \right)^{\frac{1}{p}} \leq |\mathcal{L}^n(B_n)|^{\frac{1}{q}} \, \|g - p\|_\infty ,$$

we have

$$\begin{aligned}\|f-p\|_p &\le \|f-g\|_p + \|g-p\|_p \\ &< \frac{\varepsilon}{2} + |\mathcal{L}^n(B_n)|^{\frac{1}{q}} \frac{\varepsilon}{2\,|\mathcal{L}^n(B_n)|^{\frac{1}{q}}} \\ &= \varepsilon.\end{aligned}$$

□

4.4 The Dual Space of $L^p(\mu)$

We would like to determine the dual space of $L^p(\mu)$ for $p \in (1,\infty)$. Let $g \in L^q(\mu)$ where q is conjugate to p. We define

$$\Lambda_g f = \int gf\, d\mu, \quad f \in L^p(\mu).$$

By Hölder inequality,

$$\begin{aligned}|\Lambda_g f| &\le \left|\int gf\, d\mu\right| \\ &\le \int |gf|\, d\mu \\ &\le \|g\|_q \|f\|_p, \qquad \forall f \in L^p(\mu).\end{aligned}$$

Hence, $\Lambda_g \in L^p(\mu)'$ and

$$\|\Lambda_g\| \le \|g\|_q$$

holds. On the other hand, the function $f_1 = |g|^{q-2} g$ satisfies

$$\int |f_1|^p\, d\mu = \int |g|^{(q-1)p}\, d\mu = \int |g|^q\, d\mu < \infty,$$

so $f_1 \in L^p(\mu)$. We have

$$|\Lambda_g f_1| \le \|\Lambda_g\| \|f_1\|_p,$$

that is,

$$\int |g|^q\, d\mu \le \|\Lambda_g\| \left(\int |g|^q\, d\mu\right)^{\frac{1}{p}},$$

which means

$$\|g\|_q \le \|\Lambda_g\|.$$

We conclude that the linear map $\Phi : L^q(\mu) \to L^p(\mu)'$ given by $g \mapsto \Lambda_g$ is norm-preserving. Note that norm-preserving implies that Φ is injective. We will show that it is also surjective. We have

Theorem 4.17. *Let $\Lambda \in L^p(\mu)', 1 < p < \infty$. There is a unique $g \in L^q(\mu)$ such that $\Lambda = \Lambda_g$. The correspondence between Λ and g sets up a norm-preserving linear bijection between the dual space of $\widetilde{L}^p(\mu)$ and $\widetilde{L}^q(\mu)$.*

To prepare for the proof of this theorem, we introduce a notion of uniform convexity in functional analysis. A normed space is called *uniformly convex* if for any two unit vectors satisfying $\|x - y\| \geq \varepsilon$ for some $\varepsilon \in (0, 1)$, there exists some $\theta \in (0, 1)$ depending on ε only such that $\|(x + y)/2\| \leq 1 - \theta$. This terminology comes from the shape of the unit sphere. For instance, the plane $\mathbb{R}^2$ in the Euclidean metric is uniformly convex, but is not in the maximal norm $\|(x, y)\|_\infty = \max\{|x|, |y|\}$, as its unit sphere is given by the boundary of the unit square $[-1, 1]^2$. Taking $(1, 1)$ and $(1, -1)$ on this unit sphere, $\|((1, 1)+(1, -1))/2\|_\infty = 1$. Every Hilbert space is uniformly convex, for, by the parallelogram law, for any two units vectors x and y,

$$\|x + y\|^2 + \|x - y\|^2 = 2\left(\|x\|^2 + \|y\|^2\right) = 4,$$

thus,

$$\left\|\frac{x + y}{2}\right\|^2 = 1 - \frac{1}{4}\|x - y\|^2 \leq 1 - \frac{\varepsilon^2}{4} < 1.$$

So, we can find some $\theta \in (0, 1)$ such that $\|(x + y)/2\| < 1 - \theta$. We will use uniform convexity in the following way: In case $\|(x + y)/2\| \to 1$, then $\|x - y\| \to 0$.

The projection property holds in a uniformly convex space. This is the crucial property that we will use.

Proposition 4.18. *Let X_1 be a proper, closed subspace of a uniformly convex Banach space X and x_0 a point lying outside X_1. There exists a unique $z \in X_1$ such that*

$$\|z - x_0\| \leq \|y - x_0\|, \qquad \forall y \in X_1.$$

Proof. Let $\{y_n\} \in X_1$ be minimizing the distance from x_0 to X_1, that is, $\|y_n - x_0\| \to d$ as $n \to \infty$. Noting that the distance d is positive and $\{y_n\}$ is bounded, we have

$$\begin{aligned}
& \liminf_{n\to\infty} \left\| \frac{y_n - x_0}{\|y_n - x_0\|} + \frac{y_m - x_0}{\|y_m - x_0\|} \right\| \\
\geq\ & \liminf_{n\to\infty} \left\| \left\| \frac{y_n - x_0}{\|y_n - x_0\|} + \frac{y_m - x_0}{\|y_m - x_0\|} - \left(\frac{y_n - x_0}{d} + \frac{y_m - x_0}{d} \right) \right\| - \left\| \frac{y_n - x_0}{d} + \frac{y_m - x_0}{d} \right\| \right\| \\
\geq\ & \frac{2}{d} \liminf_{n\to\infty} \|\frac{1}{2}(y_n + y_m)\| \\
\geq\ & 2,
\end{aligned}$$

after using the fact that $(y_n + y_m)/2 \in X_1$ and $\|(y_n + y_m)/2\| \geq d$. In other words,

we have

$$\liminf_{n\to\infty} \Big\| \frac{1}{2}\Big(\frac{y_n - x_0}{\|y_n - x_0\|} + \frac{y_m - x_0}{\|y_m - x_0\|} \Big) \Big\| \geq 1.$$

On the other hand, clearly we have

$$\Big\| \frac{1}{2}\Big(\frac{y_n - x_0}{\|y_n - x_0\|} + \frac{y_m - x_0}{\|y_m - x_0\|} \Big) \Big\| \leq 1,$$

so

$$\lim_{n,m\to\infty} \left\| \frac{1}{2} \left(\frac{y_n - x_0}{\|y_n - x_0\|} + \frac{y_m - x_0}{\|y_m - x_0\|} \right) \right\| = 1 \ .$$

By uniform convexity,

$$\begin{aligned} \lim_{n,m\to\infty} \|y_n - y_m\| &= d \lim_{n,m\to\infty} \left\| \frac{y_n - x_0}{d} - \frac{y_m - x_0}{d} \right\| \\ &= d \lim_{n,m\to\infty} \left\| \frac{y_n - x_0}{\|y_n - x_0\|} - \frac{y_m - x_0}{\|y_m - x_0\|} \right\| \\ &\to 0, \end{aligned}$$

as $n, m \to \infty$, that is, $\{y_n\}$ is a Cauchy sequence. By the completeness of X_1, $y_n \to z \in X_1$ for some $z \in X_1$ and $\|z - x_1\| = d$.

If there exists another $z' \in X_1$ satisfying $\|z' - x_0\| = d$, let $\|z - z'\| \geq \varepsilon$ for some $\varepsilon > 0$. For some $\theta \in (0, 1)$ determined by ε,

$$\left\| \frac{1}{2}\frac{z' - x_0}{d} + \frac{1}{2}\frac{z - x_0}{d} \right\| = \left\| \frac{\frac{1}{2}(z + z') - x_0}{d} \right\| < 1 - \theta.$$

That means the distance between x_0 and $(z + z')/2 \in X_1$ is strictly less than d, but this is impossible.

□

Theorem 4.19. *The space $L^p(\mu)$, $1 < p < \infty$, is uniformly convex. Consequently, it satisfies the projection property.*

This theorem relies on two inequalities.

Proposition 4.20 (**Clarkson's Inequalities**)**.**
Let f and g be in $L^p(\mu)$. We have

(a) For $p \geq 2$,

$$\left\| \frac{f + g}{2} \right\|_p^p + \left\| \frac{f - g}{2} \right\|_p^p \leq \frac{1}{2} \left(\|f\|_p^p + \|g\|_p^p \right).$$

(b) For $p \in (1, 2)$,

$$\|f + g\|_p^q + \|f - g\|_p^q \leq 2 \left(\|f\|_p^p + \|g\|_p^p \right)^{q-1},$$

where p and q are conjugate.

Now we prove Clarkson's inequalities.

Proof. (a) We note that (a) follows from integrating the following elementary inequality

$$\left|\frac{x+y}{2}\right|^p + \left|\frac{x-y}{2}\right|^p \le \frac{1}{2}\left(|x|^p + |y|^p\right), \quad p \ge 2,\ x, y \in \mathbb{R},$$

after substituting x and y by $f(x)$ and $g(x)$ respectively. Again, using the symmetry in this inequality, it suffices to show it under $x > y > 0$. Setting $z = y/x$, we further reduce to

$$\left(\frac{1+z}{2}\right)^p + \left(\frac{1-z}{2}\right)^p \le \frac{1}{2}(1+z^p), \quad z \in [0,1].$$

Let

$$\varphi(z) = \left(\frac{1+z}{2}\right)^p + \left(\frac{1-z}{2}\right)^p - \frac{1}{2}(1+z^p).$$

Then $\varphi(0) = 2^{-p} - 2^{-1} < 0$, $\varphi(1) = 1 - 1 = 0$. If $\varphi'(z) \ge 0$, then $\varphi(z) \le 0$ on $[0,1]$ and we are done. We compute

$$\begin{aligned}\varphi'(z) &= \frac{p}{2^p}\left[(1+z)^{p-1} - (1-z)^{p-1} - 2^{p-1}z^{p-1}\right] \\ &= \frac{p}{2^p}z^{p-1}\left[(w+1)^{p-1} - (w-1)^{p-1} - 2^{p-1}\right], \quad w = \frac{1}{z} \in [1,\infty).\end{aligned}$$

Let $h(w) = (w+1)^{p-1} - (w-1)^{p-1} - 2^{p-1}$. Then $h(1) = 0$ and $h'(w) \ge 0$ (easily seen after using $p \ge 2$), so $h(w) \ge 0$ and $\varphi'(z) \ge 0$. We have established (a).
(b) We need another inequality

$$|x+y|^q + |x-y|^q \le 2\left(|x|^p + |y|^p\right)^{q-1}, \quad x, y \in \mathbb{R}. \tag{4.3}$$

As before it reduces to

$$(1+z)^q + (1-z)^q \le 2(1+z^p)^{q-1}, \quad z \in [0,1].$$

Let

$$f(\alpha, z) = (1+\alpha^{1-q}z)(1+\alpha z)^{q-1} + (1-\alpha^{1-q}z)(1-\alpha z)^{q-1}.$$

Then $f(1,z) = (1+z)^q + (1-z)^q$ and $f(z^{p-1}, z) = 2(1+z^p)^{q-1}$. As $z^{p-1} < 1$, it

suffices to show $\partial f/\partial\alpha(\alpha, z) \le 0$ for $z, \alpha \in (0,1)$. We have

$$\begin{aligned}
\frac{\partial f}{\partial \alpha}(\alpha, z) &= (1-q)\alpha^{-q} z(1+\alpha z)^{q-1} + (1+\alpha^{1-q} z)(q-1)(1+\alpha z)^{q-2} z \\
&\quad - (1-q)\alpha^{-q} z(1-\alpha z)^{q-1} - (1-\alpha^{1-q} z)(1-\alpha z) z(q-1) \\
&= (1-q) z \left[\alpha^{-q}(1+\alpha z) - 1 - \alpha^{1-q} z\right](1+\alpha z)^{q-2} \\
&\quad - (1-q) z \left[\alpha^{-q}(1-\alpha z) - 1 + \alpha^{1-q} z\right](1-\alpha z)^{q-2} \\
&= (1-q) z(\alpha^{-q} - 1)\left[(1+\alpha z)^{q-2} - (1-\alpha z)^{q-2}\right] \\
&\le 0.
\end{aligned}$$

Using $q > 2$ and $\alpha \in (0,1)$, (4.3) holds. However, (b) doesn't come from (4.3) by integration. We need one more result, namely, for $0 < p < 1$ and non-negative f, g,

$$\|f + g\|_p \ge \|f\|_p + \|g\|_p . \tag{4.4}$$

We leave the proof of (4.4) as an exercise. Now,

$$\|f\|_p^q = \left(\int |f|^p \, d\mu\right)^{\frac{q}{p}} = \left(\int |f|^{q(p-1)} \, d\mu\right)^{\frac{1}{p-1}} = \||f|^q\|_{p-1} ,$$

and

$$\begin{aligned}
\|f+g\|_p^q + \|f-g\|_p^q &= \||f+g|^q\|_{p-1} + \||f-g|^q\|_{p-1} \\
&\le \||f+g|^q + |f-g|^q\|_{p-1} \quad \text{(use } 0 < p-1 < 1 \text{ and (4.4))} \\
&= \left[\int (|f+g|^q + |f-g|^q)^{p-1} \, d\mu\right]^{\frac{1}{p-1}} \\
&\le 2\left[\int (|f|^p + |g|^p)^{(q-1)(p-1)} \, d\mu\right]^{\frac{1}{p-1}} \quad \text{(by (4.3))} \\
&= 2\left(\int |f|^p \, d\mu + \int |g|^p \, d\mu\right)^{\frac{1}{p-1}} \\
&= 2\left(\|f\|_p^p + \|g\|_p^p\right)^{q-1} ,
\end{aligned}$$

and (b) follows after noting $(p-1)(q-1) = 1$.

□

Proof of Theorem 4.16. It remains to show that Φ is onto, i.e., for $\Lambda \in L^p(\mu)'$, there is some $g \in L^q(\mu)$ such that

$$\Lambda f = \int fg \, d\mu, \quad \forall f \in L^p(\mu).$$

We fix some $f_1 \in L^p(\mu) \setminus \ker \Lambda$. As $L^p(\mu)$ is uniformly convex, there is an

$h_0 \in \ker \Lambda$ such that

$$\|h_0 - f_1\|_p \leq \|f - f_1\|_p, \quad \forall f \in \ker \Lambda.$$

That is to say,

$$\varphi(t) = \|h_0 + tf - f_1\|_p^p$$

attains its minimum at $t = 0$ and $\varphi'(0) = 0$ (the differentiability of φ is left as an exercise). We have

$$0 = \varphi'(0) = p \int |h_0 - f_1|^{p-2} (h_0 - f_1) f \, d\mu, \quad \forall f \in \ker \Lambda.$$

Letting $g_1 = |h_0 - f_1|^{p-2} (h_0 - f_1)$, we've

$$\int |g_1|^q \, d\mu = \int |h_0 - f_1|^{(p-1)q} = \int |h_0 - f_1|^p \, d\mu < \infty,$$

so $g_1 \in L^q(\mu)$ and

$$\int g_1 f \, d\mu = 0, \quad \forall f \in \ker \Lambda. \tag{4.5}$$

On the other hand, recall that $h_0 - f_1 \notin \ker \Lambda$, so for every $f \in L^p(\mu)$, by the codimension one property of Λ we can find α and $f_2 \in \ker \Lambda$ such that

$$f = \alpha(h_0 - f_1) + f_2, \quad \alpha = \frac{\Lambda f}{\Lambda(h_0 - f_1)}.$$

Multiply both sides by g_1 and using (4.5) to get

$$\begin{aligned} \int f g_1 \, d\mu &= \alpha \int (h_0 - f_1) g_1 \, d\mu + \int f_2 g_1 \, d\mu \\ &= \alpha \int |h_0 - f_1|^p \, d\mu, \end{aligned}$$

i.e.,

$$\Lambda f = \frac{\Lambda(h_0 - f_1)}{\|h_0 - f_1\|_p^p} \int f g_1 \, d\mu, \quad \forall f \in L^p.$$

So

$$\Lambda f = \int f g \, d\mu$$

where

$$g = \frac{\Lambda(h_0 - f_1)}{\|h_0 - f_1\|_p^p} \, g_1.$$

□

So far we have left out the dual space of $L^1(\mu)$, which is given by $L^\infty(\mu)$. Now we discuss it. A μ-measurable function in $(X, \mathcal{M}, \mu)$ is called *essentially bounded* if there exist some M and a measure zero set N such that $|f(x)| \leq M$, $\forall x \in X \setminus N$. The collection of all essentially bounded functions form a vector space. For an essentially bounded function, let its *essential supremum norm* be

$$\|f\|_\infty = \inf \{\beta : \mu \{x : |f(x)| \geq \beta\} = 0\}.$$

One can show that $(L^\infty(\mu), \|\cdot\|_\infty)$ forms a Banach space after identifying f and g if they are different at a set of measure zero. The space $L^\infty(\mu)$ is not as nice as $L^p(\mu)$ for $p \in (1, \infty)$. For example, in the case of the Lebesgue measure, $L^\infty(\mathbb{R}^n, \mathcal{L}^n)$ is not separable and continuous functions are not dense in it. Nevertheless, it still comes up as the dual space of $L^1(\mu)$ for a σ-finite measure μ including the n-dimensional Lebesgue measure. Duality may not hold when the measure is not σ-finite, see [R]. On the other hand, it is clear that every function in $L^1(\mu)$ induces a bounded linear functional on n$L^\infty(\mu)$, but the dual space of $L^\infty(\mu)$ is in general larger and difficult to describe. Fortunately, it seldom comes up in applications, see [HS] for further information.

Theorem 4.21. *Let $(X, \mathcal{M}, \mu)$ be σ-finite. Then for $\Lambda \in L^1(\mu)'$, there exists some $g \in L^\infty(\mu)$ such that $\Lambda = \Lambda_g$.*

I let you provide a proof of this theorem.

4.5 Strong Convergence of L^p-sequences

Results concerning the L^1-convergence of sequences of functions are discussed in Chapter 1. Here we present two results on the convergence of L^p-functions.

Proposition 4.22 (**Brezis-Lieb Lemma**). *Consider $L^p(\mu)$, $1 \leq p < \infty$. Suppose that $\|f_n\|_p \leq M$ and $f_n \to f$ a.e.. Then*

$$\lim_{n\to\infty} \int | \, |f_n|^p - |f_n - f|^p - |f|^p \, | d\mu = 0.$$

Proof. Note the elementary inequality (see below), for $p > 1$,

$$| \, |a+b|^p - |a|^p \, | \leq \varepsilon |a|^p + C_\varepsilon |b|^p, \quad a, b \in \mathbb{R}.$$

Taking $g_n = f_n - f$ as a and f as b,

$$| \, |f + g_n|^p - |g_n|^p \, | \leq \varepsilon |g_n|^p + C_\varepsilon |f|^p,$$

or,

$$-\varepsilon |g_n|^p - C_\varepsilon |f|^p \leq |f + g_n|^p - |g_n|^p \leq \varepsilon |g_n|^p + C_\varepsilon |f|^p.$$

we have

$$-\varepsilon\,|g_n|^p - (C_\varepsilon + 1)\,|f|^p \le |f+g_n|^p - |g_n|^p - |f|^p \le \varepsilon\,|g_n|^p + (C_\varepsilon - 1)\,|f|^p\,,$$

which implies

$$|\,|f+g_n|^p - |g_n|^p - |f|^p\,| \le \varepsilon\,|g_n|^p + (1 + C_\varepsilon)\,|f|^p\,,$$

and

$$(|\,|f+g_n|^p - |g_n|^p - |f|^p\,| - \varepsilon\,|g_n|^p\,)_+ \le (1 + C_\varepsilon)\,|f|^p\,.$$

By assumption the function $\Phi_n \equiv ||f+g_n|^p - |g_n|^p - |f|^p| \to 0$. By Lebesgue's dominated convergence theorem,

$$\lim_{n\to\infty} \int (|\,|f+g_n|^p - |g_n|^p - |f|^p\,| - \varepsilon\,|g_n|^p)_+ \, d\mu = 0.$$

Using

$$\int (F-G) = \int (F-G)_+ - \int (F-G)_- \le \int (F-G)_+,$$

we have

$$\overline{\lim_{n\to\infty}} \int |\,|f+g_n|^p - |g_n|^p - |f|^p\,|\,d\mu \le \varepsilon\, \underline{\lim_{n\to\infty}} \int |g_n|^p \, d\mu.$$

As

$$\begin{aligned}\int |g_n|^p \, d\mu = \int |f_n - f|^p \, d\mu &\le 2^p \left(\int |f_n|^p \, d\mu + \int |f|^p \, d\mu \right) \\ &\le 2^{p+1} M^p,\end{aligned}$$

where Fatou's lemma is used in the last step, we conclude that

$$\overline{\lim_{n\to\infty}} \int |\,|f+g_n|^p - |g_n|^p - |f|^p\,|\,d\mu \le 2^{p+1} M^p \varepsilon,$$

and the desired result follows by letting $\varepsilon \downarrow 0$. □

I leave the case $p = 1$ as exercise. Note that this proposition tells us what is missing in Fatou's lemma.

We provide a proof of the elementary inequality used in the proof above. We have

$$\begin{aligned}|a+b|^p - |a|^p &= \int_0^1 \frac{d}{dt}\,|a+tb|^p \, dt \\ &= p \int_0^1 |a+tb|^{p-2}\,(a+tb)b\,dt\ .\end{aligned}$$

Therefore,

$$\begin{aligned} ||a+b|^p - |a|^p| &\le p\int_0^1 (|a|+|b|)^{p-1}\,|b|\;dt \\ &\le p2^{p-1}\left(|a|^{p-1}+|b|^{p-1}\right)|b| \\ &\le p2^{p-1}\,|a|^{p-1}\,|b| + p2^{p-1}\,|b|^p \\ &\le \varepsilon\,|a|^p + C_\varepsilon\,|b|^p\,, \end{aligned}$$

where in the last step we have used Young's inequality

$$xy \le \frac{x^\alpha}{\alpha} + \frac{y^\beta}{\beta}, \quad \frac{1}{\alpha}+\frac{1}{\beta}=1,\ x,y\ge 0$$

taking $x = |a|^{p-1}, y = |b|$ and suitable α, β

Next we have the Vitali convergence theorem. This theorem is used when a dominator is not available so Lebesgue's dominated convergence theorem cannot be applied.

Theorem 4.23 (**Vitali's Convergence Theorem**). *Let $\mu(X) < \infty$ and $\{f_n\} \subset L^1(\mu)$ satisfy*

1. *$f_n \to f$ a.e.,*
2. *f_n and f are finite a.e., and*
3. *$\{f_n\}$ is uniformly integrable.*

Then $f_n \to f$ in $L^1(\mu)$.

A sequence (or a set) of integrable functions is called *uniformly integrable* if for every $\varepsilon > 0$, there exists some δ such that

$$\int_E |f_n|\;d\mu < \varepsilon \qquad \forall n \text{ whenever } E \text{ is measurable with } \mu(E) < \delta.$$

Proof. By (3), there exists a $\delta \le \varepsilon$ such that the above estimate holds. For this ε, (1), (2) and $\mu(X) < \infty$ enable us to apply Egorov theorem to find some E_1, $\mu(E_1) < \varepsilon$, such that $f_n \to f$ uniformly on $X \setminus E_1$, that is,

$$\lim_{n\to\infty} \|f_n - f\|_{L^\infty(X\setminus E_1)} = 0.$$

It follows that

$$\begin{aligned} \int |f_n - f|\;d\mu &= \int_{X\setminus E_1} |f_n - f|\;d\mu + \int_{E_1} |f_n - f|\;d\mu \\ &\le \mu(X\setminus E_1)\,\|f_n - f\|_{L^\infty(X\setminus E_1)} + 2\varepsilon. \end{aligned}$$

Note that $\int_{E_1} |f|\, d\mu \le \varepsilon$ by Fatou's lemma. So

$$\begin{aligned}\overline{\lim_{n\to\infty}} \int |f_n - f|\, d\mu &\le \mu(X \setminus E_1)\, \overline{\lim_{n\to\infty}} \|f_n - f\|_{L^\infty(X\setminus E_1)} + 2\varepsilon \\ &= 2\varepsilon,\end{aligned}$$

and

$$\overline{\lim_{n\to\infty}} \int |f_n - f|\, d\mu = 0.$$

□

4.6 Weak Convergence of L^p-sequences

Weak convergence is a concept in a normed space. We call a sequence $\{x_n\}$ *converges weakly* to x in the normed space X, written as $x_n \rightharpoonup x$, if $\Lambda x_n \to \Lambda x$ for all $\Lambda \in X'$. The usual convergence in norm is sometimes called *strong convergence* in contrast with this new notion of convergence. It is clear that $x_n \to x$ implies $x_n \rightharpoonup x$ but the converse is not always true. In fact, for each $\Lambda \in X'$,

$$|\Lambda x_n - \Lambda x| = |\Lambda(x_n - x)| \le \|\Lambda\|\, \|x_n - x\| \to 0$$

as $x_n \to x$ in norm. The weak limit is also unique, for, if $x_n \rightharpoonup x_1$ and $x_n \rightharpoonup x_2$, then $z = x_1 - x_2$ satisfies $\Lambda z = 0\ \forall z \in X$. As a consequence of Hahn-Banach theorem, there always exists a bounded linear functional Λ_1, satisfying $\Lambda_1 z = \|z\|$ (for every given z), $\|z\| = \Lambda_1 z = 0$, and $x_1 = x_2$.

When adapted to the L^p-setting, by duality we know that $f_n \rightharpoonup f$ in $L^p(\mu)$, $1 < p < \infty$, if and only if

$$\int f_n g\, d\mu \to \int f g\, d\mu, \quad \forall g \in L^q(\mu).$$

It is also valid for $p = 1$ when μ is σ-finite.

Example 4.1. Let f_n be a continuous function in $[0,1]$ whose support $[a_n, b_n]$ shrinks to $x_0 \in (0,1)$ as $n \to \infty$. We also require $\|f_n\|_p = 1$. Then $f_n \rightharpoonup 0$ in $L^p(0,1)$. For, as all L^q-functions vanishing near x_0 forms a dense subset of $L^q(0,1)$ and $\int f_n g\, dx = 0$ for such g, we have $f_n \rightharpoonup 0$ in $L^p(0,1)$, $1 < p < \infty$. On the other hand, ais $\|f_n - 0\|_p = \|f_n\|_p = 1$, $\{f_n\}$ does not converge to 0 strongly.

Example 4.2. Let $f_n(x) = \sin 2n\pi x$, $x \in [0,1]$. Then $f_n \rightharpoonup$ in $L^p(0,1)$ for $1 \le p < \infty$. To see this, let $g \in C[0,1]$ first. We need to show that

$$\int_0^1 \sin 2n\pi x g(x)\, dx \to 0 \quad \text{as } n \to \infty.$$

We have

$$\begin{aligned}\int_0^1 \sin 2n\pi x g(x)\,dx &= \int_0^{2n\pi} \sin y g\left(\frac{y}{2n\pi}\right)\frac{dy}{2n\pi}\\ &= \frac{1}{2n\pi}\sum_{k=1}^{n}\int_{2(k-1)\pi}^{2k\pi} \sin y g\left(\frac{y}{2n\pi}\right)\,dy\\ &= \frac{1}{2n\pi}\sum_{k=1}^{n}\int_0^{2\pi} \sin z g\left(\frac{k-1}{n}+\frac{z}{2n\pi}\right)\,dz.\end{aligned}$$

As $g \in C[0,1]$ is uniformly continuous, for every $\varepsilon > 0$, there is some δ such that $|g(x) - g(y)| < \varepsilon$ if $|x - y| < \delta$. So, for all n large such that $\left|\frac{z}{2n\pi}\right| < \delta$, we have

$$\begin{aligned}&\left|\int_0^{2\pi} \sin z g\left(\frac{k-1}{n}+\frac{z}{2n\pi}\right)\,dz\right|\\ &= \left|\int_0^{2\pi} \sin z\left(g\left(\frac{k-1}{n}+\frac{z}{2n\pi}\right) - g\left(\frac{k-1}{n}\right)\right)\,dz\right| \quad \left(use \int_0^{2\pi}\sin z\,dz = 0\right)\\ &\le 2\pi\varepsilon\ .\end{aligned}$$

We have

$$\left|\int_0^1 \sin 2n\pi x g(x)\,dx\right| \le \frac{2\pi\varepsilon \times n}{2n\pi} = \varepsilon,$$

and the desired result holds.

As $C[0,1]$ is dense in $L^q[0,1]$, for $g \in L^q(0,1)$, and $\varepsilon > 0$, we can find a $g_1 \in C[0,1]$ such that $\|g - g_1\|_q < \varepsilon$. Then

$$\begin{aligned}\left|\int_0^1 \sin 2n\pi x g(x)\,dx\right| &\le \left|\int_0^1 \sin 2n\pi x (g-g_1)(x)\,dx\right| + \left|\int_0^1 \sin 2n\pi x g_1(x)\,dx\right|\\ &\le \|\sin 2n\pi x\|_p \|g - g_1\|_q + \left|\int_0^1 \sin 2n\pi x g_1(x)\,dx\right|.\end{aligned}$$

As $|\sin 2n\pi x| \le 1$,

$$\left|\int_0^1 \sin 2n\pi x g(x)\,dx\right| \le \|g - g_1\|_q \left|\int_0^1 \sin 2n\pi x g(x)\,dx\right|.$$

It follows that

$$\begin{aligned}\overline{\lim_{n\to\infty}}\left|\int_0^1 \sin 2n\pi x g(x)\,dx\right| &\le \varepsilon + \overline{\lim_{n\to\infty}}\left|\int_0^1 \sin 2n\pi x g_1(x)\,dx\right|\\ &\le \varepsilon.\end{aligned}$$

We conclude that $\{\sin 2n\pi x\}$ converges weakly to 0 as $n \to \infty$. Clearly the

convergence is not strong.

We examine the properties of weakly convergent L^p-sequences.

Proposition 4.24. *Let $\{f_n\}$ be a weakly convergent sequence in some $L^p(\mu)$, $1 \le p \le \infty$. Then $\{f_n\}$ is bounded in L^p-norm.*

This is in fact contained in a general result in functional analysis. Let $\{x_n\}$ be a weakly convergent sequence in the normed space X. It is always true that there is some M such that $\|x_n\| \le M$ for all n. This is called the uniform boundedness principle. In the following we give a proof which applies to $L^p(\mu)$ for $1 < p < \infty$. The cases $p = 1, \infty$ are left as exercises. If you have learnt the uniform boundedness principle, you may skip the following proof, for the proof of the general case is more transparent than this special one.

Proof. Suppose $f_n \rightharpoonup f$ for some f but $\{f_n\}$ is unbounded. By throwing away other f_n's and relabeling the indices, we may assume $\|f_n\|_p \ge 4^n$ for all n. Let $\alpha_n \in (0, 1]$ be such that $\widetilde{f}_n = \alpha_n f_n$ satisfies $\|\widetilde{f}_n\|_p = 4^n$. Let $h_n = |\widetilde{f}_n|^{p-1}\widetilde{f}_n / \|\widetilde{f}_n\|^{p-1}$. It is readily checked that $\|h_n\|_q = 1$. We let

$$h = \sum_{j=1}^{\infty} \frac{1}{3^j} \sigma_j h_j,$$

where $\sigma_j \in \{-1, 1\}$ are to be chosen later. We claim that $h \in L^q(\mu)$. For, looking at the difference of the partial sums

$$\begin{aligned} \left\| \sum_{j=1}^{n} \frac{1}{3^j} \sigma_j h_j - \sum_{j=1}^{m} \frac{1}{3^j} \sigma_j h_j \right\|_q &= \left\| \sum_{m+1}^{n} \frac{1}{3^j} \sigma_j h_j \right\|_q \\ &\le \sum_{j=m+1}^{n} \frac{1}{3^j} \|h_j\|_q \\ &= \frac{1}{2} \times \frac{1}{3^m} \to 0 \quad \text{as } m \to \infty. \end{aligned}$$

Thus the partial sums converges in L^q-norm to h. Let Λ_1 be the bounded linear functional on $L^p(\mu)$ induced by h,

$$\Lambda_1 g = \int gh \, d\mu, \quad g \in L^p(\mu).$$

We have

$$\begin{aligned}\Lambda_1 \widetilde{f}_n &= \int \widetilde{f}_n h \, d\mu \\ &= \sum_{j=1}^{\infty} \frac{1}{3^j} \sigma_j \int h_j \widetilde{f}_n \, d\mu \\ &= \sum_{j=1}^{n} \frac{1}{3_j} \sigma_j \int h_j \widetilde{f}_n \, d\mu + \sum_{n+1}^{\infty} \frac{1}{3^j} \sigma_j \int h_j \widetilde{f}_n \, d\mu.\end{aligned}$$

The second term in the right hand side is controlled by

$$\begin{aligned}\sum_{n+1}^{\infty} \frac{1}{3^j} \sigma_j \int h_j \widetilde{f}_n \, d\mu &\le \sum_{n+1}^{\infty} \frac{1}{3^j} \left\| h_j \right\|_q \left\| \widetilde{f}_n \right\| \\ &= \sum_{n+1}^{\infty} \frac{1}{3^j} 4^n = \frac{1}{2} \left(\frac{4}{3} \right)^n.\end{aligned}$$

Now, we set $\sigma_1 = 1$ and take σ_n (after $\sigma_2, \ldots, \sigma_{n-1}$ are fixed) to be 1 or -1 such that $\displaystyle\sum_{j=1}^{n-1} \frac{1}{3^j} \sigma_j \int h_j \widetilde{f}_n \, d\mu$ and $\displaystyle\sigma_n \int h_n \widetilde{f}_n \, d\mu$ have the same sign. Then

$$\sum_{j=1}^{n} \frac{1}{3^j} \sigma_j \int h_j \widetilde{f}_n \, d\mu \ge \frac{1}{3^n} \int h_n \widetilde{f}_n \, d\mu.$$

Combining these two estimates, we get

$$\begin{aligned}|\Lambda_1 f_n| &\ge \frac{1}{3^n} \int \widetilde{f}_n h_n \, d\mu - \frac{1}{2} \left(\frac{4}{3} \right)^n \\ &= \frac{1}{3^n} \left\| \widetilde{f}_n \right\|_p - \frac{1}{2} \left(\frac{4}{3} \right)^n \\ &= \frac{1}{2} \left(\frac{4}{3} \right)^n \to \infty \qquad \text{as } n \to \infty.\end{aligned}$$

So

$$\Lambda_1 f_n = \frac{1}{\alpha_n} \Lambda_1 \widetilde{f}_n \to \infty \quad \text{as } n \to \infty, \text{ too,}$$

but this is in conflict with $\Lambda_1 f_n \to \Lambda_1 f$. We conclude that $\{f_n\}$ must be bounded. □

Next, we show that the L^p-norm is lower semicontinuous with respect to weak convergence.

Proposition 4.25. *Let $\{f_n\}$ be weakly convergent to f in $L^p(\mu)$, $1 \le p < \infty$.*

Then

$$\|f\|_p \leq \varliminf_{n\to\infty} \|f_n\|_p .$$

This proposition looks like Fatou's lemma, but be cautious that the assumption of almost everywhere convergence is now replaced by weak convergence.

Proof. Let $1 < p < \infty$. As $f \in L^p(\mu)$, the function $g = |f|^{p-2} f \in L^q(\mu)$. Let Λ_1 be the bounded linear functional induced by g. We have

$$\begin{aligned} \|f\|_p^p &= \Lambda_1 f \\ &= \lim_{n\to\infty} \Lambda_1 f_n \\ &\leq \|\Lambda_1\| \varliminf_{n\to\infty} \|f_n\|_p . \end{aligned}$$

Noting that by duality,

$$\|\Lambda_1\| = \|g\|_q = \|f\|_p^{\frac{p}{q}} ,$$

so

$$\|f\|_p^p \leq \|f\|_p^{\frac{p}{q}} \varliminf_{n\to\infty} \|f_n\|_p ,$$

and the result follows. When $p = 1$, use the function $g = \operatorname{sgn} f = f/|f|$ and argue similarly. □

This proposition is still valid for $p = \infty$ when μ is σ-finite. Supply a proof for yourself.

Proposition 4.26. *Let $\{f_n\}$ be weakly convergent to f in $L^p(\mu)$, $1 \leq p \leq \infty$. There is $\{g_n\}$ in which each g_n is a convex combinations of $\{f_n\}$ such that $g_n \to f$ in $L^p(\mu)$ strongly.*

By a convex combination we mean $g = \sum_j \lambda_j f_{n_j}$ where $\lambda_j \in (0, 1]$, $j = 1, \cdots, N$, for some N satisfying $\sum_{j=1}^N \lambda_j = 1$.

This result, called Mazur's theorem, is a general one. In fact, let $\{x_n\}$ be weakly convergent to x in some normed space X. Then x belongs to the closure of the convex hull of $\{x_n\}$. The convex hull of a set $S \subset X$ is

$$Co(S) = \left\{ y : y = \sum_j^n \lambda_j x_j,\ x_j \in S,\ \lambda_j \in (0, 1],\ \sum_j^n \lambda_j = 1 \text{ for some } n \right\} .$$

The convex hull of S is the smallest convex set containing S. It is not hard to show that its closure, $\overline{Co(S)}$, is also convex. Actually it is the smallest closed, convex set containing S.

In the following we give a proof tailored to $L^p(\mu)$ for $1 < p < \infty$. Recall that $L^p(\mu)$ is uniformly convex for p in this range. In a general uniformly convex

space X, let K be a closed, convex set in X and x_0 lie outside K. The projection property asserts that there exists $z \in K$ such that

$$\|z - x_0\| \leq \|x - x_0\|, \quad \forall x \in K.$$

The proof of this fact is the same as the special case where K is a closed subspace, see Proposition 4.18.

Proof. Let $K = \overline{Co(\{f_n\})}$. We want to show that $f \in K$. Assume on the contrary that $f \notin K$. Then by uniform convexity, we can find some $h \in K$ such that

$$\|h - f\|_p \leq \|g - f\|_p, \quad \forall g \in L^p(\mu).$$

That means the function

$$\varphi(t) = \|(1-t)h + tg - f\|_p^p, \quad g \in L^p(\mu),\ t \in [0,1],$$

has a minimum at $t = 0$. So

$$\varphi'(0) = p \int |h - f|^{p-2} (h - f)(g - h)\, d\mu \geq 0.$$

Taking $g = f_n \in K$,

$$\int |h - f|^{p-2} (h - f)(f_n - h)\, d\mu \geq 0.$$

Observing that $h - f \in L^p(\mu)$ so $|h - f|^{p-2}(h - f) \in L^q(\mu)$ and $f_n \rightharpoonup f$ in $L^p(\mu)$, we have

$$\int |h - f|^{p-2} (h - f) f_n\, d\mu \to \int |h - f|^{p-2} (h - f) f\, d\mu.$$

It follows that

$$\int |h - f|^{p-2} (h - f)(f - h)\, d\mu \geq 0,$$

that is,

$$\int |h - f|^p\, d\mu \leq 0$$

which forces $f = h \in K$, contradiction holds. □

A nice property that makes weak convergence so important is the following "weak Bolzano-Weierstrass property". Recall that the Bolzano-Weierstrass property asserts that every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Theorem 4.27. *Let $\{f_n\}$ be bounded in $L^p(\mu)$, $1 < p < \infty$. There exists a subsequence $\{f_{n_k}\}$ such that $f_{n_k} \rightharpoonup f$ for some f in $L^p(\mu)$.*

Proof. We prove this theorem by further assuming that $L^q(\mu)$ is separable. The general case needs more knowledge from functional analysis, see *any* book on functional analysis. See, for instance, chapter 10 in Peter Lax "Functional Analysis".

Let $\{g_k\}$ be a dense subset of $L^q(\mu)$. By taking a Cantor diagonal process, we can extract a subsequence $\{f_{n_k}\}$ from $\{f_n\}$ such that $\lim_{k\to\infty}\int f_{n_k} g_m \, d\mu$ exists for each g_m. For any $g \in L^q(\mu)$, we can find $g_m \to g$ in $L^q(\mu)$. Then

$$\begin{aligned}\left|\int f_{n_k} g_m \, d\mu - \int f_{n_k} g_\ell \, d\mu\right| &= \left|\int f_{n_k}(g_m - g_\ell)\, d\mu\right| \\ &\le \|f_{n_k}\|_p \, \|g_m - g_\ell\|_q \\ &\le M \, \|g_m - g_\ell\|_q ,\end{aligned}$$

where $\|f_n\|_p \le M$ by assumption. So $\int f_{n_k} g_m \, d\mu$ is a Cauchy sequence for each f_{n_k} and $\lim_{k\to\infty}\int f_{n_k} g \, d\mu$ exists for each $g \in L^q(\mu)$. We define

$$\Lambda g = \lim_{k\to\infty} \int f_{n_k} g \, d\mu.$$

It is readily checked that the definition is independent of the sequence $\{g_m\}$ approximating g and $|\Lambda g| \le M\|g\|_q$, so $\Lambda \in L^q(\mu)'$. By duality, there exists an $f \in L^p(\mu)$ such that

$$\int gf \, d\mu = \lim_{k\to\infty} \int g f_{n_k} \, d\mu,$$

that is,

$$f_{n_k} \rightharpoonup f \text{ in } L^p(\mu).$$

□

We point out that this theorem is not true when $p = 1$. Consider the bump functions in Example 4.1 ($p = 1$). We have $\|f_n\|_p = 1$, $\forall n$. If $f_{n_k} \rightharpoonup f$ for some function $f \in L^1(\mu)$, f must be 0. But $g \equiv 1 \in L^\infty(0,1)$ and $\int f_{n_k} g \, dx = \int f_{n_k} \, dx = 1$ does not tend to 0.

Theorem 4.28. *For $p \in (1,\infty)$, let $\{f_n\} \subset L^p(\mu)$ satisfy (i) $\|f_n\|_p \to \|f\|_p$ and (ii) $f_n \rightharpoonup f$ as $n \to \infty$. Then $f_n \to f$ in $L^p(\mu)$.*

You should compare this result with the corollary to Brezis-Lieb lemma, which states that under (i) $\|f_n\|_p \to \|f\|_p$ and (ii) $f_n \to f$ a.e., $f_n \to f$ in $L^p(\mu)$ for $1 \le p < \infty$.

The proof of this theorem depends on the uniform convexity of $L^p(\mu)$. Recall that X is uniformly convex if for any pair of unit vectors x and y, for every

$\varepsilon > 0$, there exists some $\theta \in (0,1)$ depending only on ε such that $\|x-y\| \geq \varepsilon \Rightarrow \left\|\frac{1}{2}(x+y)\right\| < 1-\theta$. When applying to $L^p(\mu)$, in order to show $\|f_n - f\|_p \to 0$, it suffices to show $\left\|\frac{1}{2}(f_n+f)\right\|_p \to 0$.

Proof. When $f = 0$ a.e., $\|f_n - 0\|_p \to \|0\| = 0$ is contained in (i). We assume $f \neq 0$.

As $f_n \rightharpoonup f$, $\frac{1}{2}(f_n+f) \rightharpoonup f$, by Proposition 4.24,

$$\|f\|_p \leq \varliminf_{n\to\infty} \left\|\frac{1}{2}(f_n+f)\right\|_p .$$

On the other hand,

$$\left\|\frac{1}{2}(f_n+f)\right\|_p \leq \frac{1}{2}\|f_n\|_p + \frac{1}{2}\|f\|_p .$$

By (ii),

$$\varlimsup_{n\to\infty} \left\|\frac{1}{2}(f_n+f)\right\|_p \leq \varlimsup_{n\to\infty} \left(\frac{1}{2}\|f_n\|_p + \frac{1}{2}\|f\|_p\right) = \|f\|_p .$$

It forces

$$\lim_{n\to\infty} \left\|\frac{1}{2}(f_n+f)\right\|_p = \|f\|_p .$$

Using (ii) and $\|f\|_p \neq 0$, it immediately implies

$$\lim_{n\to\infty} \left\|\frac{1}{2}\left(\|f_n\|_p + \|f\|_p\right)\right\|_p = 1.$$

By uniform convexity,

$$\left\|\|f_n\|_p - \|f\|_p\right\|_p \to 0.$$

By (ii) again, we arrive at $\|f_n - f\|_p \to 0$ as $n \to \infty$. □

Comments on Chapter 4. Basic knowledge on functional analysis is needed to have a good understanding of this chapter. Apart from [R] is our treatment on the dual space of the L^p-space. Here we use the notion of uniform convexity introduced by Clarkson (1936) who showed that L^p-spaces ($p \in (1,\infty)$) are uniformly convex by the inequalities that bear his name. Our proof of Theorem 4.17, which is taken from the web, is a direct extension of the arguments that establish the self-duality of the L^2-space, see also Lieb-Loss Analysis. Another proof can be found in [HS]. Comparing with the different approach in [R], it does not require the measure to be σ-finite. The standard proof found in [R] and many other books is due to von Neumann. It is elegant but, in my opinion, is too tricky.

We will see it in next chapter. It is nice to have more than one proofs for an important theorem.

A new ingredient is weak convergence. It is not covered in [R], but, in view of its importance in applications, I decide to include it here. Many results could be proved in the more general setting of Banach spaces other than the L^p-spaces. Not to get too much involved in functional analysis, I provide self-contained proofs here or there, and sometimes run into tedious arguments, for instance, the proof of Proposition 4.24, which is just a special case of the uniform boundedness principle. Those who have learnt functional analysis can simply ignore them.

Chapter 5

Radon-Nikodym Theorem

Signed measures come up in at least two occasions. First, for any non-negative μ-measurable function f, the map $E \mapsto \int_E f d\mu$ is a measure. When f changes sign, this map is still countably additive and it should be a "signed measure". Second, in Theorem 2.8 it is shown that every positive linear functional on $C_c(X)$ (X is a locally compact Hausdorff space) comes from a regular Borel measure. From the viewpoint of functional analysis, it is desirable to characterize the dual space of $C_c(X)$ as the space of "signed measures". We introduce signed measures in Section 1. Just like the absolute value of a function is non-negative, it is shown that the " absolute value " of a signed measure, its total variation, is a measure. In Section 2 we establish the important theorem of Radon-Nikodym and discuss how to decompose a measure or a signed measure into its absolute continuous and singular parts with respect to another measure. Then we use this theorem to establish the full Riesz representation theorem in Section 3. Weak* convergence of sequences of signed measures is discussed in Section 4. As an application of the Riesz representation theorem we give a characterization of weakly convergent L^1-sequences, part of the Dunford-Pettis theorem. Finally, as another application of the Riesz representation theorem, we prove Herglotz-Riesz theorem concerning the boundary trace of a non-negative harmonic function in Section 5.

5.1 Signed Measures

Consider $\mathcal{M}$ a σ-algebra on the non-empty set X. A map $\mu : \mathcal{M} \to \mathbb{R}$ is called a *signed measure* if it satisfies

$$\mu(E) = \sum_{j=1}^{\infty} \mu(E_j), \qquad \forall \text{ partitions } \{E_j\} \text{ of } E.$$

Here $\{E_j\}$ is called a (*measurable*) *partition* of E if they are mutually disjoint, measurable and $E = \cup_j E_j$. One immediately deduces that $\mu(\phi) = 0$ and $\mu(X)$ is finite for a signed measure. The latter makes even a non-negative signed measure

different from a measure, namely it must be a finite measure.

Given a signed measure on $(X, \mathcal{M})$, its *total variation* (*measure*) is given by

$$|\mu|(E) = \sup\{\sum_j |\mu(E_j)|: \qquad \forall \text{ partitions } \{E_j\} \text{ of } E.$$

Clearly $|\mu|(E_1) \leq |\mu|(E_2)$ if $E_1 \subset E_2$, both in $\mathcal{M}$.

Proposition 5.1. *The total variation of every signed measure on $(X, \mathcal{M})$ is a finite measure on $(X, \mathcal{M})$.*

Proof. We need to show

$$|\mu|(E) = \sum_{j=1}^{\infty} |\mu|(E_j),$$

whenever $\{E_j\}$ is a partition of E.

First we establish subaddivity. Let $\{A_k\}$ be a partition of E. We have

$$\begin{aligned}
\sum_k \left|\mu(A_k)\right| &= \sum_k \left|\mu\left(A_k \cap \bigcup_j E_j\right)\right| \\
&\leq \sum_k \sum_j |\mu(A_k \cap E_j)| \\
&= \sum_j \sum_k |\mu(A_k \cap E_j)| \qquad \left(\text{use } \sum_k \sum_j \alpha_{kj} = \sum_j \sum_k \alpha_{kj},\ \forall \alpha_{kj} \geq 0\right) \\
&\leq \sum_j |\mu|(E_j). \qquad \left(\{A_k \cap E_j\}_k \text{ is a partition of } E_j\right)
\end{aligned}$$

Taking supremum over all $\{A_k\}$,

$$|\mu|(E) \leq \sum_j |\mu|(E_j).$$

Next, we show the reverse inequality. If $|\mu|(E_j) = \infty$ for some j, then $|\mu|(E) \geq |\mu|(E_j) = \infty$, the inequality holds. Let $|\mu|(E_j) < \infty\ \forall j$. For every $\varepsilon > 0$, we can find a partition $\left\{E_k^j\right\}$ of E^j for each j such that

$$|\mu|(E_j) \leq \sum_k \left|\mu\left(E_k^j\right)\right| + \frac{\varepsilon}{2^j}.$$

Now, $\{E_k^j\}_{k,j}$ forms a partition of E. We have

$$\begin{aligned}
\sum_j |\mu|(E_j) &\leq \sum_j \sum_k \left|\mu\left(E_k^j\right)\right| + \varepsilon \\
&\leq |\mu|(E) + \varepsilon
\end{aligned}$$

and we get the desired inequality after letting $\varepsilon \to 0$.

Finally, we show that $|\mu|(X)$ is finite. If on the contrary $|\mu|(X) = \infty$. By the lemma below, we can break X into A_1 and B_1 such that $|\mu(A_1)|, |\mu(B_1)| \geq 1$. As $|\mu|(A_1 + |\mu|(B_1) = |\mu|(X)$, at least one of $\mu(A_1)$ and $\mu(B_1)$ has infinite measure. Assume $|\mu|(A_1) = \infty$. Apply the same argument to A_1 to get A_2, B_2 such that $|\mu(A_2)|, |\mu(B_2)| \geq 1$. Assume again that A_2 has infinite measure. Keep doing this we get $B_{j+1} \subset A_j$, so that $B_j \cap B_k = \phi$, $j \neq k$. Letting $B = \bigcup_j B_j$, we have

$$\mu(B) = \sum_{j=1}^{\infty} \mu(B_j).$$

As $\mu(B)$ is finite, this infinite series converges, and, in particular, $\mu(B_j) \to 0$, but this is in conflict with our construction that $|\mu(B_j)| \geq 1$ for all j. □

Lemma 5.2. *If $|\mu|(E) = \infty$ for some $E \in \mathcal{M}$, then there are disjoint measurable sets A, B in E satisfying $A \cup B = E$ and $|\mu(A)|, |\mu(B)| \geq 1$.*

Proof. For, let $t > 0$, there is some partition $\{E_j\}$ of E such that $\sum_{j=1}^{\infty} |\mu(E_j)| > t$. We fix a large N such that $\sum_{j=1}^{N} |\mu(E_j)| > t$. Rearrange E_j's in the order such that $E_1, E_2, \ldots, E_m$, all $\mu(E_j) < 0$ and $E_{m+1}, \ldots, E_N$, $\mu(E_j) \geq 0$. Then $|\mu(E_1) + \cdots \mu(E_m)| + |\mu(E_{m+1}) + \cdots + \mu(E_N)| > t$, and so, either

$$|\mu(E_1) + \cdots + \mu(E_m)| > \frac{t}{2}$$

or

$$|\mu(E_{m+1}) + \cdots + \mu(E_N)| > \frac{t}{2}.$$

Assume, say, it is the former. We take $A = \bigcup_{j=1}^{m} E_j$. Then $|\mu(A)| > \frac{t}{2}$. Let $B = E \setminus A$. Then

$$\begin{aligned} |\mu(B)| &= |\mu(E) - \mu(A)| \\ &\geq |\mu(A)| - |\mu(E)| \\ &\geq \frac{t}{2} - |\mu(E)| \\ &\geq 1 \end{aligned}$$

if we choose $t > 2$ and $t > 2|\mu(E)| + 2$.

□

The following proposition yields a lot of signed measures from a single measure.

Proposition 5.3. *Let μ be a measure on $(X, \mathcal{M})$ and $f \in L^1(\mu)$. Then*

(a)

$$\lambda(E) = \int_E f d\mu, \quad \forall E \in \mathcal{M}$$

is a signed measure.

(b) Its total variation is given by

$$|\lambda|(E) = \int_E |f| d\mu.$$

Proof. (a) is immediate. To show (b) let $\{E_j\}$ be a measurable partition of E. Then

$$\begin{aligned} \sum_j |\lambda(E_j)| &= \sum_j \left| \int_{E_j} f d\mu \right| \\ &\leq \sum_j \int_{E_j} |f| d\mu \\ &= \int_E |f| d\mu. \end{aligned}$$

Taking supremum over all measurable partitions, we obtain

$$|\lambda|(E) \leq \int_E |f| d\mu.$$

On the other hand, let $A = \{x \in E : \ f(x) \geq 0\}$ and $B = \{x \in E : \ f(x) < 0\}$. A and B form a measurable partition of E. By the definition of the total variation, we have

$$|\lambda|(E) \geq |\lambda(A)| + |\lambda(B)| = \int_E |f| d\mu.$$

□

Denote by $M(X)$ or $M(X, \mathcal{M})$ the collection of all signed measures on $(X, \mathcal{M})$. It is a good exercise to verify that $M(X)$ forms a vector space and is complete under the norm

$$\|\mu\| = |\mu|(X).$$

$M(X, \mathcal{M})$ is a Banach space. Given a signed measure μ, let

$$\mu^+ = \frac{1}{2}(|\mu| + \mu), \qquad \mu^- = \frac{1}{2}(|\mu| - \mu).$$

We can write

$$\mu = \mu^+ - \mu^- \text{ and } |\mu| = \mu^+ + \mu^-,$$

where μ^+ and μ^- are finite measures. The decomposition of a signed measure into the difference of two finite measures is called the *Jordan decomposition* of the signed measure. The terminology comes from a corresponding decomposition for functions of bounded variation.

5.2 Radon-Nikodym theorem

Let μ be a measure and λ a measure or signed measure on the same σ-algebra $\mathcal{M}$. The measure λ is called *absolutely continuous* with respect to μ, $\lambda << \mu$ in notation, if every μ-null set is λ-null. λ is *concentrated* on a set $A \in \mathcal{M}$ if $\lambda(E) = \lambda(E \cap A)$, for all $E \in \mathcal{M}$. Note that the set A is not unique. When λ is concentrated on A, it is also concentrated on any set containing A and any set of the form $A \setminus N$ where N is λ-null. We call two measures/signed measures λ_1 and λ_2 *singular* to each other, $\lambda_1 \perp \lambda_2$ in notation, if λ_1 and λ_2 are concentrated respectively on A and B where A and B are disjoint. Clearly, $\lambda_1 \perp \lambda_2$ if λ_1 is concentrated on A and $\lambda_2(A) = 0$.

The following proposition can be derived easily from the definitions above and is left for you to prove.

Proposition 5.4. *Let μ be a measure and λ_i measures or signed measures, $i = 1, 2$. Then*

(a) λ *is concentrated on* $A \Rightarrow |\lambda|$ *is concentrated on* A.

(b) $\lambda_1 \perp \lambda_2 \Rightarrow |\lambda_1| \perp |\lambda_2|$.

(c) $\lambda_1 \perp \mu$, $\lambda_2 \perp \mu \Rightarrow \lambda_1 + \lambda_2 \perp \mu$.

(d) $\lambda_1 << \mu$, $\lambda_2 << \mu \Rightarrow \lambda_1 + \lambda_2 << \mu$.

(e) $\lambda << \mu \Rightarrow |\lambda| << \mu$.

(f) $\lambda_1 << \mu$, $\lambda_2 \perp \mu \Rightarrow \lambda_1 \perp \lambda_2$.

(g) $\lambda << \mu$, $\lambda \perp \mu \Rightarrow \lambda = 0$.

Example 5.1. Consider $(\mathbb{R}, \mathcal{M})$ where $\mathcal{M}$ is the σ-algebra consisting of all Lebesgue measurable sets. For $f \in L^1(\mathbb{R})$, define

$$\nu(E) = \int_E f \, d\mathcal{L}^1.$$

Then $\nu << \mathcal{L}^1$. On the other hand, consider

$$\lambda = \sum_{j=1}^{N} \alpha_j \delta_{x_j}$$

where $\{x_1, \ldots, x_N\} \subset \mathbb{R}$ and $\alpha_j \in \mathbb{R}$. Then λ is a signed measure on $(\mathbb{R}, \mathcal{M})$ (in fact, on $(\mathbb{R}, \mathcal{P}_{\mathbb{R}})$.) As $\mathcal{L}^1$ is concentrated on $\mathbb{R}$, so does it on $\mathbb{R} \setminus \{x_1, \ldots, x_N\}$. As λ is concentrated on $\{x_1, \ldots, x_N\}$, λ and $\mathcal{L}^1$ are singular. By Proposition 5.3, all ν's are singular to λ too.

Absolute continuity and mutual singularity describe two extreme relations between two measures. It is striking that they are sufficient in certain sense for the description of the relation between two measures. This result is contained in the following two theorems.

Theorem 5.5 (**Lebesgue Decomposition**). *Let μ be a σ-finite measure and λ a signed measure on $(X, \mathcal{M})$. There exist a unique pair of signed measures $(\lambda_{ac}, \lambda_s)$, $\lambda_{ac} << \mu$ and $\lambda_s \perp \mu$, such that $\lambda = \lambda_{ac} + \lambda_s$.*

Theorem 5.6 (**Radon-Nikodym Theorem**). *Let μ be a σ-finite measure and λ a signed measure on $(X, \mathcal{M})$ such that $\lambda << \mu$. There exists a unique $h \in L^1(\mu)$ such that*

$$\lambda(E) = \int_E h \, d\mu, \qquad \forall E \in \mathcal{M}.$$

The function h is called the *Radon-Nikodym derivative* of λ with respect to μ and will be denoted by $d\lambda/d\mu$.

We will prove these two theorems in one stroke. The proof is due to von Neumann.

Proof. Step 1. Assume that both μ and λ are finite measures. Setting $\rho = \mu + \lambda$ and define a functional on $L^2(\rho)$ by

$$\Lambda\varphi = \int \varphi \, d\lambda, \quad \varphi \in L^2(\rho).$$

By Cauchy-Schwarz inequality,

$$\begin{aligned} |\Lambda\varphi| &\le \int |\varphi| \, d\lambda \\ &\le \int |\varphi| \, d\rho \\ &\le \sqrt{\rho(X)} \, \|\varphi\|_{L^2(\rho)}, \end{aligned}$$

so Λ is a bounded linear functional on $L^2(\rho)$. By the self-duality of $L^2(\rho)$, there exists some $g \in L^2(\rho)$ such that

$$\int \varphi \, d\lambda = \int \varphi g \, d\rho, \qquad \forall \varphi \in L^2(\rho). \tag{5.1}$$

Taking $\varphi = \chi_E, E \in \mathcal{M}$, φ belongs to $L^2(\rho)$ and we have

$$\lambda(E) = \int_E g \, d\rho,$$

and

$$\frac{1}{\rho(E)} \int_E g \, d\rho = \frac{\lambda(E)}{\rho(E)} \in [0,1].$$

By a previous exercise, $g \in [0,1]$ ρ-a.e. By redefining g in a null set, we may assume $g(X) \subset [0,1]$. Let

$$\lambda_{ac}(E) = \lambda(E \cap A),$$

and

$$\lambda_s(E) = \lambda(E \cap B),$$

where $A = \{x \in X : g(x) \in [0,1)\}$ and $B = \{x \in X : g(x) = 1\}$. We have $\lambda = \lambda_{ac} + \lambda_s$.

We rewrite (5.1) as

$$\int \varphi(1-g) \, d\lambda = \int \varphi g \, d\mu. \tag{5.2}$$

Taking $\varphi = \chi_B$, we get

$$0 = \int_B d\mu = \mu(B).$$

We conclude that $\lambda_s \perp \mu$, since by definition λ_s is concentrated on B. Taking $\varphi = \chi_E(1 + g + \cdots + g^n)$ in (5.2), we have

$$\int \chi_E(1 - g^{n+1}) \, d\lambda = \int \chi_E(1 + g + \cdots + g^n) g \, d\mu.$$

The left hand side of this relation satisfies

$$\begin{aligned} &\int \chi_E(1 - g^{n+1}) \, d\lambda \\ &= \int_E (1 - g^{n+1}) \, d\lambda \\ &= \int_{E \cap A} (1 - g^{n+1}) \, d\lambda \\ &\to \lambda(E \cap A) = \lambda_{ac}(E), \qquad \text{as } n \to \infty, \end{aligned}$$

where the monotone convergence theorem has been used in the last step. On the other hand, the right hand side becomes

$$\begin{aligned}\int_E (1+g+\cdots+g^n)g\,d\mu &= \int_{E\cap A} g\frac{1-g^{n+1}}{1-g}\,d\mu \\ &\to \int_{E\cap A}\frac{g}{1-g}\,d\mu \qquad \text{as } n\to\infty.\end{aligned}$$

Setting $h = g/(1-g)$, h is non-negative and belongs to $\in L^1(\mu)$. We conclude that

$$\lambda_{ac}(E) = \int_E h\,d\mu, \quad \forall E\in\mathcal{M},$$

so, $\lambda_{ac} << \mu$.

Step 2. Consider the case that μ is a finite measure and λ is a signed measure. Applying what has been proved to λ^+ and λ^-, the Jordan decomposition of λ, we have

$$\lambda^+ = \lambda^+_{ac} + \lambda^+_s, \qquad \lambda^- = \lambda^-_{ac} + \lambda^-_s,$$
$$\lambda^+_{ac}, \quad \lambda^-_{ac} << \mu, \qquad \lambda^+_s, \quad \lambda^-_s \perp \mu.$$

By Proposition 5.4, $\lambda = \lambda_{ac} + \lambda_s$, $\lambda_{ac} \equiv \lambda^+_{ac} - \lambda^-_{ac}$, $\lambda_s = \lambda^+_s - \lambda^-_s$, and $\lambda_{ac} << \mu$ and $\lambda_s \perp \lambda_{ac}$. Moreover, if

$$\lambda^\pm_{ac} = \int h^\pm\,d\mu,$$

$$\lambda_{ac} = \int h\,d\mu, \qquad \text{where } h\equiv h^+ - h^- \in L^1(\mu).$$

Step 3. Let μ be a σ-finite measure and λ a signed measure. Let $\{X_j\}$ be a measurable partition of X with finite measure. Let $\mu_j = \mu\lfloor_{X_j}$ and $\lambda_j = \lambda\lfloor_{X_j}$. By the previous step,

$$\lambda_j = \lambda^j_{ac} + \lambda^j_s, \qquad \lambda^j_{ac} << \mu_j, \quad \lambda^j_s \perp \mu_j,$$

$$\lambda^j_{ac}(E) = \int_E h_j\,d\mu_j = \int_E h_j\chi_{X_j}\,d\mu, \quad h_j\in L^1(\mu_j), h_j\chi_{X_j}\in L^1(\mu).$$

Letting $\lambda_{ac} = \sum_{j=1}^\infty \lambda^j_{ac}$ and $\lambda_s = \sum_{j=1}^\infty \lambda^j_s$, we have

$$\lambda = \lambda_{ac} + \lambda_s, \quad \lambda_{ac} << \mu, \quad \lambda_s \perp \mu,$$

and

$$\lambda(E) = \int_E h \, d\mu, \quad \text{where } h = \sum_{j=1}^{\infty} h_j \chi_{X_j}.$$

By Proposition 5.3,

$$\infty > |\lambda|(X) = \int_X |h| d\mu,$$

so $h \in L^1(\mu)$.

Step 4. To finish the proof we establish uniqueness. Suppose there is a pair (λ_1, λ_2) with $\lambda_1 << \mu$ and $\lambda_2 \perp \mu$ such that

$$\lambda = \lambda_1 + \lambda_2.$$

As $\lambda = \lambda_{ac} + \lambda_s$, $\lambda_1 + \lambda_2 = \lambda_{ac} + \lambda_s$, it implies that $\lambda_1 - \lambda_{ac} = \lambda_s - \lambda_2$. As $\lambda_1 - \lambda_{ac} << \mu$ and $\lambda_s - \lambda_2 \perp \mu$, by Proposition 5.3, $\lambda_1 - \lambda_{ac} = \lambda_s - \lambda_2 = 0$, that is, $\lambda_1 = \lambda_{ac}$ and $\lambda_2 = \lambda_s$. □

We deduce a significant result from the Radon-Nikodym theorem.

Let μ be a signed measure on $(X, \mathcal{M})$. From the relation $|\mu(E)| \leq |\mu|(E)$, $\forall E \in \mathcal{M}$, we know that $\mu << |\mu|$. By Radon-Nikodym theorem, we can find some $h \in L^1(|\mu|)$ such that

$$\mu(E) = \int_E h \, d|\mu|.$$

From

$$\frac{1}{|\mu|(E)} \int_E h \, d|\mu| = \frac{\mu(E)}{|\mu|(E)} \in [-1, 1],$$

we know that $|h| \leq 1$ a.e. We claim that in fact $|h| = 1$ a.e. For, let

$$A_r = \{x \in X : |h|(x) < r\}, \quad r \in (0, 1).$$

For any partition $\{A_j\}$ of A_r,

$$\begin{aligned} \sum_j |\mu(A_j)| &= \sum_j \left| \int_{A_j} h \, d|\mu| \right| \\ &\leq \sum_j \int_{A_j} |h| \, d|\mu| \\ &\leq r \sum_j |\mu|(A_j) \\ &= r|\mu|(A_r). \end{aligned}$$

Taking supremum over all partitions $\{A_j\}$,

$$|\mu|(A_r) \le r\,|\mu|(A_r).$$

As $|\mu|(A_r) \le |\mu|(X) < \infty$, this forces A_r has $|\mu|$-measure zero. So, $|h| = 1$ $|\mu|$-a.e. After redefining h on a set of measure zero, we may assume $|h| = 1$ everywhere. Thus we have

Proposition 5.7. *Let μ be a signed measure on $(X, \mathcal{M})$.*

(a) There exists an $h \in L^1(|\mu|)$, $|h| \equiv 1$, such that $d\mu = h\,d\,|\mu|$, that is,

$$\mu(E) = \int_E h\,d\,|\mu|\,, \quad \forall E \in \mathcal{M}.$$

(b) There are disjoint measurable sets A and B such that

$$\mu^+(E) = \mu(E \cap A)$$
$$\mu^-(E) = -\mu(E \cap B), \quad \forall E \in \mathcal{M}.$$

(c) If $\mu = \lambda_1 - \lambda_2$ where λ_i are measures, then $\lambda_1 \ge \mu^+$ and $\lambda_2 \ge \mu^-$.

Proof. (a) Already done.
(b) Let $A = \{x \in X : h(x) = 1\}$ and $B = \{x \in X : h(x) = -1\}$. Then $A \dot\cup B = X$ and

$$\begin{aligned}
\mu^+(E) &= \frac{1}{2}(|\mu|(E) + \mu(E)) \\
&= \frac{1}{2}\int_E (1+h)\,d\,|\mu| \\
&= \int_{E\cap A} d\,|\mu| \quad (\text{use } 1+h = 2 \text{ on } A,\ 1+h = 0 \text{ on } B) \\
&= \int_{E\cap A} d\mu \\
&= \mu(E \cap A).
\end{aligned}$$

Similarly,

$$\begin{aligned}\mu^-(E) &= \frac{1}{2}(|\mu|(E) - \mu(E)) \\ &= \frac{1}{2}\int_E (1-h)\, d|\mu| \\ &= \int_{E\cap B} d|\mu| \\ &= -\int_{E\cap B} d\mu \\ &= -\mu(E\cap B).\end{aligned}$$

(c) As $\mu = \lambda_1 - \lambda_2 \le \lambda_1$,

$$\mu^+(E) = \mu(E\cap A) \le \lambda_1(E\cap A) \le \lambda_1(E).$$

As $-\mu = \lambda_2 - \lambda_1 \le \lambda_2$,

$$\mu^-(E) = -\mu(E\cap B) \le \lambda_2(E\cap B) \le \lambda_2(E).$$

□

The existence of disjoint A and B such that $\mu^+ = \mu\lfloor_A$ and $\mu^- = \mu\lfloor_B$ is called the *Hahn decomposition* of μ. Proposition 5.6(c) shows certain minimal property of the decomposition.

5.3 The Dual Space of $C_0(X)$

The space $C_c(X)$ may not be complete under the supnorm. For instance, the function e^{-x^2} is the uniform limit of a sequence of functions in $C_c(\mathbb{R})$, but it is positive everywhere. The space of all bounded, continuous functions on a topological space forms a Banach space where $C_c(X)$ is its subspace. We denote the closure of $C_c(X)$ in in this space by $C_0(X)$. It is a Banach space. In fact, when X is a locally compact Hausdorff space, one can show that it consists of all continuous functions that vanish at infinity. More precisely, f belongs to $C_0(X)$ if and only if, for every $\varepsilon > 0$, there is a compact set K such that $|f|$ is less than ε outside K. Of course, $C_0(X)$ is equal to $C_c(X)$ when X is compact. As $C_c(X)$ is dense in $C_0(X)$, the dual space of $C_c(X)$ can be identified with the dual space of $C_0(X)$.

From now on we take X to be a locally compact Hausdorff topological space. A signed measure $\mu \in M(X) = M(X, \mathcal{B})$ is called *regular* if every $E \in \mathcal{B}$ is inner

and outer regular, i.e.,

$$\begin{aligned} |\mu|(E) &= \inf\{|\mu|(G) : E \subset G,\ G \text{ open}\}, \quad \text{and} \\ &= \sup\{|\mu|(K) : K \subset E,\ K \text{ compact}\}. \end{aligned}$$

We use $M_r(X)$ to denote all regular Borel signed measures on $(X, \mathcal{B})$.

Proposition 5.8. *(a) $M_r(X)$ forms a closed subspace of $M(X)$ under the norm $\|\mu\| = |\mu|(X)$,*

(b) $\mu = \mu^+ - \mu^-$, $\mu \in M_r(X)$ if and only if $\mu^\pm \in M_r(X)$,

(c) For $\mu \in M_r(X)$ and $f \in L^1(|\mu|)$, the measure λ given by

$$\lambda(E) = \int_E f\, d|\mu|,$$

belongs to $M_r(X)$.

I leave the proof of this proposition as an exercise.

Each regular Borel signed measure induces a bounded linear functional on $C_c(X)$. In fact, we used to define

$$\Lambda\varphi = \int \varphi\, d\mu, \quad \forall \varphi \in C_c(X),$$

when μ is a measure. Now, for a signed measure, we could use Proposition 5.7(a) to define

$$\Lambda\varphi = \int \varphi h\, d\,|\mu| \ ,$$

or,

$$\Lambda\varphi = \int \varphi\, d\mu^+ - \int \varphi\, d\mu^-.$$

With this definition,

$$|\Lambda\varphi| \le \int |\varphi h|\ d\,|\mu| \le |\mu|(X)\, \|\varphi\|_\infty,$$

so

$$\|\Lambda\| \le |\mu|(X) = \|\mu\|, \tag{5.3}$$

holds and, in particular, it shows that $\Lambda \in C_c(X)'$.

Consider the map $\Phi : M_r(X) \to C_c(X)'$ by $\Phi : \mu \mapsto \Lambda$. We claim that that Φ is linear.

Indeed, writing $\int \varphi\, d\mu \equiv \int \varphi h\, d\,|\mu|$, we first prove that

$$\int \varphi\, d(\mu_1 + \mu_2) = \int \varphi\, d\mu_1 + \int \varphi\, d\mu_2, \quad \forall \mu_1, \mu_2 \in M_r(X)$$

and

$$\int \varphi \, d(c\mu) = c \int \varphi \, d\mu, \quad \forall c \in \mathbb{R}.$$

The second one follows directly from the definition. For the first one, we have

$$\int \varphi \, d(\mu_1 + \mu_2) = \int \varphi \, d(\mu_1 + \mu_2)^+ - \int \varphi \, d(\mu_1 + \mu_2)^- \ ,$$

$$\int \varphi \, d\mu_1 = \int \varphi \, d\mu_1^+ - \int \varphi \, d\mu_1^- \ ,$$

and

$$\int \varphi \, d\mu_2 = \int \varphi \, d\mu_2^+ - \int \varphi \, d\mu_2^- \ .$$

It suffices to show that

$$\int \varphi \, d(\mu_1 + \mu_2)^+ + \int \varphi \, d\mu_1^- + \int \varphi \, d\mu^- = \int \varphi \, d(\mu_1 + \mu_2)^- + \int \varphi \, d\mu_1^+ + \int \varphi \, d\mu_2^+. \tag{5.4}$$

Observe that for any two finite measures λ and μ on the same σ-algebra,

$$\int f \, d(\lambda + \mu) = \int f \, d\lambda + \int f \, d\mu, \qquad \forall \text{ bounded measurable} f,$$

As

$$\mu_1 + \mu_2 = (\mu_1 + \mu_2)^+ - (\mu_1 + \mu_2)^- \ ,$$
$$\mu_1 = \mu_1^+ - \mu_1^-, \text{ and} \qquad \mu_2 = \mu_2^+ - \mu_2^-,$$

we have

$$(\mu_1 + \mu_2)^+ + \mu_1^- + \mu_2^- = (\mu_1 + \mu_2)^- + \mu_1^+ + \mu_2^+ \ .$$

Therefore,

$$\int \varphi \, d(\mu_1 + \mu_2)^+ + \int \varphi \, d\mu_1^- + \int \varphi \, d\mu^- = \int \varphi \, d(\mu_1 + \mu_2)^- + \int \varphi \, d\mu_1^+ + \int \varphi \, d\mu_2^+,$$

and (5.4) follows.

Finally, let us show that the map from μ to Λ is norm-preserving. In view of (5.3), it remains to show $\|\Lambda\| \geq \|\mu\|$. Indeed, from

$$\Lambda\varphi = \int \varphi h d\mu$$

we choose a suitable φ to approximate h. Recall that $|h| \equiv 1$ and $|\mu|$ is a finite measure, by Lusin's theorem (Theorem 2.12), for each $\varepsilon > 0$, there exists some $\varphi \in C_c(X)$ satisfying (a) $|\varphi| \leq |h| = 1$ and (b) $|\mu|(A) < \varepsilon$ where $A = \{x : \ \varphi(x) \neq$

$h(x)\}$. We have

$$\begin{aligned} \|\Lambda\| &\geq |\Lambda\varphi| \\ &= \left|\int \varphi h d|\mu|\right| \\ &\geq |\mu|(X \setminus A) - \varepsilon \\ &\geq |\mu|(X) - 2\varepsilon. \end{aligned}$$

The desired inequality follows by letting $\varepsilon \to 0$.

Summarizing, we have shown that the map $\mu \mapsto \Lambda$ is a norm-preserving, linear map from $M_r(X)$ to $C_0(X)'$.

Theorem 5.9. *Let X be a locally compact Hausdorff space. The $\Phi : M_r(X, \mathcal{B}) \to C_0(X)'$ is a norm-preserving, bijective linear map. In other words, the dual space of $C_0(X)$ is equal to $M_r(X, \mathcal{B})$.*

This is the full version of Riesz representation theorem. It is different from Theorem 2.8 for two points. First, it deals only with bounded linear functionals while Theorem 2.8 deals with positive linear functionals which may not be bounded. Second, since we are only concerned with the dual of $C_c(X)$ only, we do not consider outer measures here.

Lemma 5.10. *Setting as above, there exists a positive linear functional Λ_0 on $C_c(X)$ satisfying*

$$|\Lambda f| \leq \Lambda_0(|f|) \leq \|\Lambda\| \|f\|_\infty, \quad \forall f \in C_c(X).$$

Proof. For $f \in C_c^+(X)$, we define

$$\Lambda_0 f = \sup\{|\Lambda\varphi| : |\varphi| \leq f,\ \varphi \in C_c(X)\}.$$

Clearly, $\Lambda_0 f \geq 0$, $\Lambda_0 f_2 \geq \Lambda_0 f_1$ if $f_2 \geq f_1 \geq 0$ and $\Lambda_0(cf) = c\Lambda_0 f$. We claim

$$\Lambda_0(f_1 + f_2) = \Lambda_0 f_1 + \Lambda_0 f_2, \quad f_1, f_2 \in C_c^+(X).$$

for, observe that

$$|\Lambda\varphi| \leq \|\Lambda\| \, \|\varphi\|_\infty \leq \|\Lambda\| \, \|f\|_\infty, \quad \text{if } |\varphi| \leq f.$$

Taking supremum over all these φ,

$$|\Lambda_0 f| \leq \|\Lambda\| \, \|f\|_\infty. \tag{5.5}$$

In particular, $\Lambda_0 f$ is finite. Let $f_1, f_2 \in C_c^+(X)$. For $\varepsilon > 0$, there are φ_1, φ_2,

$|\varphi_1| \le f_1$, $|\varphi_2| \le f_2$, such that

$$\Lambda_0 f_1 \le |\Lambda\varphi_1| + \varepsilon, \qquad \Lambda_0 f_2 \le |\Lambda\varphi_2| + \varepsilon.$$

Pick $\sigma_1, \sigma_2 \in \{1, -1\}$ such that $|\Lambda\varphi_1| = \sigma_1\Lambda\varphi_1$, $|\Lambda\varphi_2| = \sigma_2\Lambda\varphi_2$, then

$$\begin{aligned}\Lambda_0 f_1 + \Lambda_0 f_2 &\le \Lambda(\sigma_1\varphi_1 + \sigma_2\varphi_2) + 2\varepsilon \\ &\le \Lambda_0(f_1 + f_2) + 2\varepsilon.\end{aligned}$$

So,

$$\Lambda_0 f_1 + \Lambda_0 f_2 \le \Lambda_0(f_1 + f_2).$$

To get the reverse inequality, let $\varphi \in C_c(X)$, $|\varphi| \le f_1 + f_2$ and set

$$\varphi_1 = \frac{f_1\varphi}{f_1 + f_2}, \quad \varphi_2 = \frac{f_2\varphi}{f_1 + f_2} \quad \text{on } V,$$

and and $\varphi_1 = \varphi_2 = 0$ on $X \setminus V$ where $V = \{x \in X : (f_1 + f_2)(x) > 0\}$. Note that $\varphi_1, \varphi_2 \in C_c(X)$,$|\varphi_1| \le f_1$, $|\varphi_2| \le f_2$, and $\varphi = \varphi_1 + \varphi_2$. We have

$$\begin{aligned}|\Lambda\varphi| &= |\Lambda\varphi_1 + \Lambda\varphi_2| \\ &\le |\Lambda\varphi_1| + |\Lambda\varphi_2| \\ &\le \Lambda_0 f_1 + \Lambda_0 f_2.\end{aligned}$$

Taking supremum over all these φ,

$$\Lambda_0(f_1 + f_2) \le \Lambda_0 f_1 + \Lambda_0 f_2.$$

For $f \in C_c(X)$, we set

$$\Lambda_0 f = \Lambda_0 f^+ - \Lambda_0 f^-.$$

Using the old trick

$$(f_1 + f_2)^+ - (f_1 + f_2)^- = f_1 + f_2 = f_1^+ - f_1^- + f_2^+ - f_2^-,$$

it is routine to check that Λ_0 is our desired positive linear functional on $C_c(X)$. $\square$

Now, we can prove Theorem 5.9. To show that Φ is onto, we need to find some μ that satisfies $\Phi(\mu) = \Lambda$ for any given Λ. Indeed, applying Theorem 2.8 to Λ_0, we can find a Radon measure λ such that

$$\Lambda_0 f = \int f \, d\lambda, \quad \forall f \in C_c(X).$$

Here λ satisfies

- $\lambda(E) = \inf\{\lambda(G) : E \subset G, G \text{ open}\}, \qquad \forall E \subset X,$
- $\lambda(E) = \sup\{\lambda(K) : K \subset E, K \text{ compact}\}$ if $\lambda(E) < \infty, \qquad \forall E \in \mathcal{B}.$

By Lemma 5.10,

$$\left|\int f\, d\lambda\right| = |\Lambda_0 f| \le \|\Lambda\|\, \|f\|_\infty, \quad \forall f \in C_c(X).$$

From the proof of Theorem 2.8,

$$\lambda(X) = \sup\{|\Lambda_0 f| : 0 \le f \le 1 \text{ on } X\}.$$

In view of this, we have

$$\lambda(X) \le \|\Lambda\|,$$

i.e., λ belongs to $M_r(X)$.

From the definition of Λ_0 we also have

$$\begin{aligned} |\Lambda f| &\le \Lambda_0 f^+ + \Lambda_0 f^- \\ &= \|f\|_{L^1(\lambda)}, \qquad \forall f \in C_c(X). \end{aligned}$$

As $C_c(X)$ is dense in $L^1(\lambda)$, Λ can be extended to a bounded linear functional on $L^1(\lambda)$. Since λ is a finite measure, by L^1-L^∞ duality there exists some $h \in L^\infty(\lambda)$ such that

$$\Lambda f = \int f h\, d\lambda \quad \forall f \in L^1(\lambda),$$

and the operator norm of Λ as a linear functional on $L^1(\lambda)$ is equal to $\|h\|_\infty$. Using $|\Lambda f| \le \|f\|_{L^1(\lambda)}$, we have $\|h\|_\infty \le 1$. On the other hand, we have

$$\begin{aligned} \|\Lambda\| &= \sup\{|\Lambda\varphi|:\ \varphi \in C_c(X), |\varphi| \le 1\} \\ &\le \int |h| d\lambda \\ &\le \lambda(X) \\ &\le \|\Lambda\|, \end{aligned}$$

which forces $|h| = 1$ λ-a.e. and $\lambda(X) = \|\Lambda\|$. Now we set

$$\mu(E) = \int_E h d\lambda, \quad \forall E \in \mathcal{B}.$$

By Proposition 5.8, $\mu \in M_r(X)$. By Proposition 5.3, $|\mu| = \lambda$ and so $\|\mu\| = \|\Lambda\|$. We conclude that Φ is a norm-preserving linear bijection from $M_r(X, \mathcal{B})$ to $C_0(X)'$. The proof of Theorem 5.9 is completed.

5.4 Weak* Convergence of Measures

Let E be a normed space and E' its dual space. A sequence $\{\Lambda_k\} \subset E'$ is called *weakly* convergent* to $\Lambda \in E'$ if

$$\Lambda_k x \to \Lambda x, \quad \forall x \in E,$$

as $k \to \infty$. Consider $L^p(\mu)$, $1 < p < \infty$, we know that $L^p(\mu) = L^q(\mu)'$ where q is conjugate to p. For $f_k \in L^p$, $f_k \overset{*}{\rightharpoonup} f$ if and only if

$$\int f_k g \, d\mu \to \int f g \, d\mu, \quad \forall g \in L^q(\mu).$$

This turns out to be the same as $f_n \rightharpoonup f$. So when $1 < p < \infty$, weak* and weak convergence are the same, depending on whether to regard $L^p(\mu)$ as the "base" space or the dual of $L^q(\mu)$.

Applying to the case the space $C_0(X)$ where X is a locally compact Hausdorff space, by the representation theorem

$$\mu_n \overset{*}{\rightharpoonup} \mu \quad \text{in } M_r(X)$$

if and only if

$$\int \varphi \, d\mu_n \to \int \varphi \, d\mu, \quad \forall \varphi \in C_c(X).$$

The following result is sometimes called Helly selection theorem.

Theorem 5.11. *Let E be a separable normed space. Every bounded sequence in E' contains a weakly* convergent subsequence.*

This theorem can be proved as Theorem 5.8 and we omitted its proof. As a special case we have

Corollary 5.12. *Let $C_0(X)$ be separable. Every sequence $\{\mu_k\}$ in $M_r(X)$, $\|\mu_k\| \leq M$ for some M, contains a subsequence μ_{k_j} which $\mu_{k_j} \overset{*}{\rightharpoonup} \mu$ for some $\mu \in M_r(X)$.*

One can show that the space $C_0(X)$ is separable when X is a compact metric space or it is an open set in $\mathbb{R}^n$.

Recall that even $\|f_k\|_{L^1}$ is uniformly bounded, we cannot always pick a weakly convergent subsequence f_{n_j} in $L^1(\mu)$. As a typical example we may take $\{f_k\}$ satisfying $\operatorname{spt} f_k = [a_k, b_k]$ shrinks to $\{x_0\}$ and $\|f_k\|_{L^1} = 1$ for all k. Then no subsequence of f_k converges weakly. However, if we regard an L^1-function as a measure by setting $d\mu_k = f_k \, d\mathcal{L}^1$, μ_k now belongs to the larger space $M_r(\mathbb{R})$ and $\|\mu_k\| = \|f_k\|_{L^1} = 1$. By Corollary 5.10, it has weak* subconvergence $\mu_{k_j} \overset{*}{\rightharpoonup} \mu$. In fact, it is clear that the entire sequence converges weakly* to the Dirac measure δ_{x_0}. The lesson is, by enlarging the space from $L^1(\mathbb{R})$ to $M_r(\mathbb{R})$, we get subconvergence.

As an application of this property, we prove

Theorem 5.13. *Consider $L^1(\Omega)$ where Ω be a bounded, open set in $\mathbb{R}^n$. Let $\{f_k\}$ be a bounded sequence in $L^1(\Omega)$ which is uniformly integrable. Then $\{f_k\}$ contains a weakly convergent subsequence in $L^1(\Omega)$.*

In fact, the converse is true, namely, if $f_k \overset{*}{\rightharpoonup} f$ for some $f \in L^1(\Omega)$, then $\{f_k\}$ is bounded in $L^1(\Omega)$ and uniformly integrable. This necessary and sufficient condition for the weak subconvergence of an L^1-sequence is called Dunford-Pettis theorem.

Proof of Theorem 5.13. It suffices to consider the case $f_k \geq 0$. Let $d\mu_n = f_n \, d\mathcal{L}^n$. Then $\|\mu_k\| = \|f_k\|_{L^1}$ is uniformly bounded by some constant M. By Corollary 5.10, there exist $\{\mu_{k_j}\}$ and $\mu \in M_r(\Omega)$ such that

$$\int \varphi \, d\mu_{k_j} \to \int \varphi \, d\mu, \quad \forall \varphi \in C_c(\Omega) \qquad \text{as } j \to \infty. \tag{5.6}$$

For simplicity, we assume the entire sequence converges weakly$\hat{}*$. By Lebesgue decomposition (with respect to $\mathcal{L}^n$),

$$\mu = f\mathcal{L}^n + \nu, \quad f \in L^1(\Omega), \ \nu \perp \mathcal{L}^n.$$

It suffices to show that $\nu \equiv 0$ so that

$$\int \varphi f_k \, dx \to \int \varphi f \, dx, \quad \forall \varphi \in C_c(\overline{\Omega}) = C(\overline{\Omega}). \tag{5.7}$$

Suppose on the contrary that there is some $A \in \mathcal{B}$, $\nu(A) > 0$, such that $\mathcal{L}^n(A) = 0$. Letting $a_0 \equiv \nu(A) = \mu(A)$, by the regularity of the Lebsegue measure and μ, for $\delta > 0$, there is an open G containing A such that

$$\mu(G) < \delta,$$

and there is a compact set $K \subset A$ such that

$$\mathcal{L}^n(K) \geq \frac{a_0}{2}.$$

For $\varepsilon < a_0/2$, by uniform integrability, there is a $\delta_1 > 0$ such that

$$\int_E f_k \, dx < \varepsilon, \quad \forall E \in \mathcal{B}, \ \mathcal{L}^n(E) < \delta_1, \ \forall k \geq 1.$$

We take $\delta = \delta_1$ and pick $\varphi \in C_c(\Omega)$ such that $\varphi \equiv 1$ on K, $0 \leq \varphi \leq 1$, spt$\varphi \subset G$. Then

$$\int_G \varphi f_k \, dx \leq \int_G f_k \, dx < \varepsilon.$$

Letting $k \to \infty$, by (5.6)

$$\int_G \varphi \, d\mu \leq \varepsilon.$$

However, as $\varphi \equiv 1$ on K, we get

$$\mu(K) \leq \int_G \varphi \, d\mu \leq \varepsilon,$$

contradicting our choice of ε. So $\nu \equiv 0$.

We still have to show (5.7) holds for all $\varphi \in L^\infty(\Omega) = L^1(\Omega)'$. First, we claim that it holds for all $\varphi = \chi_G$, where G is open. For, let $\{G_j\} \uparrow G, G_j \subset\subset G_{j+1}$ and φ_j, $\overline{G_j} < \varphi_j < G_{j+1}$. The claim follows from the uniform integrability of $\{f_k\}$. Next, using outer regularity and uniform integrability we know that (5.7) holds for $\varphi = \chi_E$ for $E \in \mathcal{B}$. Consequently, it also holds for all simple functions s. Now, let $\varphi \in L^\infty(\Omega)$, say, $|\varphi| \leq M$. For $\varepsilon > 0$, let $-M - 1 = a_1 < a_2 < \cdots < a_N = M + 1$, $\Delta a_j < \varepsilon$. Define

$$s = \sum_{j=1}^{N-1} a_j \chi_{A_j}, \quad A_j = \{x \in \Omega : a_j \leq \varphi(x) < a_{j+1}\}.$$

Then $\|s - \varphi\|_\infty < \varepsilon$, so

$$\begin{aligned} \left| \int \varphi f_k - \int \varphi f \right| &\leq \left| \int (\varphi - s)(f_k - f) \right| + \left| \int s(f_k - f) \right| \\ &\leq \varepsilon \times 2M + \left| \int s(f_k - f) \right|. \end{aligned}$$

Letting $n \to \infty$,

$$\overline{\lim_{k \to \infty}} \left| \int \varphi f_k - \int \varphi f \right| \leq 2M\varepsilon,$$

and we finally conclude that (5.7) holds for all φ in $L^\infty(\Omega)$. □

5.5 Herglotz-Riesz Theorem

We present another application of the Riesz representation theorem.

Recall that a harmonic function is the real part of an analytic function and it satisfies

$$u_{xx} + u_{yy} = 0$$

in the plane. Let $D_R = \{z : |z| < R\}$, $z = x + iy$, be the disk of radius R. It is well-known that given a continuous function g on the boundary $|z| = R$, $g(\theta)$, $z = Re^{i\theta}$, $\theta \in [0, 2\pi]$, there is a unique harmonic function u in D_R which is equal to g on its boundary. In fact, this harmonic function is given by the Poisson

formula

$$u(z) = \frac{1}{2\pi} \int_{|w|=R} \frac{R^2 - |z|^2}{|w - z|^2} g(\theta)\, d\theta. \tag{5.8}$$

The question is: Instead of continuous data g, what are the most general data that can be imposed on the boundary of D_R to get solvability?

It turns out we have

Theorem 5.14. *Let u be a non-negative harmonic function defined in $D_1 = \{z : |z| < 1\}$. There exists a unique Radon measure μ on S^1 such that*

$$\int u(re^{i\theta})\varphi(\theta)\, dx \to \int \varphi(\theta)\, d\mu(\theta), \quad \forall \varphi \in C(S^1), \tag{5.9}$$

as $r \uparrow 1$. In fact,

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - r^2}{1 - 2r\cos(\alpha - \theta) + r^2}\, d\mu(\theta),\ z = re^{i\alpha},\ r \in (0,1),\ \alpha \in [0, 2\pi]. \tag{5.10}$$

Thus every non-negative harmonic function in D_1 has a boundary value which is a Radon measure. Moreover, every Radon measure on the boundary generates a harmonic function by the formula (5.10).

Proof. Let $C(S^1)$ be the space consists of all 2π–periodic, continuous functions. For $0 < r < 1$, set

$$\Lambda_r \varphi = \int_0^{2\pi} u(re^{i\theta})\varphi(\theta)\, d\theta.$$

We have

$$\begin{aligned} |\Lambda_r \varphi| &\le \int_0^{2\pi} u(re^{i\theta})\, |\varphi(\theta)|\ d\theta \\ &\le \|\varphi\|_\infty \int_0^{2\pi} u(re^{i\theta})\, d\theta. \end{aligned}$$

Setting $z = 0$ in (5.8) where now $R = r$ and $g = u(re^{i\theta})$, we obtain the mean value property of the harmonic function

$$u(0) = \frac{1}{2\pi} \int_0^{2\pi} u(re^{i\theta})\, d\theta.$$

Therefore,

$$|\Lambda_r \varphi| \le 2\pi u(0)\, \|\varphi\|_\infty,$$

and

$$\|\Lambda_r\| \le 2\pi u(0) \equiv M < \infty.$$

By Corollary 5.10 or Theorem 2.8, there exist $\{r_j\} \uparrow 1$ and a regular Borel measure μ such that

$$\Lambda_{r_j}\varphi = \int_0^{2\pi} u(r_j e^{i\theta})\varphi(\theta)\, d\theta \to \int_0^{2\pi} \varphi(\theta)\, d\mu(\theta), \quad r_j \uparrow 1.$$

We will later improve the convergence from a sequence $r_j \uparrow 1$ to for all $r \uparrow 1$.

Now, for a fixed $z = re^{i\alpha}$, $0 < r < 1$, the function

$$\varphi_R(\theta) = \frac{R^2 - r^2}{|Re^{i\theta} - re^{i\alpha}|^2} = \frac{R^2 - r^2}{1 - 2r\cos(\alpha - \theta) + r^2}$$

converges uniformly to

$$\varphi_1(\theta) = \frac{1 - r^2}{1 - 2r\cos(\alpha - \theta) + r^2}$$

as $R \uparrow 1$. It follows that

$$\Lambda_{r_j}\varphi_{r_j} \to \int_0^{2\pi} \varphi_1(\theta) d\mu(\theta), \qquad r_j \uparrow 1.$$

On the other hand, by the Poisson formula,

$$\begin{aligned}\Lambda_R\phi_R &= \int_0^{2\pi} \frac{R^2 - r^2}{1 - 2r\cos(\alpha - \theta) + r^2} u(re^{i\theta})\, d\theta \\ &= 2\pi u(z).\end{aligned}$$

Thus, we have

$$2\pi u(z) = \int_0^{2\pi} \frac{1 - r^2}{1 - 2r\cos(\alpha - \theta) + r^2}\, d\mu(\theta),$$

that is, (5.10) holds.

Now, let $g \in C(S^1)$. For a fixed $r < 1$, by Fubini's theorem and Poisson's formula,

$$\begin{aligned}\int_0^{2\pi} u(re^{i\alpha})g(\alpha)\, d\alpha &= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{2\pi} \frac{1 - r^2}{1 - 2r\cos(\alpha - \theta) + r^2} g(\alpha)\, d\alpha d\mu(\theta) \\ &= \int_0^{2\pi} h(re^{i\theta})\, d\mu(\theta),\end{aligned}$$

where h is the harmonic function determined by the boundary value g. In case

(5.10) holds for some measure μ_1 instead of μ, the same reasoning shows that

$$\int_0^{2\pi} u(re^{i\alpha})g(\alpha)\,d\alpha = \int_0^{2\pi} h(re^{i\theta})\,d\mu_1(\theta).$$

It follows that

$$\int_0^{2\pi} h(re^{i\theta})\,d\mu_1(\theta) = \int_0^{2\pi} h(re^{i\theta})\,d\mu(\theta).$$

Letting $r \uparrow 1$,

$$\int_0^{2\pi} g(\theta)\,d\mu_1(\theta) = \int_0^{2\pi} g(\theta)\,d\mu(\theta), \quad \forall g \in C(S^1)$$

so $\mu_1 = \mu$. This shows the uniqueness of the boundary trace μ. As every $r_j \uparrow 1$ contains a subsequence $\{r_{j_k}\}$ such that $\Lambda_{r_{kj}}$ converges weakly* to some boundary trace which must be μ, we conclude that (5.9) holds. □

Comments on Chapter 5. Sections 1–4 are taken from [R]. There are different proofs of Radon-Nikodym theorem, check them from the web. See [R] for a proof of $L^p - L^q$ duality based on the representation theorem when the measure is σ-finite. The discussion in Section 4 is parallel to Section 6 in Chapter 4 and one may consult Buttazzo, Gaiquinta and Hildebrandt "One-dimensional Variational Problems" for a complete proof of Dunford-Pettis theorem. Finally, Herglotz-Riesz theorem is taken from Lax "Functional Analysis".

Chapter 6

Differentiation of Measures and Functions

This chapter is concerned with the differentiation theory of Radon measures. In the first two sections we introduce the Radon measures and discuss two covering theorems, the main technical tools in differentiation theory. The Vitali's covering theorem applies to the Lebesgue measure, while the more sophisticated Besicovitch's covering theorem applies to Radon measures. In Section 3 we prove a version of Radon-Nikodym theorem for Radon measures. It differs from the version in Chapter 5 for now there is a good description of the Radon-Nikodym derivative. As application we deduce Lebsegue-Besicovitch differentiation theorem in Section 4. Next we study the differentiability properties of functions in $\mathbb{R}$. First, we show how to relate increasing functions and Radon measures on the real line in Section 5. In Section 6 the concept of functions of bounded variations and absolutely continuous functions are recalled. The former is identified to be those integrable functions whose weak derivatives are signed Radon measures and the latter are those whose weak derivatives are integrable functions.

6.1 Radon Measures

Recall that Riesz representation theorem asserts that for a positive linear functional Λ on the space $C_c(X)$ where X is a locally compact Hausdorff space, there exists an Borel outer measure λ satisfying

$$\Lambda f = \int f d\lambda,$$

for all $f \in C_c(X)$. We used to call λ the Riesz measure associated to Λ. Furthermore, λ enjoys the following regularity properties:

(a) $\lambda(E) = \inf\{\lambda(G) : \quad E \subset G,\ G \text{ open}\}$ for every $E \subset \mathbb{R}^n$.

(b) $\lambda(E) = \sup\{\lambda(K) : \quad K \subset E,\ K \text{ compact}\}$ for every λ-measurable $E \subset \mathbb{R}^n$.

Now, an outer measure μ in $\mathbb{R}^n$ is called a **Radon measure** if

(a) It is **Borel regular**, that is, for every $A \subset \mathbb{R}^n$, there is a Borel set B, $A \subset B$, such that $\mu(A) = \mu(B)$, and

(b) it is finite on compact sets.

Given a Borel measure μ, we define

$$\Lambda f = \int f d\mu, \qquad f \in C_c(X).$$

When μ is finite on compact sets, Λ is a well-defined positive functional on $C_c(X)$. By Riesz representation theorem there is an outer Borel measure λ satisfying

$$\int f d\lambda = \int f d\mu, \qquad \forall f \in C_c(\mathbb{R}^n).$$

For every compact set K sitting inside an open set G, there exists a continuous function f compactly supported in G, equals to 1 in K and bounded between 0 and 1. Plugging such functions in the relation above and passing limit by Lebsegue's dominated convergence theorem, we see that λ and μ coincide on open sets. Consequently, they are the same on all Borel sets. Now, let A be an arbitrary set in $\mathbb{R}^n$. By Borel regularity, there is a Borel set B containing A such that $\mu(B) = \mu(A)$. Therefore,

$$\begin{aligned} \mu(A) &= \mu(B) \\ &= \lambda(B) \\ &\geq \lambda(A)\ . \end{aligned}$$

On the other hand, by the regularity property of the Riesz measure, for every A and $\varepsilon > 0$, there is an open set G containing A such that $\lambda(A) + \varepsilon \geq \lambda(G)$. Therefore,

$$\begin{aligned} \lambda(A) + \varepsilon &\geq \lambda(G) \\ &= \mu(G) \\ &\geq \mu(A)\ , \end{aligned}$$

which implies $\lambda(A) \geq \mu(A)$. Summing up, we have proved that every Radon measure is a Riesz measure in $\mathbb{R}^n$.

6.2 Covering Theorems

In the following a non-degenerate closed ball means $\overline{B}_r(x) = \{y \in R^n : \ |y - x| \leq r$ for some $x \in R^n$ and $r > 0$. We will use $\overline{B}$ to denote a non-degenerate closed ball.

Theorem 6.1 (**Vitali's Covering Theorem**). *Let $\mathcal{F}$ be a collection of non-degenerate closed balls whose diameters are bounded uniformly. There exists a countable subcollection $\mathcal{F}'$ of $\mathcal{F}$ consisting of disjoint balls such that*

$$\bigcup_{\mathcal{F}} \overline{B} \subset \bigcup_{B\in\mathcal{F}'} \hat{B},$$

where $\hat{B}$ is the "5 times" of $\overline{B}$.

The "5 times" of $\overline{B}_r(x)$ is $\overline{B}_{5r}(x)$. Note that the uniform boundedness condition on the diameters cannot be removed. Take $\mathbb{R}$ and $\mathcal{F} = \{[-\alpha, \alpha] : \alpha > \prime\}$, no such $\mathcal{F}'$ can be found.

Proof. Assume that all balls in $\mathcal{F}$ are confined to a bounded set. For each $n \geq 1$, we decompose $\mathcal{F}$ into $\mathcal{F}_n$ according to the diameter of the balls: $\overline{B}$ belongs to $\mathcal{F}_n$ if and only if its diameter falls in $(2^{-n}\rho, 2^{-n+1}\rho]$, where ρ is a bound on the diameter of the balls in $\mathcal{F}$. Since $\mathcal{F}_1$ is bounded, there is an upper bound on the number of disjoint balls in $\mathcal{F}_1$ that can be chosen. Beginning from a single ball, we can add more and more disjoint balls to obtain a largest collection $\mathcal{F}_1'$ which contains finitely many disjoint balls from $\mathcal{F}_1$. Next we consider $\mathcal{F}_2$ after excluding those intersect some ball from $\mathcal{F}_1'$. Repeating the same reasoning we obtain a maximal finite collection $\mathcal{F}_2'$ of disjoint balls. In this way $\mathcal{F}_n'$ are selected for all $n \geq 1$. Then

$$\mathcal{F}' \equiv \bigcup_n \mathcal{F}_n' \ ,$$

is a subcollection of $\mathcal{F}$ consisting of countable, pairwise disjoint balls. Moreover, for any $\overline{B} \in \mathcal{F}_n$, either $\overline{B}$ intersects some ball in $\mathcal{F}_n'$ or it intersects some ball from $\mathcal{F}_m', m \leq n-1$. Letting r and r_1 be the respective radii of $\overline{B}$ and the ball $\overline{B}(z)$ it intersects, $\overline{B}$ is contained in $\overline{B}_{r_1+2r}(z)$. We have

$$\begin{aligned} r_1 + 2r &\leq r_1 + \frac{\rho}{2^{n-1}} \\ &\leq r_1 + 2 \times \frac{\rho}{2^n} \\ &\leq r_1 + 2 \times \frac{\rho}{2^m} \\ &\leq 5r_1 \ . \end{aligned}$$

We conclude that

$$\bigcup_{\mathcal{F}} \overline{B} \subset \bigcup_{\mathcal{F}'} \hat{B}.$$

When $\mathcal{F}$ is not confined to a bounded set, we decompose it as a disjoint union

$$\mathcal{F}_n = \bigcup_k \mathcal{F}_{n,k}, \quad \mathcal{F}_{n,k} = \mathcal{F}_n \bigcap B_k(0) \ .$$

To pick $\mathcal{F}'_1$ we first select a maximal finite collection from $\mathcal{F}_{1,1}$. Next consider those balls in $\mathcal{F}_{1,2}$ disjoint from this maximal finite collection and pick a maximal finite collection. By repeating this procedure we end up with a countable, maximal collection of balls from $\mathcal{F}$ called $\mathcal{F}'_1$. Next we select $\mathcal{F}'_n$ by a similar procedure making sure that the balls now are disjoint from $\mathcal{F}'_m, 1 \leq m \leq n-1$. Then $\mathcal{F}'$, the union of all these $\mathcal{F}'_n$, satisfies our requirement. □

Corollary 6.2. *Let A be a set in $\mathbb{R}^n$ and $\mathcal{F}$ a family of nondegenerate, closed balls with uniformly bounded diameters that covers A in the following sense: For each $x \in A$, there is a sequence of balls centered at x with radii going to 0 from $\mathcal{F}$. For every open set G containing A, there is a countable subfamily $\mathcal{F}'$ consisting of disjoint balls from $\mathcal{F}$ such that*

$$\bigcup_{\mathcal{F}'} \overline{B} \subset G \ ,$$

and

$$\mathcal{L}^n \left(A \setminus \bigcup_{\mathcal{F}'} \overline{B} \right) = 0 \ .$$

Proof. Assume first that A is bounded. By the outer regularity of the Lebesgue measure, we can fix an open set $G_1 \subset G$ containing A such that $\mathcal{L}^n(G_1) \leq (1+\varepsilon_0)\mathcal{L}^n(A)$ where ε_0 satisfies

$$\theta \equiv 1 + 2\varepsilon_0 - \frac{1}{5^n} \in (0,1) \ .$$

Let

$$\mathcal{F}_1 = \left\{ \overline{B}(x) : \ x \in A, \ \overline{B}(x) \in \mathcal{F}, \ \overline{B}(x) \subset G_1 \right\} \ .$$

Applying Vitali's covering theorem, there are pairwise disjoint balls $B_k, k \geq 1$, from $\mathcal{F}_1$ that satisfy

$$A \subset \bigcup_{\mathcal{F}_1} \overline{B} \subset \bigcup_k \hat{B}_k \ .$$

We have

$$\begin{aligned}
\mathcal{L}^n\left(A \setminus \bigcup_k \overline{B}_k\right) &\leq \mathcal{L}^n\left(G_1 \setminus \bigcup_k \overline{B}_k\right) \\
&= \mathcal{L}^n(G_1) - \mathcal{L}^n\left(\bigcup_k \overline{B}_k\right) \\
&\leq \mathcal{L}^n(G_1) - \sum_k \mathcal{L}^n(\overline{B}_k) \\
&= \mathcal{L}^n(G_1) - \frac{1}{5^n}\sum_k \mathcal{L}^n(\hat{B}_k) \\
&\leq (1+\varepsilon_0)\mathcal{L}^n(A) - \frac{1}{5^n}\mathcal{L}^n\left(\bigcup_k \hat{B}_k\right) \\
&\leq (1+\varepsilon_0 - \frac{1}{5^n})\mathcal{L}^n(A) \\
&< \theta\mathcal{L}^n(A)\ .
\end{aligned}$$

We can fix a large m so that

$$\mathcal{L}^n\left(A \setminus \bigcup_{k=1}^m \overline{B}_k\right) \leq \theta\mathcal{L}^n(A)\ .$$

Next choose an open set $G_2 \subset G_1$ containing

$$A_2 \equiv A \setminus \bigcup_{k=1}^m \overline{B}_k$$

such that

$$\mathcal{L}^n(G_2) \leq (1+\varepsilon_0)\mathcal{L}^n(A_2)\ ,$$

with the same ε_0 specified before. Let

$$\mathcal{F}_2 = \left\{\overline{B}(x):\ x \in A_2,\ \overline{B}(x) \in \mathcal{F},\ \overline{B}(x) \cap \overline{B}_k = \phi,\ k = 1, \cdots, m,\ \overline{B}(x) \subset G_2\right\}\ .$$

Since $\cup_{k=1}^m \overline{B}_k$ is closed, every point in A_2 belongs to some ball in $\mathcal{F}_2$. We have

$$A_2 \subset \bigcup_{\mathcal{F}_2} \overline{B}\ .$$

Repeat the argument to (A_2, G_2) instead of (A, G_1) we obtain finitely many pairwise disjoint balls from $\mathcal{F}, \overline{B}_{m+1}, \cdots, \overline{B}_l$ such that

$$\mathcal{L}^n\left(A_2 \setminus \bigcup_{k=m+1}^l \overline{B}_k\right) \leq \theta\mathcal{L}^n(A_2)\ .$$

In other words,

$$\mathcal{L}^n\left(A \setminus \bigcup_{k=1}^{l} \overline{B}_k\right) \leq \theta^2 \mathcal{L}^n(A) \ .$$

Repeating this argument, we come up with a countable, pairwise disjoint balls from $\mathcal{F}$ such that

$$\begin{aligned} \mathcal{L}^n\left(A \setminus \bigcup_{k} \overline{B}_k\right) &\leq \liminf_{p\to\infty} \mathcal{L}^n\left(A \setminus \bigcup_{k=1}^{p} \overline{B}_k\right) \\ &\leq \liminf_{p\to\infty} \theta^p \mathcal{L}^n(A) \\ &= 0 \ . \end{aligned}$$

In general, for an unbounded A, let

$$R_n = \{n < |x| < n+1\}\,, \quad A_n = A \bigcap R_n,\ n \geq 1,$$

and $A_0 = A \bigcap B_1(0)$ and apply the previous result to each $A_n,\ n \geq 0$. Note that

$$\bigcup_n \{x : |x| = n\} \cap A$$

is a null set.

□

Vitali's covering theorem is effective for the Lebesgue measure. Unfortunately, for an arbitrary Radon measure, there may be no relation between the measure of the "5 times" of a ball to the measure of the ball. In this case, we need a more powerful covering result.

Theorem 6.3 (**Besicovitch's Covering Theorem**). *Let $\mathcal{F}$ be a collection of nondegenerate closed balls whose diameters are uniformly bounded and let A be the set consisting of the centers of these balls. There exists finitely many subcollections $\mathcal{F}_1, \ldots, \mathcal{F}_N$ of $\mathcal{F}$ in which each $\mathcal{F}_j$ is composed of countably many disjoint balls such that*

$$A \subset \bigcup_{j=1}^{N} \bigcup_{\mathcal{F}_j} \overline{B}.$$

The number N depends only on the dimension n.

Corollary 6.4. *Let μ be a Borel measure and A a set in $\mathbb{R}^n$ with $\mu(A) < \infty$. Let $\mathcal{F}$ a family of nondegenerate, closed balls with uniformly bounded diameters that covers A in the following sense: For each $x \in A$, there is a sequence of balls centered at x with radii going to 0 from $\mathcal{F}$. For every open set G containing A, there is a countable subfamily $\mathcal{F}'$ from $\mathcal{F}$ such that*

$$\bigcup_k \overline{B} \subset G \ ,$$

and

$$\mu\left(A \setminus \bigcup_{\mathcal{F}'} \overline{B}\right) = 0 \ .$$

The proof of Corollary 6.4 is very much like that of Corollary 6.2 where the only difference is to replace Vitali's theorem by Besicovitch's theorem. We refer to [EG] for the details.

6.3 Radon-Nikodym Derivatives

Let μ and ν be two Radon measures on $\mathbb{R}^n$. We define the lower and upper derivatives of ν with respect to μ by

$$\underline{D}_\mu \nu(x) = \begin{cases} \underline{\lim}_{r\to 0} \dfrac{\nu(\overline{B}_r(x))}{\mu(\overline{B}_r(x))}, & \mu(\overline{B}_r(x)) > 0, \ \forall r > 0 \\ \infty, & \mu(\overline{B}_r(x)) = 0, \ \text{some } r > 0. \end{cases}$$

and

$$\overline{D}_\mu \nu(x) = \begin{cases} \overline{\lim}_{r\to 0} \dfrac{\nu(\overline{B}_r(x))}{\mu(\overline{B}_r(x))}, & \mu(\overline{B}_r(x)) > 0, \ \forall r > 0 \\ \infty, & \mu(\overline{B}_r(x)) = 0, \ \text{some } r > 0. \end{cases}$$

Here $\underline{D}_\mu \nu$ and $\overline{D}_\mu \nu$ are functions from $\mathbb{R}^n$ to $[0, \infty]$.

Lemma 6.5. *Let μ and ν be Radon measures on $\mathbb{R}^n$. For every set A and $\alpha \in (0, \infty)$,*

(a) $A \subset \{x : \underline{D}_\mu \nu(x) \le \alpha\}$ implies that $\nu(A) \le \alpha \mu(A)$.

(b) $A \subset \{x : \overline{D}_\mu \nu(x) \ge \alpha\}$ implies that $\nu(A) \ge \alpha \mu(A)$.

Proof. We only prove (a). Assume first that A is bounded. For $\varepsilon > 0$, we pick an open G containing A. For each $x \in A$ there is a sequence of closed balls $\overline{B}$ in G centering at x with radius less than 1 and going down to 0 such that $\nu(\overline{B}) \le (\alpha + \varepsilon)\mu(\overline{B})$. The collection of all these balls from a covering of A. Applying Corollary 6.4 to ν, there is a countable subcollection consisting of disjoint closed balls, $\overline{B}_j, j \ge 1$, such that $\nu(A \setminus \bigcup_j \overline{B}_j) = 0$. It

follows that

$$\begin{aligned}
\nu(A) &\leq \nu\left(A \setminus \bigcup_j \overline{B}_j\right) + \nu\left(\bigcup_j \overline{B}_j\right) \\
&= \nu\left(\bigcup_j \overline{B}_j\right) \\
&= \sum_j \nu(\overline{B}_j) \\
&\leq (\alpha+\varepsilon)\sum_j \mu(\overline{B}_j) \\
&= (\alpha+\varepsilon)\mu\left(\bigcup_j \overline{B}_j\right) \\
&\leq (\alpha+\varepsilon)\mu(G).
\end{aligned}$$

Taking infimum over all G, by the regularity property of μ we get $\nu(A) \leq (\alpha+\varepsilon)\mu(A)$ which implies the desired result after letting ε go to 0. When A is unbounded, apply the previous proof to $A \cap B_n(0)$ and then let n go to ∞. □

The technique of proof of this lemma is standard and you should understand it well.

Remark 6.1. When μ is the Lebsegue measure and $\nu << \mathcal{L}^n$, Lemma 6.5 can be proved by Corollary 6.2 instead of Corollary 6.4. In other words, Besicovitch's covering theorem could be replaced by Vitali's covering theorem in this special case. See exercise.

Theorem 6.6. *Let μ and ν be Radon measures on $\mathbb{R}^n$. Then*

(i) The Borel set $\left\{x : \overline{D}_\mu\nu(x) = \infty\right\}$ is μ-null.

(ii) The Borel set $\left\{x : \underline{D}_\mu\nu(x) < \overline{D}_\mu\nu(x) < \infty\right\}$ is μ-null.

Note that $\overline{D}_\mu\nu(x) \geq \underline{D}_\mu\nu(x)$, $\forall x$, and $\underline{D}_\mu\nu(x) \geq 0$ from definition.

Proof. For a fixed k, $\left\{x : |x| \leq k,\ \overline{D}_\mu\nu(x) = \infty\right\}$ is contained in $\left\{x : |x| \leq k, \overline{D}_\mu\nu(x) \geq \alpha\right\}$ for all $\alpha > 0$. By Lemma 6.5(ii),

$$\begin{aligned}
\mu\left(\left\{x : |x| \leq k,\ \overline{D}_\mu\nu(x) = \infty\right\}\right) &\leq \frac{1}{\alpha}\nu\left(\left\{x : |x| \leq k, \overline{D}_\mu\nu(x) \geq \alpha\right\}\right) \\
&\leq \frac{1}{\alpha}\nu(\overline{B}_k)\ ,
\end{aligned}$$

which tends to 0 as $\alpha \to \infty$.

Next, for $0 < \alpha < \beta$, let $A(\alpha,\beta) \equiv \left\{x : |x| \le k,\ \underline{D}_\mu\nu(x) < \alpha < \beta < \overline{D}_\mu\nu(x) < \infty\right\}$. By Lemma 6.5,

$$\nu(A(\alpha,\beta)) \le \alpha\mu(A(\alpha,\beta)) \ , \quad \beta\mu(A(\alpha,\beta)) \le \nu(A(\alpha,\beta)) \ ,$$

which forces $\mu(A(\alpha,\beta)) = 0$. Observing that

$$\left\{x : |x| \le k,\ \underline{D}_\mu\nu(x) < \overline{D}_\mu\nu(x) < \infty\right\} = \bigcup_{\alpha,\beta} A(\alpha,\beta) \ ,$$

which $\alpha < \beta$ run over all rational numbers, we conclude that

$$\mu\left(\left\{x : |x| \le k,\ \underline{D}_\mu\nu(x) < \overline{D}_\mu\nu(x) < \infty\right\}\right) = 0 \ .$$

□

This theorem shows that $\overline{D}_\mu\nu = \underline{D}_\mu\nu < \infty$ except on a μ-measure zero set N where $N = I \bigcup R$,

$$I = \left\{x : \overline{D}_\mu\nu(x) = \infty\right\} \ , \quad R = \left\{x : \underline{D}_\mu\nu(x) < \overline{D}_\mu\nu(x) < \infty\right\} \ .$$

To make the derivative well-defined everywhere, we set

$$\begin{aligned} D_\mu\nu(x) &= \overline{D}_\mu\nu(x), & x \in \mathbb{R}^n \setminus N \\ &= \infty, & x \in N. \end{aligned}$$

Proposition 6.7. *Let μ and ν be Radon measures on $\mathbb{R}^n$. Then $D_\mu\nu$ is μ-measurable.*

Proof. We claim that $x \mapsto \mu(\overline{B}_r(x))$ is upper semicontinuous. Let $x_n \to x$. For a fixed $\varepsilon > 0$, $\overline{B}_r(x_n)$ is contained in $\overline{B}_{r+\varepsilon}(x)$ for all large n. It follows that

$$\mu(\overline{B}_r(x_n)) \subset \mu(\overline{B}_{r+\varepsilon}(x) \ ,$$

and

$$\overline{\lim_{n\to\infty}}\, \mu(\overline{B}_r(x_n)) \le \mu(\overline{B}_{r+\varepsilon}(x)) \ .$$

Now, letting $\varepsilon \to 0$, by the monotone convergence theorem

$$\overline{\lim_{n\to\infty}}\, \mu(\overline{B}_r(x_n)) \le \mu(\overline{B}_r(x)) \ ,$$

which means that $x \mapsto \mu(\overline{B}_r(x))$ is upper semicontinuous. Similarly one can show that $x \mapsto \nu(\overline{B}_r(x))$ is upper semicontinuous. For $x \in \mathbb{R}^n \setminus N$, noting that $\mu(\overline{B}_r(x)) > 0$ for all r, we can fix a sequence $r_k \to 0$ to get

$$\begin{aligned} D_\mu\nu(x) &= \lim_{r\to 0} \frac{\nu(\overline{B}_r(x))}{\mu(\overline{B}_r(x))} \\ &= \lim_{r_k\to 0} \frac{\nu(\overline{B}_r(x))}{\mu(\overline{B}_r(x))} \ < \infty \ . \end{aligned}$$

Since N is a μ-measurable null set where $D_\mu\nu = \infty$, we conclude that $D_\mu\nu$ is μ-measurable.

□

We have the following version of Radon-Nikodym theorem. Comparing with the Radon-Nikodym theorem we proved in Chapter 5, the main difference is that we have a more concrete description of the Radon-Nikodym derivative here.

Theorem 6.8 (**Radon-Nikodym Theorem**). *Let μ and ν be Radon measures on $\mathbb{R}^n$ satisfying $\nu \ll \mu$. For every μ-measurable A,*

$$\nu(A) = \int_A D_\mu \nu \, d\mu.$$

Proof. Assume that A is bounded. We first claim that A is μ-measurable implies that A is ν-measurable too. For, as μ is a Radon measure, we can find a Borel set (in fact, a G_δ-set) A_1, $A \subset A_1$, such that $\mu(A_1 \setminus A) = 0$. As $\mu \gg \nu$, $\nu(A_1 \setminus A) = 0$. Using the fact that every ν-null set is ν-measurable, $A = A_1 \setminus (A_1 \setminus A)$ is ν-measurable.

Consider the "exceptional sets"

$$\begin{aligned} I &= \left\{x : \overline{D}_\mu \nu = \infty\right\}, \\ Z &= \left\{x : D_\mu \nu = 0\right\}, \text{ and} \\ R &= \left\{x : \underline{D}_\mu \nu < \overline{D}_\mu \nu < \infty\right\}. \end{aligned}$$

We know that $\mu(I) = \mu(R) = 0$ from Lemma 6.5 and Theorem 6.6 respectively. Although $\mu(Z)$ may not vanish, $\nu(Z) = 0$ by Lemma 6.5 too.

For a μ-measurable A, let

$$A_m = \left\{x \in A : t^m < D_\mu \nu(x) \le t^{m+1}\right\}, \quad m \in \mathbb{Z},$$

where $t > 1$ is fixed. We have

$$A \setminus \bigcup_{-\infty}^{\infty} A_m \subset I \cup Z \cup R.$$

Therefore,

$$\nu\left(A \setminus \bigcup_{-\infty}^{\infty} A_m\right) = 0, \quad \text{or} \quad \nu(A) = \sum_{-\infty}^{\infty} \nu(A_m),$$

since $\nu(I) = \nu(R) = 0$ by $\mu(I) = \mu(R) = 0$ and $\nu \ll \mu$. Note that A and A_m are

μ-measurable implies that they are also ν-measurable. We have

$$\begin{aligned}
\nu(A) &= \nu\left(\bigcup_{-\infty}^{\infty} A_m\right) \\
&= \sum_{-\infty}^{\infty} \nu(A_m) \\
&\le \sum_{-\infty}^{\infty} t^{m+1}\mu(A_m) \quad (\text{ use } A_m \subset \left\{x : D_\mu\nu(x) \le t^{m+1}\right\} \text{ and Lemma 6.5.}) \\
&= t\sum_{-\infty}^{\infty} t^{m}\mu(A_m) \\
&\le t\sum_{-\infty}^{\infty} \int_{A_m} D_\mu\nu \, d\mu \\
&\le t\int_{\bigcup A_m} D_\mu\nu \, d\mu \\
&\le t\int_{A} D_\mu\nu \, d\mu.
\end{aligned}$$

Letting $t \downarrow 1$,

$$\nu(A) \le \int_A D_\mu\nu \, d\mu.$$

On the other hand,

$$\begin{aligned}
\nu(A) &= \sum_{-\infty}^{\infty} \nu(A_m) \\
&\ge \frac{1}{t}\sum_{-\infty}^{\infty} t^{m+1}\nu(A_m) \quad (\text{ use } \; A_m \subset \{x : t^m < D_\mu\nu(x)\} \text{ and Lemma 6.5.}) \\
&\ge \frac{1}{t}\sum_{-\infty}^{\infty} \int_{A_m} D_\mu\nu \, d\mu \\
&= \frac{1}{t}\int_{\bigcup A_m} D_\mu\nu \, d\mu \\
&= \frac{1}{t}\int_{A} D_\mu\nu \, d\mu \quad \left(\text{ use } \; A \setminus \bigcup_m A_m \subset I \cup Z \cup R \text{ and } \right. \\
&\qquad \left. \int_I D_\mu\nu \, d\mu = \int_R D_\mu\nu \, d\mu = \int_Z D_\mu\nu \, d\mu = 0\right)
\end{aligned}$$

Letting $t \uparrow 1$,

$$\nu(A) \ge \int_A D_\mu\nu \, d\mu.$$

Finally, when A is unbounded, apply the above result to $A \cap B_n(0)$ and then let $n \to \infty$. □

Remark 6.2. From Remark 6.1 we see that when μ is the Lebesgue measure, the proof of the Radon-Nikodym theorem can be based on Vitali's covering theorem and is independent of Besicovitch's covering theorem.

In Theorem 6.9, the measure ν is absolutely continuous with respect to μ. More generally, for two unrelated Radon measures we have

Theorem 6.9. *Let μ and ν be two Radon measures on $\mathbb{R}^n$. There exists a Borel set $A^* \subset \mathbb{R}^n$, $\mu(\mathbb{R}^n \setminus A^*) = 0$, such that by setting*

$$\nu_{ac} = \nu\lfloor_{A^*} \qquad \text{and} \qquad \nu_s = \nu\lfloor_{\mathbb{R}^n \setminus A^*},$$

we have $\nu_{ac} \ll \mu$ and $\nu_s \perp \mu$. Moreover, $D_\mu \nu_s = 0$ μ-a.e. so that $D_\mu \nu = D_\mu \nu_{ac}$ μ-a.e.

The measures ν_{ac} and ν_s are Radon. Recall that this decomposition is called the Lebesgue decomposition of ν with respect to μ. We learnt it in Chapter 5, but here we give a second proof which contains the additional information $D_\mu \nu_s = 0$.

Proof. Let

$$\mathcal{E} = \{A \subset \mathbb{R}^n : \mu(\mathbb{R}^n \setminus A) = 0,\ A \text{ is a Borel set}\}.$$

We claim there exists an $A^* \in \mathcal{E}$ such that

$$\nu(A^*) \le \nu(A), \quad \forall A \in \mathcal{E}.$$

For, let $A_k \in \mathcal{E}$ satisfy

$$\nu(A_k) \le \inf_{\mathcal{E}} \nu(A) + \frac{1}{k}$$

and $A^* = \bigcap A_k$. Clearly, $A^* \in \mathcal{E}$ and $\nu(A^*) \le \nu(A)$.

Define ν_{ac} and ν_s as in the theorem. Clearly, $\nu_s \perp \mu$. To show $\nu_{ac} \ll \mu$, note that $E = (E \cap A^*) \cup (E \setminus A^*)$. As $\nu_{ac}(E \setminus A^*) = 0$ by definition, it suffices to assume that $E \subset A^*$ and check $\mu(E) = 0$ implies that $\nu_{ac}(E) = 0$. Suppose on the contrary there is some $E_1 \subset A^*$ such that $\mu(E_1) = 0$ but $\nu_{ac}(E_1) > 0$. Find a Borel set E_2, $E_1 \subset E_2$, such that $\mu(E_2) = 0$ and $\nu_{ac}(E_2) = \nu_{ac}(E_1) > 0$. The set $A^* \setminus E_2 \in \mathcal{E}$ but

$$\begin{aligned} \nu_{ac}(A^* \setminus E_2) &= \nu_{ac}(A^*) - \nu_{ac}(E_2) \\ &< \nu_{ac}(A^*) \\ &= \inf_{\mathcal{E}} \nu(A^*), \end{aligned}$$

contradiction holds. Hence $\nu_{ac} \ll \mu$.

To show that $D_\mu \nu_s = 0$ μ-a.e., let $E = \{x : \ D_\mu \nu_s \geq \alpha\}$ where $\alpha > 0$ is fixed. By Lemma 6.5 we have $\nu_s(E \cap A^*) \geq \alpha \mu(E \cap A^*)$. As $\nu_s(A^*) = 0$, we have $\mu(E \cap A^*) = 0$, but then $\mu(E) = \mu(E \cap A^*) + \mu(E \cap (\mathbb{R}^n \setminus A^*) = 0$, that is, $D_\mu \nu_s = 0$ μ-a.e.. It follows that $D_\mu \nu = D_\mu \nu_{ac} + D_\mu \nu_s = D_\mu \nu_{ac} = 0$ μ-a.e.. □

6.4 Lebesgue Points

Theorem 6.10 (**Lebesgue's Differentiation Theorem**). *Let μ be a Radon measure on $\mathbb{R}^n$. For $f \in L^1_{loc}(\mu)$,*

$$\lim_{r \to 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} f \, d\mu = f(x),$$

for μ-a.e. x.

This result is due to Lebesgue for $\mu = \mathcal{L}^1$ and the general case is due to Besicovitch. It is also called **Lebesgue-Besicovitch differentiation theorem.**

Proof. Clearly it is sufficient to prove the theorem by assuming $f \geq 0$ and $f \in L^1(\mathbb{R}^n)$. Define a positive linear functional Λ be

$$\Lambda \varphi = \int \varphi f d\mu, \quad \varphi \in C_c(\mathbb{R}^n) \ .$$

By Riesz representation theorem, there is a Radon measure ν satisfying

$$\int \varphi d\nu = \int \varphi f d\mu \ .$$

By a routine argument,

$$\nu(E) = \int_E f \, d\mu, \quad E \ \mu\text{-measurable} \ .$$

By Theorem 6.6,

$$\lim_{r \to 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} f(y) \, d\mu(y) = D_\mu \nu(x),$$

for μ-a.e. x. As $\nu << \mu$, by Theorem 6.8,

$$\nu(E) = \int_E f d\mu = \int_E D_\mu \nu \, d\mu, \quad E \ \mu\text{-measurable} \ ,$$

hence $f = D_\mu \nu$ μ-a.e.. Combining these we get the theorem. □

The corollary gives a stronger result.

Corollary 6.11. *Let $f \in L^1_{loc}(\mu)$ where μ is a Radon measure. There exists a set L of full measure such that for every $x \in L$, $f(x)$ is finite and*

$$\lim_{r\to 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f - f(x)| \, d\mu = 0.$$

Proof. Order all rational numbers in a sequence $\{q_j\}$ and apply Theorem 6.10 to each $g_j = |f - q_j|$ to get

$$\frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f(y) - q_j| \, d\mu(y) \to |f(x) - q_j|, \quad \forall x \notin N_j, \tag{6.1}$$

as $r \to 0$ where $\mu(N_j) = 0$. Letting $N = \bigcup_{j=1}^{\infty} N_j$, $\mu(N) = 0$ and (6.1) holds for all $x \notin N$.

Given $\varepsilon > 0$ and $x \notin N$ such that $f(x)$ is finite, pick q_j such that $|f(x) - q_j| < \varepsilon$. We have

$$\begin{aligned}\frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f(y) - f(x)| \, d\mu(y) &\le \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f(y) - q_j| \, d\mu(y) \\ &\quad + \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |q_j - f(x)| \, d\mu(y) \\ &= \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f(y) - q_j| \, d\mu(y) + |q_j - f(x)|.\end{aligned}$$

By (6.1), for $\varepsilon > 0$, we can find some r_0 such that for all $r \in (0, r_0)$,

$$\frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f(y) - q_j| \, d\mu(y) < \varepsilon.$$

Therefore, we have

$$\frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f - f(x)| \, d\mu(y) \le 2\varepsilon.$$

□

A point x at which this corollary holds is called a **Lebesgue point** for the function f and the set of all Lebesgue points forms the **Lebesgue set** of f. It is implicitly assumed that the function is finite at a Lebesgue point. Moreover, the Lebesgue point depends on the pointwise definition of f and therefore is not a concept attached to f as an equivalence class. Every point of continuity of f is a Lebesgue point, but the converse may be not true. It is not hard to construct Lebesgue points at which the function is discontinuous.

So far we have been considering taking the average of a function over balls. Now we consider taking average over other sets. Let $\{E_j\}$ be a sequence of μ-measurable sets. We call it **shrinks regularly** to x if there is $\overline{B}_{r_j}(x)$, $r_j \to 0$, such that

$$\alpha\mu(E_j) \ge \mu(\overline{B}_{r_j}(x)), \quad E_j \subset \overline{B}_{r_j}(x),$$

for some $\alpha > 0$ (depending on x).

Corollary 6.12. *Let $\{E_j\}$ shrink regularly to x, a Lebesgue point of $f \in L^1_{loc}(\mathbb{R}^n)$. Then*

$$\lim_{j\to\infty} \frac{1}{\mu(E_j)} \int_{E_j} f \, d\mu = f(x).$$

Proof. By Corollary 6.11,

$$\begin{aligned}\frac{1}{\mu(E_j)} \int_{E_j} |f - f(x)| \, d\mu &\leq \alpha \frac{1}{\mu(\overline{B}_{r_j}(x))} \int_{E_j} |f - f(x)| \, d\mu \\ &\leq \alpha \frac{1}{\mu(\overline{B}_{r_j}(x))} \int_{\overline{B}_{r_j}(x)} |f - f(x)| \, d\mu \\ &\to 0 \text{ as } r_j \to 0.\end{aligned}$$

□

Let E be a set in $\mathbb{R}^n$. A point $x \in \mathbb{R}^n$ is called a **point of density** α of E if

$$\lim_{r\to 0} \frac{\mu(E \cap \overline{B}_r(x))}{\mu(\overline{B}_r(x))} \in [0, 1] \ ,$$

exists and is equal to α.

Corollary 6.13. *Let E be μ-measurable. Every Lebesgue point of χ_E is a point of density 1. Thus the density is 1 for μ-a.e. $x \in E$.*

Proof. Let $f = \chi_E \in L^1_{loc}(\mu)$ and x a Lebesgue point of f. We have

$$\frac{\mu(E \cap \overline{B}_r(x))}{\mu(\overline{B}_r(x))} = \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} \chi_E \, d\mu \to \chi_E(x) = 1,$$

as $r \to 0$. Every Lebesgue point has density one. □

Note that for x lying outside E, applying the same argument to $\mathbb{R}^n \setminus E$, we conclude that for μ-a.e. points lying out E their density are equal to 0.

6.5 Measures and Distribution Functions

In this section general results obtained in the previous sections are applied to functions on the real line. In particular, we will establish the differentiability of functions of bounded

variations and prove the fundamental theorem of calculus for absolutely continuous functions. Since many results have been known from undergraduate real analysis, we will be sketchy here or there.

There is a close relationship between Radon measures and monotone functions in $\mathbb{R}$, and similarly for signed Radon measures and functions of bounded variation.

Let μ be a finite Radon measure on $\mathbb{R}$. Its **distribution function** is given by

$$f(x) = \mu((-\infty, x)).$$

(In some texts, $\mu(\infty, x]$ is used instead, but the difference is insignificant.) It is readily checked that f satisfies the following properties:

(a) f is increasing,

(b) f is left continuous, i.e., $\lim_{y \uparrow x} f(y) = f(x)$,

(c) $f(-\infty) = \lim\limits_{x \to -\infty} f(x) = 0$,

(d) $f(\infty) = \lim\limits_{x \to -\infty} f(x) < \infty$.

Conversely, we have

Proposition 6.14. *Let f be a function on $\mathbb{R}$ satisfying (a)–(d). There exists a unique finite Radon measure μ such that f is the distribution function of μ.*

Here we sketch a proof based on the Riemann-Stieltjes integral.

Let P be a partition of the real line and consider the Riemann-Stieltjes sum for a function φ in $C_c(\mathbb{R})$.

$$\sum_{-\infty}^{\infty} \varphi(z_j)(f(x_{j+1}) - f(x_j)),$$

where $z_j \in [x_j, x_{j+1}]$ and $P : \cdots < x_j < x_{j+1} < x_{j+1} < \cdots$, is the partition. It is not hard to show that as $\|P\| = \sup_j |x_{j+1} - x_j| \to 0$, these sums tend to a definite number called the Riemann-Stieltjes integral of φ, denoted by

$$\int \varphi \, df.$$

From the definition, it is clear the Riemann-Stieltjes integral is a positive linear functional on $C_c(\mathbb{R})$. By Riesz representation theorem, there exists a Radon measure μ such that

$$\int \varphi \, df = \int \varphi \, d\mu.$$

By suitably choosing φ, one can deduce $f(x) = \mu((-\infty, x))$, that is, F is the distribution function for μ.

Let μ_1 be another Radon measure satisfying $f(x) = \mu_1((-\infty, x))$. From $\mu_1((-\infty, x)) = \mu((-\infty, x))$ one deduces $\mu_1((x, y)) = \mu((x, y))$, $\forall x \leq y$, and μ_1 coincides with μ.

As an application of the connection between measures and functions we prove the almost everywhere differentiability of monotone functions. First of all, we recall that every monotone function defined on an interval has at most countably many discontinuity points which are jumps. Therefore, by redefining the functions at these points, one obtains a left continuous monotone function which is equal to the original function except possibly at a countable set.

Proposition 6.15. *Every monotone function on $[a, b]$ is differentiable almost everywhere.*

Proof. Let f_0 be increasing on $[a, b]$. Extend it to be constant below a and beyond b so that it is increasing on $\mathbb{R}$. Call the extension f. By the remark above we may assume f is left continuous. Moreover, by subtracting it by a constant, we may assume that it satisfies (a)-(d) above.

Let μ be the finite Radon measure taking f as its distribution function. We have the Lebesgue decomposition $\mu = \mu_{ac} + \mu_s$ with respect to the Lebesgue measure. By Theorem 6.9 $D\mu = D\mu_{ac}$, $D\mu_s = 0$ on a set of full measure E_1. Note that we have dropped the subscript $\mathcal{L}^1$. By Corollary 6.12,

$$\begin{aligned} \lim_{\delta \to 0} \frac{\mu_{ac}([x, x+\delta))}{\delta} &= \lim_{\delta \to 0} \frac{1}{\delta} \int_{[x, x+\delta)} \frac{D\mu_{ac}}{D\mathcal{L}^1} d\mathcal{L}^1 \\ &= \frac{D\mu_{ac}}{D\mathcal{L}^1}(x) \ , \end{aligned}$$

for x in a set of full measure E_2. For $x \in E_1$,

$$\begin{aligned} \overline{\lim_{\delta \to 0}} \frac{\mu_s([x, x+\delta))}{\delta} &\leq 2 \overline{\lim_{\delta \to 0}} \frac{\mu_s([x-\delta, x+\delta])}{2\delta} \\ &= 0 \ . \end{aligned}$$

Therefore, for $x \in E_1 \cap E_2$,

$$\lim_{\delta \to 0} \frac{\mu([x, x+\delta))}{\delta} = \frac{D\mu}{D\mathcal{L}^1}(x) \ .$$

Using the relation $\mu([x, x+\delta)) = f(x+\delta) - f(x)$, we conclude that

$$\frac{d^+ f}{dx}(x) = \frac{D\mu}{D\mathcal{L}^1}(x), \quad x \in E_1 \cap E_1 \ .$$

On the other hand, by a similar argument we can show that

$$\frac{d^- f}{dx}(x) = \frac{D\mu}{D\mathcal{L}^1}(x)$$

on a set of full measure. The desired result follows. □

Proposition 6.16. *Let f be the distribution function of the finite Radon measure μ. Then*

$$\int \varphi' f d\mathcal{L}^1 = -\int \varphi d\mu \ , \quad \forall \varphi \in C_c(\mathbb{R}) \cap C^1(\mathbb{R}) \ . \tag{6.2}$$

Conversely, suppose that (6.2) holds for some integrable function f and finite Radon measure μ. Then f admits a representative which is increasing and $f+c$ is the distribution function of μ for some constant c.

When $\mu << \mathcal{L}^1$, $d\mu = f' d\mathcal{L}^1$. Hence (6.2) may be interpreted as the "weak derivative" of f is the measure μ.

Proof. Looking at a sequence of partitions with length δ going to 0, we have

$$\begin{aligned}
\int \varphi d\mu &= \int \varphi df \\
&= \lim_{\delta \to 0} \sum_k \varphi(x_{k+1})(f(x_{k+1}) - f(x_k)) \\
&= -\lim_{\delta \to 0} \sum_k (\varphi(x_{k+1}) - \varphi(x_k)) f(x_k) \\
&= -\lim_{\delta \to 0} \sum_k \varphi'(z_k) f(x_k)(x_{k+1} - x_k) \\
&= -\int \varphi' f d\mathcal{L}^1 \ .
\end{aligned}$$

Conversely, let g be the distribution function of μ. By what we just proved, (6.2) holds when f is replaced by g. It follows that

$$\int \varphi'(f-g)\, d\mu = 0, \quad \forall \varphi \in C_c(\mathbb{R}) \cap C^1(\mathbb{R}) \ .$$

Let x and y be two Lebesgue points of f satisfying $x < y$. For a small $\delta > 0$, let φ be the continuous function which is equal to 1 on $[x, y]$, 0 outside $[x - \delta, y]$ and linear in between. By approximation it is legal for a test function. As a result,

$$\frac{1}{\delta} \int_{x-\delta}^{x} (f-g) d\mathcal{L}^1 - \frac{1}{\delta} \int_{y-\delta}^{y} (f-g) d\mathcal{L}^1 = 0 \ .$$

Letting $\delta \to 0$, we get $f(x) - g(x) = f(y) - g(y)$, so $g = f + c$ for some constant c. □

6.6 BV Functions and AC Functions

Recall that a function f on $[a, b]$ is of bounded variation if there is some M such that

$$\sum_{j=1}^{n} |f(x_{j+1}) - f(x_j)| \leq M$$

for all partitions $P: \ a = x_0 < x_1 < \cdots < x_n < x_{n+1} = b$ of $[a, b]$. Every monotone, real-valued function on $[a, b]$ is of bounded variation. For a function of bounded variation, its total variation (function) T_f is a function on $[a, b]$ given by

$$T_f(x) = \sup \left\{ \sum_{j=1}^{n} |f(x_{j+1}) - f(x_j)| : \ a = x_0 < x_1 < \cdots < x_{n+1} = x \text{ on } [a, x] \right\}.$$

The total variation function is increasing and of bounded variation. As a result,

$$f_2 = \frac{1}{2}(T_f + f) \qquad \text{and} \qquad f_1 = \frac{1}{2}(T_f - f)$$

are also increasing. The decomposition $f = f_2 - f_1$ is called the Jordan decomposition of f.

Theorem 6.17. *Every function f of bounded variation on $[a, b]$ is differentiable almost everywhere. Furthermore, there is a signed Radon measure μ such that*

$$\int \varphi' f d\mathcal{L}^1 = -\int \varphi d\mu \ , \quad \forall \varphi \in C_c(a, b) \cap C^1(a, b). \tag{6.3}$$

Conversely, let $f \in L^1[a, b]$ and μ a signed Radon measure μ satisfy (6.3). Then f has a representative in $BV[a, b]$.

Proof. Let $f = f_2 - f_1$ be the Jordan decomposition of f. We extend f_1 and f_2 so that they are constant on $(-\infty, a]$ and $[b, \infty)$ and let $f = f_2 - f_1$. By modifying these functions on a null set we may also assume that they are left continuous. Let μ_1 and μ_2 be the finite Radon measures associated to $f_1 - f_1(a)$ and $f_2 - f_2(a)$ respectively. From (6.2) we see that

$$\int \varphi' f d\mathcal{L}^1 = -\int \varphi d\mu \ , \quad \forall \varphi \in C_c(\mathbb{R}) \cap C^1(\mathbb{R}) \ , \quad \mu = \mu_2 - \mu_1 \ ,$$

holds.

Conversely, let $\mu = \mu_2 - \mu_1$ be the Jordan decomposition of the Radon signed measure μ. we extend μ_1, μ_2, μ to $\mathbb{R}$ by setting $\mu_1(E) = \mu_1(E \cap [a, b])$, etc. Let f_1 and f_2 be the distribution functions of μ_1 and μ_2 respectively. Then

$$\int \varphi'(f - (f_2 - f_1)) d\mathcal{L}^1 = 0 \ , \quad \forall \varphi \in C_c[a, b] \ ,$$

implies $f = f_2 - f_1 + c$ for some c almost everywhere, see the proof of Proposition 6.16 for details. □

This theorem characterizes BV-functions as those integrable functions whose weak derivatives are Radon measures. In the following we pursue the further question: What

are those BV-functions whose associated Radon measures are absolutely continuous with respect to the Lebesgue measure? The answer turns out simple; they are precisely the absolutely continuous functions. Traditionally this class of functions is characterized as those functions for which the fundamental theorem of calculus holds.

A function f on $[a,b]$ is called absolutely continuous if for every $\varepsilon > 0$, there exists a δ such that

$$\sum_k |f(x_{k+1}) - f(x_k)| < \varepsilon, \quad \text{whenever } \sum_k |x_{k+1} - x_k| < \delta$$

where $I_k = [x_k, x_{k+1}]$ are intervals in $[a,b]$ with mutually disjoint interior. The number of these intervals could be countable. An absolutely continuous function must be of bounded variation, but the converse is not always true.

Set T_f be the total variation function of f. It can be shown that T_f is absolutely continuous if f is absolutely continuous. As a result, both $f_2 = \frac{1}{2}(T_f + f)$ and $f_1 = \frac{1}{2}(T_f - f)$ are increasing and absolutely continuous.

Let $f \in BV[a,b]$. Letting $f = f_2 - f_1$ be its Jordan decomposition, we extend f, f_1 and f_2 to $\mathbb{R}$ by setting them to be constants on $(-\infty, a)$ and (b, ∞) respectively. Let $\mu = \mu_2 - \mu_1$ where μ_1 and μ_2 take f_1 and f_2 as their respective distribution functions. We will assume this in the following proof.

Theorem 6.18. *Let $f \in BV[a,b]$. The following statements are equivalent:*

(a) f is absolutely continuous on $[a,b]$.

(b) The associated Radon measure μ is absolutely continuous with respect to the Lebesgue measure.

(c) f' exists a.e., belongs to $L^1[a,b]$ and

$$f(x) = f(a) + \int_a^x f' \, d\mathcal{L}^1, \quad \forall x \in [a,b],$$

holds.

Proof. We can assume that f is increasing. The general case follows from the Jordan decomposition.

(a) $\Rightarrow$ (b). After the extension f is absolutely continuous on $\mathbb{R}$. For $\varepsilon > 0$, there is some $\delta > 0$ such that

$$\sum_k |f(b_k) - f(a_k)| < \varepsilon,$$

whenever (a_k, b_k)'s are pairwise disjoint and

$$\sum_k (b_k - a_k) < \delta \ .$$

Now, let E be a null set in Lebesgue measure. For $\delta > 0$, there are intervals (c_k, d_k) whose union covers E and satisfies $\sum_k (c_k - d_k) < \delta$. We can find pariwise disjoint $(a_k, b_k), k \geq 1$, such that

$$\bigcup_k (a_k, b_k) = \bigcup_k (c_k, d_k).$$

We have

$$\sum_k (b_k - a_k) \leq \sum_k (c_k - d_k) < \delta.$$

Therefore,

$$\begin{aligned} \mu(E) &\leq \mu\left(\bigcup_k (c_k, d_k)\right) \\ &= \mu\left(\bigcup_k (a_k, b_k)\right) \\ &= \sum_k \mu(a_k, b_k) \\ &\leq \sum_k (f(b_k) - f(a_k)) \\ &< \varepsilon \ , \end{aligned}$$

whence E is also μ-null.

(b) $\Rightarrow$ (c). From (6.2),

$$\int \varphi' f \mathcal{L}^1 = -\int \varphi f' d\mathcal{L}^1, \quad \forall \varphi \in C_c(\mathbb{R}) \cap C^1(\mathbb{R}).$$

Let $x, y, x < y$, be two Lebesgue points of f in (a, b). As in the proof of Proposition 6.16 we deduce from this identity

$$f(y) = f(x) + \int_x^y f' d\mathcal{L}^1 \ .$$

It follows that f is continuous and one can take $x = a$ and y any point in $[a, b]$.

(c)$\Rightarrow$ (a) Let us show that for every integrable function g,

$$G(x) \equiv \int_a^x g \, d\mathcal{L}^1,$$

is absolutely continuous on $[a, b]$. Recall the absolute continuity of Lebsegue integral: Letting $g \in L^1[a, b]$, for every $\varepsilon > 0$, there exists some δ such that

$$\int_E g \, d\mathcal{L}^1 < \varepsilon, \quad \text{whenever } E \text{ is measurable and } \mathcal{L}^1(E) < \delta.$$

Taking $E = \bigcup_j I_j$ where I_j are disjoint intervals, we have

$$\mathcal{L}^1(E) = \sum_j \mathcal{L}^1(I_j) = \sum_j |x_{j+1} - x_j| < \delta,$$

which implies

$$\int_E g \, d\mathcal{L}^1 = \sum_j |G(x_{j+1}) - G(x_j)| < \varepsilon,$$

so G is absolutely continuous on $[a, b]$.

□

Comments on Chapter 6. We follow [EG] closely in this chapter except on the definition of the Radon measure. If you google for Radon measure, you will find there are different definitons. Here we use one that is consistent with our terminology in the last semester. Aside from the clarification of Radon measure there are very few changes. I do not copy some of the proofs. Please look up 1.5–1.7 in [EG]. The proof of Besicovitch's covering theorem is left out. You may google for more on this theorem.

There is another version of Vitali's covering theorem where the collection of balls is a finite one. Then the number 5 can be replaced by 3, see exercise and [R1].

Lebesgue points and sets were introduced by Lebesgue in the study of Fourier series. A celebrated theorem of his asserts that the Cesàro sums of the Fourier series of a Lebesgue integrable function converge to the function at every Lebesgue point. Look up chapter 5 of [HS] for a proof.

Reflections on the definition of Lebesgue points. As pointed out already, Lebesgue points vary from representatives of the same L^1-function. In the following we let $[f]$ denote the equivalence class of functions to which f, a pointwisely defined function, belongs. For instance, consider f to be the constant function 1 and f_E the function which is equal to 1 except on the Lebesgue null set E, where it is assigned to some values not equal to 1. Let $\mathcal{L}(f)$ be the Lebesgue set of f. We have $\mathcal{L}(f) = \mathbb{R}^n$ and $\mathcal{L}(f_E) = \mathbb{R}^n \setminus E$. So the concept of Lebesgue points is not intrinsic. It is possible to obtain an intrinsic definition of a Lebesgue point. Indeed, a point x is called an **intrinsic Lebesgue point** of a locally integrable function $[f]$ in $L^1(\mu)$ if there exists some $c \in \mathbb{R}$ such that

$$\lim_{r \to 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f(y) - c| d\mu(y) = 0,$$

where f is any representative of $[f]$. It is clear that the value c, whenever exists, is the same for all representatives. Corollary 6.12 tells us that all intrinsic Lebesgue points form a set of full measure. Let us call it the **intrinsic Lebesgue set** of $[f]$, denoted by $\mathcal{L}([f])$. With this at our disposal, given $[f] \in L^1(\mu)$, we define its **precise representation** f_p to be, for $x \in \mathcal{L}([f])$, set $f_p(x)$ equal to the value c as in the above limit; and, for $x \notin \mathcal{L}([f])$, set $f_p(x) = \infty$. (You may also set it to be 0 so that f is finite everywhere.) Clearly, $f_p \in [f]$ and $\mathcal{L}(f_p) = \mathcal{L}([f])$. The precise representation of a locally integrable function has the nice property that $f_p(x)$ is finite if and only if

$$\lim_{r\to 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f(y) - f_p(x)| d\mu(y) = 0, \quad \forall f \in [f].$$

Furthermore, the precise representation has the maximal property, namely, for any representative f of $[f]$, $\mathcal{L}(f) \subset \mathcal{L}(f_p)$. An interesting question is, given a null set E, is there some $[f] \in L^1_{\text{loc}}(\mu)$ so that $\mathcal{L}([f]) = \mathbb{R}^n \setminus E$? I don't know the answer even when $\mu = \mathcal{L}^1$ and E is Borel.

One may find a detailed discussion on BV- and AC-functions in Royden's Real Analysis which is assumed to be covered in the undergraduate real analysis. Our treatment on these functions differs from those in Royden's and [R1] so that its extension to higher dimensions becomes apparent. We have restricted things to a bound interval $[a, b]$ in order to compare with the old results. In general one can define $BV(\mathbb{R})$ and $AC(\mathbb{R})$. The Sobolev space $H^1(\mathbb{R}^n)$ consists of all L^1-functions whose weak derivatives f_k defined via

$$\int f \frac{\partial \varphi}{\partial x_k} d\mathcal{L}^n = -\int f_k \varphi d\mathcal{L}^n,$$

are also in $L^1(\mathbb{R}^n)$. It reduces to $AC(\mathbb{R})$ when $n = 1$. On the other hand, $BV(\mathbb{R}^n)$ consists of all L^1-functions whose weak derivatives are Radon measures. It reduces to $BV(\mathbb{R})$ when $n = 1$. The definition is essentially the same as the relation in (6.3), see chapter 5 in [EG] for a detailed treatment.

Chapter 7

Fubini's Theorem

The product measure of two outer measures which is again an outer measure is defined in Section 1. In Section 2 Fubini's theorem which relates the integral with respect to the product measure to the iterated integrals with respect to its factor measures. Section 3, 4 and 5 contain applications of Fubini's theorem to three different topics, namely, Rademarcher's theorem on the differentiability of Lipschitz continuous functions, layer cake representation and the convolution of functions.

7.1 The Product Measure

Let μ and ν be outer measures on the non-empty sets X and Y respectively. We define the **product measure** of μ and ν on the product set $X \times Y$ as, for $E \subset X \times Y$,

$$\begin{aligned} & (\mu \times \nu)(E) \\ = \ & \inf \left\{ \sum_{j=1}^{\infty} \mu(A_j)\nu(B_j) : E \subset \bigcup_{j=1}^{\infty} A_j \times B_j, \ A_j \ \mu\text{-measurable}, \ B_j \ \nu\text{-measurable.} \right\} . \end{aligned}$$

It is understood that $\phi \times \phi = \phi$ so that $(\mu \times \nu)(\phi) = 0$. It is also straightforward to check

$$(\mu \times \nu) \left(\bigcup_{j=1}^{\infty} E_j \right) \leq \sum_{j=1}^{\infty} (\mu \times \nu)(E_j), \quad \forall E_j \subset X \times Y, j \geq 1.$$

Hence $\mu \times \nu$ is an outer measure on $X \times Y$.

In the following we study how to evaluate $\mu \times \nu$ in terms of μ and ν. Introduce the

following notations:

$$\mathcal{P}_0 = \big\{A \times B : A \ \mu\text{-measurable and } B \ \nu\text{-measurable}\big\},$$
$$\mathcal{P}_1 = \Big\{R : R = \bigcup_{j=1}^{n} A_j \times B_j,\ 1 \le n \le \infty,\ A_j \times B_j \in \mathcal{P}_0\Big\},\ \text{and}$$
$$\mathcal{P}_2 = \Big\{R : R = \bigcap_{j=1}^{n} R_j,\ 1 \le n \le \infty,\ R_j \in \mathcal{P}_1\Big\}.$$

Elements in $\mathcal{P}_0$ are called measurable rectangles. Clearly $\mathcal{P}_0 \subset \mathcal{P}_1 \subset \mathcal{P}_2$. We also set

$$\mathcal{F} = \Bigg\{R :\ \text{For } \nu\text{-a.e. } y,\ x \mapsto \chi_R(x,y) \text{ is } \mu\text{-measurable and } y \mapsto \int \chi_R(x,y)\, d\mu(x) \text{ is } \nu\text{-measurable.}\Bigg\}$$

Note that the map

$$y \mapsto \int \chi_R(x,y)\, d\mu(x)$$

is defined almost everywhere in Y. Since ν is a complete measure and so every null set is measurable, it can be extended to be a measurable function in Y. Moreover, the integral

$$\int_Y \int_X \chi_R(x,y) d\mu(x) \nu(y)$$

is independent of the extension.

For $R \in \mathcal{F}$, we can define

$$\rho(R) = \int_Y \left(\int_X \chi_R(x,y)\, d\mu(x) \right) d\nu(y).$$

We will show in a series of lemmas that $\mathcal{P}_0, \mathcal{P}_1$, and $\mathcal{P}_2 \subset \mathcal{F}$ and they are $\mu \times \nu$-measurable. Moreover,

$$(\mu \times \nu)(R) = \rho(R),$$

for $R \in \mathcal{P}_1$ or $R \in \mathcal{P}_2$ provided in the latter R satisfies $\rho(R) < \infty$.

Lemma 7.1. *$\mathcal{P}_0 \subset \mathcal{F}$ and*

$$\rho(A \times B) = \mu(A)\nu(B), \quad A \times B \in \mathcal{P}_0.$$

Proof. The key observation is

$$\chi_{A \times B}(x,y) = \chi_A(x)\chi_B(y), \quad \forall (x,y) \in X \times Y.$$

For $y \in B$,

$$x \mapsto \chi_{A\times B}(x,y) = \chi_A(x) \text{ is } \mu\text{-measurable.}$$

For $y \notin B$,

$$x \mapsto \chi_{A\times B}(x,y) \equiv 0 \text{ is } \mu\text{-measurable.}$$

Next,

$$y \mapsto \int_X \chi_{A\times B}(x,y)\, d\mu(x) = \mu(A)\chi_B(y) \text{ is } \nu\text{-measurable.}$$

We have shown that $\mathcal{P}_0 \subset \mathcal{F}$. We also have

$$\begin{aligned}\rho(A\times B) &= \int_Y \left(\int_X \chi_{A\times B}\, d\mu\right) d\nu \\ &= \int_Y \left(\int_X \chi_A(x)\chi_B(y)\, d\mu(x)\right) d\nu(y) \\ &= \mu(A)\nu(B).\end{aligned}$$

□

Lemma 7.2. *$\mathcal{P}_1 \subset \mathcal{F}$ and*

$$\rho(R) = \sum_1^\infty \mu(A_j)\nu(B_j), \quad \textit{whenever } R = \mathop{\bigcup}^{\circ} A_j \times B_j, A_j \times B_j \in \mathcal{P}_0.$$

We have put a circle on top of the union sign to indicate that this is a union of pairwise disjoint sets.

The following fact will be used several times in the subsequent development: Each $R \in \mathcal{P}_1$ can be expressed as a countable disjoint union of measurable rectangles. Indeed, it follows from the observation

$$A_2 \times B_2 \setminus A_1 \times B_1 = A_2 \times (B_2 \setminus B_1) \mathop{\bigcup}^{\circ} (A_2 \setminus A_1) \times (B_2 \cap B_1).$$

Proof. Let $R \in \mathcal{P}_1$. Then

$$R = \mathop{\bigcup_{j=1}}^{\circ} A_j \times B_j, \quad A_j \times B_j \in \mathcal{P}_0.$$

Then

$$\chi_R = \sum_{j=1}^\infty \chi_{A_j\times B_j}.$$

Let

$$\varphi_n = \sum_{j=1}^n \chi_{A_j\times B_j}.$$

From Lemma 7.1, each $\varphi_n \in \mathcal{F}$. As $\chi_R(x,y) = \lim_{n\to\infty} \varphi_n(x,y)$. For ν a.e. y, $x \mapsto \chi_R(x,y)$ is μ-measurable. By monotone convergence theorem,

$$\int_X \chi_R(x,y)\,d\mu = \lim_{n\to\infty} \int_X \varphi_n(x,y)\,d\mu.$$

As $y \mapsto \int \varphi_n(x,y)\,d\mu(x)$ is ν-measurable, $y \mapsto \int \chi_R(x,y)\,d\mu(x)$ is also ν-measurable. We have shown that $R \in \mathcal{F}$.

Moreover,

$$\begin{aligned}
\rho(R) &= \int_Y \left(\int_X \chi_R(x,y)\,d\mu(x) \right) d\nu(y) \\
&= \int_Y \left(\int_X \lim_{n\to\infty} \varphi_n\,d\mu(x) \right) d\nu(y) \\
&= \int_Y \left(\lim_{n\to\infty} \int_X \varphi_n\,d\mu(x) \right) d\nu(y) \quad \text{(Monotone convergence theorem)} \\
&= \int_Y \left(\sum_{j=1}^{\infty} \mu(A_j)\chi_{B_j}(y) \right) d\nu(y) \\
&= \sum_{j=1}^{\infty} \int_Y \mu(A_j)\chi_{B_j}(y)\,d\nu(y) \quad \text{(Monotone convergence theorem)} \\
&= \sum_{j=1}^{\infty} \mu(A_j)\nu(B_j).
\end{aligned}$$

□

Lemma 7.3. *For $E \subset X \times Y$,*

$$(\mu \times \nu)(E) = \inf \{\rho(R) : E \subset R,\ R \in \mathcal{P}_1\}.$$

In particular, for $A \times B \in \mathcal{P}_0$,

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B) = \rho(A \times B).$$

Proof. Let $R \in \mathcal{P}_1$, $E \subset R$ and express $R = \overset{\circ}{\bigcup} A_j \times B_j$ where $A_j \times B_j \in \mathcal{P}_0$. Using the definition of $\mu \times \nu$,

$$\begin{aligned}
(\mu \times \nu)(E) &\le \sum_{j=1}^{\infty} \mu(A_j)\nu(B_j) \\
&= \rho(R). \quad \text{(Lemma 7.2)}
\end{aligned}$$

Taking infimum over all these R gives

$$(\mu \times \nu)(E) \leq \inf \{\rho(R) : E \subset \mathbb{R},\ R \in \mathcal{P}_1\}.$$

On the other hand, for each n, there is some $R_n \in \mathcal{P}_1$, $E \subset R_n$, such that $R_n = \overset{\circ}{\bigcup} A_j^n \times B_j^n$, and

$$\begin{aligned}(\mu \times \nu)(E) + \frac{1}{n} &\geq \sum_{j=1}^{\infty} \mu(A_j^n)\nu(B_j^n) \\ &= \rho(R_n) \\ &\geq \inf \{\rho(R) : E \subset R,\ R \in \mathcal{P}_1\},\end{aligned}$$

and the inequality

$$(\mu \times \nu)(E) \geq \inf \{\rho(R) : E \subset \mathbb{R},\ R \in \mathcal{P}_1\}$$

follows after letting $n \to \infty$.

Now, for $A \times B \in \mathcal{P}_0$ and any $R \in \mathcal{P}_1$, $A \times B \subset R$,

$$\begin{aligned}(\mu \times \nu)(A \times B) &\leq \mu(A)\nu(B) \\ &= \rho(A \times B)(\text{Lemma 7.1}) \\ &\leq \rho(R), \quad (\text{since } \chi_{A\times B} \leq \chi_R)\end{aligned}$$

so $\rho(A \times B)$ realizes the infimum of ρ over all $R \in \mathcal{P}_1$, $A \times B \subset R$. □

Lemma 7.4. *$\mathcal{P}_1$ and $\mathcal{P}_2$ consist of $\mu \times \nu$-measurable sets. For $R \in \mathcal{P}_1$,*

$$(\mu \times \nu)(R) = \sum_j \mu(A_j)\nu(B_j) = \rho(R).$$

Proof. We claim that $A \times B \in \mathcal{P}_0$ is $\mu \times \nu$-measurable. According to the definition, we need to prove

$$(\mu \times \nu)(T) \geq (\mu \times \nu)(T \cap A \times B) + (\mu \times \nu)(T \setminus A \times B), \quad \forall T \subset X \times Y.$$

To see this, for every $R \in \mathcal{P}_1$ with $T \subset R$, we have

$$\begin{aligned}&(\mu \times \nu)(T \cap A \times B) + (\mu \times \nu)(T \setminus A \times B) \\ &\leq (\mu \times \nu)(R \cap A \times B) + (\mu \times \nu)(R \setminus A \times B) \\ &\leq \rho(R \cap A \times B) + \rho(R \setminus A \times B) \quad (\text{By Lemma 7.3 and } R \cap A \times B,\ R \setminus A \times B \in \mathcal{P}_1) \\ &= \rho(R) \quad (\text{Since } \chi_{R\cap A\times B} + \chi_{R\setminus A\times B} = \chi_R)\end{aligned}$$

Taking infimum over all these R yields the desired result.

As all $\mu\times\nu$-measurable sets form a σ-algebra, $\mathcal{P}_0$, $\mathcal{P}_1$ and $\mathcal{P}_2$ consist of $\mu\times\nu$-measurable sets.

Write $R = \overset{\circ}{\bigcup} A_j \times B_j$, $A_j \times B_j \in \mathcal{P}_0$. We have

$$\begin{aligned}
&(\mu \times \nu)(R) \\
&= \sum_j (\mu \times \nu)(A_j \times B_j) \quad (R \text{ is } \mu \times \nu\text{-measurable}) \\
&= \sum_j \rho(A_j \times B_j) \quad (\text{Lemma 7.3}) \\
&= \rho(R). \quad (\text{Lemma 7.2})
\end{aligned}$$

□

Lemma 7.5. *Let $R \in \mathcal{P}_2$. Suppose that $R = \bigcap_{j=1}^{\infty} R_j$, $R_j \in \mathcal{P}_1$, and $\rho(R_1) < \infty$. Then $R \in \mathcal{F}$ and*

$$(\mu \times \nu)(R) = \rho(R).$$

Proof. For each $n \geq 1$, let $R^n = \bigcap_{j=1}^{n} R_j \in \mathcal{P}_1$ (check!), so $R^n \in \mathcal{F}$. That means, ν-a.e. y, $x \mapsto \chi_{R^n}(x, y)$ is μ-measurable. Using

$$\chi_R(x, y) = \lim_{n \to \infty} \chi_{R^n}(x, y), \quad \forall (x, y),$$

$x \mapsto \chi_R(x, y)$ is μ-measurable, for ν-a.e. y.

Next, $\chi_{R_1} - \chi_{R^n} \uparrow \chi_{R_1} - \chi_R$ as $n \to \infty$. By Lebsegue's monotone convergence theorem, for each fixed y,

$$\int (\chi_{R_1} - \chi_{R^n})\, d\mu \to \int (\chi_{R_1} - \chi_R)\, d\mu \text{ as } n \to \infty.$$

So,

$$y \mapsto \int_X (\chi_{R_1} - \chi_R)\, d\mu(x) \text{ is } \nu\text{-measurable.}$$

Using

$$\int \chi_R \, d\mu = \int \chi_{R_1} d\mu - \int (\chi_{R_1} - \chi_R)\, d\mu,$$

$$y \mapsto \int \chi_R(x, y)\, d\mu(x) \text{ is } \nu\text{-measurable.}$$

Note that we have used the fact that $\rho(R_1) < \infty$ and hence $\int \chi_{R_1} d\mu$ is finite and measurable for ν-a.e.y. We have shown that $\mathcal{P}_2 \subset \mathcal{F}$.

By repeatedly using Lebsegue's monotone convergence theorem,

$$\int \left(\int (\chi_{R_1} - \chi_{R^n})\, d\mu \right) d\nu \to \int \left(\int (\chi_{R_1} - \chi_R)\, d\mu \right) d\nu,$$

that is,

$$\rho(R_1) - \rho(R^n) \to \rho(R_1) - \rho(R),$$

or

$$\rho(R^n) \to \rho(R) \text{ as } n \to \infty.$$

On the other hand, by applying monotone convergence theorem to $\mu \times \nu$,

$$\int_{X\times Y} (\chi_{R_1} - \chi_{R^n})\, d(\mu \times \nu) \to \int_{X\times Y} (\chi_{R_1} - \chi_R)\, d(\mu \times \nu),$$

that is,

$$(\mu \times \nu)(R_1) - (\mu \times \nu)(R^n) \to (\mu \times \nu)(R_1) - (\mu \times \nu)(R),$$

or

$$(\mu \times \nu)(R^n) \to (\mu \times \nu)(R). \quad (\text{Use } (\mu \times \nu)(R_1) \le \rho(R_1) < \infty).$$

From $\rho(R^n) = (\mu \times \nu)(R^n)$, we get $\rho(R) = (\mu \times \nu)(R)$. □

Our last lemma is concerned with a regularity property of the product measure. It shows every set can be approximated from outside by a measurable set in any product measure.

Lemma 7.6. *For $E \subset X \times Y$, $\exists R \in \mathcal{P}_2$, $E \subset R$ such that*

$$(\mu \times \nu)(E) = (\mu \times \nu)(R).$$

Proof. If $(\mu \times \nu)(E) = \infty$, take $R = X \times Y$. If $(\mu \times \nu)(E) < \infty$, for each $n \ge 1$, there exists an $R^n \in \mathcal{P}_1$ such that

$$\begin{aligned}(\mu \times \nu)(E) + \frac{1}{n} &\ge \rho(R^n) \quad (\text{Lemma 7.3})\\ &\ge \rho(R)\end{aligned}$$

if we take $R = \bigcap_{n=1}^{\infty} R^n \in \mathcal{P}_2$. Letting $n \to \infty$, $\mu \times \nu(E) \ge \rho(R) = \mu \times \nu(R)$, done. □

We point out some properties of the product measure.

- The product space $(X \times Y) \times Z$ can be identified with $X \times (Y \times Z)$ and written as $X \times Y \times Z$. For measure μ, ν and λ on X, Y and Z respectively, the product measures $(\mu \times \nu) \times \lambda$ and $\mu \times (\nu \times \lambda)$ are well-defined on $X \times Y \times Z$. It is an exercise to show that these two measures coincides and thus we can write it as $\mu \times \nu \times \lambda$. This is the distributional law for the measure product. The same property extends to the product of finitely many product as well.
- The product measure of Borel measures is again Borel. The product measure of Radon measures is again Radon.

- We have $\mathcal{L}^{n-m} \times \mathcal{L}^m = \mathcal{L}^n$ for $0 < m < n$. In [EG] the n-dimensional Lebsegue measure is defined to be the nth times product $\mathcal{L}^1 \times \cdots \times \mathcal{L}^1$. It is again an exercise to show that this definition coincides with our definition in Chapter 3.

7.2 Fubini's Theorem

Theorem 7.7 (**Fubini's Theorem**). *Let μ and ν be σ-finite outer measures on X and Y respectively.*

(a) For any non-negative $\mu \times \nu$-measurable function f,

$$x \mapsto f(x,y) \text{ is } \mu\text{-measurable for } \nu\text{-a.e.}y, \quad \text{and}$$

$$y \mapsto \int_X f(x,y)\, d\mu(x) \text{ is } \nu\text{-measurable.}$$

Moreover,

$$\int_{X\times Y} f(x,y)\, d(\mu\times\nu)(x,y) = \int_Y \left(\int_X f(x,y)\, d\mu(x) \right) d\nu(y).$$

(b) (a) holds for $f \in L^1(\mu \times \nu)$.

Part (b) was first formulated by Tonelli and is also called *Tonelli's theorem.*

Before the proof of this theorem, it is worth to look at how close we are from this goal. Taking $f = \chi_R$ where R is measurable, the integral formula in Fubini's theorem becomes

$$(\mu \times \nu)(R) = \rho(R).$$

We have shown that this formula is valid for $R \in \mathcal{P}_2$ assuming $(\mu \times \nu)(X \times Y)$ is finite. We have inches to go, namely, to improve it to all measurable R. In this regard we need Lemma 7.6. In the following we take μ and ν to be finite.

Proof of Fubini's Theorem. (a) Let R be $\mu \times \nu$-measurable. We claim that $R \in \mathcal{F}$ first. Indeed, by Lemma 7.6, there exists an $R_1 \in \mathcal{P}_2$, $R \subset R_1$, such that $(\mu \times \nu)(R) = (\mu \times \nu)(R_1)$. As R is measurable,

$$(\mu \times \nu)(R_1 \setminus R) = (\mu \times \nu)(R_1) - (\mu \times \nu)(R) = 0.$$

Fix $R_2 \in \mathcal{P}_2$, $R_1 \setminus R \subset R_2$, such that $(\mu \times \nu)(R_2) = (\mu \times \nu)(R_1 \setminus R) = 0$. By Lemma 7.5,

$$\begin{aligned} 0 &= (\mu \times \nu)(R_2) \\ &= \rho(R_2) \\ &= \int \left(\int \chi_{R_2}\, d\mu \right) d\nu. \end{aligned}$$

Thus, for ν-a.e. y,

$$\int \chi_{R_2}\, d\mu = 0,$$

so

$$\int \chi_{R_1 \setminus R}\, d\mu = 0,$$

too. It means the set $\{x : (x, y) \in R_1 \setminus R\}$ is of μ-measure zero for ν-a.e. y. As every set of measure zero is measurable here, $\chi_{R_1 \setminus R}$ is μ-measurable for ν-a.e. y. So

$$\chi_R = \chi_{R_1} - \chi_{R_1 \setminus R}$$

is μ-measurable for all ν-a.e. y. Next,

$$\begin{aligned} \int \chi_R\, d\mu &= \int \chi_{R_1}\, d\mu - \int \chi_{R_1 \setminus R}\, d\mu \\ &= \int \chi_{R_1}\, d\mu. \end{aligned}$$

$R_1 \in \mathcal{P}_2 \subset \mathcal{F}$ means

$$y \mapsto \int \chi_{R_1}\, d\mu$$

is ν-measurable, so is

$$y \mapsto \int \chi_R\, d\mu.$$

We shown that $R \in \mathcal{F}$ for every $\mu \times \nu$-measurable R.

Next,

$$\begin{aligned} \int \chi_R\, d(\mu \times \nu) &= (\mu \times \nu)(R) \\ &= (\mu \times \nu)(R_1) \quad \text{(Lemma 7.6)} \\ &= \rho(R_1) \quad \text{(Lemma 7.5)} \\ &= \int \left(\int \chi_{R_1}\, d\mu \right) d\nu \\ &= \int \left(\int \chi_R\, d\mu \right) d\nu, \end{aligned}$$

or, if you like,

$$(\mu \times \nu)(R) = \rho(R)\ , \quad R \text{ measurable}\ .$$

Starting from this formula, we can pass f for simple functions and then non-negative measurable to obtain the integral formula in (a).

(b) When $f \in L^1(\mu \times \nu)$, apply (a) to f_+ and f_- separately to get the desired result. $\square$

The theorem is usually used in this way. Given a $\mu \times \nu$-measurable function f, we apply (a) to $|f|$ and see if

$$\int_{X\times Y} |f| \, d(\mu \times \nu) = \int_Y \Big(\int_X |f| \, d\mu \Big) \, d\nu.$$

If the right hand side is finite, then $|f| \in L^1(\mu \times \nu)$ and we can now use (b) to conclude that

$$\int_{X\times Y} f \, d(\mu \times \nu) = \int_Y \left(\int_X f \, d\mu \right) \, d\nu,$$

so the double integral can be evaluated by using an iterated integral.

We discuss an example to illustrate the role of σ-finiteness in Fubini's theorem.

Example 7.1. Let $\mathcal{L}^1$ and $\mathbf{c}$, the counting measure, be defined on $\mathbb{R}$ and consider the product measure $\mathcal{L}^1 \times \mathbf{c}$ on $\mathbb{R}^2 = \mathbb{R}^1 \times \mathbb{R}^1$. Consider $f = \chi_D$ where $D = \{(x, y) : x = y\}$. We claim that f is $\mathcal{L}^1 \times \mathbf{c}$-measurable, or, D is $\mathcal{L}^1 \times \mathbf{c}$-measurable. For $k \geq 1$, the sets $A_k = \bigcup_{j\in\mathbb{Z}} \big[j/k, (j+1)/k\big] \times \big[j/k, (j+1)/k\big]$, are countable unions of measurable rectangles so are $\mathcal{L}^1 \times \mathbf{c}$-measurable. As $D = \bigcap_{k=1}^{\infty} A_k$, D is also $\mathcal{L}^1 \times \mathbf{c}$-measurable.

Suppose Fubini's theorem holds for f. We have

$$\begin{aligned} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) \, d\mathcal{L}^1(x) \right) \, d\mathbf{c}(y) &= \int_{\mathbb{R}^2} f \, d(\mathcal{L}^1 \times \mathbf{c}) \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) \, d\mathbf{c}(y) \right) \, d\mathcal{L}^1(x). \end{aligned}$$

For a fixed y, $x \mapsto f(x, y) = \chi_{\{y\}}(x)$ is $\mathcal{L}^1$-measurable and

$$\int_{\mathbb{R}} f(x, y) \, d\mathcal{L}^1(x) = \int_{\mathbb{R}} \chi_{\{y\}}(x) \, d\mathcal{L}^1(x) = 0.$$

Thus

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) \, d\mathcal{L}^1(x) \right) \, d\mathbf{c}(y) = 0.$$

On the other hand, for any fixed x,

$$y \mapsto f(x, y) = \chi_{\{x\}}(y)$$

is $\mathbf{c}$-measurable and

$$\begin{aligned} \int_{\mathbb{R}} f(x, y) \, d\mathbf{c}(y) &= \int_{\mathbb{R}} \chi_{\{x\}}(y) \, d\mathbf{c}(y) \\ &= \mathbf{c}\,\{x\} \quad \text{(definition of integral)} \\ &= 1, \end{aligned}$$

which implies

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y)\, d\mathbf{c}(y) \right) d\mathcal{L}^1(x) = \infty,$$

contradiction holds!

In this example, all assumptions Theorem 7.7(a) are satisfied except $\mathbf{c}$ is not σ-finite.

7.3 Rademacher's Theorem

Recall that a function f defined on a set $E \subset \mathbb{R}^n$ is Lipschitz continuous in E if there exists some $M > 0$ such that

$$|f(x) - f(y)| \le M\,|x - y|\,, \quad \forall x, y \in E.$$

A function f defined in an open set $G \subset \mathbb{R}^n$ is called differentiable at $x \in G$ if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}$ such that

$$\lim_{|h|\to 0} \frac{|f(x+h) - f(x) - Lh|}{|h|} = 0.$$

It is well-known that when f is differentiable at x, the partial derivatives $\dfrac{\partial f}{\partial x_j}(x)$, $j = 1, \ldots, n$, must exist and $Lh = \displaystyle\sum_{j=1}^{n} h_j \frac{\partial f}{\partial x_j}(x)$.

Theorem 7.8 (**Rademacher's Theorem**). *Every locally Lipschitz continuous function in $\mathbb{R}^n$ must be differentiable almost everywhere.*

A function is locally Lipschitz continuous in some E if it is Lipschitz continuous in every compact subset of E. Since differentiability is a local property, in the following proof we may assume that f is Lipschitz continuous, that is,

$$|f(x) - f(y)| \le M\,|x - y|\,, \quad \forall x, y \in \mathbb{R}^n.$$

Proof. We claim: For each direction v, there exists a set $S_v \subset \mathbb{R}^n$, $\mathcal{L}^n(S_v) = 0$, such that

$$D_v f(x) = \lim_{t\to 0} \frac{f(x+tv) - f(x)}{t}, \quad \forall x \in \mathbb{R}^n \setminus S_v.$$

Indeed, let

$$\overline{D}_v f(x) = \overline{\lim_{t\to 0}} \frac{f(x+tv) - f(x)}{t}, \quad \text{and}$$

$$\underline{D}_v f(x) = \underline{\lim_{t\to 0}} \frac{f(x+tv) - f(x)}{t}.$$

Then $\underline{D}_v f$ and $\overline{D}_v f$ are measurable and bounded by M. Let $S_v = \left\{x \in \mathbb{R}^n : \underline{D}_v f < \overline{D}_v f\right\}$. For any line L parallel to v, we claim that $S_v \cap L$ has $\mathcal{L}^1$-measure zero for each such L. WLOG, let $v = e_1 = (1, 0, \ldots, 0)$ and $\varphi(t) = f(x + te_1) = f(x_1 + t, x')$. For fixed (x_1, x'), $t \mapsto (x_1 + t, x')$ is the line L, φ is Lipschitz continuous and hence absolutely continuous on $\mathbb{R}$, hence is differentiable a.e. t, that is,

$$\overline{D}_v f(x + te_1) = \underline{D}_v f(x + te_1), \quad \text{a.e. } t.$$

We conclude that for a.e. x on L, $D_v f(x) = \overline{D}_v f(x) = \underline{D}_v f(x)$. Using Fubini's theorem

$$\mathcal{L}^n(S_v) = \int_{\mathbb{R}^{n-1}} \left(\int_{\mathbb{R}} \chi_{S_v}(x, x') \, d\mathcal{L}^1(x) \right) d\mathcal{L}^{n-1}(x') = 0.$$

Next, we introduce notations $D_j f = D_{e_j} f, j = 1, \ldots, n$. Let

$$T_v = \left\{ x \in \mathbb{R}^n : D_v f(x),\ D_j f(x),\ j = 1, \ldots, n, \text{ exist and } D_v f(x) = \sum_{j=1}^n v_j D_j f(x). \right\}$$

Claim: $\mathcal{L}^n(\mathbb{R}^n \setminus T_v) = 0$. Indeed, let $\varphi \in C_c^1(\mathbb{R}^n)$. We have

$$\int \frac{f(x + tv) - f(x)}{t} \varphi(x) \, d\mathcal{L}^n(x) = -\int f(x) \frac{\varphi(x - tv) - \varphi(x)}{t} \, d\mathcal{L}^n(x).$$

By Lebsegue's dominated convergence theorem,

$$\int D_v f(x) \varphi(x) \, d\mathcal{L}^n(x) = -\int f(x) D_v \varphi(x) \, d\mathcal{L}^n(x), \quad \forall v,\ |v| = 1.$$

Taking $v = e_j$,

$$\int D_j f(x) \varphi(x) \, d\mathcal{L}^n(x) = -\int f(x) \frac{\partial \varphi}{\partial x_j}(x) \, d\mathcal{L}^n(x), \quad j = 1, \ldots, n.$$

We have, for given $v = (v_1, \ldots, v_n)$,

$$\begin{aligned}
\int D_v f(x) \varphi(x) \, d\mathcal{L}^n(x) &= -\int f(x) D_v \varphi(x) \, d\mathcal{L}^n(x) \\
&= -\int f(x) \sum_j v_j \frac{\partial \varphi}{\partial x_j} \, d\mathcal{L}^n(x) \\
&= -\sum_j v_j \int f(x) \frac{\partial \varphi}{\partial x_j} \, d\mathcal{L}^n(x) \\
&= \sum_j v_j \int D_j f(x) \varphi(x) \, d\mathcal{L}^n(x) \\
&= \int \left(\sum_j v_j D_j f(x) \right) \varphi(x) \, d\mathcal{L}^n(x), \quad \forall \varphi \in C_c^1(\mathbb{R}^n),
\end{aligned}$$

which implies that $D_v f(x) = \sum_{j=1}^{n} v_j D_j f(x)$ almost everywhere in the set $\mathbb{R}^n \setminus (S_v \cup S_{e_1} \cup \cdots \cup S_{e_n})$. Hence $\mathcal{L}^n(\mathbb{R}^n \setminus T_v) = 0$ for every direction v.

For v, $|v| = 1$, $x \in \mathbb{R}^n$, $t \neq 0 \in \mathbb{R}$, set

$$Q(x, v, t) = \frac{f(x + tv) - f(x)}{t} - \sum_{j=1}^{n} v_j D_j f(x).$$

We are going to show that for every $\varepsilon > 0$, there corresponds some $\delta > 0$, such that

$$|Q(x, v, t)| < \varepsilon, \quad \text{a.e. } x, \ \forall v, \ |v| = 1, \ |t| < \delta. \tag{7.1}$$

We first fix a countable, dense set $\{v_k\}_1^\infty$ in S^{n-1}. Let

$$T = \bigcap_{k=1}^{\infty} T_{v_k}.$$

Then $\mathcal{L}^n(\mathbb{R}^n \setminus T) \leq \sum_{k=1}^{\infty} \mathcal{L}^n(\mathbb{R}^n \setminus T_{v_k}) = 0$. We claim (7.1) holds for all $x \in T$.

Given $\varepsilon > 0$, we can find finitely many points $v^1, \ldots, v^m$ on S^{n-1} such that for each v, there exists one of these points, say, v^k, satisfying $\left|v - v^k\right| < \varepsilon / 2M(1 + \sqrt{n})$. We have

$$\begin{aligned}
&\left|Q(x, v, t) - Q(x, v^k, t)\right| \\
&= \left| \frac{f(x + tv) - f(x + tv^k)}{t} - \sum_{j=1}^{n} (v_j - v_j^k) D_j f(x) \right| \\
&\leq M \left|v - v^k\right| + \sqrt{n} M \left|v - v^k\right| \\
&= M(1 + \sqrt{n}) \left|v - v^k\right| \\
&< \frac{\varepsilon}{2} \ .
\end{aligned} \tag{7.2}$$

For $x \in T$, since $Q(x, v^k, t) \to 0$ as $t \to 0$ for $k = 1, \ldots, m$, we can find a δ such that

$$\left|Q(x, v^k, t)\right| < \frac{\varepsilon}{2}, \quad \forall t, \ |t| < \delta. \tag{7.3}$$

Putting (7.2) and (7.3) together, for every $x \in T$ and $\varepsilon > 0$, there is a δ such that

$$\begin{aligned}
|Q(x, v, t)| &\leq \left|Q(x, v, t) - Q(x, v^k, t)\right| + \left|Q(x, v^k, t)\right| \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad \forall t, \ |t| < \delta, \text{ and } \forall v, |v| = 1.
\end{aligned}$$

Finally, for $x \in T$, we set $v = (y-x)/|y-x|$ for $y \neq x$, and $t = |y-x|$. By (8.1), for every ε, there is a δ such that

$$\begin{aligned} \varepsilon &> \Big|\frac{f(y)-f(x)}{|y-x|} - \sum_{j=1}^{n} \frac{(y_j-x_j)}{|y-x|} D_j f(x)\Big| \\ &= \frac{1}{|y-x|}\Big|f(u) - f(x) - \sum_{j=1}^{n}(y_j-x_j)D_j f(x)\Big|, \end{aligned}$$

for all $y, |y-x| < \delta$. The proof of Rademarcher's theorem is completed. □

7.4 The Layer Cake Representation

We begin with a lemma.

Lemma 7.9. *Let f be a non-negative, Lebsegue measurable function in $\mathbb{R}^n$. The set*

$$A = \{(x,t) \in \mathbb{R}^n \times [0,\infty) : f(x) \geq t \geq 0\},$$

is $\mathcal{L}^{n+1}$-measurable.

Proof. When $f = \chi_E$ for some measurable E,

$$A = E \times [0,1] \cup (R^n \setminus E) \times \{0\}$$

is measurable. Next, when $f = \sum_j \alpha_j \chi_{E_j}$ where E_j's are disjoint and measurable, and $\alpha_j > 0$,

$$A = \bigcup_j E_j \times [0,\alpha_j] \cup (\mathbb{R}^n \setminus \bigcup_j E_j) \times \{0\}$$

is measurable. In general, the lemma follows from using simple functions to approximate f. □

It is easy to deduce that the set

$$B = \{(x,t) \in \mathbb{R}^n \times [0,\infty) : f(x) > t > 0\}$$

is also measurable.

Let f be a non-negative measurable function in some measure space $(X, \mathcal{M}, \mu)$. We use $\{f > t\}$ to denote the set $\{x \in X : f(x) > t\}$. We are going to establish a general formula, which includes

$$\int_X f d\mu = \int_0^\infty \mu\{f > t\} d\mathcal{L}^1(t),$$

as a special case. Imagining the set $\{(x,t) : 0 \le t \le f(x)\}$ as a cake, the set $\{f > t\}$ is its layer at height t. This formula asserts that the integral of f can be computed by an integration over its layers. (Incidentally, since $\mu\{f > t\}$ is an increasing real-valued function provided it is finite and this is true, for instance, when f is integrable. As every increasing function is Riemann integrable, the integral on the right hand side of this formula is in fact an improper Riemann integral. Interestingly it shows that the abstract integral of f with respect to some μ can be defined by an Riemann integral in terms of its cake layers.)

To formulate a more general result, let φ be an increasing function in $[0,\infty]$ satisfying

(i) $\varphi(t) \to \varphi(\infty) \le \infty$ as $t \to \infty$,

(ii) $\varphi(0) = 0$,

(iii) φ is absolutely continuous on $[0,a]$, $\forall a \in (0,\infty)$.

Proposition 7.10. *Let μ be an outer measure on X and φ be given as above. For any non-negative μ-measurable function f in X,*

$$\int_X \varphi \circ f \, d\mu = \int_0^\infty \mu\{f > t\}\, \varphi'(t)\, d\mathcal{L}^1(t).$$

Note that $\varphi \circ f$ is μ-measurable and $\mu\{f > t\}\varphi'(t)$ is increasing and hence Lebsegue measurable. As a result, both integrals are well-defined.

Proof. Letting

$$A = \{(x,t) \in X \times [0,\infty) : f(x) > t\}$$

be measurable by Lemma 7.9, observe that

$$\mu\{f > t\} = \int_X \chi_A(x,t)\, d\mu(x)\ .$$

We have

$$\begin{aligned}
\int_X \varphi(f(x))\ d\mu &= \int_X \int_0^{f(x)} \varphi'(t)\, d\mathcal{L}^1(t) d\mu(x). \\
&= \int_X \int_0^\infty \chi_A(x,t) \varphi'(t)\, d\mathcal{L}^1(t) d\mu(x) \\
&= \int_0^\infty \int_X \chi_A(x,t)\, d\mu(x) \varphi'(t) d\mathcal{L}^1(t) \quad \text{(Fubini's theorem)} \\
&= \int_X \mu\{f > t\} \varphi'(t)\ d\mathcal{L}^1(t).
\end{aligned}$$

Note that in the first step, when $f(x) < \infty$,

$$\int_0^{f(x)} \varphi'(t)\, d\mathcal{L}^1(t) = \varphi(f(x)) - \varphi(0) = \varphi(f(x)) \ ,$$

by the fundamental theorem of calculus. If $f(x) = \infty$,

$$\begin{aligned} \int_0^{\infty} \varphi'(t)\, d\mathcal{L}^1(t) &= \lim_{a\to\infty} \int_0^a \varphi'(t)\, d\mathcal{L}^1(t) \quad \text{(monotone convergence theorem)} \\ &= \varphi(\infty) \quad (\text{ by (i) }) \\ &= \varphi(f(x)). \end{aligned}$$

Clearly, the proposition follows. □

Taking $\varphi(z) = z$, then $\varphi'(z) = 1$ and we cover the layer cake representation before.

We give an application of this formula to maximal functions.

Recall that for an $\mathcal{L}^n$-measurable f, its maximal function is defined to be

$$(Mf)(x) = \sup_{B\in\mathcal{B}_x} \frac{1}{\mathcal{L}^n(B)} \int_B |f| \, d\mathcal{L}^n,$$

where $\mathcal{B}_x$ is the collection of all closed balls containing x. (In fact, since the Lebesgue measure of ∂B is 0, you may take B to be an open ball.) We know that

- $Mf = M|f|$.
- Mf is $\mathcal{L}^n$-measurable.
- The weak L^1-estimate

$$\mu\{Mf > t\} \le \frac{C}{t} \|f\|_{L^1} \ , \quad \forall t > 0$$

where C is a dimensional constant, holds for $f \in L^1(\mathbb{R}^n)$.

Proposition 7.11. *Let $f \in L^p(\mathbb{R}^n)$, $p > 1$. We have*

$$\|Mf\|_{L^p} \le C \|f\|_{L^p} \ , \quad \forall f \in L^p(\mathbb{R}^n),$$

where C depends only on n and p.

Proof. We apply the formula in Proposition 7.10 by taking $\varphi(z) = z^p, p \in (1, \infty)$,

$$\int_{\mathbb{R}^n} f^p \, d\mathcal{L}^n = \int_0^{\infty} \mu\{f > t\}\, pt^{p-1}\, d\mathcal{L}^1(t)$$

where $f \geq 0$. We replace f by an L^1-function g as follows. First, for fixed $\alpha \in (0,1)$ and $t > 0$, set

$$g(x) = \begin{cases} f(x), & f(x) > \alpha t \\ 0, & f(x) \leq \alpha t, \end{cases}$$

and $h = f - g$. Then $g \in L^1(\mathbb{R}^n)$ and

$$\begin{aligned} \int g \, d\mathcal{L}^n &\leq \int_{\{f > \alpha t\}} f \, d\mathcal{L}^n \\ &= \int_{\{f > \alpha t\}} \frac{f^p}{f^{p-1}} \, d\mathcal{L}^n \\ &\leq \frac{1}{(\alpha t)^{p-1}} \int f^p \, d\mathcal{L}^n \\ &< \infty. \end{aligned}$$

Moreover, from $f = g + h$,

$$\begin{aligned} Mf &\leq Mg + Mh \\ &\leq Mg + \alpha t \end{aligned}$$

implies that if $Mf > t$ then $Mg > (1-\alpha)t$, so

$$\{Mf > t\} \subset \{Mg > (1-\alpha)t\}.$$

Using the weak L^1-estimate for $g \in L^1(\mathbb{R}^n)$

$$\begin{aligned} \mathcal{L}^n \{Mf > t\} &\leq \mathcal{L}^n \{Mg > (1-\alpha)t\} \\ &\leq \frac{C}{(1-\alpha)t} \|g\|_{L^1} \\ &= \frac{C}{(1-\alpha)t} \int_{A_{\alpha t}} f \mathcal{L}^n, \end{aligned}$$

where $A_t = \{f > \alpha t\}$. By Proposition 7.10,

$$\begin{aligned} \int (Mf)^p \, d\mathcal{L}^n &= \int_0^\infty \mathcal{L}^n \{Mf > t\} \, p t^{p-1} \, d\mathcal{L}^1(t) \\ &\leq \int_0^\infty \{Mg > (1-\alpha)t\} \, p t^{p-1} \, d\mathcal{L}^1(t) \\ &\leq \int_0^\infty \frac{Cp}{(1-\alpha)t} \int_{A_{\alpha t}} f \, d\mathcal{L}^n \, t^{p-1} \, d\mathcal{L}^1(t). \end{aligned}$$

Letting $A = \{(x,t) : f(x) > \alpha t\}$,

$$\begin{aligned}\int (Mf)^p \, d\mathcal{L}^n &\leq \frac{Cp}{1-\alpha} \int_0^\infty \int_{A_{\alpha t}} f(x) \, d\mathcal{L}^n(x) \, t^{p-2} \, d\mathcal{L}^1(t) \\ &= \frac{Cp}{1-\alpha} \int_0^\infty \int \chi_A(x,t) f(x) \, d\mathcal{L}^n(x) \, t^{p-2} \, d\mathcal{L}^1(t) \\ &= \frac{Cp}{1-\alpha} \int \int_0^\infty \chi_A(x,t) t^{p-2} \, d\mathcal{L}^1(t) \, f(x) \, d\mathcal{L}^n(x) \\ &= \frac{Cp}{1-\alpha} \int \int_0^{\frac{f(x)}{\alpha}} t^{p-2} \, d\mathcal{L}^1(t) \, f(x) \, d\mathcal{L}^n \\ &= \frac{Cp}{(1-\alpha)(p-1)\alpha^{p-1}} \int f^p(x) \, d\mathcal{L}^n.\end{aligned}$$

We conclude that

$$\|Mf\|_{L^p} \leq \left[\frac{Cp}{(1-\alpha)(p-1)\alpha^{p-1}} \right]^{\frac{1}{p}} \|f\|_{L^p}, \quad \forall \alpha \in (0,1).$$

□

In fact, by minimizing α, we get an explicit constant

$$C = \min_{\alpha \in (0,1)} \left[\frac{Cp}{(1-\alpha)(p-1)\alpha^{p-1}} \right]^{\frac{1}{p}} = (Cepq)^{\frac{1}{p}},$$

where q is conjugate to p, see [R1] for details.

7.5 Convolution

Convolution is a product between two functions. It appears in two contexts. First, it is well-known that the Fourier transform of the pointwise product of two functions equals to the pointwise product of the Fourier transform of these functions. It plays a fundamental role in harmonic analysis. Second, as it will be explained shortly, convolution can be used to construct various approximation of identity, so it is quite useful in approximation.

The definition of convolution involves an integration of the product of two integrable functions. It is not clear at all why this product makes sense. Fubini's theorem is in an essential use to justify the well-definiteness of the convolution.

Proposition 7.12. *Let $f, g \in L^1(\mathbb{R}^n)$. Then for a.e. x,*

$$\int_{\mathbb{R}^n} |f(x-y)g(y)| \, d\mathcal{L}^n(y) < \infty.$$

For those x, define the convolution of f and g by

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x-y)g(y)\, d\mathcal{L}^n(y).$$

Then $f * g \in L^1(\mathbb{R}^n)$ and

$$\|f * g\|_{L^1} \leq \|f\|_{L^1} \|g\|_{L^1}.$$

It will be understood that the integral is over $\mathbb{R}^n$ when the domain of integration is not specified in this section.

Proof. First, we claim that $(x, y) \mapsto f(x, y)g(y)$ is measurable in $\mathbb{R}^{2n}$. Recall that every measurable function is equal a.e. to a Borel function. We replace f and g by such Borel functions $\widetilde{f}$ and $\widetilde{g}$. Consider

$$(x, y) \mapsto \widetilde{f}(x-y)\widetilde{g}(y).$$

Let $\varphi(x, y) = x - y$. The map $(x, y) \mapsto \widetilde{f}(x-y) = \widetilde{f} \circ \varphi(x, y)$. As $\widetilde{f}$ is Borel and φ is continuous (hence measurable), $\widetilde{f}(x-y)$ is measurable. So is $\widetilde{f}(x-y)\widetilde{g}(y)$. As $f(x-y)g(y)$ differs from $\widetilde{f}(x-y)\widetilde{g}(y)$ on a set a of measure zero, it is also measurable.

Next, applying the first part of Fubini's theorem to the measurable function $|f(x-y)g(y)|$, we have

$$\begin{aligned}\int_{\mathbb{R}^{2n}} |f(x-y)g(y)|\, d\mathcal{L}^n(x)d\mathcal{L}^n(y) &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(x-y)|\, |g(y)|\, d\mathcal{L}^n(x) \right) d\mathcal{L}^n(y) \\ &= \|f\|_{L^1} \|g\|_{L^1} \\ &< \infty.\end{aligned}$$

We conclude that $f(x-y)g(y) \in L^1(\mathbb{R}^{2n})$. Moreover, exchanging the order of integration, we have

$$\int \left(\int |f(x-y)|\, |g(y)|\, d\mathcal{L}^n(y) \right) d\mathcal{L}^n(x) = \|f\|_{L^1} \|g\|_{L^1} < \infty,$$

$$\int |f(x-y)|\, |g(y)|\, d\mathcal{L}^n(y) < \infty$$

for a.e. x. Therefore, for a.e. x, the convolution $f * g$ is well-defined and finite. Moreover,

$$\begin{aligned}\int |f * g(x)|\, d\mathcal{L}^n(x) &= \int \left| \int f(x-y)g(y)\, d\mathcal{L}^n(y) \right| d\mathcal{L}^n(x) \\ &\leq \int \left(\int |f(x-y)|\, |g(y)|\, d\mathcal{L}^n(y) \right) d\mathcal{L}^n(x) \\ &= \int \left(\int |f(x-y)|\, |g(y)|\, d\mathcal{L}^n(x) \right) d\mathcal{L}^n(y) \\ &= \|f\|_{L^1} \|g\|_{L^1}.\end{aligned}$$

□

By the same reason, $g * f$ exists and

$$\begin{aligned}(g * f)(x) &= \int g(x-y)f(y)\,d\mathcal{L}^n(y)\\ &= \int g(y)f(x-y)\,d\mathcal{L}^n(y) \quad \text{(change of variables)}\\ &= (f * g)(x),\end{aligned}$$

whenever $f * g$ or $g * f$ is well-defined at x.

For f, g and h in $\mathcal{L}^1(\mathbb{R}^n)$, one can verify that

$$(f * g) * h(x) = f * (g * h)(x), \quad \text{for a.e. } x.$$

There are several Young's inequality. The following one is concerned with convolution of functions.

Proposition 7.13 (Young's Inequality). *Let $f \in L^1(\mathbb{R}^n)$ and $g \in L^p(\mathbb{R}^n)$, $p \in (1, \infty)$. Then a.e. x,*

$$\int |f(x-y)g(y)|\ d\mathcal{L}^n(y) < \infty,$$

and $f * g \in L^p(\mathbb{R}^n)$ with

$$\|f * g\|_{L^p} \le \|f\|_{L^1} \|g\|_{L^p}.$$

Proof. Assume $g \in L^1(\mathbb{R}^n) \cap L^p(\mathbb{R}^n)$ first. For $\varphi \in C_c(\mathbb{R}^n)$,

$$\begin{aligned}\left|\int \varphi(x) \left(\int |f(x-y)g(y)|\ d\mathcal{L}^n(y)\right) d\mathcal{L}^n(x)\right| &= \left|\int \varphi(x) \left(\int |f(y)g(x-y)|\ d\mathcal{L}^n(y)\right) d\mathcal{L}^n(x)\right|\\ &= \left|\int \left(\int \varphi(x)\,|f(y)|\,|g(x-y)|\ d\mathcal{L}^n(x)\right) d\mathcal{L}^n(y)\right|\\ &\le \int |f(y)|\,\|\varphi\|_{L^q}\,\|g\|_{L^p}\ d\mathcal{L}^n(y) \quad (\frac{1}{p}+\frac{1}{q}=1)\\ &= \|f\|_{L^1}\,\|g\|_{L^p}\,\|\varphi\|_{L^q}.\end{aligned}$$

Using the density of C_c-functions in $L^q(\mathbb{R}^n)$ and L^p-L^q duality,

$$\begin{aligned}\|f * g\|_{L^p} &= \sup\left\{\left|\int \varphi(x)(f * g)(x)\,d\mathcal{L}^n(x)\right| : \|\varphi\|_{L^q} \le 1\right\}\\ &\le \|f\|_{L^1}\,\|g\|_{L^p}.\end{aligned}$$

We have proved the proposition for $g \in L^1(\mathbb{R}^n) \cap L^p(\mathbb{R}^n)$. For $g \in L^p(\mathbb{R}^n)$, letting $g_k = \chi_{B_k(0)}\,|g|$, then $g_k \in L^1(\mathbb{R}^n)$, $g_k \uparrow |g|$. By Lebsegue's monotone convergence theorem,

$$\begin{aligned}\int \left(\int |f(x-y)|\,|g(y)|\ d\mathcal{L}^n(y)\right)^p d\mathcal{L}^n(x) &= \int \left(\lim_{k\to\infty} \int |f(x-y)|\,|g_k(y)|\ d\mathcal{L}^n(y)\right)^p d\mathcal{L}^n(x)\\ &\le \|f\|_{L^1}^p\,\|g\|_{L^p}^p\end{aligned}$$

and the conclusion follows. □

Convolution can be used to construct approximation kernel. As an illustration we prove

Theorem 7.14 (**Weierstrass Approximation Theorem**). *Let $f \in C_c(\mathbb{R}^n)$. For every $\varepsilon > 0$, there exists a polynomial p such that*

$$|f(x) - p(x)| < \varepsilon,$$

for all x in the support of f.

Proof. WLOG assume $f \in C_c(B_1(0))$. Let

$$Q_k(x) = c_k(1 - |x|^2)^k, \quad k \geq 1.$$

where the constant c_k is chosen so that

$$\int_{B_1(0)} Q_k(x)\, d\mathcal{L}^n(x) = 1,$$

and set $Q_k = 0$ outside $B_1(0)$. Define

$$p_k(x) = \int f(y) Q_k(x - y)\, d\mathcal{L}^n(y).$$

Note that

$$\begin{aligned} p_k(x) &= \int_{B_1(0)} f(y) Q_k(x - y)\, d\mathcal{L}^n(y) \\ &= c_k \int_{B_1(0)} f(y)(1 - |x - y)|^2)^k\, d\mathcal{L}^n(y) \end{aligned}$$

is a polynomial of degree $2k$. To show that p_k well-approximates f we need the following estimate on c_k,

$$c_k \leq C k^{\frac{n}{2}}, \quad \forall k \geq 1.$$

Indeed,

$$\begin{aligned} \frac{1}{c_k} &= \int_{B_1(0)} (1 - |x|^2)^k\, d\mathcal{L}^n(x) \\ &= \int_{S^{n-1}} \int_0^1 (1 - r^2)^k r^{n-1}\, dr d\theta \\ &= \mathcal{H}^{n-1}(S^{n-1}) \int_0^1 (1 - r^2)^k r^{n-1}\, dr. \end{aligned}$$

By the elementary inequality

$$(1 - r^2)^k \geq 1 - k r^2, \quad r \in (0, 1),$$

we have

$$\begin{aligned}\frac{1}{c_k} &\geq \mathcal{H}^{n-1}(S^{n-1}) \int_0^{\frac{1}{\sqrt{k}}} (1-kr^2) r^{n-1}\, dr \\ &= \mathcal{H}^{n-1}(S^{n-1}) \frac{2}{n(n+2)} \left(\frac{1}{\sqrt{k}}\right)^n,\end{aligned}$$

which implies the desired estimate. Using this, for $x \in B_1(0) \setminus B_\delta(0)$, $\delta \in (0,1)$,

$$\begin{aligned}Q_k(x) &\leq c_k(1-\delta^2)^k \\ &\leq Ck^{\frac{n}{2}}(1-\delta^2)^k \to 0 \text{ as } k \to \infty.\end{aligned}$$

Now,

$$\begin{aligned}p_k(x) &= \int f(y) Q_k(x-y)\, d\mathcal{L}^n(y) \\ &= \int f(x-y) Q_k(y)\, d\mathcal{L}^n(y) \\ &= \int_{B_1(0)} f(x-y) Q_k(y)\, d\mathcal{L}^n(y).\end{aligned}$$

We have

$$\begin{aligned}|p_k(x) - f(x)| &= \left| \int_{B_1(0)} (f(x-y) - f(x)) Q_k(y)\, d\mathcal{L}^n(y) \right| \\ &\leq \int_{B_\delta(0)} |f(x-y) - f(x)|\, Q_k(y)\, d\mathcal{L}^n(y) \\ &\quad + \int_{B_1(0)\setminus B_\delta(0)} |f(x-y) - f(x)|\, Q_k(y)\, d\mathcal{L}^n(y).\end{aligned}$$

As f is uniformly continuous in $\mathbb{R}^n$, for $\varepsilon > 0$, we can find some δ such that

$$|f(x+h) - f(x)| < \varepsilon, \quad \forall x \in \mathbb{R}^n, \text{and } h,\ |h| < \delta.$$

Therefore,

$$\int_{B_\delta(0)} |f(x-y) - f(x)|\, Q_k(y)\, d\mathcal{L}^n(y) \leq \varepsilon.$$

On the other hand,

$$\begin{aligned}\int_{B_1(0)\setminus B_\delta(0)} |f(x-y) - f(x)|\, Q_k(y)\, d\mathcal{L}^n(y) &\leq 2 \sup |f|\, |B_1(0) \setminus B_\delta(0)|\, Ck^{\frac{n}{2}}(1-\delta^2)^k \\ &< \varepsilon,\end{aligned}$$

for all sufficiently large k, $k \geq k_0$, say. Putting things together,

$$|p_k(x) - f(x)| \leq \varepsilon + \varepsilon = 2\varepsilon,$$

done. □

Corollary 7.15. *Let f be a continuous function on a compact set K in $\mathbb{R}^n$. For $\varepsilon > 0$, there exists a polynomial p such that $f(x) - p(x) < \varepsilon$, $\forall x \in K$.*

Proof. Extend f to be a function in $C_c(B_R(0))$ where R is large such that $K \subset\subset B_R(0)$. Surely there are many ways to do this. By Theorem 7.14, we can find a polynomial p such that $|f(x) - p(x)| < \varepsilon$, $\forall x \in \overline{B_R(0)}$, and this p does the job. □

Comments on Chapter 7. Our discussion on the product measure and Fubini's theorem is taken from [EG]. A treatment on the construction of the product measure from two measure spaces rather than two outer measures can be found in [R1]. Rademarcher's theorem (1919) whose proof depends crucially on Fubini's theorem is the foundation for geometric measure theory. The layer cake representation and convolution of functions are taken from [R1], except the treatment on Weierstrass approximation theorem is modified from "Principles of Mathematical Analysis" of the same author. By the way, Weierstrass approximation theorem should have been covered in some undergraduate analysis course. Indeed, it is in my lecture notes for MATH3060. However, in the past it was never taught due to some mysterious reason. In view of this, I include it here as an application of convolution. Another standard application of Fubini's theorem is the inversion formula for Fourier transform.